C*-Algebras and K-Theory

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Contents

A C^*-algebra is, by the Gelfand-Naimark theorem, nothing other than a norm-closed $*$-algebra of bounded operators on a Hilbert space; and by the commutative case of that theorem, a commutative C^*-algebra is nothing other than the algebra $C_0(X)$ of continuous functions vanishing at infinity on a locally compact Hausdorff space X. These two facts frame the whole subject. The second makes "commutative C^*-algebra" and "locally compact space" synonymous, so that a general, noncommutative C^*-algebra is to be read as a *noncommutative space*; the first guarantees the dictionary is concrete, every C^*-algebra being realized faithfully by operators. The theory of C^*-algebras is the geometry of noncommutative spaces, written in the language of operators.

The organizing question of this book is how the invariants of topology extend to the noncommutative world, and the cleanest answer is K-theory. To a C^*-algebra A one attaches groups $K_0(A)$ and $K_1(A)$, built from the projections and unitaries of its matrix algebras; they are 2-periodic (a six-term exact sequence and Bott periodicity, exactly as for topological K-theory) and they package operator-theoretic data, most vividly the Fredholm index, as in the Toeplitz extension. The Serre-Swan theorem completes the identification: for X compact, projections over $C(X)$ are vector bundles over X, so $K_0(C(X)) = K^0(X)$, and the operator theory recovers the topological K-theory of [10] exactly.

This book develops C^*-algebras from their axioms through Gelfand-Naimark duality, states and the GNS construction, and representations, and then through the K-theory just described. It is the operator-algebraic member of the K-theory family, alongside [10] (topological) and [2] (algebraic). Beyond that role it is meant as a foundation: the index theory of elliptic operators, noncommutative geometry, and the harmonic analysis of group C^*-algebras all begin here.

We assume the functional analysis of [9] (Banach and Hilbert spaces, the spectral theorem for bounded operators) and the measure theory of [11]; no prior operator-algebra theory is presumed. A continuous function on a compact space and the Toeplitz algebra serve as the two examples to keep in mind throughout, the first commutative and geometric, the second noncommutative and analytic.

Part I

C^*-Algebra Theory

1

Banach Algebras and C^*-Algebras

The book's guiding dictionary, in which commutative C^*-algebras are function algebras and every C^*-algebra is an algebra of operators, presupposes that a C^*-algebra is first of all a *Banach algebra*: a complete normed algebra in which the norm controls multiplication. This chapter develops the algebraic and spectral foundation on which everything later rests. The spectral theory of Banach algebras, namely the spectrum of an element, its nonemptiness and compactness, and the spectral radius formula, is what makes the norm computable from the algebra alone; it already suffices to prove that the only Banach division algebra over $\mathbb{C}$ is $\mathbb{C}$ itself.

The passage from Banach algebras to C^*-algebras is a single axiom, the C^*-identity $\|a^*a\| = \|a\|^2$. Slight as it looks, it rigidifies the theory completely: it forces every $*$-homomorphism to be contractive and every injective one to be isometric, so the norm of a C^*-algebra is fixed by its algebra structure and cannot be chosen. The chapter closes with the two devices that let the theory work without a unit, approximate units and the unitization $\tilde{A}$; these recur throughout, since the function algebras $C_0(X)$ on non-compact spaces are the motivating non-unital examples. The Banach- and Hilbert-space theory these arguments rest on, including the spectral theorem for bounded operators, is taken from [9].

1.1 Banach Algebras and Spectral Theory

A C^*-algebra is, before anything else, a Banach algebra, and the spectral theory this book uses is developed for arbitrary Banach algebras in [9]. This section recalls the facts we need and cites their proofs there. Each uses only the Banach-space structure and complex analysis, never an involution, which is precisely why it is prior to the C^*-theory that begins in the next section.

Definition 1.1.1: Banach Algebra

A *Banach algebra* is an associative algebra A over $\mathbb{C}$ with a complete submultiplicative norm, $\|ab\| \le \|a\|\,\|b\|$. It is *unital* if it has a multiplicative identity 1 satisfying $\|1\| = 1$.

Every C^*-algebra is a Banach algebra, but the converse fails widely, since the definition asks for no adjoint: the disk algebra of functions holomorphic on the open unit disk and continuous on its closure, and the convolution algebra $L^1(\mathbb{R})$, are Banach algebras with no natural C^*-structure. When no unit is present we pass silently to the unitization $\tilde{A}$ of proposition 1.4.5, in which spectra are computed.

Definition 1.1.2: Spectrum and Spectral Radius

For a in a unital Banach algebra A, the *spectrum* is

$$\sigma(a) = \{\lambda \in \mathbb{C} : a - \lambda 1 \text{ is not invertible}\},$$

the *resolvent set* is $\rho(a) = \mathbb{C} \setminus \sigma(a)$, the *resolvent* is $\lambda \mapsto (a - \lambda 1)^{-1}$ on $\rho(a)$, and the *spectral radius* is $r(a) = \sup\{|\lambda| : \lambda \in \sigma(a)\}$.

Theorem 1.1.3: Spectrum and the Spectral Radius Formula

Let a be an element of a unital Banach algebra over $\mathbb{C}$. Then $\sigma(a)$ is a nonempty compact subset of $\{\lambda : |\lambda| \le \|a\|\}$, and

$$r(a) = \lim_{n\to\infty} \|a^n\|^{1/n}$$

***Proof*:**
Both statements are proved in [9, Spectrum Is Compact]: nonemptiness by applying Liouville's theorem to the resolvent, compactness because $\rho(a)$ is open and $\sigma(a) \subseteq \{|\lambda| \le \|a\|\}$, and the spectral radius formula by the uniform boundedness principle applied to the Laurent expansion of the resolvent. No involution enters any of these arguments.

That the spectrum is nonempty holds only over $\mathbb{C}$. It forces the division-algebra rigidity below, and, through the continuous functional calculus developed next, it is what lets a C^*-algebra be studied one commutative piece at a time.

Theorem 1.1.4: Gelfand-Mazur

A unital Banach algebra over $\mathbb{C}$ that is a division algebra equals $\mathbb{C} \cdot 1$.

***Proof*:**
Given a, choose $\lambda \in \sigma(a)$ by theorem 1.1.3; then $a - \lambda 1$ is not invertible, so in a division algebra $a - \lambda 1 = 0$, that is $a = \lambda 1$. See [9, Gelfand-Mazur].

1.2 C^*-Algebras and the C^*-Identity

A Banach algebra carries a norm compatible with its multiplication, but nothing yet ties that norm to any notion of adjoint. The passage to a C^*-algebra adds two ingredients: an involution $a \mapsto a^*$ modelled on the Hilbert-space adjoint, and a single normalizing axiom, the C^*-identity, welding the involution to the norm. It forces the involution to be isometric, pins the norm down uniquely from the algebraic structure, and (as later sections show) makes every $*$-homomorphism automatically contractive. This section states the axioms, extracts their first consequence, and checks them against the concrete algebras that recur throughout the book.

Definition 1.2.1: C^*-Algebra

A *C^*-algebra* is a Banach algebra A over $\mathbb{C}$ equipped with a map $a \mapsto a^*$, called an *involution*, satisfying for all $a, b \in A$ and $\lambda \in \mathbb{C}$

1. $(a + b)^* = a^* + b^*$ and $(\lambda a)^* = \overline{\lambda}\, a^*$ (conjugate-linearity),
2. $(ab)^* = b^* a^*$ (anti-multiplicativity),
3. $(a^*)^* = a$ (involutivity),

together with the *C^*-identity*

$$\|a^* a\| = \|a\|^2 \qquad \text{for all } a \in A$$

An element a with $a = a^*$ is *self-adjoint*. If A has a multiplicative unit 1 (necessarily $1^* = 1$), the algebra is *unital*; the general theory admits non-unital algebras, and the unit is adjoined when needed. A C^*-algebra is *separable* if it is separable as a metric space, that is, if it has a countable dense subset.

The three algebraic axioms say only that A is a Banach $*$-algebra: an involution making $a \mapsto a^*$ a conjugate-linear anti-automorphism of period two. Many Banach $*$-algebras exist, and most are analytically unremarkable. The C^*-identity is the entire content of the definition. It is a curious constraint at first sight, coupling the norm of the product a^*a to the square of the norm of a, and its power comes precisely from being so much stronger than the submultiplicative inequality $\|a^*a\| \leq \|a^*\|\,\|a\|$ that a Banach algebra guarantees for free. Where submultiplicativity is an inequality that any reasonable operator norm satisfies, the C^*-identity is an equality, and equalities rigidify. The unital and separable qualifiers flag the two structural refinements used repeatedly later: unitality is the algebraic analogue of a compact (rather than merely locally compact) underlying space, and separability is the countability hypothesis under which the K-theory of the book's second part is computable.

The prototype to keep in mind is the algebra $B(H)$ of bounded operators on a Hilbert space H, with a^* the Hilbert-space adjoint and $\|a\|$ the operator norm. There the identity $\|a^*a\| = \|a\|^2$ is the familiar relation between an operator and its adjoint, and the abstract definition simply axiomatizes the features of this example that survive passage to norm-closed $*$-subalgebras. The commutative counterpart is $C_0(X)$ with pointwise operations, complex conjugation as involution, and the supremum norm; there the identity reduces to $\sup_x |\overline{f(x)}f(x)| = \sup_x |f(x)|^2$, which is immediate. These two examples, the operator algebra and the function algebra, recur throughout.

Before cataloguing examples we record the first structural consequence of the identity, since it is used silently in nearly every later computation: the involution changes nothing in norm.

Proposition 1.2.2: The Involution is Isometric

Let A be a C^*-algebra. Then $\|a^*\| = \|a\|$ for every $a \in A$; that is, the involution is an isometry. Consequently the C^*-identity also reads $\|aa^*\| = \|a\|^2$.

***Proof*:**
If $a = 0$ the claim is trivial, so assume $a \neq 0$. Submultiplicativity of the norm in a Banach algebra gives $\|a^*a\| \leq \|a^*\| \, \|a\|$, and the C^*-identity rewrites the left-hand side as $\|a\|^2$, so

$$\|a\|^2 = \|a^*a\| \leq \|a^*\| \, \|a\|$$

Dividing by $\|a\| \neq 0$ yields $\|a\| \leq \|a^*\|$. This inequality holds for every element of A; applying it to a^* in place of a and using involutivity $(a^*)^* = a$ gives $\|a^*\| \leq \|(a^*)^*\| = \|a\|$. The two inequalities combine to $\|a^*\| = \|a\|$, so the involution is isometric.

For the final assertion, replace a by a^* in the C^*-identity: $\|(a^*)^*a^*\| = \|a^*\|^2$, which is $\|aa^*\| = \|a^*\|^2 = \|a\|^2$ by the isometry just established, as we sought to show.

The short argument repays reading, because it exhibits the mechanism the C^*-identity gives throughout the theory. Submultiplicativity $\|a^*a\| \leq \|a^*\| \, \|a\|$ is a one-directional bound available in any Banach algebra; the C^*-identity converts the left side into $\|a\|^2$, and the surplus strength of an equality forces the reverse bound $\|a\| \leq \|a^*\|$. Symmetry between a and a^* then closes the loop. The involution and the norm are thus not independent data: the identity binds them so tightly that isometry of the involution is not an extra axiom but a theorem. This is the first instance of a recurring pattern, that norm-analytic facts about a C^*-algebra are forced by its algebraic structure, and it is why the norm on a C^*-algebra turns out to be unique (a point taken up when $*$-homomorphisms are shown contractive in section 1.3). The identity $\|aa^*\| = \|a\|^2$ recorded at the end shows the axiom is left-right symmetric even though it was stated on one side only.

With the axioms in hand, the definition shows what it does only against concrete algebras. The following example verifies the axioms in each of the recurring cases, checking the C^*-identity explicitly where it is not immediate.

Example 1.2.3: The Standard C^*-Algebras
Each of the following is a C^*-algebra.

1. *The scalars* $\mathbb{C}$. With ordinary multiplication, absolute value as norm, and complex conjugation $\lambda^* = \overline{\lambda}$ as involution, $\mathbb{C}$ is a unital C^*-algebra. The identity is $|\overline{\lambda}\,\lambda| = |\lambda|^2$, which is the definition of the modulus. This is the smallest C^*-algebra and the ground field over which every other one is built.

2. *Continuous functions vanishing at infinity,* $C_0(X)$. Let X be a locally compact Hausdorff space and $C_0(X)$ the algebra of continuous $f\colon X \to \mathbb{C}$ vanishing at infinity, with pointwise operations, the supremum norm $\|f\| = \sup_{x\in X} |f(x)|$, and involution $f^*(x) = \overline{f(x)}$. The three algebraic axioms hold pointwise from the corresponding facts in $\mathbb{C}$. For the C^*-identity,

$$\|f^*f\| = \sup_{x\in X} \left|\overline{f(x)}\, f(x)\right| = \sup_{x\in X} |f(x)|^2 = \Big(\sup_{x\in X} |f(x)|\Big)^2 = \|f\|^2$$

where the third equality uses that $t \mapsto t^2$ is increasing on $[0, \infty)$. This algebra is unital exactly when X is compact, in which case $C_0(X) = C(X)$ contains the constant function 1; when X is only locally compact there is no unit. It is the commutative prototype, and Gelfand-Naimark duality will show that every commutative C^*-algebra arises this way.

3. *Bounded operators, $B(H)$.* Let H be a complex Hilbert space and $B(H)$ the algebra of bounded linear operators on H under composition, with the operator norm and a^* the Hilbert-space adjoint, characterized by $\langle a\xi, \eta\rangle = \langle \xi, a^*\eta\rangle$ for all $\xi, \eta \in H$. That the adjoint is a conjugate-linear anti-automorphism of period two is standard operator theory (see [9, Existence Of Adjoint]). For the C^*-identity, submultiplicativity of the operator norm gives $\|a^*a\| \le \|a^*\|\,\|a\| = \|a\|^2$ once the involution is known isometric; the reverse bound is the substance. For any unit vector $\xi \in H$, the Cauchy-Schwarz inequality gives

$$\|a\xi\|^2 = \langle a\xi, a\xi\rangle = \langle a^*a\,\xi, \xi\rangle \le \|a^*a\,\xi\|\,\|\xi\| \le \|a^*a\|$$

Taking the supremum over unit vectors ξ yields $\|a\|^2 \le \|a^*a\|$, and with the opposite inequality $\|a^*a\| = \|a\|^2$. Thus $B(H)$ is a unital C^*-algebra; it is the noncommutative prototype, and the Gelfand-Naimark theorem will later realize every C^*-algebra as a norm-closed $*$-subalgebra of some $B(H)$.

4. *The compact operators, $\mathcal{K}(H)$.* The compact operators on H form a norm-closed two-sided ideal of $B(H)$ that is closed under the adjoint (see [9, $B_0(\mathbb{H})$ Is A Closed Self-Adjoint Ideal]), so the operations and norm of $B(H)$ restrict to $\mathcal{K}(H)$ and the C^*-identity is inherited. Hence $\mathcal{K}(H)$ is a C^*-algebra. It is unital precisely when H is finite-dimensional (the identity operator is compact only then); for infinite-dimensional H it is the leading example of a non-unital C^*-algebra, and it is separable exactly when H is. More generally any norm-closed $*$-subalgebra of a C^*-algebra is again a C^*-algebra under the restricted structure, since the identity, being an equation among norms, passes to subsets closed under the operations.

5. *Matrix algebras, $M_n(\mathbb{C})$.* The algebra $M_n(\mathbb{C})$ of complex $n \times n$ matrices, with involution the conjugate transpose $a^* = \overline{a}^{\mathsf{T}}$ and norm the operator norm from the action on $\mathbb{C}^n$, is the case $H = \mathbb{C}^n$ of $B(H)$. It is a finite-dimensional unital C^*-algebra, and the conjugate transpose is exactly the Hilbert-space adjoint in an orthonormal basis, so the C^*-identity holds by the computation in the previous item. These are the simplest noncommutative C^*-algebras and the building blocks of the matrix amplifications $M_n(A)$ used throughout the K-theory of the book's second part.

6. *Direct sums and products.* Given C^*-algebras A and B, the direct sum $A \oplus B$ with coordinatewise operations, involution $(a, b)^* = (a^*, b^*)$, and norm $\|(a, b)\| = \max(\|a\|, \|b\|)$ is a C^*-algebra: the C^*-identity holds because

$$\|(a,b)^*(a,b)\| = \max(\|a^*a\|, \|b^*b\|) = \max(\|a\|^2, \|b\|^2) = \big(\max(\|a\|, \|b\|)\big)^2 = \|(a,b)\|^2$$

The same computation, with a supremum in place of the maximum, makes the product $\prod_i A_i$ of a family of C^*-algebras a C^*-algebra under the supremum norm $\|(a_i)_i\| = \sup_i \|a_i\|$, provided one restricts to the bounded elements (those with $\sup_i \|a_i\| < \infty$) so that the norm is finite.

These examples span the range of behaviour. The commutative pole is $C_0(X)$, whose only source of noncommutativity, the failure of $fg = gf$, is absent because scalars commute; the noncommutative pole is $B(H)$, where composition of operators need not commute. Every other example sits between them: $\mathcal{K}(H)$ and $M_n(\mathbb{C})$ are subalgebras of $B(H)$, while direct sums and products build new algebras from old and, applied to copies of $\mathbb{C}$, recover the finite- and infinite-dimensional commutative algebras $\mathbb{C}^n = C(\{1, \dots, n\})$ and their bounded infinite analogues. The verification of the C^*-identity was routine in $C_0(X)$, where it reduces to the modulus, but genuinely used Cauchy-Schwarz and the Hilbert-space inner product in $B(H)$, which is the mechanism by which the identity encodes positivity of a^*a. That positivity, latent in the computation $\langle a^*a\,\xi, \xi\rangle = \|a\xi\|^2 \geq 0$, is the seed of the order theory and the functional calculus developed in the next chapter.

1.3 ∗-Homomorphisms and Subalgebras

Having fixed the objects, the maps between them are next. A map of C^*-algebras must respect all three pieces of structure at once: the algebra operations, the norm, and the involution. Only two of these need be imposed. A map preserving the algebra structure and the involution automatically preserves the norm as tightly as it possibly could, because the C^*-identity pins the norm to the algebra. This section makes that precise: a $*$-homomorphism is automatically continuous, indeed contractive, and is isometric exactly when it is injective. The same rigidity forces the image of a $*$-homomorphism to be closed, unlike a bounded injection of Banach spaces, which can fail to have closed range. Along the way the notion of a sub-C^*-algebra and of the C^*-algebra generated by a set of elements is set down, the vocabulary in which every later construction is phrased.

The right notion of morphism preserves multiplication, addition, and the involution. Continuity is deliberately *not* required, since it will turn out to be free.

Definition 1.3.1: ∗-Homomorphism

Let A and B be C^*-algebras. A *$*$-homomorphism* is a linear map $\varphi\colon A \to B$ that is multiplicative and involution-preserving, that is,

$$\varphi(ab) = \varphi(a)\varphi(b) \qquad \text{and} \qquad \varphi(a^*) = \varphi(a)^*$$

for all $a, b \in A$. If A and B are unital and $\varphi(1_A) = 1_B$, then φ is called *unital*. A bijective $*$-homomorphism is a *$*$-isomorphism*, and A and B are then *$*$-isomorphic*, written $A \cong B$.

The definition asks only for algebraic compatibility. No norm condition appears, and this is the point: the analytic content is a theorem, not an axiom. A $*$-homomorphism between abstract C^*-algebras is the correct notion of "the same up to relabelling" when it is bijective, and the correct notion of "A sits inside B" when it is injective. Because the involution is part of the data, a $*$-homomorphism carries self-adjoint elements to self-adjoint elements and unitaries to unitaries (when unital), so all of the structure built from the involution transports along φ.

The most familiar $*$-homomorphisms come from maps of the underlying spaces, in the commutative case, and from restriction or amplification, in general.

Example 1.3.2: Pullback, Evaluation, and Inclusion

Three sources of $*$-homomorphisms recur throughout.

1. *Pullback along a continuous map.* Let X and Y be compact Hausdorff spaces and let

$h: X \to Y$ be continuous. The map

$$h^*: C(Y) \to C(X) \qquad h^*(f) = f \circ h$$

is a unital $*$-homomorphism: it is linear and multiplicative because these operations are pointwise, and it preserves the involution because $\overline{f(h(x))} = \overline{f}(h(x))$, so $h^*(\overline{f}) = \overline{h^*(f)}$. It is unital because the constant function 1 pulls back to the constant function 1. This is the arrow-reversing functoriality that Gelfand-Naimark duality will make an equivalence of categories.

2. *Evaluation at a point.* For a fixed $x_0 \in X$, evaluation $\varepsilon_{x_0}: C(X) \to \mathbb{C}$, $\varepsilon_{x_0}(f) = f(x_0)$, is a unital $*$-homomorphism onto $\mathbb{C}$. It is the special case $Y = \{x_0\}$ of a pullback.

3. *Inclusion of an invariant subspace.* If $H_0 \subseteq H$ is a closed subspace invariant under a $*$-subalgebra $A \subseteq B(H)$ (meaning $aH_0 \subseteq H_0$ and $a^*H_0 \subseteq H_0$ for all $a \in A$), then $a \mapsto a|_{H_0}$ is a $*$-homomorphism $A \to B(H_0)$; invariance of H_0 under both a and a^* is exactly what makes the restriction of a^* equal the adjoint of the restriction of a.

The first example is the template for the whole space thread of the book: continuous maps of spaces become $*$-homomorphisms of function algebras, in reverse. The third is the template for the operator thread: representations, developed later in the book, are $*$-homomorphisms into some $B(H)$, and passing to an invariant subspace is how one representation is built inside another.

Now to the analytic payoff. That a $*$-homomorphism is bounded is not automatic for algebra maps: a mere algebra homomorphism of Banach algebras can be discontinuous. The C^*-identity removes that freedom. The mechanism is that the norm of a self-adjoint element is its spectral radius, the spectral radius is controlled by the spectrum, and a homomorphism can only shrink spectra. The next result runs this chain and gets the sharp constant 1.

Theorem 1.3.3: $*$-Homomorphisms Are Contractive

Let $\varphi: A \to B$ be a $*$-homomorphism between C^*-algebras. Then φ is contractive:

$$\|\varphi(a)\| \leq \|a\| \qquad \text{for all } a \in A$$

In particular φ is continuous. If φ is injective, then it is isometric, $\|\varphi(a)\| = \|a\|$ for all $a \in A$.

***Proof*:**
Reduce first to the unital self-adjoint case, then remove each restriction.

Step 1: The unital case. Assume for the moment that A and B are unital and that φ is unital. Fix $a \in A$ and set $x = a^*a$, a self-adjoint element of A; then $\varphi(x) = \varphi(a^*a) = \varphi(a)^*\varphi(a)$ is the corresponding self-adjoint element of B. The key observation is that φ can only shrink the spectrum:

$$\sigma(\varphi(x)) \subseteq \sigma(x) \tag{1.1}$$

Indeed, if $\lambda \notin \sigma(x)$ then $x - \lambda 1_A$ is invertible in A, say with inverse y; applying φ to $(x - \lambda 1_A)y = 1_A = y(x - \lambda 1_A)$ and using that φ is a unital homomorphism gives $(\varphi(x) - \lambda 1_B)\varphi(y) = 1_B = \varphi(y)(\varphi(x) - \lambda 1_B)$, so $\varphi(x) - \lambda 1_B$ is invertible and $\lambda \notin \sigma(\varphi(x))$. This proves (1.1), hence $r(\varphi(x)) \leq r(x)$ for the spectral radius, since the spectral radius is the largest modulus attained on the spectrum by theorem 1.1.3.

Now apply the C^*-identity at both ends. Since $\varphi(x) = \varphi(a)^*\varphi(a)$ and $x = a^*a$ are self-adjoint, and the norm of a self-adjoint element equals its spectral radius (again theorem 1.1.3, since $\|z\|^2 = \|z^*z\| = \|z^2\|$ for self-adjoint z forces $\|z\| = r(z)$), the chain

$$\|\varphi(a)\|^2 = \|\varphi(a)^*\varphi(a)\| = \|\varphi(a^*a)\| = r(\varphi(a^*a)) \le r(a^*a) = \|a^*a\| = \|a\|^2$$

gives $\|\varphi(a)\| \le \|a\|$.

Step 2: Removing unitality. In general A and B need not be unital, and even if they are, φ need not preserve the unit. Pass to the unitizations $\tilde{A}$ and $\tilde{B}$ (proposition 1.4.5) and extend φ to the unital $*$-homomorphism

$$\tilde{\varphi} \colon \tilde{A} \to \tilde{B} \qquad \tilde{\varphi}(a + \lambda 1) = \varphi(a) + \lambda 1$$

which is unital by construction. Step 1 applies to $\tilde{\varphi}$, so $\|\tilde{\varphi}(y)\| \le \|y\|$ for all $y \in \tilde{A}$. The norm on $\tilde{A}$ restricts to the norm on A (proposition 1.4.5), and $\tilde{\varphi}$ restricts to φ on A, so $\|\varphi(a)\| \le \|a\|$ for every $a \in A$. Contractivity gives continuity, with $\|\varphi\| \le 1$.

Step 3: Injective implies isometric. Suppose φ is injective; it suffices to show $\|\varphi(a)\| \ge \|a\|$, and again the self-adjoint reduction applies, since $\|a\|^2 = \|a^*a\|$ and $\|\varphi(a)\|^2 = \|\varphi(a^*a)\|$ with a^*a self-adjoint. So take $x = a^*a$ self-adjoint and pass to unitizations as in Step 2; $\tilde{\varphi}$ is injective because φ is and $\tilde{A}/A \cong \mathbb{C}$. For a self-adjoint element the norm is the spectral radius, so it is enough to prove $\sigma(\tilde{\varphi}(x)) = \sigma(x)$, that is, that an injective unital $*$-homomorphism does not shrink spectra. Suppose it did, so that some $\lambda \in \sigma(x)$ has $\lambda \notin \sigma(\tilde{\varphi}(x))$. Choose a continuous function $f \colon \sigma(x) \to [0, \infty)$ that vanishes on $\sigma(\tilde{\varphi}(x)) \cap \sigma(x)$ but not at λ, so f is not identically zero on $\sigma(x)$ while $f = 0$ on $\sigma(\tilde{\varphi}(x)) \cap \sigma(x)$. Applying f to x through the continuous functional calculus produces a nonzero element $f(x) \in \tilde{A}$ with $\sigma(f(x)) = f(\sigma(x)) \neq \{0\}$; on the other hand $\tilde{\varphi}(f(x)) = f(\tilde{\varphi}(x))$, whose spectrum is $f(\sigma(\tilde{\varphi}(x))) = \{0\}$, so $f(\tilde{\varphi}(x)) = 0$ because it is self-adjoint with spectral radius zero. Thus $f(x)$ is a nonzero element of the kernel of $\tilde{\varphi}$, contradicting injectivity. Hence $\sigma(\tilde{\varphi}(x)) = \sigma(x)$, so $\|\tilde{\varphi}(x)\| = \|x\|$, and unwinding the self-adjoint reduction gives $\|\varphi(a)\| = \|a\|$, as we sought to show.

A $*$-homomorphism carries no metric information beyond its algebra: the norm is determined by the algebraic structure, so respecting the algebra and the involution already respects the norm. Two consequences are worth isolating. First, there is at most one C^*-norm on a given $*$-algebra that makes it a C^*-algebra, because the identity map would be a $*$-isomorphism between the two normed completions and hence isometric; the norm is an invariant of the $*$-algebraic structure, not extra data. Second, an injective $*$-homomorphism is an isometric embedding, so "A is a sub-C^*-algebra of B" and "A embeds into B by an injective $*$-homomorphism" say the same thing. The injective-isometric step used the continuous functional calculus and spectral permanence, both developed in the following chapter; the reader may take Step 3 on the understanding that an injective $*$-homomorphism does not shrink the spectrum of a self-adjoint element, which is the content used.

A single instance shows the rigidity: the pullback of example 1.3.2 is contractive, and injective exactly when it is isometric, in a way one can read off the topology.

Example 1.3.4: The Pullback of a Surjection Is Isometric

Let $h \colon X \to Y$ be continuous with X, Y compact Hausdorff, and consider $h^* \colon C(Y) \to C(X)$ from example 1.3.2. For $f \in C(Y)$,

$$\|h^*(f)\|_\infty = \sup_{x \in X} |f(h(x))| = \sup_{y \in h(X)} |f(y)|$$

which is $\le \|f\|_\infty = \sup_{y\in Y}|f(y)|$, confirming contractivity directly. If h is surjective then $h(X) = Y$, the two suprema coincide, and h^* is isometric; and h^* is injective precisely when $h(X)$ is all of Y, since a function supported off $h(X)$ pulls back to zero. Thus for pullbacks the abstract equivalence "injective $\iff$ isometric" of theorem 1.3.3 is visibly the statement "h has dense (here, full) image". That h^* is unital and $*$-preserving was checked in example 1.3.2.

With morphisms in hand, subobjects are next. Inside a C^*-algebra one wants the sub-objects that are again C^*-algebras in their own right. Closure under the involution and under limits are both required: an involution-closed subalgebra that is not norm-closed is only a dense skeleton, while a norm-closed subalgebra that is not involution-closed is not stable under the involution.

Definition 1.3.5: Sub-C^*-Algebra

Let A be a C^*-algebra. A *$*$-subalgebra* of A is a linear subspace $B \subseteq A$ that is closed under multiplication and under the involution ($b, b' \in B$ imply $bb' \in B$ and $b^* \in B$). A *sub-C^*-algebra* of A is a $*$-subalgebra that is closed in the norm topology of A.

A sub-C^*-algebra is a C^*-algebra in its own right: the operations and norm of A restrict to it, the C^*-identity is inherited pointwise, and norm-closedness in the complete space A makes it complete. So a sub-C^*-algebra is exactly a subset that both is a $*$-subalgebra and is itself a C^*-algebra under the ambient structure. Note that a sub-C^*-algebra of a unital A need not contain the unit 1 of A, and even when it is itself unital its unit may differ from 1: the notion is not intrinsically unital. This is not a defect but a feature the theory needs, since the compact operators $\mathcal{K}(H)$ form a nonunital sub-C^*-algebra of $B(H)$.

The reason closure is the right requirement, rather than a nuisance, is that it is automatic when one takes closures.

Proposition 1.3.6: The Closure of a $*$-Subalgebra Is a Sub-C^*-Algebra

Let A be a C^*-algebra and $S \subseteq A$ a $*$-subalgebra. Then its norm closure $\overline{S}$ is a sub-C^*-algebra of A.

***Proof*:**
The set $\overline{S}$ is closed by construction, so only its algebraic stability is at issue, and this follows from continuity of the operations. The addition, scalar multiplication, and multiplication of A are continuous, and the involution is isometric hence continuous (proposition 1.2.2). Let $a, b \in \overline{S}$ and choose sequences $a_n, b_n \in S$ with $a_n \to a$ and $b_n \to b$. Then $a_n + b_n \to a + b$, $\lambda a_n \to \lambda a$, $a_n b_n \to ab$, and $a_n^* \to a^*$, and each left-hand side lies in S because S is a $*$-subalgebra. The limits therefore lie in $\overline{S}$, so $\overline{S}$ is a $*$-subalgebra, and being closed it is a sub-C^*-algebra, completing the proof.

That taking closures produces sub-C^*-algebras is what lets one *generate* a C^*-algebra from raw material. Given any elements of A, there is a smallest sub-C^*-algebra containing them, obtained by throwing in all algebraic combinations and their adjoints and then closing up. This is the standard way C^*-algebras are presented: not by an abstract axiom list but by naming generators and relations inside a known ambient algebra.

Definition 1.3.7: C^*-Algebra Generated by a Set

Let A be a C^*-algebra and $S \subseteq A$ a subset. The *sub-C^*-algebra generated by S*, written $C^*(S)$, is the smallest sub-C^*-algebra of A containing S, equivalently the intersection of all sub-C^*-algebras of A containing S. Concretely, $C^*(S) = \overline{S_0}$, where S_0 is the $*$-subalgebra of all noncommutative polynomials without constant term in the elements of S and their adjoints. When A is unital and one wishes to include the unit, one writes $C^*(S, 1)$ for the sub-C^*-algebra generated by $S \cup \{1\}$. For a single element a one writes $C^*(a)$ and $C^*(a, 1)$.

The two descriptions agree because the intersection of sub-C^*-algebras is a sub-C^*-algebra, so the smallest one exists, and it must contain every polynomial in $S \cup S^*$ (by algebraic closure) and hence their limits (by norm-closure), which is exactly $\overline{S_0}$; conversely $\overline{S_0}$ is a sub-C^*-algebra containing S by proposition 1.3.6. The single-generator algebra $C^*(a, 1)$ is the object the continuous functional calculus will identify with $C(\sigma(a))$ when a is normal. The most important case fixes ideas.

Example 1.3.8: The Algebra Generated by a Self-Adjoint Operator

Let $a = a^* \in B(H)$ be a bounded self-adjoint operator on a Hilbert space. Because a is self-adjoint, the $*$-subalgebra it generates (with the identity) is spanned by the powers $1, a, a^2, \ldots$ alone, no adjoints being needed since $a^* = a$, so S_0 is the algebra of polynomials $p(a)$ with complex coefficients. Thus

$$C^*(a, 1) = \overline{\{\, p(a) \big| p \in \mathbb{C}[t] \,\}}$$

the norm closure of the polynomials in a. For the concrete operator of multiplication by the coordinate function t on $H = L^2[0, 1]$, one has $\sigma(a) = [0, 1]$, and $p(a)$ is multiplication by $p(t)$; the norm $\|p(a)\|$ equals $\sup_{t\in[0,1]} |p(t)|$. The closure of the polynomials in the supremum norm on $[0, 1]$ is, by the Stone-Weierstrass theorem, all of $C[0, 1]$, so $C^*(a, 1) \cong C[0, 1]$. This is Gelfand-Naimark in miniature and the reason $C^*(a, 1)$ is the natural home of the functional calculus.

The rigidity of theorem 1.3.3 has a structural consequence that is false for general Banach spaces: the image of a $*$-homomorphism is automatically closed, hence a sub-C^*-algebra of the target. For a bounded injection of Banach spaces the range can be a dense proper subspace, so completeness of the domain says nothing about closedness of the image. Here the isometry forced by injectivity, applied after quotienting out the kernel, rescues closedness for free.

Proposition 1.3.9: The Image of a $*$-Homomorphism Is a Sub-C^*-Algebra

Let $\varphi\colon A \to B$ be a $*$-homomorphism between C^*-algebras. Then $\varphi(A)$ is a sub-C^*-algebra of B.

***Proof*:**

That $\varphi(A)$ is a $*$-subalgebra of B is immediate from the definitions: it is a linear subspace because φ is linear, closed under multiplication because $\varphi(a)\varphi(a') = \varphi(aa')$, and closed under the involution because $\varphi(a)^* = \varphi(a^*)$. Only closedness in norm requires the analytic input.

Step 1: Factor through the kernel. The kernel $I = \ker\varphi$ is a closed two-sided ideal of A: it is closed because φ is continuous (theorem 1.3.3), a two-sided ideal because φ is a homomorphism, and self-adjoint because $\varphi(a^*) = \varphi(a)^*$ shows $a \in I \iff a^* \in I$. The quotient A/I is again a

C^*-algebra (see theorem 1.4.4), and φ factors as

$$A \xrightarrow{q} A/I \xrightarrow{\overline{\varphi}} B$$

where q is the quotient $*$-homomorphism and $\overline{\varphi}$ is the induced map $\overline{\varphi}(a+I) = \varphi(a)$, which is a well-defined injective $*$-homomorphism with the same image, $\overline{\varphi}(A/I) = \varphi(A)$.

Step 2: The injective factor is isometric. Since $\overline{\varphi}$ is injective, theorem 1.3.3 makes it isometric: $\|\overline{\varphi}(x)\| = \|x\|$ for all $x \in A/I$.

Step 3: Isometry forces closed range. Let $\varphi(a_n) = \overline{\varphi}(x_n)$ be a sequence in $\varphi(A)$ converging in B to some b, with $x_n = a_n + I \in A/I$. A convergent sequence is Cauchy, and $\overline{\varphi}$ is isometric, so $\|x_n - x_m\| = \|\overline{\varphi}(x_n) - \overline{\varphi}(x_m)\| \to 0$; thus (x_n) is Cauchy in the complete space A/I and converges to some x. By continuity of $\overline{\varphi}$, $\overline{\varphi}(x) = \lim \overline{\varphi}(x_n) = b$, so $b = \overline{\varphi}(x) \in \varphi(A)$. Hence $\varphi(A)$ is closed, and being a $*$-subalgebra it is a sub-C^*-algebra, as we sought to show.

The proposition rests on nothing beyond the isometry of injective $*$-homomorphisms: once the induced map on the quotient is isometric, its range is complete, and a complete subspace of a Banach space is closed. This is the reason one rarely has to check closedness by hand in this subject. Ranges of $*$-homomorphisms, quotients, and (as the sequel will show) images under the functional calculus are all closed automatically, so the sub-C^*-algebras of a given C^*-algebra are plentiful and easy to produce. The single input driving all of it is the C^*-identity, transmitted through theorem 1.3.3: the norm is so tightly bound to the algebra that maps preserving the algebra cannot help but preserve the metric structure the norm carries.

1.4 Ideals, Approximate Units, and the Unitization

A commutative C^*-algebra $C_0(X)$ has a unit exactly when X is compact, namely the constant function 1. When X is only locally compact the constant function is not in $C_0(X)$, and the algebra has no unit. Non-unital C^*-algebras are therefore the generic case, and the tools of the previous sections which quietly used a unit (the spectrum, the spectral radius, functions of an element) need a substitute. This section gives two of them. An approximate unit is a net that behaves like a unit in the limit, and every C^*-algebra has one; it is what proves that closed ideals of C^*-algebras are automatically $*$-closed and that their quotients are again C^*-algebras. The unitization $\tilde{A}$ then adjoins a unit outright, embedding any A as a closed ideal in a unital C^*-algebra, so that every question about a non-unital algebra can be pushed into the unital world where the machinery of the previous sections applies.

The one input drawn from ahead is the theory of positive elements, which chapter 2 develops in full. An element a of a C^*-algebra is *positive*, written $a \geq 0$, when $a = a^*$ and $\sigma(a) \subseteq [0, \infty)$; this is equivalent to $a = b^*b$ for some b, and to $a = c^2$ for a self-adjoint $c \geq 0$. The positive elements form a cone, and $a \leq b$ means $b - a \geq 0$. These facts are used below only to name and order the elements of an approximate unit; their proofs belong to the functional calculus and are cited forward.

The defining feature of a unit 1 is that $a1 = 1a = a$ for every a. A net of elements e_λ that multiplies each fixed a back to itself in the limit will serve the same purpose for the constructions that follow, provided the e_λ are uniformly bounded and increase toward 1 in the order on positive elements.

Definition 1.4.1: Approximate Unit

Let A be a C^*-algebra. An *approximate unit* for A is a net $(e_\lambda)_{\lambda\in\Lambda}$ of positive elements of the closed unit ball, so $0 \le e_\lambda$ and $\|e_\lambda\| \le 1$, that is *increasing* in the order on positive elements, $e_\lambda \le e_\mu$ whenever $\lambda \le \mu$, and satisfies

$$\|ae_\lambda - a\| \to 0 \qquad \text{and} \qquad \|e_\lambda a - a\| \to 0$$

for every $a \in A$. When A is separable the net may be taken to be a sequence $(e_n)_{n\in\mathbb{N}}$, and the approximate unit is called *sequential*.

An approximate unit is thus a directed family of positive elements, each of norm at most one, that acts as an identity in the limit from both sides. Two comments fix its meaning. First, the net is directed by the index set Λ, and the increasing condition ties the algebraic order $e_\lambda \le e_\mu$ to that direction, so the e_λ climb monotonically inside the unit ball. Second, the two limit conditions are separate requirements in a general algebra, but in a C^*-algebra either one implies the other by taking adjoints, since $\|e_\lambda a - a\| = \|(a^*e_\lambda - a^*)^*\| = \|a^*e_\lambda - a^*\|$ and a^* ranges over A as a does. A left approximate unit is automatically a two-sided one.

Such a net always exists.

Theorem 1.4.2: Existence of Approximate Units

Every C^*-algebra A has an approximate unit. Concretely, let

$$\Lambda = \{\, e \in A \big| e \ge 0,\ \|e\| < 1 \,\}$$

be the positive elements of the open unit ball, ordered by $e \le f$ iff $f - e \ge 0$. Then Λ is a directed set, and the net $(e)_{e\in\Lambda}$ indexed by itself, with $e_\lambda = \lambda$, is an approximate unit for A.

***Proof*:**
The argument has two parts: that Λ is directed, and that the net it indexes multiplies each a back to itself. Both rest on a single device, the map $x \mapsto x(1+x)^{-1}$ computed inside the unitization, which converts the additive order into a form where positivity is transparent.

Step 1: Λ is directed. Work inside the unitization $\tilde{A}$ constructed in proposition 1.4.5 below, so that $1 + x$ is invertible for every positive x (its spectrum lies in $[1, \infty)$, away from 0). Define

$$f(x) = x(1+x)^{-1} = 1 - (1+x)^{-1}$$

for $x \ge 0$, computed through the functional calculus of chapter 2. The scalar function $t \mapsto t/(1+t)$ is a strictly increasing bijection of $[0, \infty)$ onto $[0, 1)$ and is operator monotone, so f carries positive elements to positive elements of norm below 1, is order-preserving, and has order-preserving inverse $g \mapsto g(1-g)^{-1}$ on the positive part of the open unit ball. In particular $f(x) \in \Lambda$ for every $x \ge 0$, and f maps the positive cone bijectively onto Λ. Given $e_1, e_2 \in \Lambda$, write $e_i = f(x_i)$ with $x_i = e_i(1-e_i)^{-1} \ge 0$; then $x_1 + x_2 \ge x_i$ for each i, so $f(x_1 + x_2) \ge f(x_i) = e_i$ by monotonicity, and $f(x_1 + x_2) \in \Lambda$ is an upper bound for both e_1 and e_2. The set Λ is directed.

Step 2: the net acts as an identity. Fix $a \in A$ and set $h = a^*a \ge 0$. It suffices to show $\|a - ae_\lambda\| \to 0$, equivalently $\|a(1 - e_\lambda)\| \to 0$ computed in $\tilde{A}$; the other side then follows by the adjoint remark after definition 1.4.1. Since Λ is directed and each e_λ lies in it, it is enough to exhibit, for every $\varepsilon > 0$, a single index $e_0 \in \Lambda$ such that $\|a(1-e_\lambda)\| \le \varepsilon$ for *all* $\lambda \ge e_0$; that is exactly convergence of the net.

Fix $n \in \mathbb{N}$ and take the specific element $e_n = f(nh) = nh(1+nh)^{-1} \in \Lambda$, for which $1 - e_n = (1+nh)^{-1}$. We claim

$$\|a(1-e_\lambda)\|^2 \le \frac{1}{n} \qquad \text{for every } \lambda \ge e_n. \tag{1.2}$$

Granting (1.2), the proof is complete: given $\varepsilon > 0$, choose $n > 1/\varepsilon^2$ and set $e_0 = e_n$; then $\|a(1-e_\lambda)\| \le 1/\sqrt{n} < \varepsilon$ for all $\lambda \ge e_0$, so $\|a(1-e_\lambda)\| \to 0$ along the whole net.

To prove (1.2), fix $\lambda \ge e_n$ and write $c = 1 - e_\lambda$, $d = 1 - e_n$, so that $0 \le c \le d \le 1$ (the order $e_n \le e_\lambda$ reverses under $x \mapsto 1 - x$, and both lie in the unit ball). The C^*-identity applied to $x = h^{1/2}c$ gives $\|a(1-e_\lambda)\|^2 = \|chc\| = \|h^{1/2}c\|^2 = \|h^{1/2}c^2h^{1/2}\|$, using $\|x^*x\| = \|xx^*\|$ for the last equality. Now $0 \le c \le 1$ forces $c^2 \le c$, and $c \le d$, so $c^2 \le d$; conjugating this order relation by $h^{1/2}$ preserves it, and passing to norms of positive elements gives

$$\|a(1-e_\lambda)\|^2 = \|h^{1/2}c^2h^{1/2}\| \le \|h^{1/2}d\,h^{1/2}\| = \|d^{1/2}h\,d^{1/2}\|$$

where the final equality again uses $\|xx^*\| = \|x^*x\|$ with $x = h^{1/2}d^{1/2}$. Working inside the commutative C^*-algebra $C^*(h) \cong C_0(\sigma(h) \setminus \{0\})$ generated by the single positive element h (chapter 3), the element $d^{1/2}h\,d^{1/2} = (1+nh)^{-1/2}\,h\,(1+nh)^{-1/2}$ corresponds under the functional calculus to the function

$$t \mapsto \frac{t}{1+nt}$$

on $\sigma(h) \subseteq [0, \|h\|]$, whose supremum over $t \ge 0$ is $1/n$. Hence $\|a(1-e_\lambda)\|^2 \le \|d^{1/2}h\,d^{1/2}\| \le 1/n$, which is (1.2). The estimate is uniform over the entire tail $\{\lambda : \lambda \ge e_n\}$, not merely along the sequence (e_n), so the crucial monotone domination is genuinely established rather than assumed.

The construction is canonical: the approximate unit is nothing but the entire positive part of the open unit ball, directed upward by its own order. No choices are made, which is why the assignment is compatible with $*$-homomorphisms and why the same net serves every purpose below. The one genuinely C^*-algebraic step is the reduction $\|a(1-e)\|^2 = \|(1-e)a^*a(1-e)\|$: it converts a norm estimate about an arbitrary a into a scalar computation inside the commutative algebra generated by the single positive element a^*a, where everything reduces to the calculus of a real function. For a separable algebra one can thin the large index set to a sequence by choosing a countable cofinal family, which is the sequential case named in definition 1.4.1.

Example 1.4.3: Approximate Units in $C_0(X)$ and $\mathcal{K}(H)$

Two concrete cases show the definition at work.

1. For $A = C_0(X)$ with X locally compact Hausdorff and non-compact, an approximate unit is given by a net of compactly supported functions rising to 1. Order the compact subsets $K \subseteq X$ by inclusion, and for each K pick $g_K \in C_0(X)$ with $0 \le g_K \le 1$, $g_K \equiv 1$ on K, and g_K of compact support (Urysohn's lemma produces such a function). To make the family increasing, set $e_K = \max_{j \le m} g_{K_j}$ where $K_1 \subseteq \ldots \subseteq K_m = K$ runs over any finite chain of previously chosen indices below K (equivalently, invoke the existence theorem directly, which produces an increasing net with no manual monotonization); each e_K still lies in $C_0(X)$, satisfies $0 \le e_K \le 1$, and equals 1 on K. For $a \in C_0(X)$ and $\varepsilon > 0$ the set $\{\,x \in X \,|\, |a(x)| \ge \varepsilon\,\}$ is compact, so once K contains it $\|ae_K - a\|_\infty \le \varepsilon$. The net multiplies

each a back to itself, matching definition 1.4.1. The unit is being switched on over larger and larger compact sets, which is exactly why it exists only in the limit when X is non-compact.

2. For $A = \mathcal{K}(H)$ the compact operators on a separable Hilbert space, fix an orthonormal basis (ξ_n) and let p_n be the orthogonal projection onto the span of $\xi_1, \ldots, \xi_n$. Then (p_n) is an increasing sequence of projections, $0 \leq p_n \leq p_{n+1} \leq 1$, each of norm 1, and for a compact operator T one has $\|Tp_n - T\| \to 0$ and $\|p_n T - T\| \to 0$ because T is a norm-limit of finite-rank operators, each of which is eventually fixed by the p_n. This is a sequential approximate unit, as $\mathcal{K}(H)$ is separable.

In both cases the algebra has no unit (X non-compact; H infinite-dimensional), yet the approximate unit recovers the identity in the limit.

The reason approximate units are introduced here, rather than as an afterthought, is that they force two structural facts about ideals that have no analogue in a bare Banach algebra. Throughout, an *ideal* of A means a closed two-sided ideal: a norm-closed subspace $I \subseteq A$ with $AI \subseteq I$ and $IA \subseteq I$. In a general Banach $*$-algebra there is no reason for such an I to be closed under the involution, nor for the quotient norm to satisfy the C^*-identity. In a C^*-algebra both hold, and the bridge in each case is an approximate unit of I itself.

Theorem 1.4.4: Ideals are Self-Adjoint and Quotients are C^*-Algebras

Let A be a C^*-algebra and $I \subseteq A$ a closed two-sided ideal. Then:

1. I is self-adjoint, $I^* = I$, hence is itself a C^*-algebra; and
2. the quotient A/I, with the quotient norm $\|a + I\| = \inf_{x \in I} \|a - x\|$, is a C^*-algebra under the involution $(a + I)^* = a^* + I$.

***Proof*:**
The ideal I is a closed two-sided ideal in its own right, so by theorem 1.4.2 it has an approximate unit (e_λ); this net does all the work.

1. *Self-adjointness.* Let $a \in I$. Since I is an ideal and each $e_\lambda \in I$, the products $e_\lambda a^*$ lie in I. Their limit is a^*: using that (e_λ) is an approximate unit for I applied to a, and self-adjointness $e_\lambda = e_\lambda^*$,
$$\|a^* - e_\lambda a^*\| = \|(a - ae_\lambda)^*\| = \|a - ae_\lambda\| \to 0$$
so $a^* = \lim_\lambda e_\lambda a^*$ is a norm-limit of elements of I. As I is closed, $a^* \in I$. Thus $I^* \subseteq I$, and applying this to I^* gives the reverse inclusion, so $I^* = I$. A norm-closed self-adjoint subalgebra of a C^*-algebra is a C^*-algebra, establishing (1).

2. *The quotient is a C^*-algebra.* The quotient of a Banach algebra by a closed two-sided ideal is a Banach algebra under the quotient norm, and by (1) the involution descends: $(a + I)^* = a^* + I$ is well-defined since $x \in I$ implies $x^* \in I$, and it is an isometric conjugate-linear involution on A/I because $\|(a + I)^*\| = \|a^* + I\| = \inf_{x \in I} \|a^* - x\| = \inf_{y \in I} \|(a - y)^*\| = \|a + I\|$. It remains to verify the C^*-identity
$$\|(a + I)^*(a + I)\| = \|a + I\|^2$$

The inequality $\|(a+I)^*(a+I)\| \le \|(a+I)^*\|\,\|a+I\| = \|a+I\|^2$ holds in any Banach $*$-algebra with isometric involution, so only $\|a+I\|^2 \le \|(a+I)^*(a+I)\|$ needs proof.

Key claim. The quotient norm is computed by the approximate unit:

$$\|a+I\| = \lim_\lambda \|a - ae_\lambda\| = \lim_\lambda \|a(1-e_\lambda)\| \tag{1.3}$$

the last expression read in $\tilde{A}$. Granting (1.3), the C^*-identity follows. Writing $u_\lambda = 1 - e_\lambda$, which satisfies $0 \le u_\lambda \le 1$ and hence $\|u_\lambda\| \le 1$ in $\tilde{A}$, the C^*-identity in A gives

$$\|au_\lambda\|^2 = \|u_\lambda a^* a u_\lambda\| \le \|a^*a\,u_\lambda\| = \|a^*a - a^*ae_\lambda\|$$

where the inequality uses $\|u_\lambda y u_\lambda\| \le \|u_\lambda\|\,\|yu_\lambda\| \le \|yu_\lambda\|$ with $y = a^*a$, and the final equality is $a^*au_\lambda = a^*a - a^*ae_\lambda$. Since $a^*ae_\lambda \in I$, taking $\lim_\lambda$ and applying (1.3) to both a and a^*a yields

$$\|a+I\|^2 = \lim_\lambda \|au_\lambda\|^2 \le \lim_\lambda \|a^*a - a^*ae_\lambda\| = \|a^*a + I\| = \|(a+I)^*(a+I)\|$$

which is the missing inequality.

Proof of (1.3). On one hand $ae_\lambda \in I$ gives $\|a+I\| \le \|a - ae_\lambda\| = \|a(1-e_\lambda)\|$ for every λ, so $\|a+I\| \le \liminf_\lambda \|a(1-e_\lambda)\|$. On the other hand, fix $x \in I$ and $\varepsilon > 0$ with $\|a - x\| \le \|a+I\| + \varepsilon$; then, using $\|1 - e_\lambda\| \le 1$ in $\tilde{A}$,

$$\|a(1-e_\lambda)\| \le \|(a-x)(1-e_\lambda)\| + \|x(1-e_\lambda)\| \le \|a-x\| + \|x - xe_\lambda\|$$

and since $x \in I$ the term $\|x - xe_\lambda\| \to 0$, so $\limsup_\lambda \|a(1-e_\lambda)\| \le \|a-x\| \le \|a+I\| + \varepsilon$. As $\varepsilon > 0$ was arbitrary, $\limsup_\lambda \|a(1-e_\lambda)\| \le \|a+I\|$. The two bounds give (1.3), completing the proof.

Both statements are automatic consequences of positivity that fail in the purely Banach setting: a closed ideal of a Banach algebra need be neither self-adjoint nor have a Banach-$*$-algebra quotient satisfying the C^*-identity, and it is precisely the approximate unit, itself a consequence of the C^*-identity through the functional calculus, that repairs both. In the commutative model the content is familiar. For $A = C_0(X)$ a closed ideal is $I = C_0(U)$ for an open set $U \subseteq X$ (the functions vanishing off U), self-adjointness of I is closure under complex conjugation, and the quotient $C_0(X)/C_0(U) \cong C_0(X \setminus U)$ is again a function algebra on the closed complement, visibly a C^*-algebra. The theorem says this geometric picture persists with no commutativity: ideals correspond to open subspaces and quotients to closed ones, and the noncommutative A/I is as good a C^*-algebra as the commutative $C_0(X \setminus U)$. This is what makes the short exact sequences $0 \to I \to A \to A/I \to 0$ of the K-theory chapters sequences of genuine C^*-algebras.

An approximate unit substitutes for a unit in a limiting sense. Often one wants an actual unit, if only to invoke the spectral theory of chapter 2, which was stated for unital algebras, or simply to write $1 - e_\lambda$ as above. The remedy is to adjoin one. The naive construction $A \oplus \mathbb{C}$ with coordinatewise operations is a unital algebra with unit $(0,1)$, but the obvious coordinate norm $\|(a,\lambda)\| = \|a\| + |\lambda|$ is not a C^*-norm: it fails the C^*-identity. The fix is to norm $A \oplus \mathbb{C}$ not by its coordinates but by how it acts on A by multiplication, embedding $\tilde{A}$ into the bounded operators on the Banach space A.

Proposition 1.4.5: The Unitization

Let A be a C^*-algebra. On the vector space $\tilde{A} = A \oplus \mathbb{C}$, define

$$(a, \lambda)(b, \mu) = (ab + \lambda b + \mu a,\ \lambda\mu)$$
$$(a, \lambda)^* = (a^*,\ \overline{\lambda})$$

with unit $(0, 1)$. When A is non-unital, the norm

$$\|(a, \lambda)\| = \sup_{\substack{b \in A \\ \|b\| \le 1}} \|ab + \lambda b\|$$

the operator norm of left multiplication on A, makes $\tilde{A}$ a C^*-algebra; when A is unital, $\tilde{A}$ carries instead the direct-sum norm making it the product C^*-algebra $A \oplus \mathbb{C}$. In either case $\tilde{A}$ is a unital C^*-algebra, the map $a \mapsto (a, 0)$ is an isometric $*$-isomorphism of A onto a closed two-sided ideal of $\tilde{A}$, and the quotient $\tilde{A}/A \cong \mathbb{C}$ via $(a, \lambda) \mapsto \lambda$. The assignment $A \mapsto \tilde{A}$ is functorial: a $*$-homomorphism $\varphi\colon A \to B$ extends uniquely to a unital $*$-homomorphism $\tilde{\varphi}\colon \tilde{A} \to \tilde{B}$ with $\tilde{\varphi}(a, \lambda) = (\varphi(a), \lambda)$, and $\widetilde{\psi \circ \varphi} = \tilde{\psi} \circ \tilde{\varphi}$, $\widetilde{\mathrm{id}_A} = \mathrm{id}_{\tilde{A}}$.

***Proof*:**
The multiplication is designed so that $(0, 1)$ acts as a unit and so that the map

$$L\colon \tilde{A} \to B(A), \qquad L(a, \lambda)(b) = ab + \lambda b$$

sending (a, λ) to left multiplication by $a + \lambda$ on the Banach space A is an algebra homomorphism; the norm is by definition $\|(a, \lambda)\| = \|L(a, \lambda)\|_{B(A)}$, the operator norm. The proof verifies that this is a C^*-norm and computes the stated structure.

Step 1: A embeds isometrically. For $a \in A$, $\|(a, 0)\| = \sup_{\|b\| \le 1} \|ab\|$. Taking $b = a^*/\|a\|$ for $a \neq 0$ gives $\|aa^*\|/\|a\| = \|a\|^2/\|a\| = \|a\|$ by the C^*-identity, so $\|(a, 0)\| \ge \|a\|$; and $\|ab\| \le \|a\|\,\|b\| \le \|a\|$ gives the reverse. Thus $\|(a, 0)\| = \|a\|$, so $a \mapsto (a, 0)$ is an isometric embedding onto $A \oplus 0$, which is a closed two-sided ideal of $\tilde{A}$ since $(a, 0)(b, \mu) = (ab + \mu a, 0)$ and $(b, \mu)(a, 0) = (ba + \mu a, 0)$ both lie in $A \oplus 0$.

Step 2: L is a homomorphism, injective when A is non-unital, and the formula is a norm. That L is linear and $L((a, \lambda)(b, \mu)) = L(a, \lambda)L(b, \mu)$ is a direct expansion of the multiplication, so $\|(a, \lambda)\| = \|L(a, \lambda)\|$ is a submultiplicative seminorm with $\|(0, 1)\| = \|\mathrm{id}_A\| = 1$. Suppose $\|(a, \lambda)\| = 0$, so $ab + \lambda b = 0$ for all $b \in A$. If $\lambda = 0$ then $ab = 0$ for all b; taking $b = a^*$ gives $\|a\|^2 = \|aa^*\| = 0$, so $a = 0$, and $(a, \lambda) = 0$. If $\lambda \neq 0$ then $b = -\lambda^{-1}ab$ for all b, so $u = -\lambda^{-1}a$ is a left unit for A; taking adjoints in $b = ub$ gives $b^* = b^*u^*$, so u^* is a right unit and A is unital with unit $u = u^*$. Thus when A is *non-unital* the second case cannot occur, L has trivial kernel, and $\|\cdot\|$ is a genuine norm. When A is already unital the map L does have a kernel, spanned by $(-1_A, 1)$, and the norm degenerates; in that case one instead *defines* $\tilde{A}$ to be the direct-sum C^*-algebra $A \oplus \mathbb{C}$ of example 1.2.3, which already has a unit and needs no adjunction. The construction below therefore assumes A non-unital, and the unital case is recorded in example 1.4.6.

Step 3: completeness. The image $L(\tilde{A}) \subseteq B(A)$ is the sum of $L(A \oplus 0)$, isometric to the complete space A by Step 1, and the one-dimensional space $\mathbb{C} \cdot \mathrm{id}_A$; a finite-dimensional extension of a complete subspace is closed in $B(A)$, so $L(\tilde{A})$ is complete. As L is an isometric bijection onto its image, $\tilde{A}$ is complete, hence a Banach $*$-algebra once the involution is shown isometric in Step 4.

Step 4: the C^*-identity. Write $x = (a, \lambda)$, so $x^* = (a^*, \overline{\lambda})$ and $L(x^*) = L(x)^*$ when $B(A)$ is given the adjoint coming from the C^*-structure of A acting on itself. Submultiplicativity gives $\|x^*x\| \leq \|x^*\|\,\|x\|$. For the reverse, take any $b \in A$ with $\|b\| \leq 1$; using the C^*-identity in A and $L(x^*)L(x) = L(x^*x)$,

$$\|L(x)b\|^2 = \|(L(x)b)^*(L(x)b)\| = \|b^*\,L(x^*x)\,b\| \leq \|L(x^*x)\| = \|x^*x\|$$

since $\|b^*db\| \leq \|b\|^2\|d\| \leq \|d\|$ for $\|b\| \leq 1$. Taking the supremum over $\|b\| \leq 1$ gives $\|x\|^2 = \|L(x)\|^2 \leq \|x^*x\|$, the substantive inequality; combined with the reverse bound, $\|x^*x\| = \|x\|^2$. In particular the involution is isometric ($\|x\|^2 = \|x^*x\| \leq \|x^*\|\,\|x\|$ gives $\|x\| \leq \|x^*\|$, and symmetry gives equality), so $\tilde{A}$ is a unital C^*-algebra.

Step 5: the quotient and functoriality. The map $q\colon \tilde{A} \to \mathbb{C}$, $q(a, \lambda) = \lambda$, is a surjective unital $*$-homomorphism with kernel $A \oplus 0$, so $\tilde{A}/A \cong \mathbb{C}$ by the first isomorphism theorem. For functoriality, given a $*$-homomorphism $\varphi\colon A \to B$, the map $\tilde{\varphi}(a, \lambda) = (\varphi(a), \lambda)$ is a unital $*$-homomorphism: it respects the multiplication and involution because φ does and fixes the $\mathbb{C}$-coordinate, and it restricts to φ on A. It is the unique unital extension, since its values on A and on the unit $(0, 1)$ determine it on all of $\tilde{A} = A + \mathbb{C} \cdot (0, 1)$. The identities $\widetilde{\psi \circ \varphi} = \tilde{\psi} \circ \tilde{\varphi}$ and $\widetilde{\mathrm{id}_A} = \mathrm{id}_{\tilde{A}}$ are immediate from the formula, so unitization is a functor, completing the proof.

The operator norm does the work: $\tilde{A}$ is realized as an algebra of operators on the Banach space A, and the C^*-identity is inherited from that concrete realization rather than imposed by fiat. It is also forced, not chosen: a C^*-algebra carries at most one norm satisfying the C^*-identity, because that norm is determined by the algebraic structure through $\|x\|^2 = \|x^*x\| = r(x^*x)$ and the spectral radius of theorem 1.1.3. The operator-norm formula is simply the concrete presentation of that unique norm. When A is already unital the construction does not return A: the original unit 1_A becomes the idempotent $(1_A, 0)$, the adjoined $(0, 1)$ is a *different* unit for the larger algebra, and $\tilde{A}$ strictly enlarges A by one dimension. The next example makes this precise and identifies the non-unital case with a compactification.

Example 1.4.6: Unitization of $C_0(X)$ and of a Unital Algebra

Two computations locate $\tilde{A}$ concretely.

1. *One-point compactification.* Let X be a locally compact Hausdorff space and $X^+ = X \cup \{\infty\}$ its one-point compactification, the compact Hausdorff space whose open sets are the open subsets of X together with the complements in X^+ of compact subsets of X. Then

$$\widetilde{C_0(X)} \cong C(X^+)$$

as unital C^*-algebras. The isomorphism sends $(f, \lambda) \in \widetilde{C_0(X)}$ to the function $g \in C(X^+)$ with $g|_X = f + \lambda$ and $g(\infty) = \lambda$; that is, λ is the value at the point at infinity and $f = g - g(\infty)$ is the part vanishing at infinity, which lies in $C_0(X)$. This g is continuous on X^+: continuity at points of X is inherited from f, and continuity at ∞ says $g(x) \to \lambda$ as $x \to \infty$, which holds because $f \in C_0(X)$ vanishes at infinity. The map is a bijective $*$-homomorphism, and since an injective $*$-homomorphism between C^*-algebras is isometric (theorem 1.3.3), it is an isometric $*$-isomorphism. Adjoining a unit to $C_0(X)$ is thus exactly compactifying X by a single point: the unit is the constant function 1 on X^+, and $C_0(X)$ sits inside as the maximal ideal of functions vanishing at ∞, matching $\widetilde{C_0(X)}/C_0(X) \cong \mathbb{C}$ as evaluation at ∞.

2. *The unital case.* If A is already unital with unit 1_A, then as an algebra $\tilde{A} \cong A \oplus \mathbb{C}$ with the direct-sum (coordinatewise) multiplication, the isomorphism being $(a, \lambda) \mapsto (a + \lambda 1_A,\ \lambda)$. Under it the adjoined unit $(0, 1)$ maps to $(1_A, 1)$, so $\tilde{A}$ splits as the product C^*-algebra $A \oplus \mathbb{C}$: the unitization of a unital algebra is A with a disjoint copy of $\mathbb{C}$ glued on. Geometrically, for $A = C(Y)$ with Y compact, $X = Y$ is already compact, so $X^+ = Y \sqcup \{\infty\}$ is Y with an isolated point, and $C(Y \sqcup \{\infty\}) \cong C(Y) \oplus \mathbb{C}$ recovers the algebraic splitting.

The unitization is what lets later chapters reduce a non-unital statement to the unital case: to prove something about A, embed it in $\tilde{A}$, run the unital theory of chapter 2 and its successors there, and read the conclusion back along the ideal $A \subseteq \tilde{A}$. The spectrum of an element $a \in A$, for instance, is defined via $\tilde{A}$, and the K-theory of chapter 6 defines the invariants of a non-unital A through $\tilde{A}$ and the splitting $\tilde{A}/A \cong \mathbb{C}$.

2

Spectral Theory and the Continuous Functional Calculus

With the C^*-identity in hand, the spectrum of section 1.1 acquires the rigidity that turns it from an existence statement into a computational tool. A self-adjoint element has real spectrum, a normal element has spectral radius equal to its norm, and the spectrum of an element is the same whether it is computed in the whole algebra or in any C^*-subalgebra containing it. These facts, assembled in the first section, are what let the second carry out the construction the rest of the book runs on: the continuous functional calculus, which turns a continuous function on the spectrum of a normal element a into an element of the algebra and gives an isometric identification $C(\sigma(a)) \cong C^*(a, 1)$.

The functional calculus is the single-element form of the dictionary between algebra and geometry: it reads a normal element a as the space $\sigma(a)$ together with the coordinate function on it. The last two sections put it to work on the order structure. An element is positive when it is self-adjoint with spectrum in $[0, \infty)$, and the functional calculus shows this to be the same as being of the form b^*b; the positive elements then form a closed cone that orders the self-adjoint part of the algebra, and every positive element has a unique positive square root, hence an absolute value $|a| = (a^*a)^{1/2}$.

2.1 Spectrum in a C^*-Algebra

The spectrum $\sigma(a)$ of an element of a Banach algebra was defined in section 1.1: it is the set of scalars $\lambda \in \mathbb{C}$ for which $a - \lambda 1$ fails to be invertible. In that generality the spectrum is a nonempty compact subset of $\mathbb{C}$, and little more can be said. The involution and the C^*-identity of definition 1.2.1 sharpen this picture decisively. Two facts underlie the functional calculus. First, the geometry of the spectrum is pinned down by the algebraic type of the element: a self-adjoint element has spectrum on the real line, a unitary element on the unit circle. Second, for the well-behaved (normal) elements the spectral radius, a purely spectral quantity, coincides exactly with the norm, an analytic quantity. Together these say that for a normal element the spectrum sees the norm and the norm sees the spectrum, which is what makes the continuous functional calculus of the next section possible.

A third question is one of well-posedness. The spectrum of a is defined by invertibility, and invertibility is a statement about the ambient algebra: an element can be non-invertible in a small algebra yet become invertible in a larger one containing more elements to serve as its inverse. So $\sigma(a)$ could in principle depend on which C^*-algebra one computes it in. This section closes by showing it does not: the spectrum is the same in every unital C^*-subalgebra containing a. This *spectral permanence* is what lets us speak of $\sigma(a)$ without naming an ambient algebra, and it is indispensable once the functional calculus is used to pass between an element and the small algebra it generates.

Throughout, A is a unital C^*-algebra. Recall that $a \in A$ is *self-adjoint* if $a = a^*$, *normal* if $aa^* = a^*a$, and *unitary* if $a^*a = aa^* = 1$. The self-adjoint elements form the real subspace A_{sa}, and the unitaries the group $\mathcal{U}(A)$.

The first constraint is the one self-adjointness places on the spectrum. In the scalar case the analogue is elementary: a real number has real spectrum, and a complex number of modulus one lies on the unit circle. The content of the next result is that these scalar facts persist in a noncommutative C^*-algebra, and the mechanism is the C^*-identity applied to the imaginary translate $a + ir1$.

Proposition 2.1.1: Spectrum of Self-Adjoint and Unitary Elements

Let A be a unital C^*-algebra.

1. If $a \in A$ is self-adjoint, then $\sigma(a) \subseteq \mathbb{R}$.
2. If $u \in A$ is unitary, then $\sigma(u) \subseteq \{\lambda \in \mathbb{C} \big| |\lambda| = 1\}$.

***Proof*:**
Part (1): self-adjoint elements. Let $a = a^*$ and suppose $\lambda = s + it \in \sigma(a)$ with $s, t \in \mathbb{R}$; the claim is $t = 0$. The device is to translate a by an arbitrary imaginary scalar $ir1$, bound the modulus of the shifted spectral value by the norm of the shift via the C^*-identity, and then let r run to infinity, where the bound can hold only if $t = 0$.

Fix $r \in \mathbb{R}$. Adding the scalar $ir1$ translates the spectrum, so from $\lambda \in \sigma(a)$ one gets $\lambda + ir = s + i(t + r) \in \sigma(a + ir1)$. A spectral value never exceeds the norm in modulus, so

$$s^2 + (t + r)^2 = |\lambda + ir|^2 \leq \|a + ir1\|^2 \tag{2.1}$$

Bound the right side with the C^*-identity. Because a is self-adjoint, $(a + ir1)^* = a - ir1$, and therefore

$$(a + ir1)^*(a + ir1) = (a - ir1)(a + ir1) = a^2 + r^2 1$$

The C^*-identity of definition 1.2.1 gives $\|a + ir1\|^2 = \|(a + ir1)^*(a + ir1)\| = \|a^2 + r^2 1\|$, which the triangle inequality bounds by $\|a\|^2 + r^2$. Substituting into (2.1),

$$s^2 + (t + r)^2 \leq \|a\|^2 + r^2$$

Expanding $(t + r)^2 = t^2 + 2rt + r^2$ and cancelling the common r^2,

$$s^2 + t^2 + 2rt \leq \|a\|^2 \tag{2.2}$$

The inequality (2.2) holds for every $r \in \mathbb{R}$, yet its left side is linear in r with slope $2t$. If $t > 0$ the left side tends to $+\infty$ as $r \to +\infty$, and if $t < 0$ it does so as $r \to -\infty$; in either case (2.2) fails for $|r|$ large. Hence $t = 0$, so $\lambda = s \in \mathbb{R}$, giving $\sigma(a) \subseteq \mathbb{R}$.

Part (2): unitary elements. Let $u^*u = uu^* = 1$. A unitary has norm one: the C^*-identity gives $\|u\|^2 = \|u^*u\| = \|1\| = 1$. Its inverse $u^{-1} = u^*$ is again unitary, so $\|u^{-1}\| = 1$ as well. Let $\lambda \in \sigma(u)$. Since u is invertible, $\lambda \neq 0$, and $|\lambda| \leq \|u\| = 1$. For the reverse bound, factor

$$u - \lambda 1 = -\lambda u(u^{-1} - \lambda^{-1}1)$$

whose left factor $-\lambda u$ is invertible; hence $u - \lambda 1$ is invertible if and only if $u^{-1} - \lambda^{-1}1$ is, and non-invertibility of the former (from $\lambda \in \sigma(u)$) forces $\lambda^{-1} \in \sigma(u^{-1})$. Therefore $|\lambda^{-1}| \leq \|u^{-1}\| = 1$, that is $|\lambda| \geq 1$. Combining the two bounds, $|\lambda| = 1$, so $\sigma(u)$ lies on the unit circle, as we sought to show.

The two statements are the noncommutative form of the observation that a self-adjoint operator on a Hilbert space has real eigenvalues and a unitary operator has eigenvalues of modulus one. No Hilbert space appears here: the conclusions are read off the abstract C^*-axioms alone, via the single identity $\|x\|^2 = \|x^*x\|$. This is the first place the involution does work that the norm and the algebra structure could not do by themselves, and the pattern, choose an auxiliary element whose x^*x is computable and apply the C^*-identity, recurs throughout the chapter.

Example 2.1.2: The Cayley Transform in a C^*-Algebra

The two parts of proposition 2.1.1 are linked by an explicit construction. Let $a = a^*$ be self-adjoint. By part (1), $\sigma(a) \subseteq \mathbb{R}$, so $a - i1$ and $a + i1$ are invertible (their spectra miss 0), and the *Cayley transform*

$$u = (a - i1)(a + i1)^{-1}$$

is defined. A direct check shows u is unitary: since a commutes with $a \pm i1$ and with their inverses, all four factors below commute, so

$$u^*u = (a + i1)^{-1}(a^* + i1)(a^* - i1)(a - i1)^{-1} = (a + i1)^{-1}(a^2 + 1)(a - i1)^{-1} = 1$$

using $a^* = a$ and $(a^* + i1)(a^* - i1) = a^2 + 1 = (a + i1)(a - i1)$. The same computation gives $uu^* = 1$. Concretely, take $A = M_2(\mathbb{C})$ and $a = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right)$; then $\sigma(a) = \{0, 1\} \subseteq \mathbb{R}$, and the scalar Cayley map $t \mapsto (t - i)/(t + i)$ sends $0 \mapsto -1$ and $1 \mapsto (1 - i)/(1 + i) = -i$, so $\sigma(u) = \{-1, -i\}$, two points on the unit circle, as proposition 2.1.1(2) requires.

The next result is where the C^*-identity makes the spectral radius computable. In a general Banach algebra the spectral radius $r(a)$ is computed by the spectral radius formula $r(a) = \lim_n \|a^n\|^{1/n}$ of theorem 1.1.3, and it satisfies $r(a) \leq \|a\|$ but generally nothing better; a nilpotent element has $r(a) = 0$ while $\|a\|$ is arbitrary. For a normal element the inequality becomes an equality. This is the analytic heart of the functional calculus: it says the norm on the C^*-subalgebra generated by a normal element is determined by the spectrum, so functions of that element can be normed by their supremum on $\sigma(a)$.

Proposition 2.1.3: Spectral Radius of a Normal Element

Let A be a unital C^*-algebra and $a \in A$ normal. Then

$$r(a) = \|a\|$$

In particular this holds for self-adjoint and for unitary elements.

***Proof*:**
The argument is in three steps: first establish the norm-doubling identity $\|x^2\| = \|x\|^2$ for self-adjoint x, bootstrap it to $\|a^{2^k}\| = \|a\|^{2^k}$ for normal a, and then feed the resulting sequence into the spectral radius formula.

Step 1: $\|x^2\| = \|x\|^2$ for self-adjoint x. Let $x = x^*$. The C^*-identity of definition 1.2.1 applied to x gives $\|x\|^2 = \|x^*x\| = \|x^2\|$, since $x^*x = x^2$. This is the identity claimed.

Step 2: $\|a^{2^k}\| = \|a\|^{2^k}$ for normal a. Let a be normal, so $a^*a = aa^*$. The element a^*a is self-adjoint, and more generally powers of a interact cleanly with the involution because a and a^* commute: $(a^n)^*a^n = (a^*)^n a^n = (a^*a)^n$. Apply the C^*-identity to $x = a^n$:

$$\|a^n\|^2 = \|(a^n)^*a^n\| = \|(a^*a)^n\| \tag{2.3}$$

The element a^*a is self-adjoint, so Step 1 applies to it and its self-adjoint powers. Taking $n = 2^k$ in (2.3) and using that a^*a is self-adjoint with $\|(a^*a)^{2^k}\| = \|(a^*a)^{2^{k-1}}\|^2$ obtained by iterating Step 1,

$$\|a^{2^k}\|^2 = \|(a^*a)^{2^k}\| = \|a^*a\|^{2^k} = \|a\|^{2^{k+1}}$$

where the last equality is the C^*-identity $\|a^*a\| = \|a\|^2$ raised to the power 2^k. Taking square roots,

$$\|a^{2^k}\| = \|a\|^{2^k} \tag{2.4}$$

for every $k \geq 0$.

Step 3: apply the spectral radius formula. By theorem 1.1.3 the spectral radius is $r(a) = \lim_{n\to\infty} \|a^n\|^{1/n}$, and the limit exists and equals the infimum. It may therefore be computed along any subsequence, in particular along $n = 2^k$. Using (2.4),

$$r(a) = \lim_{k\to\infty} \|a^{2^k}\|^{1/2^k} = \lim_{k\to\infty} \left(\|a\|^{2^k}\right)^{1/2^k} = \|a\|$$

This proves $r(a) = \|a\|$ for normal a. Self-adjoint and unitary elements are normal, so the equality holds for them as well, completing the proof.

The equality $r(a) = \|a\|$ is what makes a C^*-algebra rigid where a general Banach algebra is slack. In a Banach algebra one may renorm within bounds without changing the algebra, and the norm carries information the spectrum cannot see. Here, for the normal elements, that slack is gone: the norm is a spectral invariant. A first structural consequence is uniqueness of the C^*-norm. If $\|\cdot\|_1$ and $\|\cdot\|_2$ are two norms making the same $*$-algebra a C^*-algebra, then for self-adjoint a both equal $r(a)$, which is computed from the algebraic spectrum alone, and for general a the C^*-identity gives $\|a\|_j^2 = \|a^*a\|_j = r(a^*a)$; the two norms agree. The other consequence is the one the next section exploits: on the commutative subalgebra generated by a single normal element, every element is normal, so its norm is the supremum of the modulus of the corresponding function on the spectrum.

Example 2.1.4: The Nilpotent Jordan Block

The normality hypothesis is not decorative. In $A = M_2(\mathbb{C})$ take the non-normal element $a = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, a single nilpotent Jordan block. It satisfies $a^2 = 0$, so $\sigma(a) = \{0\}$ and $r(a) = 0$, yet the operator norm is $\|a\| = 1$. Here $a^*a = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right) \neq \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right) = aa^*$, so a is not normal and proposition 2.1.3 does not apply; indeed $r(a) = 0 < 1 = \|a\|$. Contrast the normal element $a = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ of example 2.1.2: there $\sigma(a) = \{0, 1\}$, so $r(a) = 1 = \|a\|$, in agreement with the proposition.

The remaining question is whether $\sigma(a)$ depends on the ambient algebra. The definition through invertibility shows the concern. If $B \subseteq A$ is a unital C^*-subalgebra (sharing the unit of A) and $a \in B$, an element $a - \lambda 1$ might have no inverse inside B yet acquire one inside the larger algebra A. A priori, then, $\sigma_B(a)$, the spectrum computed in B, could be strictly larger than $\sigma_A(a)$. That the two coincide is the content of *spectral permanence*. The obstruction to overcome is genuine, and it is instructive to see where it does *not* vanish: for a general (non-self-adjoint) Banach subalgebra the spectrum really can shrink upon enlargement, and the extra hypothesis making it stable is that we are in a C^*-algebra, where invertibility is controlled by the self-adjoint element a^*a.

The engine is the following lemma, which isolates the self-adjoint case, where the real spectrum forces the resolvent set to be connected and thus prevents the spectrum from shrinking.

Lemma 2.1.5: Invertibility of Self-Adjoint Elements is Subalgebra-Independent

Let $B \subseteq A$ be a unital C^*-subalgebra of the unital C^*-algebra A, sharing the unit. If $b \in B$ is self-adjoint and b is invertible in A, then $b^{-1} \in B$; equivalently b is invertible in B.

***Proof*:**

Since $b = b^*$, proposition 2.1.1 gives $\sigma_A(b) \subseteq \mathbb{R}$. The element b is invertible in A, so $0 \notin \sigma_A(b)$. The claim is that b^{-1}, formed in A, already lies in B.

Consider, for $t \in \mathbb{R} \setminus \{0\}$, the element $b - it1 \in B$. Its adjoint is $b + it1$, and because $\sigma_A(b) \subseteq \mathbb{R}$ the scalar $it \notin \sigma_A(b)$, so $b - it1$ is invertible in A. The point is that its inverse is manufactured from b by a convergent series that stays in B. Indeed, $b - it1 = -it(1 + ib/t)$, and $\|(i/t)b\| = \|b\|/|t| < 1$ once $|t| > \|b\|$; for such t the Neumann series

$$(b - it1)^{-1} = \frac{-1}{it} \sum_{n=0}^{\infty} \left(\frac{-i}{t} b\right)^n$$

converges in A. Every partial sum is a polynomial in b with the unit as constant term, hence lies in B (which contains 1 and b, so all their products and sums), and B is norm-closed, so the limit lies in B. Thus $(b - it1)^{-1} \in B$ for all $|t| > \|b\|$.

Now let $R_A = \{\lambda \in \mathbb{C} \,|\, b - \lambda 1 \text{ invertible in } A\}$ be the resolvent set in A and R_B the resolvent set in B. The set G_B of elements invertible in B is open in B, and inversion is continuous on it, so R_B is open; the same holds for R_A, and always $R_B \subseteq R_A$. The topological fact the argument turns on is that R_A is *connected* here: its complement $\sigma_A(b)$ is a compact subset of $\mathbb{R}$, and the complement in $\mathbb{C}$ of a compact subset of the real line is path-connected (any two points are joined by a path routed through the upper half-plane, which meets $\mathbb{R}$ nowhere). On the connected set R_A the resolvent $\lambda \mapsto (b - \lambda 1)^{-1}$ is A-valued and continuous.

Consider the subset $S = \{\lambda \in R_A \,|\, (b - \lambda 1)^{-1} \in B\} \subseteq R_A$. It is nonempty: it contains every

$\lambda = it$ with $|t| > \|b\|$ by the previous paragraph. It is open in R_A: if $(b - \lambda_0 1)^{-1} \in B$, then for λ near λ_0 the Neumann series

$$(b - \lambda 1)^{-1} = \big((b - \lambda_0 1) - (\lambda - \lambda_0)1\big)^{-1} = (b - \lambda_0 1)^{-1} \sum_{n=0}^{\infty} \big((\lambda - \lambda_0)(b - \lambda_0 1)^{-1}\big)^n$$

converges (for $|\lambda - \lambda_0| < \|(b - \lambda_0 1)^{-1}\|^{-1}$) with all terms in B, so its sum is in B. It is closed in R_A: if $\lambda_j \to \lambda \in R_A$ with $(b - \lambda_j 1)^{-1} \in B$, then by continuity of the resolvent on R_A the inverses converge in A to $(b - \lambda 1)^{-1}$, and B is norm-closed, so the limit lies in B. A nonempty subset of the connected set R_A that is both open and closed is all of R_A; hence $S = R_A$. In particular $0 \in R_A$ lies in S, so $b^{-1} = (b - 0 \cdot 1)^{-1} \in B$, as we sought to show.

The argument needs the spectrum to be real for a precise reason: connectedness of R_A is what let the clopen set S propagate from the far-off values $\lambda = it$ all the way in to $\lambda = 0$. For a non-self-adjoint element the spectrum can be a full disc or annulus and R_A can be disconnected, breaking the propagation; this is exactly the loophole through which the spectrum of a general element can shrink in a larger algebra. The C^*-identity closes the loophole by routing every invertibility question through the self-adjoint element a^*a, whose spectrum is real by proposition 2.1.1.

Theorem 2.1.6: Spectral Permanence

Let $B \subseteq A$ be a unital C^*-subalgebra of the unital C^*-algebra A, sharing the unit, and let $a \in B$. Then

$$\sigma_B(a) = \sigma_A(a)$$

The spectrum of a is the same computed in B or in A; in particular it depends only on a and not on the ambient C^*-algebra.

***Proof*:**

The inclusion $\sigma_A(a) \subseteq \sigma_B(a)$ is automatic and holds for any pair of unital algebras: if $a - \lambda 1$ is not invertible in the larger algebra A, then a fortiori it has no inverse in the smaller $B \subseteq A$ (an inverse in B would be an inverse in A). The content is the reverse inclusion $\sigma_B(a) \subseteq \sigma_A(a)$, equivalently: if $a - \lambda 1$ is invertible in A, it is already invertible in B. Translating by λ, it suffices to prove the case $\lambda = 0$: if $a \in B$ is invertible in A, then $a^{-1} \in B$.

So suppose $a \in B$ is invertible in A. Then $a^* \in B$ is invertible in A (with inverse $(a^{-1})^*$), and hence so is $a^*a \in B$, with inverse $a^{-1}(a^*)^{-1} \in A$. The element a^*a is self-adjoint, so lemma 2.1.5 applies: its inverse lies in B, that is $(a^*a)^{-1} \in B$. But then

$$a^{-1} = (a^*a)^{-1} a^*$$

because $(a^*a)^{-1}a^* \cdot a = (a^*a)^{-1}(a^*a) = 1$, and a is invertible in A so this left inverse is the inverse. The right side lies in B, since $(a^*a)^{-1} \in B$ and $a^* \in B$. Therefore $a^{-1} \in B$, which is the reverse inclusion. Combining the two inclusions gives $\sigma_B(a) = \sigma_A(a)$, completing the proof.

Spectral permanence removes the ambient algebra from the notation $\sigma(a)$ and is a licence used constantly in what follows. The functional calculus of the next section is built inside the small algebra $C^*(a, 1)$ generated by a single normal element; permanence guarantees that the spectrum computed there is the same as in any larger algebra, so passing a into or out of $C^*(a, 1)$ never changes $\sigma(a)$. It also settles a matter of bookkeeping that would otherwise recur: statements about $\sigma(a)$ proved in a convenient representation transfer verbatim to any other. The result is special to C^*-algebras. A closed unital *Banach* subalgebra can carry a strictly larger spectrum, the standard example being the disc algebra $A(\mathbb{D})$ of functions holomorphic on the open unit disc and continuous on its closure, sitting inside $C(\partial\mathbb{D})$: the coordinate function z is invertible in $C(\partial\mathbb{D})$ but not in $A(\mathbb{D})$, so its spectrum is the full closed disc in $A(\mathbb{D})$ and only the circle in $C(\partial\mathbb{D})$. The involution and the C^*-identity are exactly what rule this out.

2.2 The Continuous Functional Calculus

The previous section made the spectrum of a normal element computable: its spectral radius equals its norm (proposition 2.1.3), and the answer does not depend on which unital C^*-subalgebra the computation is carried out in (theorem 2.1.6). These two facts are exactly what is needed to promote polynomials in a to arbitrary continuous functions on the spectrum. For a normal element a, every $f \in C(\sigma(a))$ will name an element $f(a) \in C^*(a, 1)$, and the assignment $f \mapsto f(a)$ will be an isometric $*$-isomorphism onto the whole subalgebra $C^*(a, 1)$. This is the continuous functional calculus; the remaining sections apply it directly, since positivity, square roots, and the absolute value are all continuous functions applied to a.

The construction proceeds in two steps. First the polynomial functional calculus, where f is a polynomial in the two commuting variables a and a^*; here everything is explicit and the key isometry is a direct consequence of proposition 2.1.3. Then the passage to a general continuous f by approximating it uniformly with polynomials, using the Stone-Weierstrass theorem, and extending the isometric map by continuity.

Fix a normal element a of a unital C^*-algebra A, so that $aa^* = a^*a$. Because a and a^* commute, they generate a *commutative* unital $*$-subalgebra: the set of finite sums

$$p(a, a^*) = \sum_{j,k\geq 0} c_{jk}\, a^j (a^*)^k, \qquad c_{jk} \in \mathbb{C}$$

is closed under multiplication (the factors commute, so monomials multiply by adding exponents) and under the involution (which sends $a^j(a^*)^k$ to $a^k(a^*)^j$). Its norm closure is $C^*(a, 1)$.

To make this precise, let $\mathbb{C}[z, \overline{z}]$ denote the algebra of complex polynomials in two commuting indeterminates z and $\overline{z}$, equipped with the involution $z \mapsto \overline{z}$ extended conjugate-linearly, that is, $\overline{p}(z, \overline{z}) = \overline{p(\overline{z}, z)}$ with the coefficients conjugated. A polynomial is written $p(z, \overline{z}) = \sum_{j,k} c_{jk} z^j \overline{z}^k$, and as a function on $\mathbb{C}$ it takes the value $p(\lambda, \overline{\lambda})$ at λ; every polynomial in the real and imaginary parts of λ arises this way, since $\operatorname{Re}\lambda = (\lambda + \overline{\lambda})/2$ and $\operatorname{Im}\lambda = (\lambda - \overline{\lambda})/2i$.

Proposition 2.2.1: The Polynomial Functional Calculus

Let a be a normal element of a unital C^*-algebra A. The map

$$\Phi_0 \colon \mathbb{C}[z,\overline{z}] \to C^*(a,1), \qquad p(z,\overline{z}) \mapsto p(a,a^*)$$

obtained by substituting $z = a$, $\overline{z} = a^*$, is a unital $*$-homomorphism, and for every $p \in \mathbb{C}[z,\overline{z}]$ it satisfies the isometry

$$\|p(a,a^*)\| = \sup_{\lambda\in\sigma(a)} \left|p(\lambda,\overline{\lambda})\right| \tag{2.5}$$

***Proof*:**
Step 1: Φ_0 is a unital $*$-homomorphism. Substitution of commuting elements into polynomials respects addition and scalar multiplication automatically. It respects multiplication because a and a^* commute: a monomial $z^j\overline{z}^k$ maps to $a^j(a^*)^k$, and since these factors commute, $(z^j\overline{z}^k)(z^m\overline{z}^n) = z^{j+m}\overline{z}^{k+n}$ maps to $a^{j+m}(a^*)^{k+n} = (a^j(a^*)^k)(a^m(a^*)^n)$; linearity extends this to all products. The constant polynomial 1 maps to the unit, so Φ_0 is unital. Finally Φ_0 intertwines the involutions: the monomial $z^j\overline{z}^k$ has involution $\overline{z}^jz^k = z^k\overline{z}^j$, which maps to $a^k(a^*)^j = (a^j(a^*)^k)^*$, and conjugate-linearity extends this to all of $\mathbb{C}[z,\overline{z}]$.

Step 2: $p(a,a^*)$ is normal, and its spectrum is $\{p(\lambda,\overline{\lambda}) : \lambda \in \sigma(a)\}$. Set $b = p(a,a^*)$. Since b lies in the commutative $*$-algebra generated by a and a^*, it commutes with its own adjoint $b^* = \overline{p}(a,a^*)$, so b is normal. All spectra below may be computed in $B := C^*(a,1)$ rather than in A, since theorem 2.1.6 makes them agree, and B is a *commutative* unital C^*-algebra. The goal is

$$\sigma\big(p(a,a^*)\big) = \big\{p(\lambda,\overline{\lambda}) : \lambda \in \sigma(a)\big\} =: V \tag{2.6}$$

The engine is a single elementary fact about a commutative unital Banach algebra B: an element $c \in B$ is non-invertible if and only if it lies in some maximal ideal $\mathfrak{m}$, and the quotient $B/\mathfrak{m}$ is a Banach division algebra, hence equal to $\mathbb{C}$ by the Gelfand-Mazur theorem [9, Gelfand-Mazur]. The quotient map is therefore a nonzero algebra homomorphism $\pi \colon B \to \mathbb{C}$ with $\pi(c) = 0$; conversely, for any nonzero homomorphism $\pi \colon B \to \mathbb{C}$, $\pi(1) = 1$ and $\pi(c - \pi(c)) = 0$ forces $c - \pi(c)$ into $\ker\pi$, a proper ideal, so $\pi(c) \in \sigma(c)$. This does not require the identification of B with a function algebra, the Gelfand-Naimark theorem of the next chapter; only the one-ideal-at-a-time consequence of Gelfand-Mazur is used. Two features of such a π are needed. First, π is $*$-preserving: for self-adjoint $s \in B$ one has $\pi(s) \in \sigma(s) \subseteq \mathbb{R}$ by proposition 2.1.1, and writing an arbitrary element as $c = s_1 + is_2$ with s_1, s_2 self-adjoint gives $\pi(c^*) = \pi(s_1) - i\pi(s_2) = \overline{\pi(c)}$. Second, since $b = p(a,a^*)$ is a polynomial in a and a^*, multiplicativity and the previous point give

$$\pi\big(p(a,a^*)\big) = p\big(\pi(a), \overline{\pi(a)}\big) \tag{2.7}$$

$V \subseteq \sigma(b)$. Fix $\lambda_0 \in \sigma(a)$. Then $a - \lambda_0$ is non-invertible, so it lies in a maximal ideal, yielding a homomorphism $\pi \colon B \to \mathbb{C}$ with $\pi(a) = \lambda_0$. By (2.7), $\pi(b) = p(\lambda_0, \overline{\lambda}_0)$, and since $\pi(b) \in \sigma(b)$, the value $p(\lambda_0,\overline{\lambda}_0)$ lies in $\sigma(b)$.

$\sigma(b) \subseteq V$. Fix $\mu \in \sigma(b)$. Then $b - \mu$ is non-invertible, so a maximal ideal containing it gives a homomorphism $\pi \colon B \to \mathbb{C}$ with $\pi(b) = \mu$. Setting $\lambda := \pi(a)$, the point λ lies in $\sigma(a)$ (as

$\pi(a) \in \sigma(a)$), and (2.7) gives $\mu = \pi(b) = p(\lambda, \overline{\lambda}) \in V$. This establishes (2.6).

Step 3: the isometry. The element $b = p(a, a^*)$ is normal by Step 1, so proposition 2.1.3 gives $\|b\| = r(b)$, the spectral radius. By definition $r(b) = \sup\{|\mu| : \mu \in \sigma(b)\}$, and by (2.6) the spectrum of b is exactly $V = \{p(\lambda, \overline{\lambda}) : \lambda \in \sigma(a)\}$. Therefore

$$\|p(a, a^*)\| = r(b) = \sup_{\mu \in \sigma(b)} |\mu| = \sup_{\lambda \in \sigma(a)} \left|p(\lambda, \overline{\lambda})\right|$$

which is (2.5), completing the proof.

The right-hand side of (2.5) is the supremum norm of p regarded as a continuous function on the compact set $\sigma(a) \subseteq \mathbb{C}$; write it $\|p\|_{\sigma(a)}$. So the polynomial functional calculus is an *isometry* from the $*$-algebra of polynomial functions on $\sigma(a)$ (with the supremum norm) onto a dense $*$-subalgebra of $C^*(a, 1)$. Two features of this isometry drive the extension to continuous functions. First, $\|p(a, a^*)\|$ depends only on the values of p on $\sigma(a)$: two polynomials agreeing there produce the same element, so $p(a, a^*)$ depends on the *function* $\lambda \mapsto p(\lambda, \overline{\lambda})$, not on the formal expression. Second, an isometry is uniformly continuous, so it extends uniquely to the completion of its domain. The completion of the polynomial functions in the supremum norm is a closed $*$-subalgebra of $C(\sigma(a))$; the content of the next theorem is that this completion is all of $C(\sigma(a))$.

That the polynomials fill out $C(\sigma(a))$ is precisely the Stone-Weierstrass theorem. Recall its statement: a subalgebra of $C(X)$, for X compact Hausdorff, that contains the constants, separates points, and is closed under complex conjugation is dense in the supremum norm [6]. The algebra of functions $\lambda \mapsto p(\lambda, \overline{\lambda})$ contains the constants (take p constant), separates points of $\sigma(a) \subseteq \mathbb{C}$ (the coordinate function $\lambda \mapsto \lambda$ already does), and is closed under conjugation (the conjugate of $p(\lambda, \overline{\lambda})$ is $\overline{p}(\lambda, \overline{\lambda})$, again a polynomial). The presence of $\overline{z}$ as an independent variable is exactly what gives conjugation-closure; a calculus built from polynomials in a alone would not be dense unless $\sigma(a)$ happened to lie in a set where $\overline{\lambda}$ is already a polynomial in λ, such as the real line.

Theorem 2.2.2: The Continuous Functional Calculus

Let a be a normal element of a unital C^*-algebra A. There is a unique unital $*$-homomorphism

$$\Phi\colon C(\sigma(a)) \to A, \qquad f \mapsto \Phi(f)$$

carrying the identity function $\mathrm{id}_{\sigma(a)}\colon \lambda \mapsto \lambda$ to a. This Φ is isometric, its image is $C^*(a, 1)$, and it carries the constant function 1 to the unit. Writing $f(a) := \Phi(f)$, one has

$$\|f(a)\| = \sup_{\lambda \in \sigma(a)} |f(\lambda)|$$

for every $f \in C(\sigma(a))$, and $f(a)$ is normal, with $f(a)^* = \overline{f}(a)$.

***Proof*:**

Step 1: existence. Let $P \subseteq C(\sigma(a))$ be the subalgebra of functions of the form $\lambda \mapsto p(\lambda, \overline{\lambda})$ for $p \in \mathbb{C}[z, \overline{z}]$. By proposition 2.2.1 the assignment $p(\cdot, \bar{\cdot}) \mapsto p(a, a^*)$ is a well-defined isometry $P \to C^*(a, 1)$, since (2.5) shows the image depends only on the function and its norm equals the supremum norm. Call this isometry $\Phi_0 \colon P \to A$; it is a unital $*$-homomorphism onto a dense subalgebra of $C^*(a, 1)$.

By Stone-Weierstrass [6], P is dense in $C(\sigma(a))$: it contains the constants, separates points, and is closed under conjugation, as noted above. Given $f \in C(\sigma(a))$, choose polynomials p_n with $\|p_n(\cdot, \bar{\cdot}) - f\|_{\sigma(a)} \to 0$. The sequence $p_n(\cdot, \bar{\cdot})$ is Cauchy in $C(\sigma(a))$, hence, by the isometry, $\big(\Phi_0(p_n)\big)_n = \big(p_n(a, a^*)\big)_n$ is Cauchy in A; since A is complete it converges. Define

$$\Phi(f) := \lim_{n\to\infty} p_n(a, a^*)$$

The limit is independent of the approximating sequence: if $q_n(\cdot, \bar{\cdot}) \to f$ as well, then $\|p_n(a, a^*) - q_n(a, a^*)\| = \|p_n(\cdot, \bar{\cdot}) - q_n(\cdot, \bar{\cdot})\|_{\sigma(a)} \to 0$. Passing to the limit in $\|\Phi_0(p_n)\| = \|p_n(\cdot, \bar{\cdot})\|_{\sigma(a)}$ gives $\|\Phi(f)\| = \|f\|_{\sigma(a)}$, so Φ is isometric. The algebraic operations and the involution are continuous, so passing to limits in the identities for Φ_0 shows Φ is a unital $*$-homomorphism; in particular $\Phi(1) = 1$ and $\Phi(\mathrm{id}_{\sigma(a)}) = a$, the latter because $\mathrm{id}_{\sigma(a)}$ is the function $\lambda \mapsto \lambda$, itself a polynomial, sent by Φ_0 to a. Being an isometric $*$-homomorphism, Φ has closed image containing the dense subalgebra $\Phi_0(P)$ of $C^*(a, 1)$, so its image is exactly $C^*(a, 1)$. Finally $f(a)^* = \Phi(f)^* = \Phi(\overline{f}) = \overline{f}(a)$, and $f(a)$ commutes with $\overline{f}(a)$ because $C^*(a, 1)$ is commutative, so $f(a)$ is normal.

Step 2: uniqueness. Suppose $\Psi \colon C(\sigma(a)) \to A$ is any unital $*$-homomorphism with $\Psi(\mathrm{id}_{\sigma(a)}) = a$. A unital $*$-homomorphism sends the identity function to a and its conjugate $\overline{\mathrm{id}}_{\sigma(a)} \colon \lambda \mapsto \overline{\lambda}$ to a^*; being multiplicative and linear, it therefore sends every polynomial function $p(\cdot, \bar{\cdot})$ to $p(a, a^*) = \Phi(p(\cdot, \bar{\cdot}))$. So Ψ and Φ agree on the dense subalgebra P. A $*$-homomorphism between C^*-algebras is contractive (theorem 1.3.3), hence continuous, so two continuous maps agreeing on a dense set are equal: $\Psi = \Phi$, as we sought to show.

The theorem justifies the notation $f(a)$ for $\Phi(f)$ and makes it consistent with every prior meaning of the symbol. When f is a polynomial in z and $\overline{z}$, $f(a)$ is the explicit expression $p(a, a^*)$; when $f(\lambda) = \lambda$, $f(a) = a$; when $f(\lambda) = 1$, $f(a) = 1$. Because Φ is a $*$-homomorphism, the calculus is compatible with all the algebra: $(f + g)(a) = f(a) + g(a)$, $(fg)(a) = f(a)g(a)$, and $\overline{f}(a) = f(a)^*$. And because Φ is isometric, it is injective, so $f(a) = 0$ if and only if f vanishes on $\sigma(a)$; a relation among elements of $C^*(a, 1)$ holds if and only if the corresponding relation among functions holds on the spectrum. This is the sense in which a "is" the coordinate function on its own spectrum: the abstract element and the concrete function $\mathrm{id}_{\sigma(a)}$ are interchangeable inside $C^*(a, 1)$.

The uniqueness clause says there is nothing arbitrary in $f(a)$: any reasonable way of assigning elements to functions, so long as it is a unital $*$-homomorphism and sends $\lambda \mapsto \lambda$ to a, must produce the same $f(a)$. This is what lets one compute $f(a)$ by whatever means is convenient, most often by approximating f with polynomials, and be sure the answer does not depend on the method.

A concrete instance grounds the abstraction before it is used in earnest.

Example 2.2.3: The Calculus in $C(X)$

Let X be a compact Hausdorff space and $A = C(X)$, the commutative unital C^*-algebra of continuous functions with the supremum norm and pointwise involution $g^*(x) = \overline{g(x)}$. Every element $g \in C(X)$ is normal. Its spectrum is its range,

$$\sigma(g) = g(X) = \{g(x) : x \in X\}$$

since $g-\mu$ is invertible in $C(X)$ precisely when it vanishes nowhere, that is, when $\mu \notin g(X)$. For $f \in C(\sigma(g)) = C(g(X))$, the element $f(g) \in C(X)$ produced by the functional calculus is the composite

$$f(g) = f \circ g \colon x \mapsto f(g(x))$$

To verify this, note that $f \mapsto f \circ g$ is a unital $*$-homomorphism $C(g(X)) \to C(X)$ (composition preserves sums, products, and conjugates) and sends $\mathrm{id}_{g(X)}$ to g; by the uniqueness clause of theorem 2.2.2 it must be Φ. So in the commutative case the continuous functional calculus is nothing but composition with g, and the isometry $\|f(g)\|_\infty = \sup_x |f(g(x))| = \sup_{\lambda \in g(X)} |f(\lambda)|$ is the statement that the range of $f \circ g$ is $f(g(X))$. This is the model every noncommutative instance imitates.

The two governing properties of the calculus that remain are how it interacts with the spectrum and how repeated applications compose. Both are inherited from the polynomial case by continuity, and both are constantly used: the spectral mapping theorem is how one reads off the spectrum of $f(a)$, and the composition rule is what makes iterated calculus (a function of $|a|$, a function of a spectral projection) unambiguous.

Theorem 2.2.4: Spectral Mapping and Composition

Let a be a normal element of a unital C^*-algebra A.

1. For every $f \in C(\sigma(a))$,
$$\sigma\big(f(a)\big) = f\big(\sigma(a)\big) = \{f(\lambda) : \lambda \in \sigma(a)\}$$
2. For every $f \in C(\sigma(a))$ and every $g \in C\big(\sigma(f(a))\big) = C\big(f(\sigma(a))\big)$,
$$g\big(f(a)\big) = (g \circ f)(a)$$

***Proof*:**

1. *Spectral mapping.* The functional calculus $\Phi \colon C(\sigma(a)) \to C^*(a,1)$ is an isometric $*$-isomorphism onto $C^*(a,1)$ by theorem 2.2.2. An isometric $*$-isomorphism of unital C^*-algebras preserves spectra: for $f \in C(\sigma(a))$, the element $f(a) - \mu$ is invertible in $C^*(a,1)$ if and only if $f - \mu$ is invertible in $C(\sigma(a))$, since Φ carries inverses to inverses and is onto. By theorem 2.1.6 the spectrum of $f(a)$ in $C^*(a,1)$ equals its spectrum in A, so it is immaterial which algebra the invertibility is tested in. Now $f - \mu$ is invertible in $C(\sigma(a))$ exactly when it vanishes nowhere on $\sigma(a)$, that is, when $\mu \notin f(\sigma(a))$, as in example 2.2.3. Therefore
$$\mu \in \sigma\big(f(a)\big) \iff \mu \in f\big(\sigma(a)\big)$$
which is the claim. (The set $f(\sigma(a))$ is compact, being the continuous image of the compact set $\sigma(a)$, so it is closed and the spectrum comes out closed as it must.)
2. *Composition.* Fix f and set $b = f(a)$, a normal element with $\sigma(b) = f(\sigma(a))$ by part (1). Both sides of the asserted identity are unital $*$-homomorphisms $C(\sigma(b)) \to A$ in the variable g: the left side $g \mapsto g(b) = g(f(a))$ is the continuous functional calculus for b, and the right side $g \mapsto (g \circ f)(a)$ is the composite of the $*$-homomorphism $g \mapsto g \circ f$ (from $C(\sigma(b))$

to $C(\sigma(a))$, well-defined since $g \circ f$ makes sense on $\sigma(a)$ because $f(\sigma(a)) = \sigma(b))$ with the calculus Φ for a. To conclude they agree, by the uniqueness clause of theorem 2.2.2 it suffices to check both send $\mathrm{id}_{\sigma(b)}$ to b. The left side sends $\mathrm{id}_{\sigma(b)}$ to $\Phi_b(\mathrm{id}) = b$ by construction. The right side sends $\mathrm{id}_{\sigma(b)}$, the function $\mu \mapsto \mu$, to $(\mathrm{id}_{\sigma(b)} \circ f)(a) = f(a) = b$, since $\mathrm{id}_{\sigma(b)} \circ f = f$. The two homomorphisms agree on $\mathrm{id}_{\sigma(b)}$, hence are equal, so $g(f(a)) = (g \circ f)(a)$, completing the proof.

The spectral mapping theorem computes the spectrum of $f(a)$ is computed by simply applying f to the spectrum of a, no operator theory required. It reduces questions about the element $f(a)$ to questions about a function on a compact subset of $\mathbb{C}$. The composition rule says the calculus is functorial in the obvious way: applying g to $f(a)$ is the same as applying $g \circ f$ to a. Together they make expressions like $g(f(a))$ unambiguous and computable, and they guarantee that the elements one builds from a by successive continuous functions all live in the commutative algebra $C^*(a, 1)$, where they can be manipulated as functions on $\sigma(a)$.

The calculus is only as useful as the functions one is willing to feed it. The following examples exhibit the constructions the next two sections depend on, each obtained by naming a specific continuous function on $\sigma(a)$.

Example 2.2.5: Absolute Value, Spectral Projections, and Exponentials

Fix a normal element a of a unital C^*-algebra A.

1. *The absolute value.* The element a^*a is self-adjoint, so $\sigma(a^*a) \subseteq [0, \infty)$ (this uses positivity, developed in section 2.3; for now take the spectrum to be a compact subset of $[0, \infty)$). The function $t \mapsto \sqrt{t}$ is continuous on $[0, \infty)$, hence on $\sigma(a^*a)$, so the calculus produces
$$|a| := (a^*a)^{1/2} = f(a^*a), \qquad f(t) = \sqrt{t}$$
By spectral mapping, $\sigma(|a|) = \sqrt{\sigma(a^*a)} \subseteq [0, \infty)$, so $|a|$ is again a self-adjoint element with nonnegative spectrum. Applied to $A = C(X)$ this recovers the pointwise absolute value $|g|(x) = |g(x)|$, consistent with example 2.2.3, since $\sqrt{\overline{g(x)}g(x)} = |g(x)|$.

2. *Spectral projections.* Suppose $a = a^*$ is self-adjoint with $\sigma(a) \subseteq \mathbb{R}$, and suppose the spectrum splits as a disjoint union $\sigma(a) = K_0 \sqcup K_1$ of two nonempty *closed* (hence, by compactness, positively separated) sets, so that $\sigma(a)$ has an isolated piece. The indicator function
$$\chi\colon \sigma(a) \to \mathbb{C}, \qquad \chi(\lambda) = \begin{cases} 1 & \lambda \in K_1 \\ 0 & \lambda \in K_0 \end{cases}$$
is continuous on $\sigma(a)$, because K_0 and K_1 are separated, so it is a legitimate input to the calculus. The element
$$p := \chi(a)$$
satisfies $p = p^*$ (since $\overline{\chi} = \chi$) and $p^2 = p$ (since $\chi^2 = \chi$ as functions), so p is a self-adjoint idempotent, that is, a *projection* in A. By spectral mapping $\sigma(p) = \chi(\sigma(a)) = \{0, 1\}$ (both values are attained, as $K_0, K_1 \neq \emptyset$). This p is the spectral projection onto the part of a living over K_1: on $C^*(a, 1)$ it is the function that is 1 on K_1 and 0 on K_0, and $pa = ap$ is the "restriction" of a to that piece of its spectrum. Projections obtained this way are the raw material of the later K-theory, whose group K_0 is built from equivalence classes of projections. The construction requires the isolation of K_1: without a gap, no continuous function equals a nonconstant indicator, and the projection need not exist inside $C^*(a, 1)$.

3. *Exponentials.* The function $t \mapsto e^t$ is entire, so for any normal a the element

$$e^a := \exp(a) = f(a), \qquad f(\lambda) = e^\lambda$$

is defined by the calculus (with f restricted to $\sigma(a)$). Because the calculus is a $*$-homomorphism and $\lambda \mapsto e^\lambda$ is the uniform limit on the compact set $\sigma(a)$ of its Taylor polynomials $\sum_{k=0}^n \lambda^k/k!$, the element e^a agrees with the norm-convergent series $\sum_{k\geq 0} a^k/k!$; the functional-calculus definition and the power-series definition coincide. Spectral mapping gives $\sigma(e^a) = \{e^\lambda : \lambda \in \sigma(a)\} = \exp(\sigma(a))$. When $a = a^*$ is self-adjoint, $\sigma(a) \subseteq \mathbb{R}$ forces $\sigma(e^a) \subseteq (0, \infty)$, so e^a has strictly positive spectrum. Now take $u = e^{ia}$ with a self-adjoint, that is, $u = f(a)$ for $f(\lambda) = e^{i\lambda}$. Since λ is real, $\overline{f(\lambda)} = e^{-i\lambda}$, so $u^* = \overline{f}(a) = e^{-ia}$, and the calculus being a $*$-homomorphism gives $u^*u = e^{-ia}e^{ia} = e^0 = 1 = uu^*$, exhibiting u as a unitary. This is the operator analogue of writing a unit-modulus complex number as $e^{i\theta}$.

Each of these examples is a single continuous function evaluated at a, and each will reappear as a working tool. The absolute value $|a| = (a^*a)^{1/2}$ and the square root are the subject of section 2.4, where their uniqueness and their inequalities are established through the calculus. The spectral projections make the order structure of self-adjoint elements visible and are the bridge to the projections whose equivalence classes form K_0. The exponential realizes unitaries as e^{ia} and is the germ of the one-parameter groups $t \mapsto e^{ita}$ that organize the dynamics of self-adjoint elements. In every case the computation reduces, by spectral mapping, to reading a function off the compact set $\sigma(a) \subseteq \mathbb{C}$, so the element a is traded for the space $\sigma(a)$.

2.3 Positive Elements

The functional calculus turns a self-adjoint element into a function on its spectrum, and functions on a subset of $\mathbb{R}$ carry an order. This section transports that order back into the algebra. An element is positive when its spectrum sits in $[0, \infty)$; the content of the section is that this spectral condition, which mentions no product, is the same as the algebraic condition $a = b^*b$, and that the positive elements assemble into a cone whose induced order is the one on which every later positivity estimate rests.

The starting point is the spectral definition. It is the cleanest of the three equivalent conditions to state, and by spectral permanence it does not depend on which C^*-algebra one views a inside.

Definition 2.3.1: Positive Element

Let A be a unital C^*-algebra. An element $a \in A$ is *positive*, written $a \geq 0$, if $a = a^*$ and $\sigma(a) \subseteq [0, \infty)$. The set of positive elements is denoted A_+.

Positivity is intrinsic to the element, not an artifact of an ambient algebra. If $B \subseteq A$ is a unital C^*-subalgebra containing a, then spectral permanence (theorem 2.1.6) gives $\sigma_B(a) = \sigma_A(a)$, so whether $\sigma(a) \subseteq [0, \infty)$ is decided by a alone. In particular one may test positivity inside the commutative algebra $C^*(a, 1)$, where by the functional calculus (theorem 2.2.2) the element a is literally the function $\mathrm{id}_{\sigma(a)}$ and $a \geq 0$ says exactly that this function is nonnegative.

Example 2.3.2: Positive Functions and Positive Matrices
Two concrete algebras make the definition transparent.

1. In $C(X)$ for X compact Hausdorff, an element is a continuous function $f\colon X \to \mathbb{C}$, and $f = f^*$ means f is real-valued. The spectrum is the range, $\sigma(f) = f(X)$, so $f \geq 0$ in the C^*-algebra sense is exactly $f(x) \geq 0$ for every $x \in X$: positivity is pointwise positivity.
2. In $M_n(\mathbb{C}) = B(\mathbb{C}^n)$, a matrix a is self-adjoint when $a = a^*$, and then $\sigma(a) \subseteq \mathbb{R}$ consists of its eigenvalues. Thus $a \geq 0$ means every eigenvalue is nonnegative, which for a Hermitian matrix is the familiar condition $\langle a\xi, \xi\rangle \geq 0$ for all $\xi \in \mathbb{C}^n$. The abstract definition recovers the classical notion of a positive semidefinite matrix.

The two examples already display the pattern the main theorem generalizes. A positive function is a square, $f = (\sqrt{f})^2$, and a positive matrix is b^*b for $b = \sqrt{a}$; conversely b^*b has nonnegative eigenvalues because $\langle b^*b\,\xi, \xi\rangle = \|b\xi\|^2 \geq 0$. The following theorem says these coincidences are the general rule: the spectral, the factored, and the squared descriptions of positivity are one and the same. The first two implications are the functional calculus doing bookkeeping; the closing implication, that b^*b always has spectrum in $[0, \infty)$, is where the real work lies.

Theorem 2.3.3: Characterizations of Positivity

Let A be a unital C^*-algebra and $a \in A$. The following are equivalent.

1. $a \geq 0$.
2. $a = b^*b$ for some $b \in A$.
3. $a = c^2$ for some self-adjoint $c \geq 0$.

***Proof*:**
The argument runs (i)⇒(iii)⇒(ii)⇒(i); only the last step requires more than the functional calculus.

Step 1: (i)⇒(iii). Suppose $a \geq 0$. Then $a = a^*$ and $\sigma(a) \subseteq [0, \infty)$, so the function $t \mapsto \sqrt{t}$ is continuous on $\sigma(a)$. Apply the functional calculus (theorem 2.2.2) and set $c = f(a)$ where $f(t) = \sqrt{t}$. Since f is real-valued, $c = c^*$, and by the spectral mapping theorem (theorem 2.2.4)

$$\sigma(c) = f(\sigma(a)) = \{\sqrt{t} : t \in \sigma(a)\} \subseteq [0, \infty)$$

so $c \geq 0$. The functional calculus is a homomorphism sending the product $f \cdot f = \mathrm{id}_{\sigma(a)}$ to $c^2 = a$, since $\mathrm{id}_{\sigma(a)} \mapsto a$. Thus $a = c^2$ with c self-adjoint and positive.

Step 2: (iii)⇒(ii). If $a = c^2$ with $c = c^*$, then $a = c^*c$, so (ii) holds with $b = c$.

Step 3: (ii)⇒(i). This is the substantive direction. Suppose $a = b^*b$. First, a is self-adjoint, since $a^* = (b^*b)^* = b^*b = a$. The claim is that $\sigma(a) \subseteq [0, \infty)$.

Because a is self-adjoint, its spectrum is real (proposition 2.1.1). Split it by sign through the functional calculus. Define continuous functions on $\mathbb{R}$ by $f_+(t) = \max(t, 0)$ and $f_-(t) = \max(-t, 0)$, so that $f_+, f_- \geq 0$, $f_+(t) - f_-(t) = t$, and $f_+(t)f_-(t) = 0$ for all t. Set $a_+ = f_+(a)$ and $a_- = f_-(a)$. The functional calculus is a unital $*$-homomorphism on $C(\sigma(a))$, so these relations pass to the algebra:

$$a_+ \geq 0, \qquad a_- \geq 0, \qquad a = a_+ - a_-, \qquad a_+ a_- = 0$$

where $a_\pm \geq 0$ follows because $f_\pm \geq 0$ on $\sigma(a)$ (as in Step 1), and $a_+ a_- = 0$ because $f_+ f_- = 0$ on $\sigma(a)$. The two pieces are the positive and negative parts of a, and they annihilate each other. It suffices to show $a_- = 0$, for then $a = a_+ \geq 0$.

Consider the element $-a_-$ measured through $a = b^*b$. Put $d = b\,a_-$. Then, using $a_+ a_- = 0$ and a_- self-adjoint,

$$d^*d = a_- b^* b\, a_- = a_-\, a\, a_- = a_-(a_+ - a_-)a_- = -a_-^3$$

since $a_- a_+ = (a_+ a_-)^* = 0$. Now $a_- \geq 0$, so $a_-^3 = g(a_-)$ for $g(t) = t^3$, and by spectral mapping $\sigma(a_-^3) = \{t^3 : t \in \sigma(a_-)\} \subseteq [0, \infty)$; hence $a_-^3 \geq 0$ and therefore $d^*d = -a_-^3$ has $\sigma(d^*d) \subseteq (-\infty, 0]$.

On the other hand, write $d = x + iy$ with $x = (d + d^*)/2$ and $y = (d - d^*)/(2i)$ self-adjoint. Then

$$d^*d + dd^* = 2x^2 + 2y^2$$

Both x^2 and y^2 are positive: $x^2 = (x^*)^2$ has $\sigma(x^2) = \{s^2 : s \in \sigma(x)\} \subseteq [0, \infty)$ by spectral mapping, and likewise for y^2. The positive elements are closed under sums and nonnegative scalars (this is proposition 2.3.5, whose proof below does not use the present result), so $2x^2 + 2y^2 \geq 0$, and hence $dd^* = 2x^2 + 2y^2 - d^*d \geq 0$ as a difference $2x^2 + 2y^2 - d^*d = 2x^2 + 2y^2 + a_-^3$ of positive elements. Thus $\sigma(dd^*) \subseteq [0, \infty)$.

The two conclusions meet at a general spectral fact: for any elements u, v of a unital algebra, $\sigma(uv) \cup \{0\} = \sigma(vu) \cup \{0\}$, since $1 - uv$ is invertible if and only if $1 - vu$ is (a two-line identity for the candidate inverse). Applying this to $u = d^*$, $v = d$ gives $\sigma(d^*d) \cup \{0\} = \sigma(dd^*) \cup \{0\}$. But $\sigma(d^*d) \subseteq (-\infty, 0]$ and $\sigma(dd^*) \subseteq [0, \infty)$, so both spectra are forced into $\{0\}$; in particular $\sigma(d^*d) = \{0\}$. Since $d^*d = -a_-^3$ is self-adjoint, its norm equals its spectral radius (theorem 1.1.3 together with proposition 2.1.3), so $\|a_-^3\| = \|d^*d\| = r(d^*d) = 0$, whence $a_-^3 = 0$. Finally $a_- \geq 0$ gives $\|a_-\|^3 = \|a_-^3\| = 0$ by the same spectral-radius identity applied to a_-, so $a_- = 0$. Therefore $a = a_+ \geq 0$, completing the proof.

Positivity was defined spectrally, a condition about the location of $\sigma(a)$ that says nothing about multiplication; the equivalence with $a = b^*b$ ties it to the algebra's product and, through it, to the involution. This is why positivity is stable under the operations that preserve C^*-structure: a $*$-homomorphism π sends b^*b to $\pi(b)^*\pi(b)$, so it carries positive elements to positive elements, a fact that would be far from obvious from the spectral definition alone. The decomposition $a = a_+ - a_-$ produced inside the proof is itself worth keeping: every self-adjoint element splits canonically into orthogonal positive and negative parts, the operator-algebra analogue of writing a real function as the difference of its positive and negative parts.

Example 2.3.4: Positive and Negative Parts of a Self-Adjoint Operator

Let $a = a^*$ act on $\mathbb{C}^2$ as the diagonal matrix with entries 3 and -2, so $\sigma(a) = \{3, -2\}$. The functions $f_+(t) = \max(t, 0)$ and $f_-(t) = \max(-t, 0)$ evaluate on the spectrum to $f_+(3) = 3$,

$f_+(-2) = 0$ and $f_-(3) = 0$, $f_-(-2) = 2$, so

$$a_+ = \begin{pmatrix} 3 & 0 \\ 0 & 0 \end{pmatrix}, \qquad a_- = \begin{pmatrix} 0 & 0 \\ 0 & 2 \end{pmatrix}$$

Then $a_+, a_- \geq 0$, their product $a_+ a_- = 0$ vanishes because they are supported on complementary eigenspaces, and $a = a_+ - a_-$. Writing $a = b^*b$ requires b with $\sigma(b^*b) \subseteq [0, \infty)$, which is impossible here since $-2 \in \sigma(a)$; the obstruction is exactly the nonzero negative part a_-, the element the proof of theorem 2.3.3 shows must vanish when a is a square b^*b.

With positivity characterized, its global shape can be described. The positive elements are closed under the two operations one expects of an order, addition and scaling by nonnegative reals, and they form a norm-closed set. A subset with these properties is a closed cone, and a cone is precisely the data needed to define a translation-invariant partial order. The order it defines on the self-adjoint part A_{sa} is the order every positivity estimate in the sequel is phrased in.

Proposition 2.3.5: The Positive Cone and the Order on A_{sa}

Let A be a unital C^*-algebra.

1. A_+ is a closed cone: if $a, b \in A_+$ and $\lambda \geq 0$ then $a + b \in A_+$ and $\lambda a \in A_+$, and A_+ is closed in the norm topology. Moreover $A_+ \cap (-A_+) = \{0\}$.
2. The relation $a \leq b \iff b - a \in A_+$ is a partial order on the real vector space A_{sa} of self-adjoint elements, invariant under translation and under scaling by nonnegative reals.
3. For self-adjoint a one has $-\|a\|\,1 \leq a \leq \|a\|\,1$.

***Proof*:**
Part 1: closed cone. Closure under nonnegative scaling is immediate: if $a \geq 0$ and $\lambda \geq 0$ then λa is self-adjoint and, by spectral mapping applied to $t \mapsto \lambda t$, $\sigma(\lambda a) = \lambda\,\sigma(a) \subseteq [0, \infty)$.

For closure under addition, characterize positivity through the norm. If $a = a^*$ and $\|a\| = r$, then $\sigma(a) \subseteq [-r, r]$, and

$$a \geq 0 \iff \sigma(a) \subseteq [0, r] \iff \sigma(r \cdot 1 - a) \subseteq [0, r] \iff \|r \cdot 1 - a\| \leq r$$

the last equivalence because $r \cdot 1 - a$ is self-adjoint with spectrum $r - \sigma(a) \subseteq [0, 2r]$, so its norm equals its spectral radius (proposition 2.1.3), and that radius is $\leq r$ exactly when $\sigma(a) \subseteq [0, r]$. Thus for self-adjoint a with $\|a\| \leq r$,

$$a \geq 0 \iff \|r \cdot 1 - a\| \leq r$$

Now take $a, b \geq 0$; set $r = \|a\|$ and $s = \|b\|$. Then $\|r \cdot 1 - a\| \leq r$ and $\|s \cdot 1 - b\| \leq s$, so

$$\|(r + s) \cdot 1 - (a + b)\| \leq \|r \cdot 1 - a\| + \|s \cdot 1 - b\| \leq r + s$$

Since $a + b$ is self-adjoint with $\|a + b\| \leq r + s$, the criterion gives $a + b \geq 0$.

Closure in the norm topology follows from the same criterion. Let $a_n \geq 0$ with $a_n \to a$. Each a_n is self-adjoint and the involution is isometric, so $a = a^*$. With $r_n = \|a_n\|$ and $r = \|a\|$, the norms converge $r_n \to r$, and

$$\|r \cdot 1 - a\| = \lim_n \|r_n \cdot 1 - a_n\| \leq \lim_n r_n = r$$

so $a \geq 0$ by the criterion. Hence A_+ is norm-closed.

Finally, if $a \in A_+ \cap (-A_+)$, then both $a \geq 0$ and $-a \geq 0$, so $\sigma(a) \subseteq [0, \infty)$ and $\sigma(-a) = -\sigma(a) \subseteq [0, \infty)$, forcing $\sigma(a) = \{0\}$. As a is self-adjoint, $\|a\| = r(a) = 0$ (proposition 2.1.3), so $a = 0$.

Part 2: partial order. On A_{sa}, if a, b are self-adjoint then so is $b - a$, so the relation $a \leq b \iff b - a \in A_+$ is defined on all of A_{sa}. Reflexivity is $0 \in A_+$. For transitivity, if $b - a \geq 0$ and $c - b \geq 0$ then $c - a = (c - b) + (b - a) \geq 0$ by the cone property. For antisymmetry, if $a \leq b$ and $b \leq a$ then $b - a \in A_+ \cap (-A_+) = \{0\}$, so $a = b$. Translation invariance ($a \leq b \Rightarrow a + c \leq b + c$ for c self-adjoint) and scaling invariance ($a \leq b \Rightarrow \lambda a \leq \lambda b$ for $\lambda \geq 0$) are immediate from $(b + c) - (a + c) = b - a$ and $\lambda b - \lambda a = \lambda(b - a)$ together with the cone property.

Part 3: the norm bound. Let a be self-adjoint with $\|a\| = r$. Then $\sigma(a) \subseteq \mathbb{R}$ and $r(a) = r$, so $\sigma(a) \subseteq [-r, r]$. The element $r \cdot 1 - a$ is self-adjoint with $\sigma(r \cdot 1 - a) = r - \sigma(a) \subseteq [0, 2r] \subseteq [0, \infty)$, hence $r \cdot 1 - a \geq 0$, which is $a \leq \|a\| 1$. Applying the same to $-a$ gives $r \cdot 1 + a \geq 0$, that is $-\|a\| 1 \leq a$, as we sought to show.

The order defined here is the one the whole positivity theory is phrased in, and part 3 is the reason it is useful rather than merely formal. Every self-adjoint element is squeezed between $-\|a\| 1$ and $\|a\| 1$, so the order and the norm constrain each other: an inequality $0 \leq a \leq b$ will let one bound $\|a\|$ by $\|b\|$, and the two-sided bound is the first instance of that mechanism. The requirement $A_+ \cap (-A_+) = \{0\}$ is what makes $\leq$ a genuine partial order rather than a preorder; it says the only element that is both positive and negative is 0, the algebraic form of the fact that a real number in $[0, \infty) \cap (-\infty, 0]$ can only be zero. This order does not make A_{sa} a lattice in general, since two self-adjoint elements need not have a least upper bound, but it is exactly the structure the square root and absolute value are built to respect.

2.4 Square Roots and the Absolute Value

The order on A_{sa} from the previous section is inert until it is put to work, and the tool that activates it is the continuous functional calculus applied to a single nonnegative function. Two functions do most of the labour: $\sqrt{t}$ on $[0, \infty)$ produces the square root of a positive element, and the composite $\sqrt{a^* a}$ produces an absolute value $|a|$ that measures the "size" of an arbitrary element the way $|z| = \sqrt{\bar{z} z}$ measures a complex number. With these in hand the order relation $a \leq b$ starts to behave like the order on the real line: it is preserved by the norm, by conjugation $c^*(-)c$, and inverted by passage to inverses. These inequalities are what prove approximate units, the polar decomposition, and the positivity arguments that run through the rest of the theory.

The square root is where the abstract order meets a concrete construction. A positive element a has $\sigma(a) \subseteq [0, \infty)$, so $\sqrt{t}$ is a genuine continuous function on the spectrum, and feeding it to the functional calculus manufactures an element whose square is a. What makes the construction more than a formal convenience is that the square root produced this way is the *only* positive element squaring to a: uniqueness is what lets one speak of *the* square root and build $|a|$ unambiguously.

Theorem 2.4.1: Positive Square Root

Let A be a unital C^*-algebra and $a \in A$ with $a \geq 0$. Then there is a unique positive element $a^{1/2} \in A$ with $(a^{1/2})^2 = a$, namely

$$a^{1/2} = f(a), \qquad f\colon \sigma(a) \to \mathbb{R}, \quad f(t) = \sqrt{t}$$

obtained from the continuous functional calculus (theorem 2.2.2). Moreover $a^{1/2} \in \overline{C^*(a,1)}$ lies in the closure of the polynomials in a, so it commutes with every element of A that commutes with a.

***Proof*:**

Since $a \geq 0$, definition 2.3.1 gives $\sigma(a) \subseteq [0,\infty)$, so $f(t) = \sqrt{t}$ is a well-defined continuous real-valued function on $\sigma(a)$. Set $a^{1/2} = f(a)$ via theorem 2.2.2.

Existence and positivity. The functional calculus is a unital $*$-homomorphism, so $f = \bar{f}$ gives $(a^{1/2})^* = \bar{f}(a) = f(a) = a^{1/2}$, and $f(t)^2 = t$ on $\sigma(a)$ gives $(a^{1/2})^2 = f(a)^2 = (f^2)(a) = \mathrm{id}_{\sigma(a)}(a) = a$. By the spectral mapping theorem (theorem 2.2.4), $\sigma(a^{1/2}) = f(\sigma(a)) = \{\sqrt{t} | t \in \sigma(a)\} \subseteq [0,\infty)$, so $a^{1/2}$ is self-adjoint with spectrum in $[0,\infty)$, that is, $a^{1/2} \geq 0$.

Location and commutation. By construction $a^{1/2}$ is the image of f under the isomorphism $C(\sigma(a)) \cong C^*(a,1)$, so $a^{1/2} \in C^*(a,1)$. Stone-Weierstrass makes f a uniform limit of polynomials p_n on the compact set $\sigma(a)$; because the functional calculus is isometric, $a^{1/2} = \lim_n p_n(a)$ in norm. Each $p_n(a)$ is a polynomial in a and 1, hence commutes with any $c \in A$ satisfying $ca = ac$, and this property passes to the norm limit: $c\,a^{1/2} = a^{1/2}c$.

Uniqueness. Suppose $b \geq 0$ also satisfies $b^2 = a$. Then b commutes with $b^2 = a$, so by the commutation property just proved b commutes with $a^{1/2}$. Let $B \subseteq A$ be the unital C^*-subalgebra generated by the two commuting positive elements b and $a^{1/2}$; since they commute and are self-adjoint, B is commutative, and it contains $a = b^2$. The strategy is to recognize both b and $a^{1/2}$ as $g(a)$ for the single function $g(t) = \sqrt{t}$, computed by the functional calculus inside B. Spectral permanence (theorem 2.1.6) guarantees the spectrum of a computed in B equals $\sigma(a) \subseteq [0,\infty)$, so g is continuous on it. For $a^{1/2}$ this is the definition. For b: the function $t \mapsto t^2$ carries $\sigma(b) \subseteq [0,\infty)$ onto $\sigma(b^2) = \sigma(a)$ by the spectral mapping theorem (theorem 2.2.4), and $g(t^2) = t$ for every $t \geq 0$, so $g(a) = g(b^2) = b$ by functoriality of the functional calculus under composition. Hence $b = g(a) = a^{1/2}$, both computed in B and therefore in A. Thus $b = a^{1/2}$, as we sought to show.

The uniqueness clause is the substance of the theorem, and the commutativity trick behind it recurs throughout the subject: to compare two positive elements built from the same a, collect them into a commutative subalgebra, where the functional calculus reduces every question to one about continuous functions on a compact subset of $\mathbb{R}$. There the statement is transparent, since $t \mapsto \sqrt{t}$ is the only continuous nonnegative function on $[0,\infty)$ that squares to t. Uniqueness also explains why $a^{1/2}$ deserves an unqualified name: no choice of "branch" is involved, unlike the situation for square roots of a general element, where a positive element such as 1 has many non-positive square roots (for instance -1, or any self-adjoint unitary).

Example 2.4.2: Square Roots in $C(X)$ and in $M_n(\mathbb{C})$

Two computations pin down what $a^{1/2}$ is.

1. In $A = C(X)$ for X compact Hausdorff, a positive element is a function $a\colon X \to [0,\infty)$, and the functional calculus is composition: $a^{1/2}$ is the pointwise square root $x \mapsto \sqrt{a(x)}$. Uniqueness among positive elements is the pointwise statement that $\sqrt{a(x)}$ is the only nonnegative real number squaring to $a(x)$. The non-positive square roots correspond to sign choices $x \mapsto \pm\sqrt{a(x)}$; only the all-plus choice is a positive element.

2. In $A = M_n(\mathbb{C})$ a positive element is a positive semidefinite matrix a. Diagonalizing $a = u\,\mathrm{diag}(\lambda_1, \ldots, \lambda_n)\,u^*$ with u unitary and each $\lambda_i \geq 0$, the functional calculus applies $\sqrt{\cdot}$ to the eigenvalues:
$$a^{1/2} = u\,\mathrm{diag}\Big(\sqrt{\lambda_1}, \ldots, \sqrt{\lambda_n}\Big)\,u^*$$
For instance $a = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ has eigenvalues 3 and 1 with orthonormal eigenvectors $\frac{1}{\sqrt{2}}(1,1)$ and $\frac{1}{\sqrt{2}}(1,-1)$, so $a^{1/2} = \frac{1}{2}\left(\begin{smallmatrix} \sqrt{3}+1 & \sqrt{3}-1 \\ \sqrt{3}-1 & \sqrt{3}+1 \end{smallmatrix}\right)$, and one checks directly that its square is a.

With a well-defined square root available for every positive element, and every a^*a positive by theorem 2.3.3, one can attach to an *arbitrary* element a positive "modulus". The model is the scalar case: for $z \in \mathbb{C}$ the modulus is $|z| = \sqrt{\bar{z}z}$, the unique nonnegative real whose square recovers $\bar{z}z$. Replacing $\bar{z}z$ by a^*a and the scalar square root by the functional-calculus square root gives the operator absolute value.

Definition 2.4.3: Absolute Value

Let A be a unital C^*-algebra and $a \in A$. The *absolute value* of a is the positive element
$$|a| = (a^*a)^{1/2}$$
the positive square root (theorem 2.4.1) of a^*a, which is positive by theorem 2.3.3.

The definition uses a^*a rather than aa^*, and the two generally differ: $|a| = (a^*a)^{1/2}$ and $|a^*| = (aa^*)^{1/2}$ need not agree when a is not normal. The choice a^*a is the one that measures the length of a acting on the right, as the norm identity below records; on a Hilbert space it is the operator whose action on a vector ξ satisfies $\||a|\xi\| = \|a\xi\|$, so $|a|$ has the same "stretching" behaviour as a but is positive.

Proposition 2.4.4: Norm Identities for the Absolute Value

Let A be a unital C^*-algebra and $a, x \in A$. Then

1. $\big|\,|a|\,\big| = |a|$ and $|a| \geq 0$;
2. $\big\||a|\big\| = \|a\|$;
3. $\big\||a|\,x\big\| = \|ax\|$.

***Proof*:**
Part (1) is immediate: $|a| = (a^*a)^{1/2}$ is a positive square root, hence self-adjoint and positive by theorem 2.4.1.

For (2), the defining relation $|a|^2 = a^*a$ together with the C^*-identity gives

$$\big\||a|\big\|^2 = \big\||a|^*|a|\big\| = \big\||a|^2\big\| = \|a^*a\| = \|a\|^2$$

where the first equality is the C^*-identity applied to the self-adjoint element $|a|$ and the last is the C^*-identity for a (definition 1.2.1). Taking square roots gives $\big\||a|\big\| = \|a\|$.

For (3), apply the C^*-identity to $|a|x$ and to ax and compare. Since $|a|$ is self-adjoint with $|a|^2 = a^*a$,

$$\big\||a|\,x\big\|^2 = \big\|(|a|x)^*(|a|x)\big\| = \big\|x^*|a|^2x\big\| = \|x^*a^*ax\| = \big\|(ax)^*(ax)\big\| = \|ax\|^2$$

Taking square roots yields $\big\||a|\,x\big\| = \|ax\|$, completing the proof.

The three identities say that $|a|$ is a faithful stand-in for a as far as norms are concerned: it has the same norm, and it moves any x to an element of the same size that a would. This is the algebraic form of the geometric fact that a and $|a|$ differ only by a partial isometry, the content of the polar decomposition $a = v|a|$. That decomposition requires either the extra room of a von Neumann algebra or a limiting argument, and lies past the scope of this chapter; the norm identities here are exactly the estimates one needs to build v when the decomposition is taken up. Setting $x = 1$ in (3) recovers (2), so the third identity is the general statement and the second is its unital instance.

Example 2.4.5: Absolute Value of a Weighted Shift
On $H = \mathbb{C}^2$ take $a = \left(\begin{smallmatrix} 0 & 2 \\ 0 & 0 \end{smallmatrix}\right)$, a nilpotent (hence non-normal) operator. Then

$$a^*a = \left(\begin{smallmatrix} 0 & 0 \\ 2 & 0 \end{smallmatrix}\right)\left(\begin{smallmatrix} 0 & 2 \\ 0 & 0 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 4 \end{smallmatrix}\right), \qquad |a| = (a^*a)^{1/2} = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 2 \end{smallmatrix}\right)$$

while $aa^* = \left(\begin{smallmatrix} 4 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ gives $|a^*| = \left(\begin{smallmatrix} 2 & 0 \\ 0 & 0 \end{smallmatrix}\right)$. Thus $|a| \neq |a^*|$: the absolute value built from a^*a records that a stretches the second basis vector by 2, whereas $|a^*|$ records the stretching of the first. Both have norm $2 = \|a\|$, illustrating proposition 2.4.4(2). Writing $v = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ (a partial isometry), one checks $v|a| = \left(\begin{smallmatrix} 0 & 2 \\ 0 & 0 \end{smallmatrix}\right) = a$, exhibiting the polar decomposition in this finite-dimensional case.

The order relation now interacts cleanly with the norm, with conjugation, and with inversion. Each of the three statements below is proved by pushing the inequality through the functional calculus and the positive cone of proposition 2.3.5; each is used constantly. The first says the order dominates the norm on positives, so norm bounds can be read off from order bounds. The second says conjugation $x \mapsto c^*xc$ preserves positivity, the mechanism by which positivity is transported along $*$-homomorphisms and states. The third is an order-reversal for inverses, the analogue of $0 < s \leq t \Rightarrow t^{-1} \leq s^{-1}$ for real numbers.

Proposition 2.4.6: Positivity Inequalities

Let A be a unital C^*-algebra.

1. If $0 \le a \le b$ then $\|a\| \le \|b\|$.
2. If $a \ge 0$ and $c \in A$ then $c^*ac \ge 0$. More generally, $a \le b$ implies $c^*ac \le c^*bc$ for every $c \in A$.
3. If $0 \le a \le b$ and a is invertible, then b is invertible, $0 \le b^{-1} \le a^{-1}$.

***Proof*:**
Part (1). Because b is positive, its norm equals its spectral radius: $\|b\| = r(b) = \max \sigma(b)$, using proposition 2.1.3 for the self-adjoint (hence normal) element b together with $\sigma(b) \subseteq [0,\infty)$. The element $\|b\| \cdot 1 - b$ has spectrum $\{\|b\| - t \big| t \in \sigma(b)\} \subseteq [0,\infty)$ by the spectral mapping theorem (theorem 2.2.4), so $\|b\| \cdot 1 - b \ge 0$, that is $b \le \|b\| \cdot 1$. Combined with $a \le b$ this gives $a \le \|b\| \cdot 1$, hence $\|b\| \cdot 1 - a \ge 0$, so every point of $\sigma(a)$ is at most $\|b\|$; as $a \ge 0$ this says $\|a\| = \max \sigma(a) \le \|b\|$.

Part (2). If $a \ge 0$ then $a = d^*d$ with $d = a^{1/2}$ self-adjoint (theorem 2.3.3), and

$$c^*ac = c^*d^*dc = (dc)^*(dc) \ge 0$$

since any element of the form y^*y is positive, again by theorem 2.3.3. For the monotonicity statement, $a \le b$ means $b - a \ge 0$, so applying the case just proved to $b - a$ gives $c^*(b-a)c \ge 0$; since $c^*(b-a)c = c^*bc - c^*ac$ and the positive cone is closed under the resulting difference (proposition 2.3.5), $c^*ac \le c^*bc$.

Part (3). First, $a \ge 0$ invertible forces $\sigma(a) \subseteq [\varepsilon, \infty)$ for some $\varepsilon > 0$: the spectrum is a compact subset of $[0,\infty)$ that avoids 0 precisely because a is invertible, so it is a compact subset of $(0,\infty)$ and $\varepsilon = \min \sigma(a) > 0$. The functional calculus applied to the continuous function $t \mapsto t^{-1}$ on $\sigma(a)$ then produces $a^{-1} \ge 0$ with $\|a^{-1}\| = \varepsilon^{-1}$.

Invertibility of b. From $a \le b$ and $a \ge \varepsilon \cdot 1$ (which holds since $\min \sigma(a) = \varepsilon$) one gets $b \ge a \ge \varepsilon \cdot 1$, so $b - \varepsilon \cdot 1 \ge 0$ and $\sigma(b) \subseteq [\varepsilon, \infty)$. In particular $0 \notin \sigma(b)$, so b is invertible and $b^{-1} \ge 0$.

The reversed inequality. Conjugate $a \le b$ by $c = a^{-1/2}$, which is positive and self-adjoint. By part (2),

$$a^{-1/2} a\, a^{-1/2} \le a^{-1/2} b\, a^{-1/2}$$

and the left side is $a^{-1/2}a^{1/2}a^{1/2}a^{-1/2} = 1$, so $1 \le a^{-1/2}b\,a^{-1/2}$. Write $u = a^{-1/2}b\,a^{-1/2}$; then $u \ge 1$, so u is invertible with $\sigma(u) \subseteq [1,\infty)$, whence $\sigma(u^{-1}) \subseteq (0,1]$ by the spectral mapping theorem (theorem 2.2.4) and therefore $u^{-1} \le 1$. Unwinding the definition,

$$u^{-1} = \left(a^{-1/2}b\,a^{-1/2}\right)^{-1} = a^{1/2}b^{-1}a^{1/2}$$

so $a^{1/2}b^{-1}a^{1/2} \le 1$. Conjugating this last inequality by $a^{-1/2}$ (part (2) once more) gives

$$a^{-1/2}\left(a^{1/2}b^{-1}a^{1/2}\right)a^{-1/2} \le a^{-1/2} \cdot 1 \cdot a^{-1/2}$$

that is $b^{-1} \le a^{-1}$, completing the proof.

Part (3) records that $t \mapsto t^{-1}$ reverses the order on invertible positives, and the conjugation trick that proves it is the standard device for turning an inequality $a \le b$ into a statement about $a^{-1/2} b\, a^{-1/2}$, where the scalar fact $s \ge 1 \Rightarrow s^{-1} \le 1$ can be applied through the functional calculus. Which power functions respect the order and which do not is a question with a clean answer. The map $t \mapsto t^{1/2}$ is *operator monotone*: $0 \le a \le b$ implies $a^{1/2} \le b^{1/2}$, a fact of the same family as part (3). By contrast $t \mapsto t^2$ is *not* operator monotone: there exist positive $a \le b$ with $a^2 \not\le b^2$. A concrete witness lives already in $M_2(\mathbb{C})$, with

$$a = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right), \qquad b = a + \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$$

both positive semidefinite with $a \le b$, yet $b^2 - a^2$ fails to be positive semidefinite (its determinant is negative). The failure is not an artifact: squaring mixes the noncommuting parts of a and b in a way the order cannot see, whereas the square root, being a norm-limit of the "spreading" functions $\sqrt{t}$, respects it. This asymmetry, that $t^{1/2}$ is operator monotone but t^2 is not, is one of the first signs that operator inequalities are finer than their scalar counterparts, and it is exactly what makes part (3) available while the analogous statement for squares fails.

3

Commutative C^*-Algebras: Gelfand-Naimark and Serre-Swan

The continuous functional calculus of chapter 2 identified one normal element with a space, its spectrum. This chapter promotes that identification to a theorem about whole algebras: a commutative C^*-algebra is, up to isometric isomorphism, the algebra $C_0(X)$ of continuous functions vanishing at infinity on a locally compact Hausdorff space X, and X is recovered from the algebra as its space of characters. This is the commutative Gelfand-Naimark theorem, the precise sense in which "commutative C^*-algebra" and "locally compact Hausdorff space" name the same thing; the functional calculus is its one-generator case.

The correspondence is contravariant and functorial, an equivalence of categories, so each topological construction on spaces has an algebraic counterpart, and it is this dictionary that a noncommutative C^*-algebra is read as extending. The chapter closes with the module-level refinement the second part of the book needs: the Serre-Swan theorem, that for X compact the vector bundles over X are exactly the finitely generated projective modules over $C(X)$. Projections over $C(X)$ thereby become bundles, and the K-theory assembled from them becomes the topological K-theory of X.

3.1 Characters and the Gelfand Transform

To realize a commutative C^*-algebra A as an algebra of functions, the first task is to produce the underlying space on which those functions live. That space is assembled out of A itself: its points are the scalar-valued multiplicative functionals on A, and a point φ evaluates an element a by returning $\varphi(a)$. This section builds the space, shows it is (locally) compact Hausdorff, and packages the evaluation pairing as the Gelfand transform $a \mapsto \hat{a}$. The Gelfand transform lands in the continuous functions on the space, and it is the map that the next section proves to be an isometric

$*$-isomorphism.

Throughout, A is a commutative C^*-algebra, unital unless stated otherwise, with a fixed multiplicative identity 1. The starting object is a homomorphism from A to the scalars.

Definition 3.1.1: Character

A *character* of a C^*-algebra A is a nonzero algebra homomorphism $\varphi\colon A \to \mathbb{C}$; that is, φ is $\mathbb{C}$-linear and satisfies $\varphi(ab) = \varphi(a)\varphi(b)$ for all $a, b \in A$, and φ is not identically zero.

No boundedness or continuity is built into the definition, and no compatibility with the involution is demanded. Both turn out to be automatic on a C^*-algebra, which is what makes characters so well-behaved. The mechanism is the spectrum: on a unital algebra a character sends a to a point of $\sigma(a)$, and the C^*-identity forces that point to reflect the involution. The following proposition records these automatic properties.

Proposition 3.1.2: Characters Are Contractive $*$-Homomorphisms

Let A be a commutative C^*-algebra and φ a character of A.

1. If A is unital, then $\varphi(1) = 1$, and $\varphi(a) \in \sigma(a)$ for every $a \in A$.
2. φ is a $*$-homomorphism, that is, $\varphi(a^*) = \overline{\varphi(a)}$ for all $a \in A$.
3. φ is bounded with $\|\varphi\| \le 1$, and $\|\varphi\| = 1$ when A is unital.

***Proof*:**
Part (1): the unit and the spectrum. Since φ is nonzero, choose b with $\varphi(b) \neq 0$. Then $\varphi(b) = \varphi(b \cdot 1) = \varphi(b)\varphi(1)$, and cancelling $\varphi(b)$ gives $\varphi(1) = 1$. Fix $a \in A$ and set $\lambda = \varphi(a)$. Applying φ to $a - \lambda 1$ gives $\varphi(a - \lambda 1) = \lambda - \lambda\varphi(1) = 0$, so $a - \lambda 1$ lies in $\ker \varphi$. A character is a homomorphism onto the field $\mathbb{C}$, so its kernel is a proper ideal; a proper ideal contains no invertible element, since an ideal containing a unit is the whole algebra. Hence $a - \lambda 1$ is not invertible, which is precisely the statement $\lambda \in \sigma(a)$. Thus $\varphi(a) \in \sigma(a)$.

Part (2): compatibility with the involution. First consider a self-adjoint element $a = a^*$. In a C^*-algebra the spectrum of a self-adjoint element is real, so by part (1) the value $\varphi(a) \in \sigma(a)$ is a real number, giving $\varphi(a) = \overline{\varphi(a)}$. For a general $b \in A$, write $b = x + iy$ with $x = \frac{1}{2}(b + b^*)$ and $y = \frac{1}{2i}(b - b^*)$ self-adjoint, so $b^* = x - iy$. Since $\varphi(x)$ and $\varphi(y)$ are real,

$$\varphi(b^*) = \varphi(x) - i\varphi(y) = \overline{\varphi(x) + i\varphi(y)} = \overline{\varphi(b)}$$

which is the asserted $*$-compatibility. (The decomposition uses only that A is a C^*-algebra, not that it is unital: on a non-unital A apply the argument in the unitization $\tilde{A}$, into which φ extends by $\varphi(a, \lambda) = \varphi(a) + \lambda$.)

Part (3): the norm bound. Suppose first that A is unital. By part (1), $\varphi(a) \in \sigma(a)$, so $|\varphi(a)| \le \sup\{|\mu| : \mu \in \sigma(a)\} = r(a)$, the spectral radius. Since $r(a) \le \|a\|$ always holds (theorem 1.1.3), we get $|\varphi(a)| \le \|a\|$, so $\|\varphi\| \le 1$. Equality holds because $\varphi(1) = 1 = \|1\|$. When A is non-unital, extend φ to the character $\tilde{\varphi}$ on $\tilde{A}$ as above; the unital case gives $\|\tilde{\varphi}\| = 1$, and restricting to $A \subseteq \tilde{A}$ gives $\|\varphi\| \le 1$, completing the proof.

The content of the proposition is that a purely algebraic object, a homomorphism to $\mathbb{C}$ with no continuity hypothesis, is automatically continuous, contractive, and $*$-preserving. Two features of C^*-algebras account for this: invertibility is controlled by the spectrum, which pins $\varphi(a)$ inside $\sigma(a)$ and hence inside a disc of radius $\|a\|$; and self-adjoint elements have real spectrum, which is what makes φ respect the involution. Neither holds for a general Banach algebra, and both are consequences of the C^*-identity established in section 1.1 and chapter 2. Because every character is contractive, the characters all live in the closed unit ball of the continuous dual A', which is what will make their collection compact.

Example 3.1.3: Evaluation Characters on $C(X)$

Let X be a compact Hausdorff space and $A = C(X)$ the C^*-algebra of continuous complex functions on X with the supremum norm. For each point $x \in X$, evaluation at x,

$$\mathrm{ev}_x \colon C(X) \to \mathbb{C}, \qquad \mathrm{ev}_x(f) = f(x)$$

is $\mathbb{C}$-linear and multiplicative, since $(fg)(x) = f(x)g(x)$, and it is nonzero because $\mathrm{ev}_x(1) = 1$. So ev_x is a character. The automatic properties are visible directly: $\mathrm{ev}_x(f) = f(x)$ lies in the range of f, which is $\sigma(f)$ for $A = C(X)$; $\mathrm{ev}_x(\overline{f}) = \overline{f(x)} = \overline{\mathrm{ev}_x(f)}$; and $|\mathrm{ev}_x(f)| = |f(x)| \leq \|f\|_\infty$, with equality for a suitable f. Distinct points give distinct characters: if $x \neq y$, Urysohn's lemma ([7, Urysohn, Compact Hausdorff Spaces Are Normal]) produces $f \in C(X)$ with $f(x) \neq f(y)$, so $\mathrm{ev}_x \neq \mathrm{ev}_y$. That every character of $C(X)$ arises this way is proved in example 3.1.10, and is the model for the general theorem.

Having fixed the individual points, the collection of all of them becomes a space once it carries a topology. The natural topology is the one making each evaluation $\varphi \mapsto \varphi(a)$ continuous, which is the weak-$*$ topology inherited from the dual A'.

Definition 3.1.4: Character Space

The *character space* (or *Gelfand spectrum*) of a commutative C^*-algebra A is the set

$$\hat{A} = \{\varphi \colon A \to \mathbb{C} \big| \varphi \text{ is a character of } A\}$$

equipped with the weak-$*$ topology it inherits as a subset of the continuous dual A': a net $\varphi_i \to \varphi$ in $\hat{A}$ precisely when $\varphi_i(a) \to \varphi(a)$ in $\mathbb{C}$ for every $a \in A$.

The weak-$*$ topology is exactly the topology of pointwise convergence of the functionals: two characters are close when they agree, to within a small tolerance, on each of finitely many fixed elements of A. This is the coarsest topology for which every evaluation map $\hat{a} \colon \varphi \mapsto \varphi(a)$ is continuous, which is what one wants, since those evaluation maps are the functions that will represent the elements of A. The choice of topology is thus forced by the goal: make the elements of A into continuous functions on $\hat{A}$.

Example 3.1.5: The Character Space of $C(X)$ as a Set

Continuing example 3.1.3 with X compact Hausdorff and $A = C(X)$, the assignment $x \mapsto \mathrm{ev}_x$ maps X into $\widehat{C(X)}$. It is injective, as shown above. The weak-$*$ topology on the image is the topology of pointwise convergence: $\mathrm{ev}_{x_i} \to \mathrm{ev}_x$ means $f(x_i) \to f(x)$ for every $f \in C(X)$, and since $C(X)$ separates points and X is compact Hausdorff, this holds if and only if $x_i \to x$ in X.

So $x \mapsto \mathrm{ev}_x$ is a continuous injection $X \to \widehat{C(X)}$, and it is a homeomorphism onto its image. Surjectivity, that this exhausts $\widehat{C(X)}$, is example 3.1.10; granting it, $\widehat{C(X)} \cong X$.

The character space is not merely a topological space but a compact one, at least in the unital case. This is what turns A into an algebra of functions on a compact Hausdorff space, and it is the topological input to Gelfand duality. The proof identifies $\hat{A}$ as a weak-$*$-closed subset of the unit ball of A' and invokes Banach-Alaoglu.

Proposition 3.1.6: The Character Space Is Locally Compact Hausdorff

Let A be a commutative C^*-algebra.

1. If A is unital, then $\hat{A}$ is compact Hausdorff.
2. If A is non-unital, then $\hat{A}$ is locally compact Hausdorff and noncompact; explicitly $\hat{A}$ is homeomorphic to $\hat{\tilde{A}} \setminus \{\varphi_\infty\}$, where $\tilde{A}$ is the unitization and φ_∞ is the unique character of $\tilde{A}$ vanishing on A.

***Proof*:**
Part (1): the unital case. Let $B = \{\psi \in A' \big| \|\psi\| \leq 1\}$ be the closed unit ball of the dual. By proposition 3.1.2(3) every character satisfies $\|\varphi\| \leq 1$, so $\hat{A} \subseteq B$. The Banach-Alaoglu theorem ([9, Alaoglu's Theorem]) states that B is compact in the weak-$*$ topology. A closed subset of a compact space is compact, so it suffices to show $\hat{A}$ is weak-$*$ closed in B, and Hausdorffness of $\hat{A}$ is inherited from the weak-$*$ topology on A', which is Hausdorff since the evaluations $\{\psi \mapsto \psi(a)\}_{a \in A}$ separate points.

For closedness, describe $\hat{A}$ inside B by the equations defining a unital homomorphism. A functional $\psi \in B$ is a character precisely when

$$\psi(ab) = \psi(a)\psi(b) \quad \text{for all } a, b \in A, \qquad \psi(1) = 1$$

(multiplicativity together with $\psi(1) = 1$ forces $\psi \neq 0$, and linearity is already built into $\psi \in A'$). For each fixed pair a, b, the set $\{\psi \big| \psi(ab) = \psi(a)\psi(b)\}$ is weak-$*$ closed: the maps $\psi \mapsto \psi(ab)$ and $\psi \mapsto \psi(a)\psi(b)$ are both weak-$*$ continuous on A', the first by definition of the weak-$*$ topology and the second as a product of two such continuous maps, so their coincidence set is closed. Likewise $\{\psi \big| \psi(1) = 1\}$ is weak-$*$ closed. Now $\hat{A}$ is the intersection over all pairs (a, b) of these closed sets with $\{\psi \big| \psi(1) = 1\}$, an intersection of weak-$*$ closed sets, hence weak-$*$ closed. Therefore $\hat{A}$ is a closed subset of the weak-$*$ compact ball B, so $\hat{A}$ is compact Hausdorff.

Part (2): the non-unital case. The unitization $\tilde{A} = A \oplus \mathbb{C}1$ is a commutative unital C^*-algebra, so by part (1) its character space $\hat{\tilde{A}}$ is compact Hausdorff. Every character φ of A extends to a character $\tilde{\varphi}$ of $\tilde{A}$ by $\tilde{\varphi}(a + \lambda 1) = \varphi(a) + \lambda$, which is multiplicative and unital; conversely, a character ψ of $\tilde{A}$ restricts to a homomorphism $\psi|_A \colon A \to \mathbb{C}$, and this restriction is either a character of A or the zero map. There is exactly one character of $\tilde{A}$ whose restriction to A is zero, namely $\varphi_\infty(a + \lambda 1) = \lambda$, the projection onto the scalar part. Thus restriction gives a bijection

$$\hat{\tilde{A}} \setminus \{\varphi_\infty\} \longrightarrow \hat{A}, \qquad \psi \mapsto \psi|_A$$

with inverse $\varphi \mapsto \tilde{\varphi}$. Both maps are weak-$*$ continuous: restriction is continuous because for fixed

$a \in A$ the map $\psi \mapsto \psi(a)$ is continuous, and extension is continuous because $\varphi \mapsto \tilde{\varphi}(a + \lambda 1) = \varphi(a) + \lambda$ depends weak-$*$ continuously on φ. So the bijection is a homeomorphism.

It remains to show $\hat{A}$ is noncompact. Since $\widehat{\tilde{A}}$ is compact Hausdorff and $\hat{A}$ is homeomorphic to $\widehat{\tilde{A}} \setminus \{\varphi_\infty\}$, this amounts to showing φ_∞ is not weak-$*$ isolated: removing an isolated point of a compact Hausdorff space leaves a compact space, while removing a non-isolated point leaves a noncompact one, because $\hat{A}$ is then not weak-$*$ closed in the compact $\widehat{\tilde{A}}$. So suppose, toward a contradiction, that φ_∞ were isolated. Then some basic weak-$*$ neighbourhood

$$U = \{\psi \in \widehat{\tilde{A}} \big| |\psi(a_i + \lambda_i 1) - \lambda_i| < \epsilon,\ i = 1, \ldots, n\}, \qquad a_i \in A$$

contains no character other than φ_∞. Since $\psi(a_i + \lambda_i 1) - \lambda_i = \psi(a_i)$ for every character ψ (using $\psi(1) = 1$), membership in U says $|\psi(a_i)| < \epsilon$ for all i. Set $b = \sum_{i=1}^n a_i^* a_i \in A$, a self-adjoint element; because each ψ is a $*$-homomorphism (proposition 3.1.2(2)), $\psi(b) = \sum_i |\psi(a_i)|^2$, so $\psi(b) < \epsilon^2$ forces $\psi \in U$. As $U \cap \widehat{\tilde{A}} = \{\varphi_\infty\}$ and $\varphi_\infty(b) = 0$, every character $\psi \neq \varphi_\infty$ satisfies $\psi(b) \geq \epsilon^2$.

By proposition 3.1.9 applied to $\tilde{A}$, a scalar μ lies in $\sigma_{\tilde{A}}(b)$ precisely when $b - \mu 1$ lies in some maximal ideal, that is, when $\psi(b) = \mu$ for some character ψ; hence $\sigma_{\tilde{A}}(b) = \{\psi(b) \big| \psi \in \widehat{\tilde{A}}\} \subseteq \{0\} \cup [\epsilon^2, \infty)$. Thus 0 is an isolated point of the spectrum, and the indicator $f = \chi_{\{0\}}$ is continuous on $\sigma_{\tilde{A}}(b)$. The continuous functional calculus (theorem 2.2.2) produces $p = f(b) \in \tilde{A}$, a projection since $f = f^2 = \overline{f}$, with $\psi(p) = f(\psi(b))$ for every character ψ; so $\psi(p) = 1$ when $\psi(b) = 0$ and $\psi(p) = 0$ when $\psi(b) \geq \epsilon^2$. In particular $\varphi_\infty(p) = 1$, so $e := 1 - p$ satisfies $\varphi_\infty(e) = 0$, whence $e \in \ker \varphi_\infty = A$, while $\psi(e) = 1$ for every character $\psi \neq \varphi_\infty$.

This element $e \in A$ is a unit for A. Fix $x \in A$ and put $y = xe - x \in A$. For every character ψ of $\tilde{A}$, either $\psi = \varphi_\infty$, giving $\psi(x) = 0$ and so $\psi(y) = 0$, or $\psi \neq \varphi_\infty$, giving $\psi(e) = 1$ and again $\psi(y) = \psi(x)(\psi(e) - 1) = 0$. Hence $\psi(y^* y) = |\psi(y)|^2 = 0$ for all ψ, so $\sigma_{\tilde{A}}(y^* y) = \{0\}$ by the spectral description above, and $y^* y$ is self-adjoint with $\|y^* y\| = r(y^* y) = 0$ (proposition 2.1.3). The C^*-identity gives $\|y\|^2 = \|y^* y\| = 0$, so $y = 0$ and $xe = x$. As $x \in A$ was arbitrary and A is commutative, e is a two-sided identity for A, contradicting that A is non-unital. Therefore φ_∞ is not weak-$*$ isolated, so $\widehat{\tilde{A}} \setminus \{\varphi_\infty\}$, and with it $\hat{A}$, is locally compact Hausdorff and noncompact, as we sought to show.

The proposition gives $\hat{A}$ its topological standing, and it fixes the shape of Gelfand duality: unital algebras pair with compact spaces, non-unital algebras with locally compact noncompact spaces. The non-unital picture is the topological analogue of one-point compactification. Adjoining a unit to A corresponds to adjoining a point φ_∞ to $\hat{A}$, and $\widehat{\tilde{A}}$ is the one-point compactification $\hat{A}^+$ with φ_∞ the point at infinity. This matches the function-algebra intuition: $C_0(X)$ has no unit when X is noncompact, and adjoining the constant function 1 produces $C(X^+)$ on the one-point compactification. For $A = C(X)$ with X compact, part (1) recovers that $\widehat{C(X)}$ is compact, consistent with $\widehat{C(X)} \cong X$ from example 3.1.5.

With the space in hand, each element of A becomes a function on it by the evaluation pairing, and the assignment $a \mapsto \hat{a}$ is the Gelfand transform. The name $\hat{A}$ for the character space and $\hat{a}$ for the transformed element are chosen to display this pairing: $\hat{a}$ is the function on $\hat{A}$ given by $\hat{a}(\varphi) = \varphi(a)$.

Definition 3.1.7: Gelfand Transform

Let A be a commutative C^*-algebra with character space $\hat{A}$. The *Gelfand transform* of $a \in A$ is the function

$$\hat{a}\colon \hat{A} \to \mathbb{C}, \qquad \hat{a}(\varphi) = \varphi(a)$$

The *Gelfand transform* of A is the map $\Gamma\colon A \to C_0(\hat{A})$, $\Gamma(a) = \hat{a}$.

Three claims are folded into this definition and must be checked: that each $\hat{a}$ is continuous, that each $\hat{a}$ vanishes at infinity so that $\hat{a} \in C_0(\hat{A})$, and that Γ is a $*$-homomorphism of algebras. The next proposition verifies them.

Proposition 3.1.8: The Gelfand Transform Is a $*$-Homomorphism

Let A be a commutative C^*-algebra. Then $\Gamma\colon A \to C_0(\hat{A})$, $\Gamma(a) = \hat{a}$, is a well-defined homomorphism of $*$-algebras. That is, for all $a, b \in A$ and $\lambda \in \mathbb{C}$:

1. $\hat{a} \in C_0(\hat{A})$, and Γ is $\mathbb{C}$-linear;
2. $\widehat{ab} = \hat{a}\,\hat{b}$;
3. $\widehat{a^*} = \overline{\hat{a}}$.

When A is unital, Γ maps into $C(\hat{A}) = C_0(\hat{A})$ and $\hat{1} = 1$, the constant function.

***Proof*:**

Continuity and vanishing at infinity. By the very definition of the weak-$*$ topology on $\hat{A}$, the map $\hat{a}\colon \varphi \mapsto \varphi(a)$ is continuous. When A is unital, $\hat{A}$ is compact by proposition 3.1.6(1), so every continuous function on it lies in $C(\hat{A}) = C_0(\hat{A})$ automatically. When A is non-unital, extend a to $(a, 0) \in \tilde{A}$ and let $\widehat{\tilde{A}} = \hat{A} \cup \{\varphi_\infty\}$ be the one-point compactification from proposition 3.1.6(2). The function $\widehat{(a,0)}$ is continuous on the compact space $\widehat{\tilde{A}}$ and takes the value $\varphi_\infty(a, 0) = 0$ at the point at infinity. A continuous function on a one-point compactification $\hat{A} \cup \{\infty\}$ vanishing at ∞ restricts to a function in $C_0(\hat{A})$, so $\hat{a} = \widehat{(a,0)}|_{\hat{A}} \in C_0(\hat{A})$.

Algebraic properties. These are pointwise consequences of φ being a character. For each $\varphi \in \hat{A}$,

$$\widehat{ab}(\varphi) = \varphi(ab) = \varphi(a)\varphi(b) = \hat{a}(\varphi)\hat{b}(\varphi) = (\hat{a}\,\hat{b})(\varphi)$$

giving (2); linearity of Γ follows the same way from linearity of each φ, giving the linear part of (1). For (3), proposition 3.1.2(2) gives $\varphi(a^*) = \overline{\varphi(a)}$, so

$$\widehat{a^*}(\varphi) = \varphi(a^*) = \overline{\varphi(a)} = \overline{\hat{a}(\varphi)}$$

which is $\widehat{a^*} = \overline{\hat{a}}$. When A is unital, $\hat{1}(\varphi) = \varphi(1) = 1$ for all φ by proposition 3.1.2(1), so $\hat{1}$ is the constant function 1, completing the proof.

The Gelfand transform is thus a canonical $*$-homomorphism from any commutative C^*-algebra into the continuous functions on its character space, built with no choices beyond the algebra itself. It is exactly the pairing $\langle a, \varphi \rangle = \varphi(a)$ read as a function of φ for fixed a. Nothing so far controls its kernel or its image; the assertions that Γ is injective, isometric, and surjective are the content of the commutative Gelfand-Naimark theorem in section 3.2, and each rests on the C^*-identity in an essential way. What the present section establishes is that the target space exists and that the transform is a morphism of $*$-algebras into it. For $A = C(X)$ the transform is the identification of a function with itself: under $X \cong \widehat{C(X)}$ from example 3.1.5, $\hat{f}(\mathrm{ev}_x) = \mathrm{ev}_x(f) = f(x)$, so $\hat{f}$ and f are the same function on the same space.

The character space carries a second, purely algebraic, description that is often the more convenient one: characters are the same data as maximal ideals. This is the vantage point from which $\hat{A}$ is called the maximal-ideal space, and it is the description that connects the analytic construction here to the ideal theory of commutative algebra.

Proposition 3.1.9: Characters and Maximal Ideals

Let A be a commutative unital C^*-algebra. The map $\varphi \mapsto \ker\varphi$ is a bijection from the character space $\hat{A}$ onto the set of maximal ideals of A. Its inverse sends a maximal ideal M to the character given by the composite $A \to A/M \cong \mathbb{C}$.

***Proof*:**

Kernels are maximal ideals. Let $\varphi \in \hat{A}$. Its kernel $\ker\varphi$ is an ideal, and since φ is a nonzero homomorphism onto the field $\mathbb{C}$, the first isomorphism theorem gives $A/\ker\varphi \cong \mathbb{C}$. A quotient of A isomorphic to a field has the quotient by a maximal ideal, so $\ker\varphi$ is maximal. Different characters have different kernels: if $\ker\varphi = \ker\psi = M$, then both φ and ψ factor as $A \to A/M \cong \mathbb{C}$, and the two identifications $A/M \cong \mathbb{C}$ can differ only by an automorphism of the field $\mathbb{C}$ fixing 1; the induced maps agree on 1 (both send it to 1 by proposition 3.1.2(1)) and are $\mathbb{C}$-linear, hence agree, so $\varphi = \psi$. Thus $\varphi \mapsto \ker\varphi$ is injective.

Every maximal ideal is a kernel. Let $M \subsetneq A$ be a maximal ideal. A maximal ideal in a commutative unital ring is closed under the topology here: its closure $\overline{M}$ is again an ideal, and $\overline{M} \neq A$ because the invertible elements form an open set disjoint from M (an element within distance $\|a^{-1}\|^{-1}$ of an invertible a is invertible, so no ideal avoiding the units can approach a unit), so $\overline{M}$ is a proper ideal containing the maximal ideal M, forcing $\overline{M} = M$. Hence M is closed, and A/M is a commutative unital C^*-algebra (a quotient of a C^*-algebra by a closed ideal is a C^*-algebra). Since M is maximal, A/M is a field, and it is also a unital C^*-algebra; by Gelfand-Mazur (section 1.1) the only C^*-algebra that is a field is $\mathbb{C}$ itself, so $A/M \cong \mathbb{C}$. The composite $\varphi_M : A \to A/M \cong \mathbb{C}$ is then a character with $\ker\varphi_M = M$. This exhibits every maximal ideal as a kernel and provides the stated inverse, so $\varphi \mapsto \ker\varphi$ is a bijection, as we sought to show.

Two descriptions of the same space are now available: analytically $\hat{A}$ is the space of characters with the weak-$*$ topology, and algebraically it is the set of maximal ideals of A. This is why $\hat{A}$ is called the *maximal-ideal space*. The algebraic side makes the functorial behaviour transparent, since a unital $*$-homomorphism $A \to B$ pulls maximal ideals back and hence induces $\hat{B} \to \hat{A}$, while the analytic side gives the topology and, through proposition 3.1.6, the compactness that a bare ideal-theoretic construction does not see. The correspondence also shows that A has more than one character whenever $A \neq \mathbb{C}$. If A is not spanned by 1, choose $a \in A$ with $a \notin \mathbb{C}1$; since a is normal, $\sigma(a)$ contains two distinct points $\mu \neq \nu$ (a single-point spectrum would force $a = \mu 1$ by proposition 2.1.3). Then $a - \mu 1$ and $a - \nu 1$ are non-invertible, and they lie in no common maximal ideal, since their difference $(\nu - \mu)1$ is invertible. So A has at least two distinct maximal ideals, and $\hat{A}$ has at least two points.

Example 3.1.10: Characters of $C(X)$ Are Evaluations

Let X be a compact Hausdorff space and $A = C(X)$. Every character of $C(X)$ is evaluation at some point, so $\widehat{C(X)} \cong X$. To see this through proposition 3.1.9, it is enough to show that every maximal ideal of $C(X)$ has the form

$$M_x = \{f \in C(X) \big| f(x) = 0\} = \ker \mathrm{ev}_x$$

for some $x \in X$, since M_x is the kernel of the character ev_x. Each M_x is a maximal ideal, being the kernel of the surjective homomorphism $\mathrm{ev}_x \colon C(X) \to \mathbb{C}$ onto a field. Conversely, let M be a proper ideal of $C(X)$; the claim is that the functions in M have a common zero. If not, then for each $x \in X$ there is $f_x \in M$ with $f_x(x) \neq 0$, so $|f_x|^2 = \overline{f_x} f_x \in M$ is a nonnegative function positive at x; by continuity it is positive on an open neighbourhood U_x of x. Compactness of X extracts a finite subcover $U_{x_1}, \ldots, U_{x_n}$, and then

$$g = |f_{x_1}|^2 + \cdots + |f_{x_n}|^2 \in M$$

is strictly positive on all of X, hence invertible in $C(X)$, contradicting that M is proper. So the functions in M share a common zero x_0, giving $M \subseteq M_{x_0}$; maximality forces $M = M_{x_0}$. Every maximal ideal is therefore M_x for a unique x (uniqueness because Urysohn's lemma ([7, Urysohn, Compact Hausdorff Spaces Are Normal]) separates distinct points, so $M_x \neq M_y$ for $x \neq y$), and $\varphi \mapsto \ker \varphi$ identifies $\widehat{C(X)}$ with X. Together with the homeomorphism of example 3.1.5, this gives $\widehat{C(X)} \cong X$ as topological spaces.

The computation for $C(X)$ is the concrete case of the general theorem: for the algebra of functions on a space, the characters recover the space, point for point and topology for topology. Gelfand-Naimark asserts that this is not special to function algebras, that *every* commutative C^*-algebra is C_0 of its character space, so that the passage $A \mapsto \hat{A}$ loses no information.

3.2 The Commutative Gelfand-Naimark Theorem

The Gelfand transform of definition 3.1.7 sends a commutative C^*-algebra A into the function algebra $C_0(\hat{A})$ on its character space. In general this transform is only a $*$-homomorphism, and for a Banach algebra it can be badly degenerate: it may fail to be injective, and its image need not be closed or all of $C_0(\hat{A})$. The task of this section is to show that the C^*-axioms remove every such defect at once. For a commutative C^*-algebra the transform is an isometry, so injective with closed image, and its image exhausts $C_0(\hat{A})$, so it is an isomorphism onto the whole function algebra. The mechanism has two parts, each already in hand: isometry comes from the spectral radius equality for normal elements of proposition 2.1.3, which applies to every element here because a commutative algebra has only normal elements, and surjectivity comes from the Stone-Weierstrass theorem. Assembling these gives the identification of a commutative C^*-algebra with a function algebra, upgraded from a slogan to a theorem, and its functorial form makes the correspondence a contravariant equivalence of categories.

The isometry is the point at which the C^*-identity enters. In a commutative algebra $ab = ba$ for all

a, b, so in particular $aa^* = a^*a$: every element is normal. The equality $r(a) = \|a\|$ of proposition 2.1.3 therefore holds without restriction, and it is exactly what is needed, because the supremum norm of the Gelfand transform $\hat{a}$ is the spectral radius of a.

Theorem 3.2.1: Commutative Gelfand-Naimark

Let A be a commutative C^*-algebra and $\hat{A}$ its character space. The Gelfand transform

$$\Gamma\colon A \to C_0(\hat{A}), \qquad \Gamma(a) = \hat{a}, \quad \hat{a}(\varphi) = \varphi(a)$$

is an isometric $*$-isomorphism of A onto $C_0(\hat{A})$. If A is unital, then $\hat{A}$ is compact and Γ is an isometric $*$-isomorphism onto $C(\hat{A})$.

***Proof*:**

That Γ is a $*$-homomorphism into $C_0(\hat{A})$ is definition 3.1.7, and $\hat{A}$ is compact Hausdorff when A is unital and locally compact Hausdorff otherwise by proposition 3.1.6. Two things remain: Γ is isometric, and Γ is surjective. The proof treats the unital case first and reduces the non-unital case to it.

Step 1: Γ is isometric, unital case. Assume A unital. For $a \in A$ the supremum norm of $\hat{a}$ over the compact space $\hat{A}$ is

$$\|\hat{a}\|_\infty = \sup_{\varphi \in \hat{A}} |\hat{a}(\varphi)| = \sup_{\varphi \in \hat{A}} |\varphi(a)|$$

The set of values $\{\varphi(a) : \varphi \in \hat{A}\}$ is exactly the spectrum $\sigma(a)$: by proposition 3.1.9 the characters correspond to the maximal ideals of A, and a scalar λ lies in $\sigma(a)$ if and only if $a - \lambda 1$ is non-invertible, that is, lies in a maximal ideal, that is, $\varphi(a) = \lambda$ for some character φ. Hence

$$\{\varphi(a) : \varphi \in \hat{A}\} = \sigma(a) \tag{3.1}$$

and taking suprema of moduli gives $\|\hat{a}\|_\infty = \sup_{\lambda \in \sigma(a)} |\lambda| = r(a)$, the spectral radius. Now A is commutative, so a is normal ($aa^* = a^*a$), and proposition 2.1.3 gives $r(a) = \|a\|$. Combining,

$$\|\hat{a}\|_\infty = r(a) = \|a\|$$

so Γ is isometric. An isometry is injective, and its image is complete, hence closed in $C(\hat{A})$.

Step 2: Γ is surjective, unital case. The image $\Gamma(A) \subseteq C(\hat{A})$ is a subalgebra, since Γ is an algebra homomorphism, and it is closed by Step 1. It satisfies the three hypotheses of the Stone-Weierstrass theorem for the compact Hausdorff space $\hat{A}$ [7].

First, $\Gamma(A)$ contains the constant functions: $\hat{1}(\varphi) = \varphi(1) = 1$ for every character φ, so $\hat{1}$ is the constant function 1, and scalar multiples give all constants. Second, $\Gamma(A)$ separates the points of $\hat{A}$. If $\varphi \neq \psi$ are distinct characters, they differ at some $a \in A$, that is $\varphi(a) \neq \psi(a)$, which reads $\hat{a}(\varphi) \neq \hat{a}(\psi)$, so $\hat{a}$ separates φ from ψ. Third, $\Gamma(A)$ is closed under complex conjugation: Γ is a $*$-homomorphism, and the involution on $C(\hat{A})$ is pointwise conjugation, so $\overline{\hat{a}} = \widehat{a^*} = \Gamma(a^*) \in \Gamma(A)$.

By Stone-Weierstrass, a subalgebra of $C(\hat{A})$ that contains the constants, separates points, and is closed under conjugation is dense in the supremum norm [6]. So $\Gamma(A)$ is dense; being also closed by Step 1, it is all of $C(\hat{A})$. Thus Γ is an isometric $*$-isomorphism of A onto $C(\hat{A})$, establishing the unital case.

Step 3: the non-unital case. Let A be commutative but non-unital, and pass to the unitization $\tilde{A}$ of proposition 1.4.5, a commutative unital C^*-algebra containing A as a maximal ideal with $\tilde{A}/A \cong \mathbb{C}$. By Step 2 applied to $\tilde{A}$, the Gelfand transform $\Gamma_{\tilde{A}} \colon \tilde{A} \to C(\widehat{\tilde{A}})$ is an isometric $*$-isomorphism onto $C(\widehat{\tilde{A}})$.

The characters of $\tilde{A}$ are the characters of A each extended by sending the adjoined unit to 1, together with one additional character φ_∞, the composite $\tilde{A} \to \tilde{A}/A \cong \mathbb{C}$, which kills A. Concretely $\varphi_\infty(a, \lambda) = \lambda$, and a character φ of A extends to $\tilde{\varphi}(a, \lambda) = \varphi(a) + \lambda$. This identifies $\widehat{\tilde{A}}$, as a set, with $\hat{A} \cup \{\varphi_\infty\}$, and the weak-$*$ topology makes it the one-point compactification $(\hat{A})^+$ of the locally compact space $\hat{A}$, with φ_∞ the point at infinity: a net φ_i in $\hat{A}$ converges to φ_∞ in $\widehat{\tilde{A}}$ precisely when $\varphi_i(a) \to 0$ for every $a \in A$, which is the criterion for escaping to infinity in $\hat{A}$.

Under the isomorphism $C(\widehat{\tilde{A}}) = C\big((\hat{A})^+\big)$, the functions vanishing at the point φ_∞ are exactly $C_0(\hat{A})$. For $a \in A$ the transform $\Gamma_{\tilde{A}}(a)$ vanishes at φ_∞, since $\varphi_\infty(a) = 0$, and its restriction to $\hat{A}$ is $\hat{a} = \Gamma(a)$. So $\Gamma_{\tilde{A}}$ carries the ideal $A \subseteq \tilde{A}$ isometrically onto the ideal $C_0(\hat{A}) \subseteq C\big((\hat{A})^+\big)$ of functions vanishing at infinity, and this restriction is exactly $\Gamma \colon A \to C_0(\hat{A})$. An isometric $*$-isomorphism of $\tilde{A}$ onto $C\big((\hat{A})^+\big)$ restricts to an isometric $*$-isomorphism between corresponding ideals, so Γ is an isometric $*$-isomorphism of A onto $C_0(\hat{A})$, completing the proof.

The theorem is the precise form of the identification "commutative C^*-algebra = locally compact Hausdorff space." A commutative C^*-algebra A is not merely abstractly isomorphic to some function algebra; it is its own function algebra, on the space $\hat{A}$ that the algebra manufactures from its own characters, and the isomorphism Γ is canonical, built with no choices. The unital case recovers the compact spaces and the algebras $C(X)$; the non-unital case recovers the general locally compact spaces and the algebras $C_0(X)$ of functions vanishing at infinity, with unitization on the algebra side matching one-point compactification on the space side, exactly as the two examples of example 1.4.6 anticipated.

Two features of the proof are worth isolating. The isometry is where the C^*-identity is spent: for a general commutative Banach algebra the Gelfand transform is a contraction, $\|\hat{a}\|_\infty = r(a) \leq \|a\|$, and can strictly contract, as it does for the disk algebra, where the transform loses the boundary values. The equality $r(a) = \|a\|$ that fails there holds here because proposition 2.1.3 makes it hold for normal elements and commutativity makes every element normal. The surjectivity is where Stone-Weierstrass is spent, and it uses the $*$-structure through conjugation-closure: without the involution, a point-separating unital subalgebra of $C(X)$ need not be dense, the disk algebra again being the witness. The two C^*-ingredients, the norm equality and the involution, are exactly the two Stone-Weierstrass hypotheses beyond the algebra structure.

Example 3.2.2: Recovering X from $C(X)$

Let X be a compact Hausdorff space and $A = C(X)$. Each point $x \in X$ gives the evaluation character $\varphi_x \colon f \mapsto f(x)$, a nonzero algebra homomorphism $C(X) \to \mathbb{C}$. The map $x \mapsto \varphi_x$ is a homeomorphism $X \to \widehat{C(X)}$: it is injective because Urysohn's lemma produces, for distinct $x \neq y$, a continuous function with $f(x) \neq f(y)$ ([7, Urysohn, Compact Hausdorff Spaces Are Normal]), so $\varphi_x \neq \varphi_y$; it is surjective because every character of $C(X)$ is an evaluation, its kernel being a maximal ideal, which for $C(X)$ on a compact space is the ideal of functions vanishing at a single point; and it is a homeomorphism because the weak-$*$ topology on characters pulls back to the topology of X, both being the topology of pointwise convergence of the functions f. Under this homeomorphism the Gelfand transform becomes the identity: $\hat{f}(\varphi_x) = \varphi_x(f) = f(x)$, so $\hat{f}$ "is" f read on $\widehat{C(X)} = X$. Thus $\Gamma\colon C(X) \to C(\widehat{C(X)})$ is, up to the homeomorphism $X \cong \widehat{C(X)}$, the identity map on $C(X)$, and the theorem says nothing was lost: the space X is reconstructed from the bare algebra $C(X)$ as its character space, and the functions are reconstructed as their own transforms.

That X is recovered from $C(X)$ is one half of a duality; the other half is that a map of spaces is recovered from the induced map of algebras, and the recovery is functorial. A continuous map $\varphi\colon X \to Y$ of compact Hausdorff spaces induces the pullback $\varphi^*\colon C(Y) \to C(X)$, $\varphi^*(f) = f \circ \varphi$, a unital $*$-homomorphism running in the opposite direction. This contravariance, together with the reconstruction of the space, organizes the whole correspondence into an equivalence of categories, which is the theorem's structural form.

Theorem 3.2.3: Gelfand-Naimark Duality

Let CommC^*_1 be the category of commutative unital C^*-algebras with unital $*$-homomorphisms, and $\mathsf{CptHaus}$ the category of compact Hausdorff spaces with continuous maps. The assignments

$$A \mapsto \hat{A}, \qquad X \mapsto C(X)$$

extend to a contravariant equivalence between $\mathbf{CommC}^*_1$ and **CptHaus**. On morphisms, a unital $*$-homomorphism $\theta\colon A \to B$ induces the continuous map $\hat{\theta}\colon \hat{B} \to \hat{A}$, $\hat{\theta}(\varphi) = \varphi \circ \theta$, and a continuous map $\varphi\colon X \to Y$ induces the pullback $\varphi^*\colon C(Y) \to C(X)$, $\varphi^*(f) = f \circ \varphi$. The natural isomorphisms $A \cong C(\hat{A})$ (the Gelfand transform) and $X \cong \widehat{C(X)}$ (evaluation) witness the equivalence.

***Proof*:**
Step 1: the two assignments are contravariant functors. Sending X to $C(X)$ and a continuous $\varphi\colon X \to Y$ to $\varphi^*\colon C(Y) \to C(X)$ is functorial and contravariant: φ^* is a unital $*$-homomorphism because precomposition preserves pointwise sums, products, conjugates, and the constant 1; and $(\psi \circ \varphi)^* = \varphi^* \circ \psi^*$ with $(\mathrm{id}_X)^* = \mathrm{id}_{C(X)}$, from associativity of composition. Sending A to $\hat{A}$ and a unital $*$-homomorphism $\theta\colon A \to B$ to $\hat{\theta}\colon \hat{B} \to \hat{A}$, $\hat{\theta}(\varphi) = \varphi \circ \theta$, is likewise functorial and contravariant: if φ is a character of B then $\varphi \circ \theta$ is a homomorphism $A \to \mathbb{C}$, nonzero because θ is unital so $(\varphi \circ \theta)(1) = 1$, hence a character of A; the map $\hat{\theta}$ is continuous for the weak-$*$ topologies, since $\varphi \mapsto (\varphi \circ \theta)(a) = \varphi(\theta(a))$ is continuous for each a; and $\widehat{\eta \circ \theta} = \hat{\theta} \circ \hat{\eta}$ with $\widehat{\mathrm{id}_A} = \mathrm{id}_{\hat{A}}$.

Step 2: the composite $A \mapsto C(\hat{A})$ is naturally isomorphic to the identity. For each A the Gelfand transform $\Gamma_A\colon A \to C(\hat{A})$ is an isometric $*$-isomorphism by theorem 3.2.1. Naturality is the assertion that for every unital $*$-homomorphism $\theta\colon A \to B$ the square

$$(\hat{\theta})^* \circ \Gamma_A = \Gamma_B \circ \theta$$

commutes as maps $A \to C(\hat{B})$. Evaluating both sides at $a \in A$ and at a character $\varphi \in \hat{B}$: the right side gives $\Gamma_B(\theta(a))(\varphi) = \varphi(\theta(a))$, and the left side gives $(\hat{\theta})^*(\Gamma_A(a))(\varphi) = \Gamma_A(a)(\hat{\theta}(\varphi)) = (\varphi \circ \theta)(a) = \varphi(\theta(a))$. The two agree, so Γ is a natural isomorphism from the identity functor to $A \mapsto C(\hat{A})$.

Step 3: the composite $X \mapsto \widehat{C(X)}$ is naturally isomorphic to the identity. For each compact Hausdorff X the evaluation map $\varepsilon_X\colon X \to \widehat{C(X)}$, $\varepsilon_X(x) = \varphi_x$ with $\varphi_x(f) = f(x)$, is a homeomorphism by example 3.2.2. Naturality is that for every continuous $\varphi\colon X \to Y$ the square

$$\widehat{\varphi^*} \circ \varepsilon_X = \varepsilon_Y \circ \varphi$$

commutes as maps $X \to \widehat{C(Y)}$. Evaluating at $x \in X$ and $f \in C(Y)$: the right side gives $\varphi_{\varphi(x)}(f) = f(\varphi(x))$, and the left side gives $\widehat{\varphi^*}(\varphi_x)(f) = (\varphi_x \circ \varphi^*)(f) = \varphi_x(f \circ \varphi) = (f \circ \varphi)(x) = f(\varphi(x))$. The two agree, so ε is a natural isomorphism from the identity functor to $X \mapsto \widehat{C(X)}$.

The two natural isomorphisms of Steps 2 and 3 exhibit the two functors as inverse up to natural isomorphism, which is the definition of a contravariant equivalence, completing the proof.

The equivalence is the reason a commutative C^*-algebra and its character space carry the same information: any construction on compact Hausdorff spaces has a mirror image on commutative unital C^*-algebras, with arrows reversed, and conversely. A point of X is a character of $C(X)$; a continuous map is a unital $*$-homomorphism the other way; a closed subspace $Z \subseteq X$ is a quotient $C(X) \to C(Z)$, and an open subset $U \subseteq X$ is an ideal $C_0(U) \subseteq C(X)$. This dictionary is what permits the reading of a general, noncommutative C^*-algebra as a "noncommutative space," since every notion on spaces has been rewritten in a purely algebraic form that no longer refers to points and so continues to make sense when the algebra is noncommutative and has too few characters to reconstruct any space at all. The non-unital algebras extend the dictionary to locally compact spaces and proper maps, with $C_0(-)$ contravariant on the category of locally compact Hausdorff spaces and proper continuous maps.

The functional calculus of chapter 2 is the one-generator instance of this duality, and the corollary below makes the link exact. A single normal element a generates a commutative unital C^*-algebra $C^*(a, 1)$, whose character space is a compact Hausdorff space; the assertion is that this space is nothing other than the spectrum $\sigma(a)$, the same compact set the functional calculus was already built over.

Corollary 3.2.4: The Character Space of a Singly-Generated Algebra

Let a be a normal element of a unital C^*-algebra A, and $C^*(a,1)$ the unital C^*-subalgebra it generates. The map

$$\hat{a}\colon \widehat{C^*(a,1)} \to \sigma(a), \qquad \varphi \mapsto \varphi(a)$$

is a homeomorphism. Under it the Gelfand transform $\Gamma\colon C^*(a,1) \to C\big(\widehat{C^*(a,1)}\big)$ becomes an isometric $*$-isomorphism $C^*(a,1) \cong C(\sigma(a))$.

***Proof*:**
Write $B = C^*(a,1)$, a commutative unital C^*-algebra since a is normal, so a and a^* commute and generate a commutative algebra. By (3.1) applied inside B, the range of the transform $\hat{a}$ is the spectrum of a computed in B, and by spectral permanence this equals $\sigma(a)$; so $\hat{a}$ maps $\hat{B}$ onto $\sigma(a)$. The map $\hat{a}$ is continuous, being $\varphi \mapsto \varphi(a)$ in the weak-$*$ topology, and $\hat{B}$ is compact by proposition 3.1.6, so it suffices to show $\hat{a}$ is injective; a continuous bijection from a compact space to a Hausdorff space is a homeomorphism.

For injectivity, suppose $\varphi, \psi \in \hat{B}$ have $\varphi(a) = \psi(a)$. Then $\varphi(a^*) = \overline{\varphi(a)} = \overline{\psi(a)} = \psi(a^*)$, since a character is $*$-preserving. A character is an algebra homomorphism, so φ and ψ agree on every polynomial in a and a^*; these polynomials are dense in $B = C^*(a,1)$, and characters are continuous (norm ≤ 1 by definition 3.1.1), so $\varphi = \psi$. Hence $\hat{a}$ is a homeomorphism $\hat{B} \cong \sigma(a)$.

Precomposing the Gelfand isomorphism $\Gamma\colon B \to C(\hat{B})$ of theorem 3.2.1 with the homeomorphism $\hat{a}$ yields an isometric $*$-isomorphism $B \cong C(\sigma(a))$, completing the proof.

The corollary identifies the two constructions the book has produced for a normal element. The functional calculus of theorem 2.2.2 gave an isometric $*$-isomorphism $C(\sigma(a)) \cong C^*(a,1)$ sending the coordinate function to a; the commutative Gelfand-Naimark theorem gives an isometric $*$-isomorphism $C^*(a,1) \cong C(\widehat{C^*(a,1)})$; and the corollary says the two spaces $\sigma(a)$ and $\widehat{C^*(a,1)}$ agree, so the two isomorphisms are inverse to one another. The correspondence $\widehat{C^*(a,1)} \cong \sigma(a)$ also explains why the spectrum was the natural domain for the calculus in the first place: a character of $C^*(a,1)$ is determined by its value on the generator a, that value ranges over $\sigma(a)$, and evaluating a function of a at the character amounts to evaluating the corresponding continuous function at the spectral point.

3.3 The Functional Calculus as Gelfand Duality

The continuous functional calculus of theorem 2.2.2 and the commutative Gelfand-Naimark theorem of theorem 3.2.1 were built by different routes: the first was assembled by hand from Stone-Weierstrass on the single algebra $C^*(a,1)$ generated by one normal element, while the second is the structural statement that every commutative C^*-algebra is a function algebra. Applied to $A = C^*(a,1)$, these are the same isomorphism. The task here is to make that identification precise and to read off what it explains.

Recall the two isomorphisms in play. Theorem 2.2.2 furnishes an isometric $*$-isomorphism

$$C(\sigma(a)) \cong C^*(a,1), \qquad f \mapsto f(a)$$

carrying the coordinate function $z \mapsto z$ to a and the constant 1 to the unit. Theorem 3.2.1 furnishes the Gelfand transform, an isometric $*$-isomorphism $\Gamma\colon C^*(a,1) \to C(\widehat{C^*(a,1)})$, $b \mapsto \hat{b}$. Corollary 3.2.4 identifies the character space $\widehat{C^*(a,1)}$ with the spectrum $\sigma(a)$ by evaluating a character on the generator, $\varphi \mapsto \varphi(a)$. Composing Γ with this homeomorphism turns the Gelfand transform into an isometric $*$-isomorphism $C^*(a,1) \to C(\sigma(a))$ pointing the opposite way to the functional calculus. The claim is that the two are mutually inverse.

Proposition 3.3.1: The Functional Calculus is the Inverse Gelfand Transform

Let A be a unital C^*-algebra and $a \in A$ normal, and write $B = C^*(a,1)$ for the unital C^*-subalgebra it generates. Let

$$h\colon \widehat{B} \to \sigma(a), \qquad h(\varphi) = \varphi(a)$$

be the homeomorphism of corollary 3.2.4, and let $\Gamma\colon B \to C(\widehat{B})$ be the Gelfand transform. Then the continuous functional calculus $\Phi\colon C(\sigma(a)) \to B$, $\Phi(f) = f(a)$, of theorem 2.2.2 satisfies

$$\Phi(f) = \Gamma^{-1}(f \circ h) \qquad \text{for all } f \in C(\sigma(a))$$

Equivalently, under the identification $C(\widehat{B}) \cong C(\sigma(a))$ induced by h, the functional calculus Φ is the inverse of the Gelfand transform: $\Gamma \circ \Phi = \mathrm{id}$ and $\Phi \circ \Gamma = \mathrm{id}$.

***Proof*:**
Both $\Phi\colon C(\sigma(a)) \to B$ and the map $\Psi\colon C(\sigma(a)) \to B$ defined by $\Psi(f) = \Gamma^{-1}(f \circ h)$ are unital $*$-homomorphisms. For Φ this is theorem 2.2.2; for Ψ it is a composite of $*$-homomorphisms, since $f \mapsto f \circ h$ is a $*$-isomorphism $C(\sigma(a)) \to C(\widehat{B})$ (pullback along the homeomorphism h) and Γ^{-1} is a $*$-isomorphism by theorem 3.2.1. To show $\Phi = \Psi$ it suffices to check that they agree on a generating set of the C^*-algebra $C(\sigma(a))$, and by Stone-Weierstrass, applied as in theorem 2.2.2 [7], the constant 1 together with the coordinate function

$$\iota\colon \sigma(a) \to \mathbb{C}, \qquad \iota(z) = z$$

generate $C(\sigma(a))$ as a C^*-algebra.

On the unit both agree: $\Phi(1) = 1$ and $\Psi(1) = \Gamma^{-1}(1 \circ h) = \Gamma^{-1}(1) = 1$, since Γ is unital.

It remains to check agreement on ι. By construction of the functional calculus, $\Phi(\iota) = a$. For Ψ, unwind the pullback: for every character $\varphi \in \widehat{B}$,

$$(\iota \circ h)(\varphi) = \iota(h(\varphi)) = \iota(\varphi(a)) = \varphi(a) = \hat{a}(\varphi)$$

so $\iota \circ h = \hat{a} = \Gamma(a)$ as elements of $C(\widehat{B})$. Applying Γ^{-1} gives $\Psi(\iota) = \Gamma^{-1}(\Gamma(a)) = a = \Phi(\iota)$.

Two unital $*$-homomorphisms out of $C(\sigma(a))$ that agree on the generating set $\{1, \iota\}$ agree on the $*$-subalgebra it generates, and this subalgebra is dense; both maps are continuous (a $*$-homomorphism between C^*-algebras is contractive), so they agree everywhere. Hence $\Phi = \Psi$, that is, $\Phi(f) = \Gamma^{-1}(f \circ h)$ for all f. The two displayed equations $\Gamma \circ \Phi = \mathrm{id}$ and $\Phi \circ \Gamma = \mathrm{id}$ follow because Φ and $f \mapsto f \circ h$ are inverse to Γ and its precomposition, as we sought to show.

The proposition says the two constructions, the hand-built calculus of chapter 2 and the Gelfand transform of this chapter, are one map seen from two sides. Reading it from left to right, $f(a) =$

$\Gamma^{-1}(f \circ h)$ evaluates a continuous function f on the spectrum by transporting it to a function on characters and inverting the Gelfand transform. Reading it from right to left, the Gelfand transform of $b = f(a)$ is $\hat{b} = f \circ h$: a character φ sends $f(a)$ to $f(\varphi(a))$, so knowing the value $\varphi(a) \in \sigma(a)$ of a character on the generator determines its value on every element of B. This is why $\widehat{B}$ is exactly $\sigma(a)$ and no larger: a character of B is pinned down by one complex number, the point of the spectrum where it evaluates.

The identification also settles the uniqueness clause of theorem 2.2.2 at a structural level. That theorem's functional calculus is the *unique* unital $*$-homomorphism $C(\sigma(a)) \to A$ sending ι to a; the hand-built proof obtained uniqueness from the density of polynomials in z and $\overline{z}$. Gelfand duality gives the same conclusion without touching polynomials: a unital $*$-homomorphism $C(\sigma(a)) \to A$ with image in B is, after transporting along h, a $*$-homomorphism into B inverse to Γ on the generator, and Γ has exactly one inverse. Uniqueness of the functional calculus is thus uniqueness of the inverse of an isomorphism.

The same lens gives the spectral mapping theorem a second proof. Theorem 2.2.4 was established alongside the functional calculus; from the Gelfand picture it is a statement about function algebras. Under the isomorphism $B \cong C(\sigma(a))$, the element $f(a)$ corresponds to the function f, and the spectrum of an element of a commutative unital C^*-algebra $C(Y)$ is its range (a character of $C(Y)$ is evaluation at a point of Y, so $\sigma_{C(Y)}(g) = \{g(y) : y \in Y\} = g(Y)$). Applying this with $Y = \sigma(a)$ and $g = f$ gives $\sigma(f(a)) = f(\sigma(a))$, which is the spectral mapping theorem. The functional calculus reduces spectral computations in B to reading off the range of a function, and the range is what Gelfand duality identifies the spectrum to be.

A concrete instance makes the two viewpoints coincide on paper.

Example 3.3.2: The Functional Calculus for a Self-Adjoint Matrix

Let $A = M_2(\mathbb{C})$ and take the self-adjoint element

$$a = \begin{pmatrix} 3 & 0 \\ 0 & -1 \end{pmatrix}$$

Its spectrum is $\sigma(a) = \{3, -1\}$, a two-point space, and $B = C^*(a, 1)$ is the algebra of diagonal matrices $\{\mathrm{diag}(\lambda, \mu) : \lambda, \mu \in \mathbb{C}\}$, isomorphic to $\mathbb{C} \oplus \mathbb{C}$.

The functional calculus side. For $f \in C(\{3, -1\})$, the calculus of theorem 2.2.2 gives

$$f(a) = \begin{pmatrix} f(3) & 0 \\ 0 & f(-1) \end{pmatrix}$$

since f is a limit of polynomials in $z, \overline{z}$ and each such polynomial is applied entrywise to the diagonal matrix a. A continuous function on the two points $\{3, -1\}$ is a pair of complex numbers $(f(3), f(-1))$, so $C(\{3, -1\}) \cong \mathbb{C} \oplus \mathbb{C}$ and the calculus $f \mapsto f(a)$ is the identification $\mathbb{C} \oplus \mathbb{C} \cong B$.

The Gelfand side. The characters of $B \cong \mathbb{C} \oplus \mathbb{C}$ are the two coordinate projections $\varphi_1(\lambda, \mu) = \lambda$ and $\varphi_2(\lambda, \mu) = \mu$, so $\widehat{B} = \{\varphi_1, \varphi_2\}$. Evaluating on the generator, $\varphi_1(a) = 3$ and $\varphi_2(a) = -1$, so the homeomorphism $h \colon \widehat{B} \to \sigma(a)$ of corollary 3.2.4 sends $\varphi_1 \mapsto 3$, $\varphi_2 \mapsto -1$. The Gelfand transform of $b = \mathrm{diag}(\lambda, \mu)$ is $\hat{b}(\varphi_1) = \lambda$, $\hat{b}(\varphi_2) = \mu$.

Now check proposition 3.3.1 directly. Given f, the function $f \circ h$ on $\widehat{B}$ takes the values $f(h(\varphi_1)) = f(3)$ and $f(h(\varphi_2)) = f(-1)$. The unique diagonal matrix whose Gelfand transform is this function is $\mathrm{diag}(f(3), f(-1))$, so

$$\Gamma^{-1}(f \circ h) = \mathrm{diag}(f(3), f(-1)) = f(a)$$

matching the functional-calculus value. The spectral mapping theorem reads off as $\sigma(f(a)) = \{f(3), f(-1)\} = f(\sigma(a))$, the range of f.

The matrix example shows the mechanism with no analysis: the diagonal matrix a has as many characters as it has distinct eigenvalues, each character records one eigenvalue, and applying f is applying it eigenvalue by eigenvalue. The general normal element replaces the finite eigenvalue set by the compact spectrum $\sigma(a)$ and the entrywise action by the functional calculus, but the content is unchanged: B is the function algebra $C(\sigma(a))$, characters are points of $\sigma(a)$, and $f(a)$ is the element whose value at each point $z \in \sigma(a)$ is $f(z)$. Gelfand-Naimark is what promotes the single-generator computation of chapter 2 to this equivalence of the whole commutative theory with topology.

3.4 The Serre-Swan Theorem

The vector bundles over a compact Hausdorff space X are read off $C(X)$ from its module theory. Fix throughout a compact Hausdorff space X and write $C(X)$ for the C^*-algebra of continuous complex-valued functions on it. A complex vector bundle $E \to X$ has a set of continuous global sections

$$\Gamma(E) = \{s\colon X \to E \big| s \text{ continuous, } s(x) \in E_x \text{ for all } x\}$$

and this set is a module over $C(X)$: sections add pointwise, and a function $f \in C(X)$ scales a section by $(f \cdot s)(x) = f(x)\, s(x)$, the scaling being continuous because multiplication in each fiber is. The task is to identify the modules that arise this way. They turn out to be exactly the finitely generated projective ones, and the assignment $E \mapsto \Gamma(E)$ is an equivalence of categories onto them. The one topological input is that a bundle over a compact Hausdorff space is a summand of a trivial bundle, proved in [10, Vector Bundle Over Compact Are Projective]; the one algebraic input is the characterization of finitely generated projective modules as summands of free modules, from [3, Equivalencies for Projective Modules].

A trivial bundle grounds the correspondence before any theorem. The product bundle $X \times \mathbb{C}^n \to X$ has a section given by an n-tuple of continuous functions, so

$$\Gamma(X \times \mathbb{C}^n) \cong C(X)^n$$

is the free module of rank n. This is the model case, and the content of the theorem is that a general bundle deviates from it in exactly the way a general projective module deviates from a free one: both are direct summands, cut out by an idempotent.

Example 3.4.1: Sections of the Tautological Bundle over $\mathbb{C}\mathrm{P}^1$

Let $X = \mathbb{C}\mathrm{P}^1 = S^2$, the space of complex lines through the origin in $\mathbb{C}^2$, and let $L \to \mathbb{C}\mathrm{P}^1$ be the tautological line bundle, whose fiber over a point $\ell \in \mathbb{C}\mathrm{P}^1$ is the line $\ell \subseteq \mathbb{C}^2$ it names. A continuous section is a continuous choice $s(\ell) \in \ell$ of a vector on each line. This module is not free of rank 1: a free generator would be a nowhere-vanishing section, hence a continuous map assigning to each line ℓ a nonzero vector $s(\ell)$ lying on ℓ; normalizing $s(\ell)/\|s(\ell)\|$ would then give a continuous section of the associated unit-sphere bundle, which is the Hopf fibration $S^3 \to S^2$. A continuous section of the Hopf fibration would split it as a product, forcing $S^3 \cong S^2 \times S^1$, which is false on homotopy grounds ($\pi_2(S^3) = 0$ while $\pi_2(S^2 \times S^1) = \mathbb{Z}$); see [8, Connectivity and Bottom Homotopy of Spheres, Homotopy Groups of a Product, The Homotopy of S^2 via the Hopf Fibration]. So L has no nowhere-vanishing section, and $\Gamma(L)$ is a rank-one module with no free generator, projective but not free, the module-level record of the fact that L is a nontrivial line bundle.

The example shows the phenomenon the theorem organizes: a nontrivial bundle produces a projective-but-not-free module, and the failure to be free is the failure of the bundle to be trivial. To state the equivalence, $\mathrm{Vect}_{\mathbb{C}}(X)$ denotes the category whose objects are complex vector bundles over X and whose morphisms are continuous bundle maps covering the identity on X, and finitely generated projective $C(X)$-modules form a category under $C(X)$-linear maps.

Theorem 3.4.2: Serre-Swan

Let X be a compact Hausdorff space. The global-sections functor

$$\Gamma\colon \mathrm{Vect}_{\mathbb{C}}(X) \to \{\text{finitely generated projective } C(X)\text{-modules}\}$$

sending a bundle E to its section module $\Gamma(E)$ and a bundle map to the induced map on sections is an equivalence of categories.

***Proof*:**
The proof has three parts: that $\Gamma(E)$ is finitely generated projective, that every such module is Γ of some bundle, and that Γ is a bijection on morphism sets.

Step 1: $\Gamma(E)$ is finitely generated projective. Since X is compact, E is a direct summand of a trivial bundle: there is an n and a bundle $E^\perp$ with $E \oplus E^\perp \cong X \times \mathbb{C}^n$, by [10, Vector Bundle Over Compact Are Projective]. Applying Γ and using that sections of a direct sum of bundles are pairs of sections, $\Gamma(E) \oplus \Gamma(E^\perp) \cong \Gamma(X \times \mathbb{C}^n) \cong C(X)^n$. Thus $\Gamma(E)$ is a direct summand of the free module $C(X)^n$, which is the definition of a finitely generated projective module in [3, Equivalencies for Projective Modules].

Step 2: essential surjectivity. Let P be a finitely generated projective $C(X)$-module. By [3, Equivalencies for Projective Modules] it is a direct summand of some $C(X)^n$, so there is an idempotent $p \in \mathrm{End}_{C(X)}(C(X)^n) = M_n(C(X))$ with $P \cong \mathrm{im}(p)$. Write $p = (p_{ij})$ with entries $p_{ij} \in C(X)$. Evaluating at a point $x \in X$ gives a matrix $p(x) = (p_{ij}(x)) \in M_n(\mathbb{C})$, and $p^2 = p$ in $M_n(C(X))$ says $p(x)^2 = p(x)$ for every x, so each $p(x)$ is an idempotent on $\mathbb{C}^n$. Set

$$E_p = \{(x, v) \in X \times \mathbb{C}^n \big| p(x)v = v\}$$

the union over x of the images $\operatorname{im} p(x) \subseteq \mathbb{C}^n$. This is a subbundle of the trivial bundle $X \times \mathbb{C}^n$: the fibers $\operatorname{im} p(x)$ have locally constant dimension because $\operatorname{rank} p(x) = \operatorname{tr} p(x)$ for an idempotent and $x \mapsto \operatorname{tr} p(x) = \sum_i p_{ii}(x)$ is a continuous integer-valued function, hence locally constant. Local trivializations come from p itself: fix x_0 and choose vectors $w_1, \ldots, w_k \in \mathbb{C}^n$ whose images $p(x_0)w_1, \ldots, p(x_0)w_k$ form a basis of $\operatorname{im} p(x_0)$; the vectors $p(x)w_1, \ldots, p(x)w_k$ vary continuously in x and are independent at x_0, hence remain independent on a neighborhood, where they span the k-dimensional fiber $\operatorname{im} p(x)$ and so furnish a continuous frame identifying E_p with a trivial bundle. A section of E_p is a continuous $s\colon X \to X \times \mathbb{C}^n$ with $p(x)s(x) = s(x)$, that is, an element of $C(X)^n$ fixed by p; and the fixed submodule of p acting on $C(X)^n$ is exactly $\mathrm{im}(p)$. Therefore $\Gamma(E_p) \cong \mathrm{im}(p) \cong P$.

Step 3: full faithfulness. A bundle map $\Phi\colon E \to F$ covering the identity restricts to a linear map $\Phi_x\colon E_x \to F_x$ on each fiber and sends a section s of E to the section $x \mapsto \Phi_x(s(x))$ of F; this is $C(X)$-linear because Φ_x is linear and commutes with scaling by $f(x)$. So Γ is a functor, and $\Gamma(\Phi) = 0$ forces $\Phi_x(s(x)) = 0$ for all sections s and all x; since the values $s(x)$ of sections at a fixed x exhaust the fiber E_x (a vector in E_x extends to a global section, E being a summand of a trivial bundle), $\Phi_x = 0$ for all x, so $\Phi = 0$ and Γ is injective on morphisms. For surjectivity, let $T\colon \Gamma(E) \to \Gamma(F)$ be $C(X)$-linear. Evaluation at x is well defined on the level of values: if two sections agree at x, then their difference lies in the submodule $I_x \cdot \Gamma(E)$, where $I_x = \{f \in C(X) \big| f(x) = 0\}$ is the maximal ideal at x, because a section vanishing at x is a $C(X)$-combination of sections weighted by functions in I_x; and T carries $I_x \cdot \Gamma(E)$ into $I_x \cdot \Gamma(F)$, whose sections vanish at x. So T descends to a linear map $T_x\colon E_x \to F_x$ on fibers, $T_x(s(x)) = (Ts)(x)$, and these assemble into a continuous bundle map Φ with $\Gamma(\Phi) = T$; continuity of Φ follows because Ts is continuous for every continuous s and the $s(x)$ span the fibers. Thus Γ is bijective on morphism sets. Steps 1 and 2 give that Γ is essentially surjective and Step 3 that it is fully faithful, so Γ is an equivalence of categories, completing the proof.

The three parts say the same thing in three registers. Geometrically a bundle is a summand of a trivial bundle; algebraically its module of sections is a summand of a free module; and the two summand decompositions are the same, both recorded by the idempotent $p \in M_n(C(X))$ whose fiberwise value $p(x)$ projects $\mathbb{C}^n$ onto E_x. The functor Γ trades the geometric picture for the algebraic one without loss, which is what an equivalence of categories asserts. Note that Γ is nowhere contravariant, unlike the Gelfand correspondence $X \mapsto C(X)$ of theorem 3.2.3: bundles and their section modules face the same way because a bundle map and its map on sections point in the same direction, whereas passing from a space to its function algebra reverses arrows.

Compactness is essential, and it enters at exactly one point: Step 1's assertion that E is a summand of a trivial bundle. On a compact base a finite trivializing cover and a partition of unity subordinate to it patch the local trivializations into a global embedding $E \hookrightarrow X \times \mathbb{C}^n$ with a complementary bundle, so every bundle is a summand of a trivial one of some finite rank n. On a noncompact space this fails: an infinite-dimensional ambient trivial bundle may be forced, and the section module of a bundle need not be finitely generated. The right statement over a locally compact base replaces $C(X)$ by $C_0(X)$, replaces finitely generated projective modules by a larger class allowing infinite rank, and is more delicate; the compact case is where bundles and finitely generated projective modules correspond on the nose, and it is the case K-theory uses. The finiteness on the algebraic side and the compactness on the geometric side are the same hypothesis seen from the two ends of Γ.

The equivalence has an immediate consequence in the currency K-theory is built from. A projection in a matrix algebra $M_n(C(X))$ is a self-adjoint idempotent $p = p^* = p^2$, and the idempotents produced in Step 2 can always be taken self-adjoint: an idempotent in $M_n(C(X))$ is similar to a projection with the same image, replacing p by the orthogonal projection onto $\operatorname{im} p$ along its orthogonal complement, which is fiberwise the Gram-Schmidt projection and so continuous in x. So the bundles over X are recorded not by idempotents alone but by projections, and isomorphic bundles by equivalent projections.

Corollary 3.4.3: Projections over $C(X)$ Are Vector Bundles

Let X be compact Hausdorff. The map sending a projection $p \in M_n(C(X))$, over all n, to the vector bundle $E_p \subseteq X \times \mathbb{C}^n$ of theorem 3.4.2 induces a bijection

$$\{\text{projections in } M_\infty(C(X))\}\big/\sim \;\longleftrightarrow\; \{\text{isomorphism classes of vector bundles over } X\}$$

where two projections are identified when they are Murray-von Neumann equivalent, and $M_\infty(C(X)) = \bigcup_n M_n(C(X))$ under the corner embeddings $q \mapsto \left(\begin{smallmatrix} q & 0 \\ 0 & 0 \end{smallmatrix}\right)$.

***Proof*:**
A projection $p \in M_n(C(X))$ is in particular an idempotent, so theorem 3.4.2 attaches to it the bundle E_p with $\Gamma(E_p) \cong \mathrm{im}(p)$, and every bundle arises this way by essential surjectivity together with the reduction of an idempotent to a projection above. Two projections p, q have isomorphic image modules, hence isomorphic bundles $E_p \cong E_q$, precisely when they are Murray-von Neumann equivalent, that is, when $p = v^*v$ and $q = vv^*$ for some $v \in M_\infty(C(X))$: such a v restricts fiberwise to a partial isometry implementing $E_p \cong E_q$. Conversely a bundle isomorphism $u\colon E_p \to E_q$ gives on each fiber a linear isomorphism $u_x\colon \operatorname{im} p(x) \to \operatorname{im} q(x)$, not yet a unitary; equip $\mathbb{C}^n$ with its standard Hermitian inner product and pass to the fiberwise polar part $w_x = u_x(u_x^*u_x)^{-1/2}$, where $u_x^*u_x$ is the positive invertible operator on $\operatorname{im} p(x)$ and its inverse square root is defined by the functional calculus. Because u_x is invertible and varies continuously in x, so does $u_x^*u_x$ and hence $(u_x^*u_x)^{-1/2}$, so w_x is a continuous field of unitaries $\operatorname{im} p(x) \to \operatorname{im} q(x)$. Extending each w_x by zero off $\operatorname{im} p(x)$ gives a continuous field of partial isometries on $\mathbb{C}^n$, whose continuous matrix entries assemble the element $v \in M_n(C(X))$ with $v^*v = p$ and $vv^* = q$. The corner embeddings make the union $M_\infty(C(X))$ well defined and identify E_p with $E_{p\oplus 0}$, so the correspondence is independent of the ambient n, giving the stated bijection, as we sought to show.

This is the geometric fact that the K-theory of Part II converts into a group. The group $K_0(C(X))$ is assembled from Murray-von Neumann classes of projections in $M_\infty(C(X))$, and the corollary identifies those classes with isomorphism classes of vector bundles over X; the identification of the resulting groups, $K_0(C(X)) \cong K^0(X)$, is what makes operator K-theory recover the topological K-theory of X, and it is developed in the K_0 chapter (chapter 6), not here. What the present section gives is the object-level dictionary underneath that isomorphism: over a compact space, a projection over the function algebra is a vector bundle, and equivalence of projections is isomorphism of bundles.

4

States and the GNS Construction

Gelfand-Naimark identified the commutative C^*-algebras as function algebras; this chapter gives the noncommutative half of the dictionary, realizing every C^*-algebra concretely as an algebra of Hilbert-space operators. The bridge is the notion of a state, a positive linear functional of norm one, which plays on a C^*-algebra the role a probability measure plays on a commutative one. Out of a single state the Gelfand-Naimark-Segal construction manufactures a representation on a Hilbert space, and assembling the representations of all states embeds the algebra faithfully and isometrically into $B(H)$. This is the second Gelfand-Naimark theorem, the exact sense in which every C^*-algebra is a concrete algebra of operators.

The chapter proceeds in that order: states and their basic calculus, the GNS construction, the universal representation, and finally the pure states, the extreme points of the state space, from which the irreducible representations of the following chapter are built.

4.1 Positive Functionals and States

A linear functional φ is positive when it sends every element of the form a^*a to a nonnegative number, exactly the algebraic form of $\langle T\xi, T\xi\rangle = \|T\xi\|^2 \geq 0$. Normalizing a positive functional gives a state, and this section establishes the two facts that make states usable. First, positivity alone forces a Cauchy-Schwarz inequality and boundedness, so a state is automatically a norm-one functional with no extra hypothesis. Second, there are enough states to separate elements by norm: for any self-adjoint a some state attains $|\varphi(a)| = \|a\|$. The commutative model to hold in mind throughout is $C(X)$.

Definition 4.1.1: Positive Functional

Let A be a C^*-algebra. A linear functional $\varphi\colon A \to \mathbb{C}$ is *positive* if

$$\varphi(a^*a) \geq 0 \qquad \text{for every } a \in A$$

Equivalently, by theorem 2.3.3, $\varphi(a) \geq 0$ for every positive element $a \geq 0$ (definition 2.3.1).

The two formulations agree because the positive elements are exactly those of the form a^*a (theorem 2.3.3): writing $a \geq 0$ as $a = b^*b$ shows $\varphi(a) = \varphi(b^*b) \geq 0$, and conversely every a^*a is positive. Positivity is a purely order-theoretic condition, a linear functional that respects the positive cone of A_{sa}, and this compatibility with the order is what the GNS construction of section 4.2 uses. A state is the normalized version.

Definition 4.1.2: State

A *state* on a C^*-algebra A is a positive functional φ with $\|\varphi\| = 1$. The set of all states is the *state space* $S(A)$.
When A is unital, a positive functional φ is a state precisely when $\varphi(1) = 1$; this normalization by the value at the identity replaces the norm condition in the unital case, as proposition 4.1.3 shows.

Both conditions in the definition of a state, positivity and $\|\varphi\| = 1$, are needed to fix the functional, but they interact: the next result shows that positivity by itself already forces φ to be bounded with $\|\varphi\| = \varphi(1)$, so in the unital setting the single equation $\varphi(1) = 1$ pins down a state among positive functionals. The engine behind this is that a positive functional makes A into a semi-inner-product space, and every semi-inner product obeys Cauchy-Schwarz.

Proposition 4.1.3: Properties of Positive Functionals

Let φ be a positive functional on a unital C^*-algebra A. Then:

1. (Cauchy-Schwarz) For all $a, b \in A$,

$$|\varphi(b^*a)|^2 \leq \varphi(a^*a)\,\varphi(b^*b)$$

2. φ is self-adjoint: $\varphi(a^*) = \overline{\varphi(a)}$ for all $a \in A$.
3. φ is bounded, with $\|\varphi\| = \varphi(1)$.

***Proof*:**
Step 1: The sesquilinear form. Define $\langle a, b\rangle_\varphi = \varphi(b^*a)$ for $a, b \in A$. Since φ is linear and $a \mapsto b^*a$, $b \mapsto b^*a$ are respectively linear and conjugate-linear (the map $b \mapsto b^*$ is conjugate-linear), the form is linear in the first variable and conjugate-linear in the second. It is positive semidefinite: $\langle a, a\rangle_\varphi = \varphi(a^*a) \geq 0$ by positivity. The general Cauchy-Schwarz inequality for a positive semidefinite sesquilinear form ([9, Cauchy-Schwarz For Positive Semidefinite Forms]) gives

$$|\langle a, b\rangle_\varphi|^2 \leq \langle a, a\rangle_\varphi\,\langle b, b\rangle_\varphi$$

which is exactly $|\varphi(b^*a)|^2 \leq \varphi(a^*a)\,\varphi(b^*b)$, proving (1). Positivity already forces the form to be conjugate-symmetric, $\langle b, a\rangle_\varphi = \overline{\langle a, b\rangle_\varphi}$, since a positive semidefinite sesquilinear form over $\mathbb{C}$ is automatically Hermitian (by polarization), so this uses no property of φ beyond positivity. We recall the short argument for completeness: for $t \in \mathbb{C}$ and any a, b, positivity gives $0 \leq \langle a + tb, a + tb\rangle_\varphi = \langle a, a\rangle_\varphi + 2\operatorname{Re}\big(\bar{t}\langle a, b\rangle_\varphi\big) + |t|^2\langle b, b\rangle_\varphi$, and choosing t to minimize the right-hand side yields the inequality.

Step 2: Self-adjointness. First, φ is real on self-adjoint elements. Let $a = a^*$, and apply positivity to $b = 1 + a$: then $b^*b = (1 + a)^2 = 1 + 2a + a^2$ is self-adjoint with $\varphi(b^*b) \geq 0$ real, so

$$\varphi(1) + 2\varphi(a) + \varphi(a^2) \in \mathbb{R}$$

Both $\varphi(1) = \varphi(1^*1)$ and $\varphi(a^2) = \varphi(a^*a)$ are values of φ on elements of the form c^*c, hence nonnegative reals, so $\varphi(a) \in \mathbb{R}$. Now for arbitrary $a \in A$, write $a = h + ik$ with $h = \frac{1}{2}(a + a^*)$ and $k = \frac{1}{2i}(a - a^*)$ self-adjoint. Then $a^* = h - ik$, and since $\varphi(h), \varphi(k) \in \mathbb{R}$,

$$\varphi(a^*) = \varphi(h) - i\varphi(k) = \overline{\varphi(h) + i\varphi(k)} = \overline{\varphi(a)}$$

proving (2).

Step 3: Boundedness and the norm. Apply Cauchy-Schwarz with $b = 1$:

$$|\varphi(a)|^2 = |\varphi(1^*a)|^2 \leq \varphi(a^*a)\,\varphi(1) = \varphi(1)\,\varphi(a^*a)$$

For a with $\|a\| \leq 1$ we have $\|a^*a\| \leq 1$, so $a^*a \leq \|a^*a\|\,1 \leq 1$ in the order of A_{sa} (using $\|c\|\,1 - c \geq 0$ for self-adjoint c, a consequence of the spectrum lying in $[-\|c\|, \|c\|]$). Positivity applied to $1 - a^*a \geq 0$ gives $\varphi(a^*a) \leq \varphi(1)$, hence

$$|\varphi(a)|^2 \leq \varphi(1)^2$$

so $|\varphi(a)| \leq \varphi(1)$ whenever $\|a\| \leq 1$. Thus φ is bounded with $\|\varphi\| \leq \varphi(1)$. The reverse inequality is immediate, since $\|1\| = 1$ forces $\|\varphi\| \geq |\varphi(1)| = \varphi(1)$, the last equality because $\varphi(1) = \varphi(1^*1) \geq 0$ is real and nonnegative. Therefore $\|\varphi\| = \varphi(1)$, as we sought to show.

The proposition dissolves the apparent redundancy in the definition of a state. Positivity is the substantive hypothesis; boundedness and the value $\|\varphi\| = \varphi(1)$ come for free. For a unital algebra this means the state space is

$$S(A) = \{\varphi\colon A \to \mathbb{C} \,\big|\, \varphi \text{ positive},\ \varphi(1) = 1\}$$

a set cut out by linear positivity constraints plus one normalization, which is why it turns out to be convex and weak-$*$ compact (developed in section 4.4). Self-adjointness, part (2), says a state is determined by its values on self-adjoint elements and takes real values there, matching the intuition that a state is a noncommutative expectation: real observables have real expected values.

Example 4.1.4: A State on $M_2(\mathbb{C})$

On the C^*-algebra $M_2(\mathbb{C})$ of 2×2 complex matrices, fix a unit vector $\xi = (\xi_1, \xi_2)^\top \in \mathbb{C}^2$ and set

$$\varphi(a) = \langle a\xi, \xi\rangle = \sum_{i,j} a_{ij}\,\xi_j\,\overline{\xi_i}$$

This φ is linear in a, and $\varphi(a^*a) = \langle a^*a\,\xi, \xi\rangle = \|a\xi\|^2 \geq 0$, so it is positive. It is normalized: $\varphi(1) = \langle \xi, \xi\rangle = \|\xi\|^2 = 1$, so φ is a state. Concretely, for $\xi = (1, 0)^\top$ one gets $\varphi(a) = a_{11}$, the upper-left entry. A different normalized functional, the normalized trace $\tau(a) = \frac{1}{2}(a_{11} + a_{22})$, is also a state: $\tau(a^*a) = \frac{1}{2}\big(|a_{11}|^2 + |a_{21}|^2 + |a_{12}|^2 + |a_{22}|^2\big) \geq 0$ and $\tau(1) = 1$. The vector states $\langle \cdot\, \xi, \xi\rangle$ and the trace already exhibit the two kinds of states, pure and mixed, that section 4.4 separates.

Having established that states are automatically bounded, the second task is to guarantee there are enough of them. A single state sees only a sliver of the algebra; to embed A faithfully into operators one needs states that collectively detect the norm of every element. The following proposition gives the key case: a self-adjoint element whose norm is witnessed exactly by some state. The construction runs entirely through the commutative theory, taking a character of the abelian C^*-algebra generated by a and extending it by Hahn-Banach.

Proposition 4.1.5: States Detect the Norm

Let A be a unital C^*-algebra and $a \in A$ a self-adjoint element. Then there is a state $\varphi \in S(A)$ with

$$|\varphi(a)| = \|a\|$$

***Proof*:**
Step 1: A character on the commutative subalgebra. Let $B = C^*(a, 1)$ be the unital commutative C^*-algebra generated by a (commutative because a is self-adjoint). By commutative Gelfand-Naimark (theorem 3.2.1), the Gelfand transform is an isometric $*$-isomorphism $B \cong C(\Omega)$ where Ω is the character space of B, and by corollary 3.2.4 evaluation identifies Ω with the spectrum: $\Omega \cong \sigma(a)$, with a character ω corresponding to a point $\lambda \in \sigma(a)$ via $\omega(a) = \lambda$. Since a is self-adjoint its spectral radius equals its norm, $r(a) = \|a\|$, so there is a point $\lambda \in \sigma(a)$ with $|\lambda| = \|a\|$. Choose the character $\omega_0 \in \Omega$ with $\omega_0(a) = \lambda$; then $|\omega_0(a)| = \|a\|$.

Step 2: ω_0 is a state on B. A character is a unital $*$-homomorphism $B \to \mathbb{C}$, and characters are automatically positive: for $b \in B$,

$$\omega_0(b^*b) = \omega_0(b^*)\,\omega_0(b) = \overline{\omega_0(b)}\,\omega_0(b) = |\omega_0(b)|^2 \geq 0$$

where $\omega_0(b^*) = \overline{\omega_0(b)}$ because ω_0 is a $*$-homomorphism. And $\omega_0(1) = 1$, so ω_0 is a state on B by proposition 4.1.3. In particular $\|\omega_0\| = \omega_0(1) = 1$.

Step 3: Hahn-Banach extension to A. By the Hahn-Banach theorem ([9, Complex Hahn-Banach Theorem]) extend ω_0 to a linear functional φ on all of A with $\|\varphi\| = \|\omega_0\| = 1$. It remains to check φ is positive; extension preserves the norm but not obviously positivity, and this is where the norm normalization does the work. Since A is unital and φ extends ω_0, we have $\varphi(1) = \omega_0(1) = 1 = \|\varphi\|$. A norm-one functional on a unital C^*-algebra with $\varphi(1) = 1$ is positive, and the argument has two parts: first that φ is real on self-adjoint elements, then that it is nonnegative on positive ones.

Fix a self-adjoint $c \in A$ and write $\varphi(c) = x + iy$ with $x, y \in \mathbb{R}$. For real r the element $c + ir\,1$ satisfies $(c + ir\,1)^*(c + ir\,1) = c^2 + r^2\,1$, so

$$\|c + ir\,1\|^2 = \|(c + ir\,1)^*(c + ir\,1)\| = \|c^2 + r^2\,1\| \leq \|c\|^2 + r^2$$

using the C^*-identity. On the other hand $\|\varphi\| = 1$ gives $|\varphi(c + ir\,1)| \leq \|c + ir\,1\|$, and $\varphi(c + ir\,1) = \varphi(c) + ir\varphi(1) = x + i(y + r)$, so

$$x^2 + (y + r)^2 = |\varphi(c + ir\,1)|^2 \leq \|c\|^2 + r^2$$

The r^2 terms cancel, leaving $x^2+y^2+2ry \le \|c\|^2$ for every real r. Letting $r \to \pm\infty$ forces $y = 0$, so $\varphi(c) = x \in \mathbb{R}$: thus φ is real on self-adjoint elements. Now if $\sigma(c) \subseteq [\alpha, \beta] \subseteq \mathbb{R}$, apply the norm bound to the real shift $c - t\,1$: since $\|\varphi\| = 1$ and $\varphi(c) \in \mathbb{R}$, $|\varphi(c) - t| = |\varphi(c - t\,1)| \le \|c - t\,1\| = \max(|\alpha - t|, |\beta - t|)$. Taking t to the midpoint $(\alpha+\beta)/2$ gives $|\varphi(c) - t| \le (\beta - \alpha)/2$, which for the real number $\varphi(c)$ forces $\varphi(c) \in [\alpha, \beta]$. In particular a positive element (spectrum in $[0, \infty)$) has $\varphi(c) \ge 0$. Hence $\varphi(a^*a) \ge 0$ for all a, so φ is a positive functional, and with $\|\varphi\| = 1$ it is a state.

Finally $|\varphi(a)| = |\omega_0(a)| = \|a\|$, completing the proof.

The proposition establishes norm separation for a single self-adjoint element. It says the spectrum of a is not just an abstract set attached to a but is realized by functionals on A: each point of $\sigma(a)$ is $\omega_0(a)$ for a character of $C^*(a, 1)$, and each such character extends to a state on the ambient algebra. Because $\|a\| = \max_{\lambda\in\sigma(a)} |\lambda|$ for self-adjoint a, choosing the extremal point produces a state that reads off the norm. For a general (not self-adjoint) x, applying the result to the self-adjoint x^*x recovers $\|x\|^2 = \|x^*x\|$ from a state, which is what section 4.3 needs to make the universal representation isometric. The extension step also isolates a fact worth recording: on a unital C^*-algebra, a functional with $\|\varphi\| = \varphi(1) = 1$ is automatically a state, the converse of proposition 4.1.3(3).

The commutative case is not just a tool in the proof; it is the picture every reader should carry. For $A = C(X)$ the states are precisely the probability measures on X, and this identification makes "state = noncommutative probability measure" literal rather than metaphorical.

Example 4.1.6: States of $C(X)$ Are Probability Measures

Let X be a compact Hausdorff space and $A = C(X)$ the commutative unital C^*-algebra of continuous complex functions on X. A positive functional φ on $C(X)$ is exactly a positive linear functional in the sense of the Riesz representation theorem: $\varphi(f) \ge 0$ whenever $f \ge 0$, since the positive elements of $C(X)$ are the pointwise-nonnegative real functions (an $f \in C(X)$ satisfies $f = g^*g = |g|^2$ for some g iff $f \ge 0$ pointwise). By the Riesz representation theorem ([11, Riesz-Markov Representation]) there is a unique regular Borel (Radon) measure μ on X with

$$\varphi(f) = \int_X f\,d\mu \qquad \text{for all } f \in C(X)$$

and μ is a positive measure precisely because φ is positive. The normalization $\varphi(1) = 1$ becomes

$$\mu(X) = \int_X 1\,d\mu = \varphi(1) = 1$$

so the states of $C(X)$ are exactly the Radon probability measures on X. The extreme case is a point mass: for $x \in X$ the Dirac measure δ_x corresponds to the evaluation state

$$\mathrm{ev}_x(f) = \int_X f\,d\delta_x = f(x)$$

which is the character of $C(X)$ at x. Thus the vertices of the state space of $C(X)$ are the point evaluations, mixing them by a probability measure produces a general state, and the noncommutative theory replaces "probability measure on the space X" by "state on the algebra A" with X dissolved into A.

4.2 The GNS Construction

A state φ is a rule for taking expectations, a positive normalized functional on A with no geometry attached. The construction of this section attaches the geometry: it manufactures from φ alone a Hilbert space and a representation of A as operators on it, in which φ becomes the diagonal matrix coefficient at a single distinguished vector. The recipe is the one Gelfand, Naimark, and Segal isolated, and it is the exact operator analogue of the passage from a probability measure μ on a space X to the Hilbert space $L^2(\mu)$ on which $C(X)$ acts by multiplication. What plays the role of $L^2(\mu)$ is the completion of A itself under the inner product $\langle a, b\rangle = \varphi(b^*a)$ that the state gives, and what plays the role of multiplication is the left regular action of A on this completion.

The output is a representation with a cyclic vector, so the section begins by naming that notion, then carries out the construction, and closes by showing the triple it produces is unique. The single input the construction needs from the previous section is that $\langle a, b\rangle = \varphi(b^*a)$ is a positive semidefinite sesquilinear form obeying the Cauchy-Schwarz inequality (proposition 4.1.3); everything else is built here.

A representation is a $*$-homomorphism into the bounded operators on a Hilbert space, the concrete realization the abstract axioms of definition 1.2.1 were designed to admit. A vector is cyclic when the algebra, acting on it, sweeps out a dense subspace, so that the whole representation is generated by moving that one vector around.

Definition 4.2.1: Representation and Cyclic Vector

Let A be a C^*-algebra. A *representation* of A is a $*$-homomorphism $\pi\colon A \to B(H)$ into the bounded operators on a Hilbert space H. A vector $\xi \in H$ is *cyclic* for π if the orbit

$$\pi(A)\xi = \{\pi(a)\xi : a \in A\}$$

is dense in H. A *cyclic representation* is a triple (π, H, ξ) in which π is a representation on H and ξ is a cyclic vector.

The dense orbit is what makes a cyclic representation manageable: the single vector ξ determines everything, because any element of H is a limit of vectors $\pi(a)\xi$, and any bounded operator commuting with $\pi(A)$ is pinned down by its value on ξ. A general representation need not have a cyclic vector, but it always decomposes into cyclic pieces, one for each vector and its orbit closure, so the cyclic case is the building block from which representations are assembled. The construction below produces exactly a cyclic representation, and its cyclic vector is the image of the unit of A.

Example 4.2.2: The Multiplication Representation of $C(X)$

Let X be a compact Hausdorff space, $A = C(X)$, and μ a finite regular Borel measure on X with full support (every nonempty open set has positive measure). Take $H = L^2(X, \mu)$ and let $\pi(f)$ be multiplication by f,

$$\pi(f)g = fg, \qquad g \in L^2(X, \mu)$$

which is bounded with $\|\pi(f)\| \le \|f\|_\infty$, and $\pi(f^*) = \pi(f)^*$ since $\langle fg, h\rangle = \int fg\overline{h}\, d\mu = \int g\overline{\overline{f}h}\, d\mu = \langle g, \overline{f}h\rangle$. The constant function $\xi = 1 \in L^2(X, \mu)$ is cyclic: its orbit $\pi(C(X))\xi = \{f \cdot 1 : f \in C(X)\} = C(X)$ is dense in $L^2(X, \mu)$, since continuous functions are dense in L^2 of a regular Borel measure [11, $C_c(X)$ Dense in L^p]. The matrix coefficient at ξ recovers integration against μ,

$$\langle \pi(f)\xi, \xi\rangle = \int_X f \cdot 1 \cdot \overline{1}\, d\mu = \int_X f\, d\mu$$

so the state whose data this triple encodes is the probability measure μ, viewed as a state of $C(X)$ as in example 4.1.6. The full-support hypothesis is exactly what makes ξ cyclic rather than generating a proper subspace; without it the multiplication action would fix the functions supported off the support of μ.

The example is the commutative model the general construction imitates: a state φ of $C(X)$ is a measure μ, the Hilbert space is $L^2(\mu)$, the algebra acts by multiplication, and the constant function is the cyclic vector reproducing φ as a matrix coefficient. The content of the GNS theorem is that this picture requires no commutativity and no ambient measure space: the state alone builds the Hilbert space out of the algebra, and the multiplication action becomes the left regular action of A on itself.

The construction rests on the sesquilinear form that a state defines on the algebra. Positivity of φ says $\langle a, a\rangle = \varphi(a^*a) \geq 0$, so the form is positive semidefinite but generally not definite: the elements a with $\varphi(a^*a) = 0$ are the obstruction that must be quotiented away before an inner product survives. The first step identifies these elements as a subspace, indeed a left ideal, which is what lets the algebra continue to act after the quotient.

Theorem 4.2.3: The GNS Construction

Let A be a unital C^*-algebra and φ a state of A. Then there is a cyclic representation $(\pi_\varphi, H_\varphi, \xi_\varphi)$ of A, the *GNS representation* of φ, with

$$\varphi(a) = \langle \pi_\varphi(a)\xi_\varphi, \xi_\varphi\rangle \qquad \text{for all } a \in A$$

and $\|\xi_\varphi\| = 1$. It is built as follows. The set

$$N_\varphi = \{a \in A : \varphi(a^*a) = 0\}$$

is a closed left ideal of A; the quotient A/N_φ carries the definite inner product $\langle a+N_\varphi, b+N_\varphi\rangle = \varphi(b^*a)$; its completion is the Hilbert space H_φ; left multiplication descends to a $*$-representation π_φ with $\pi_\varphi(a)(b + N_\varphi) = ab + N_\varphi$; and $\xi_\varphi = 1 + N_\varphi$ is a cyclic vector.

***Proof*:**
Write $\langle a, b\rangle = \varphi(b^*a)$ for the sesquilinear form on A; it is linear in a, conjugate-linear in b, and positive semidefinite, $\langle a, a\rangle = \varphi(a^*a) \geq 0$, because φ is positive. The Cauchy-Schwarz inequality (proposition 4.1.3),

$$|\varphi(b^*a)|^2 \leq \varphi(a^*a)\,\varphi(b^*b) \tag{4.1}$$

is the one inequality the construction leans on.

Step 1: N_φ is a closed left ideal. By (4.1), for every $b \in A$,

$$|\varphi(b^*a)|^2 \leq \varphi(a^*a)\,\varphi(b^*b)$$

so if $a \in N_\varphi$, meaning $\varphi(a^*a) = 0$, then $\varphi(b^*a) = 0$ for all b. Thus

$$N_\varphi = \{a \in A : \varphi(b^*a) = 0 \text{ for all } b \in A\} \tag{4.2}$$

the left kernel of the form. This description makes N_φ a linear subspace: it is the intersection over b of the kernels of the linear functionals $a \mapsto \varphi(b^*a)$. For the left-ideal property, let $a \in N_\varphi$ and $c \in A$; then $ca \in N_\varphi$, since for every b,

$$\varphi(b^*(ca)) = \varphi((c^*b)^*a) = 0$$

by (4.2) applied with c^*b in place of b. Finally N_φ is closed: each functional $a \mapsto \varphi(b^*a)$ is continuous because φ is bounded and $a \mapsto b^*a$ is continuous, so N_φ is an intersection of closed sets by (4.2).

Step 2: the quotient is a pre-Hilbert space. On the quotient vector space A/N_φ, define

$$\langle a + N_\varphi, b + N_\varphi\rangle = \varphi(b^*a)$$

This is well-defined: if $a' = a + m$ and $b' = b + n$ with $m, n \in N_\varphi$, then $\varphi(b'^*a') = \varphi(b^*a) + \varphi(n^*a) + \varphi(b^*m) + \varphi(n^*m)$, and the three added terms vanish by (4.2) (the terms $\varphi(b^*m)$ and $\varphi(n^*m)$ because $m \in N_\varphi$ sits in the right slot, and $\varphi(n^*a) = \overline{\varphi(a^*n)}$ because $n \in N_\varphi$ sits in the right slot of $\varphi(a^*n)$). The form inherited on A/N_φ is sesquilinear and positive semidefinite, and it is now definite: if $\langle a + N_\varphi, a + N_\varphi\rangle = \varphi(a^*a) = 0$ then $a \in N_\varphi$, so $a + N_\varphi = 0$. Thus A/N_φ is an inner product space.

Step 3: complete to H_φ. Let H_φ be the Hilbert space completion of the inner product space A/N_φ [9, The Quotient Inner Product, Hilbert Space Completion]. The quotient map composed with the inclusion into the completion is the linear map $A \to H_\varphi$, $a \mapsto \hat{a} := a + N_\varphi$, with dense image (the image is A/N_φ, dense in its completion by construction) and $\langle \hat{a}, \hat{b}\rangle = \varphi(b^*a)$.

Step 4: left multiplication is bounded and descends. Fix $a \in A$. The map $b \mapsto ab$ on A carries N_φ into N_φ, since N_φ is a left ideal, so it descends to a linear map $\pi_\varphi(a)$ on A/N_φ with $\pi_\varphi(a)\hat{b} = \widehat{ab}$. The claim is that $\pi_\varphi(a)$ is bounded with $\|\pi_\varphi(a)\| \le \|a\|$. The estimate is the point at which the C^*-structure enters. The element a^*a is positive by theorem 2.3.3, and proposition 2.3.5(3) gives $a^*a \le \|a^*a\|\,1 = \|a\|^2 1$, the last equality by the C^*-identity of definition 1.2.1. Hence $\|a\|^2 1 - a^*a \ge 0$, and conjugating a positive element by b keeps it positive (proposition 2.4.6(2)),

$$b^*\big(\|a\|^2 1 - a^*a\big)b \ge 0$$

Applying the state φ, which is positive, to this positive element gives $\varphi\big(b^*(\|a\|^2 1 - a^*a)b\big) \ge 0$, that is

$$\varphi\big((ab)^*(ab)\big) = \varphi(b^*a^*ab) \le \|a\|^2\,\varphi(b^*b) \tag{4.3}$$

In terms of the norm on A/N_φ, the inequality (4.3) reads

$$\|\pi_\varphi(a)\hat{b}\|^2 = \langle \widehat{ab}, \widehat{ab}\rangle = \varphi(b^*a^*ab) \le \|a\|^2\,\varphi(b^*b) = \|a\|^2\,\|\hat{b}\|^2$$

so $\pi_\varphi(a)$ is bounded on the dense subspace A/N_φ with $\|\pi_\varphi(a)\| \leq \|a\|$. A bounded operator on a dense subspace extends uniquely to a bounded operator on the completion of the same norm [9, Defining Operator using Dense Sets], so $\pi_\varphi(a)$ extends to $\pi_\varphi(a) \in B(H_\varphi)$ with $\|\pi_\varphi(a)\| \leq \|a\|$.

Step 5: π_φ is a $*$-homomorphism. On the dense subspace A/N_φ, the maps agree: $\pi_\varphi(aa')\hat{b} = \widehat{aa'b} = \pi_\varphi(a)\widehat{a'b} = \pi_\varphi(a)\pi_\varphi(a')\hat{b}$, and $\pi_\varphi(\lambda a + a')\hat{b} = \pi_\varphi(\lambda a)\hat{b} + \pi_\varphi(a')\hat{b}$ by linearity of left multiplication, so π_φ is a unital algebra homomorphism into $B(H_\varphi)$ (unital because $\pi_\varphi(1)\hat{b} = \hat{b}$). For the involution, compute the matrix coefficients on A/N_φ: for all $b, c \in A$,

$$\langle \pi_\varphi(a)\hat{b}, \hat{c}\rangle = \langle \widehat{ab}, \hat{c}\rangle = \varphi(c^*ab) = \varphi((a^*c)^*b) = \langle \hat{b}, \widehat{a^*c}\rangle = \langle \hat{b}, \pi_\varphi(a^*)\hat{c}\rangle$$

so $\pi_\varphi(a)^* = \pi_\varphi(a^*)$ on the dense subspace, hence on H_φ by continuity. Thus π_φ is a $*$-homomorphism, a representation of A on H_φ.

Step 6: the cyclic vector reproduces φ. Set $\xi_\varphi = \hat{1} = 1 + N_\varphi$. Its orbit is $\pi_\varphi(A)\xi_\varphi = \{\pi_\varphi(a)\hat{1} : a \in A\} = \{\hat{a} : a \in A\} = A/N_\varphi$, which is dense in H_φ by Step 3, so ξ_φ is cyclic. Its norm is $\|\xi_\varphi\|^2 = \langle \hat{1}, \hat{1}\rangle = \varphi(1^*1) = \varphi(1) = 1$, since φ is a state (definition 4.1.2). Finally, for every $a \in A$,

$$\langle \pi_\varphi(a)\xi_\varphi, \xi_\varphi\rangle = \langle \hat{a}, \hat{1}\rangle = \varphi(1^*a) = \varphi(a)$$

which is the asserted reproduction of the state, completing the proof.

The construction is a machine that turns expectation values into geometry. Every ingredient is forced by the state: the inner product is $\varphi(b^*a)$, the vectors are the elements of A themselves (modulo the null directions the state cannot see), and the operators are left multiplications. The role of the C^*-identity is confined to a single estimate, (4.3), where positivity of $\|a\|^2 1 - a^*a$ bounds the operator norm of $\pi_\varphi(a)$ by $\|a\|$; without it the left regular action would not be by bounded operators and no representation would result. The name $L^2(\varphi)$ for H_φ is apt: in the commutative case φ is a measure μ, the null space N_φ consists of the functions vanishing μ-almost everywhere, and H_φ is literally $L^2(\mu)$, recovering example 4.2.2. This is the noncommutative Riesz representation theorem: where Riesz builds the measure underlying a positive functional on $C(X)$, GNS builds the Hilbert space underlying a state of an arbitrary C^*-algebra.

Two features of the output deserve to be recorded before uniqueness. First, the representation π_φ is contractive, $\|\pi_\varphi(a)\| \leq \|a\|$, as every $*$-homomorphism of C^*-algebras must be; the construction gives this bound directly rather than by appeal to the general theorem. Second, the state need not be recoverable from π_φ alone: it is the pair $(\pi_\varphi, \xi_\varphi)$ that carries φ, through the matrix coefficient at ξ_φ. Different states can give unitarily equivalent representations that differ only in the choice of cyclic vector.

A worked instance shows the quotient doing visible work, collapsing the algebra onto a space smaller than itself.

Example 4.2.4: The GNS Space of a Vector State on $M_n(\mathbb{C})$

Let $A = M_n(\mathbb{C})$ and let φ be the vector state $\varphi(a) = \langle ae_1, e_1\rangle = a_{11}$, the $(1,1)$ entry, where $e_1, \ldots, e_n$ is the standard basis of $\mathbb{C}^n$. This is a state: $\varphi(a^*a) = (a^*a)_{11} = \sum_k |a_{k1}|^2 \geq 0$ and $\varphi(1) = 1$. The null space is

$$N_\varphi = \{a \in M_n(\mathbb{C}) : \varphi(a^*a) = 0\} = \left\{a : \sum_k |a_{k1}|^2 = 0\right\} = \{a : a_{k1} = 0 \text{ for all } k\}$$

the matrices whose first column vanishes, a left ideal of codimension n. The quotient A/N_φ is thus n-dimensional, a matrix a being determined modulo N_φ by its first column $(a_{11}, \ldots, a_{n1})$. The inner product is $\langle a + N_\varphi, b + N_\varphi\rangle = \varphi(b^*a) = (b^*a)_{11} = \sum_k \overline{b_{k1}} a_{k1}$, the standard inner product on the first columns. So $H_\varphi \cong \mathbb{C}^n$ under $a + N_\varphi \mapsto (a_{k1})_k$, the map $\pi_\varphi(a)$ becomes multiplication of column vectors by a, and $\xi_\varphi = 1 + N_\varphi$ is the first column of the identity, namely e_1. The GNS representation of the vector state $\varphi(a) = a_{11}$ is thus the defining representation of $M_n(\mathbb{C})$ on $\mathbb{C}^n$ with cyclic vector e_1, and $\langle \pi_\varphi(a)e_1, e_1\rangle = a_{11} = \varphi(a)$ as required.

The GNS triple was constructed by definite choices, quotient by N_φ, completion, left multiplication, and one might worry that a different route to a cyclic representation reproducing φ could yield something different. It cannot. Any cyclic representation whose matrix coefficient at the cyclic vector is φ is unitarily equivalent to the GNS triple, so the construction is canonical: φ determines its cyclic representation up to the only ambiguity a Hilbert space representation admits, a unitary change of coordinates.

Proposition 4.2.5: Uniqueness of the GNS Triple

Let A be a unital C^*-algebra and φ a state. Suppose (π, H, ξ) is any cyclic representation of A with $\|\xi\| = 1$ and

$$\varphi(a) = \langle \pi(a)\xi, \xi\rangle \qquad \text{for all } a \in A$$

Then there is a unique unitary $U\colon H_\varphi \to H$ with $U\xi_\varphi = \xi$ and $U\pi_\varphi(a) = \pi(a)U$ for all $a \in A$. Thus (π, H, ξ) is unitarily equivalent to the GNS triple $(\pi_\varphi, H_\varphi, \xi_\varphi)$.

***Proof*:**

The two representations are compared through their cyclic vectors. Define a map on the dense orbit $\pi_\varphi(A)\xi_\varphi = A/N_\varphi \subseteq H_\varphi$ by

$$U_0\colon \pi_\varphi(a)\xi_\varphi \mapsto \pi(a)\xi, \qquad \text{that is } U_0(\hat{a}) = \pi(a)\xi$$

Step 1: U_0 is well-defined and isometric. For $a, b \in A$, the inner products in the two spaces agree,

$$\langle \pi(a)\xi, \pi(b)\xi\rangle = \langle \pi(b^*a)\xi, \xi\rangle = \varphi(b^*a) = \langle \pi_\varphi(a)\xi_\varphi, \pi_\varphi(b)\xi_\varphi\rangle$$

using that π is a $*$-homomorphism ($\pi(b)^*\pi(a) = \pi(b^*a)$) in the first equality, the hypothesis on ξ in the second, and the same two facts for π_φ and ξ_φ in the third (which is the reproduction property of the GNS triple). Taking $a = b$ shows $\|\pi(a)\xi\| = \|\hat{a}\|$, so U_0 carries a vector to zero only when its source is zero; hence U_0 is well-defined on A/N_φ (if $\hat{a} = \hat{b}$ then $\|\pi(a)\xi - \pi(b)\xi\| = \|\widehat{a-b}\| = 0$) and preserves the inner product on that subspace.

Step 2: extend to a unitary. The domain A/N_φ is dense in H_φ, and U_0 is a linear isometry onto the orbit $\pi(A)\xi$, which is dense in H because ξ is cyclic for π. An isometry defined on a dense subspace with dense range extends uniquely to a surjective isometry $U\colon H_\varphi \to H$ [9, Unitary Extension Of A Dense Isometry], and a surjective linear isometry between Hilbert spaces is unitary.

Step 3: U intertwines the representations and matches the cyclic vectors. On the dense subspace A/N_φ, for $a, b \in A$,

$$U\pi_\varphi(a)\hat{b} = U\widehat{ab} = \pi(ab)\xi = \pi(a)\pi(b)\xi = \pi(a)U\hat{b}$$

so $U\pi_\varphi(a) = \pi(a)U$ on the dense subspace, hence everywhere by continuity. Taking $a = 1$ in the definition of U_0 gives $U\xi_\varphi = U\hat{1} = \pi(1)\xi = \xi$. For uniqueness, any unitary V with $V\xi_\varphi = \xi$ and $V\pi_\varphi(a) = \pi(a)V$ satisfies $V\hat{a} = V\pi_\varphi(a)\xi_\varphi = \pi(a)V\xi_\varphi = \pi(a)\xi = U\hat{a}$ on the dense subspace A/N_φ, so $V = U$, as we sought to show.

Uniqueness up to unitary equivalence is what entitles one to speak of *the* GNS representation of a state, and it identifies the abstractly constructed H_φ with any concrete cyclic model at hand: the defining representation of $M_n(\mathbb{C})$ was recognized as the GNS representation of a vector state in example 4.2.4 precisely by matching a cyclic vector and its matrix coefficient, and the proposition guarantees the match is canonical. The correspondence runs in both directions. A state produces a cyclic representation, and conversely a cyclic representation (π, H, ξ) with $\|\xi\| = 1$ produces the state $a \mapsto \langle \pi(a)\xi, \xi\rangle$, so states and (equivalence classes of) cyclic representations with a distinguished unit cyclic vector are the same data. This is the dictionary the next section exploits: assembling the GNS representations over all states at once produces a single faithful representation, embedding A isometrically into the bounded operators on a Hilbert space.

4.3 The Gelfand-Naimark Theorem

The GNS construction turns a single state into a cyclic representation. Running it over every state at once, and adding the resulting representations together, produces a single representation that sees all of A. The task of this section is to check that this combined representation is faithful and isometric.

The one input beyond the GNS construction is that states are plentiful enough to detect the norm. Proposition 4.1.5 gives exactly this: for each self-adjoint element there is a state attaining $\|a\|$. Applied to a^*a, it will let a single GNS representation reproduce the norm of a itself, and the direct sum over all states then loses nothing.

Before assembling the universal representation, recall how the pieces fit. Each state $\varphi \in S(A)$ gives a GNS triple $(\pi_\varphi, H_\varphi, \xi_\varphi)$ with $\varphi(a) = \langle \pi_\varphi(a)\xi_\varphi, \xi_\varphi\rangle$ (theorem 4.2.3). The individual π_φ need not be faithful: a state can be blind to part of the algebra, as a point evaluation ev_x on $C(X)$ ignores every function vanishing at x. The remedy is to use all states simultaneously, so that no element escapes detection. The direct sum of Hilbert spaces and the direct sum of representations are the tools for combining them.

Definition 4.3.1: Direct Sum of Representations

Let $\{(\pi_i, H_i)\}_{i\in I}$ be a family of representations of a C^*-algebra A. Their *direct sum* is the representation $\pi = \bigoplus_{i\in I} \pi_i$ on the Hilbert-space direct sum $H = \bigoplus_{i\in I} H_i$ defined by

$$\pi(a)\big((\xi_i)_{i\in I}\big) = \big(\pi_i(a)\xi_i\big)_{i\in I}$$

for $a \in A$ and $(\xi_i)_{i\in I} \in H$.

The Hilbert-space direct sum $\bigoplus_{i\in I} H_i$ consists of those families $(\xi_i)_{i\in I}$ with $\sum_{i\in I} \|\xi_i\|^2 < \infty$, under the inner product $\langle (\xi_i), (\eta_i)\rangle = \sum_{i\in I}\langle \xi_i, \eta_i\rangle$ ([9, Hilbert Space Direct Sum, The Direct Sum Is A Hilbert Space]). For $\pi(a)$ to be a bounded operator on this space, the operators $\pi_i(a)$ must be uniformly bounded in i; this holds because each π_i is a $*$-homomorphism and hence contractive, so $\|\pi_i(a)\| \le \|a\|$ for every i (theorem 1.3.3). The estimate

$$\sum_{i\in I} \|\pi_i(a)\xi_i\|^2 \le \sum_{i\in I} \|a\|^2\|\xi_i\|^2 = \|a\|^2 \sum_{i\in I} \|\xi_i\|^2$$

shows $\pi(a)$ maps H into H with $\|\pi(a)\| \le \|a\|$. That π is linear, multiplicative, and $*$-preserving follows coordinatewise from the corresponding properties of each π_i, so π is a representation.

Example 4.3.2: A Direct Sum Detecting a Diagonal Matrix

Take $A = M_2(\mathbb{C})$ and the two vector states $\varphi_1(a) = \langle ae_1, e_1\rangle$ and $\varphi_2(a) = \langle ae_2, e_2\rangle$, where e_1, e_2 is the standard basis of $\mathbb{C}^2$. Each φ_j is a state, and the GNS representation of either one is unitarily equivalent to the standard action of $M_2(\mathbb{C})$ on $\mathbb{C}^2$, cyclic with cyclic vector e_j. Individually each already sees all of $M_2(\mathbb{C})$ here, but the diagonal element $a = \operatorname{diag}(\lambda, \mu)$ shows the detection: $\varphi_1(a^*a) = |\lambda|^2$ and $\varphi_2(a^*a) = |\mu|^2$, so

$$\|\pi_{\varphi_1}(a)\xi_{\varphi_1}\|^2 + \|\pi_{\varphi_2}(a)\xi_{\varphi_2}\|^2 = |\lambda|^2 + |\mu|^2$$

equals $\|a\|_{\mathrm{HS}}^2$, the squared Hilbert-Schmidt norm. The direct sum $\pi_{\varphi_1} \oplus \pi_{\varphi_2}$ acts on $\mathbb{C}^2 \oplus \mathbb{C}^2$; it is faithful because $a = 0$ is forced once both λ and μ vanish. The operator norm $\|a\| = \max(|\lambda|, |\mu|)$ is recovered by the larger of the two summands, which is the mechanism the general theorem exploits.

The direct sum over *all* states is the universal representation. Its faithfulness is where proposition 4.1.5 enters: for each nonzero a, the positive element a^*a has a state attaining its norm, and the corresponding summand reproduces $\|a\|$ exactly.

Theorem 4.3.3: The Gelfand-Naimark Theorem

Every C^*-algebra A is isometrically $*$-isomorphic to a norm-closed $*$-subalgebra of $B(H)$ for some Hilbert space H. Concretely, the *universal representation*

$$\pi = \bigoplus_{\varphi\in S(A)} \pi_\varphi \colon A \longrightarrow B(H), \qquad H = \bigoplus_{\varphi\in S(A)} H_\varphi$$

is an injective $*$-homomorphism with $\|\pi(a)\| = \|a\|$ for all $a \in A$, and its image $\pi(A)$ is a norm-closed $*$-subalgebra of $B(H)$.

***Proof*:**
Assume first that A is unital; Step 4 adapts the argument to the non-unital case.

Step 1: π is a $*$-homomorphism, hence contractive. The universal representation is the direct sum $\bigoplus_{\varphi\in S(A)} \pi_\varphi$ of the GNS representations over all states (definition 4.3.1). Each summand π_φ is a representation of A (theorem 4.2.3), so π is a $*$-homomorphism into $B(H)$ by the coordinatewise computation preceding example 4.3.2. As a $*$-homomorphism it is contractive: $\|\pi(a)\| \le \|a\|$ for every $a \in A$ (theorem 1.3.3).

Step 2: π is isometric. The reverse inequality $\|\pi(a)\| \ge \|a\|$ is where the states are used. Fix $a \in A$. The element a^*a is self-adjoint, so proposition 4.1.5 provides a state $\varphi \in S(A)$ with

$$\varphi(a^*a) = \|a^*a\| = \|a\|^2$$

the last equality being the C^*-identity (definition 1.2.1). By the defining property of the GNS triple (theorem 4.2.3),

$$\varphi(a^*a) = \langle \pi_\varphi(a^*a)\xi_\varphi, \xi_\varphi\rangle = \langle \pi_\varphi(a)\xi_\varphi, \pi_\varphi(a)\xi_\varphi\rangle = \|\pi_\varphi(a)\xi_\varphi\|^2$$

where the middle equality uses $\pi_\varphi(a^*a) = \pi_\varphi(a)^*\pi_\varphi(a)$. Since $\|\xi_\varphi\|^2 = \varphi(1) = 1$ for the unital state, this gives

$$\|\pi_\varphi(a)\| \ge \frac{\|\pi_\varphi(a)\xi_\varphi\|}{\|\xi_\varphi\|} = \|\pi_\varphi(a)\xi_\varphi\| = \|a\|$$

The norm of the direct sum dominates the norm of any single summand: for the unit vector $\eta \in H$ supported in the H_φ coordinate at ξ_φ, one has $\|\pi(a)\eta\| = \|\pi_\varphi(a)\xi_\varphi\| = \|a\|$, so $\|\pi(a)\| \ge \|a\|$. Combined with Step 1, $\|\pi(a)\| = \|a\|$.

Step 3: π is injective and its image is closed. An isometric linear map has trivial kernel, so π is injective. Its image $\pi(A)$ is a $*$-subalgebra of $B(H)$ because π is a $*$-homomorphism. To see that $\pi(A)$ is norm-closed, let $\pi(a_n)$ be a Cauchy sequence in $B(H)$. Isometry gives $\|a_n - a_m\| = \|\pi(a_n) - \pi(a_m)\|$, so (a_n) is Cauchy in A; by completeness of A it converges to some $a \in A$, and then $\pi(a_n) \to \pi(a)$ by continuity of π. Thus every limit point of $\pi(A)$ lies in $\pi(A)$. Consequently $\pi\colon A \to \pi(A)$ is an isometric $*$-isomorphism onto a norm-closed $*$-subalgebra of $B(H)$.

Step 4: The non-unital case. Suppose A has no unit. The claim is again that A's own universal representation $\pi = \bigoplus_{\varphi\in S(A)} \pi_\varphi$ is isometric; Steps 1 and 3 already apply verbatim, since neither used the unit, so only the isometry of Step 2 must be re-established. Two facts of the general (possibly non-unital) theory carry it through. First, for any state $\varphi \in S(A)$ the GNS cyclic vector satisfies $\|\xi_\varphi\|^2 = \|\varphi\| = 1$ (theorem 4.2.3); this is the substitute for the identity $\varphi(1) = 1$ used in Step 2, and it agrees with it when a unit is present. Second, the states persist. Fix $a \in A$ and pass to the unitization $\tilde{A}$, a unital C^*-algebra containing A as a closed ideal with $A \hookrightarrow \tilde{A}$ an isometric $*$-embedding (proposition 1.4.5); the self-adjoint element a^*a lies in A and has the same norm in $\tilde{A}$. By proposition 4.1.5 applied in $\tilde{A}$ there is a state $\psi \in S(\tilde{A})$ with $\psi(a^*a) = \|a^*a\| = \|a\|^2$. Its restriction $\varphi = \psi|_A$ is positive with $\|\varphi\| \le \|\psi\| = 1$, and $\varphi(a^*a) = \|a\|^2 = \|a^*a\|$ forces $\|\varphi\| \ge 1$; hence $\varphi \in S(A)$ and $\varphi(a^*a) = \|a\|^2$. The computation of Step 2 now runs unchanged in A: $\|\pi_\varphi(a)\xi_\varphi\|^2 = \varphi(a^*a) = \|a\|^2$ with $\|\xi_\varphi\| = 1$, so $\|\pi(a)\| \ge \|\pi_\varphi(a)\xi_\varphi\| = \|a\|$, and with Step 1 this gives $\|\pi(a)\| = \|a\|$. Thus π is isometric, injective, and has norm-closed image in the non-unital case as well, completing the proof.

The theorem removes the distinction between abstract and concrete C^*-algebras. An abstract

C^*-algebra is defined by axioms on a norm and an involution (definition 1.2.1), with no reference to operators; a concrete one is a norm-closed $*$-subalgebra of some $B(H)$, an algebra of actual Hilbert-space operators. The universal representation shows the two notions coincide, and it does so canonically, using no choice beyond the state space itself. The Hilbert space it produces is enormous, since $S(A)$ is typically uncountable, so the universal representation is a proof device rather than a computational one; for concrete work one uses a single well-chosen faithful representation on a manageable space. What matters is the existence of some faithful representation, which the theorem guarantees.

A $*$-homomorphism into $B(H)$ that failed to preserve the norm would still realize A as operators, but it could collapse distinct elements or distort distances; the C^*-identity forces the norm, and proposition 4.1.5 certifies that enough states exist to recover it. This is the operator-algebra counterpart of the commutative statement that the Gelfand transform is isometric on a commutative C^*-algebra (theorem 3.2.1): there the norm was recovered from characters, here from states.

Example 4.3.4: The Universal Representation of $C(X)$

Let X be a compact Hausdorff space and $A = C(X)$. The states of A are the probability measures on X (example 4.1.6). For a probability measure μ, the GNS construction produces the representation of $C(X)$ on $L^2(X,\mu)$ by multiplication operators, $\pi_\mu(f)g = fg$, with cyclic vector the constant function $\xi_\mu = 1$; indeed

$$\langle \pi_\mu(f)\xi_\mu, \xi_\mu\rangle = \int_X f\, d\mu = \mu(f)$$

recovers the state. The universal representation is the direct sum $\bigoplus_\mu L^2(X,\mu)$ over all probability measures μ. A single measure already often suffices for faithfulness: if μ has full support (every nonempty open set has positive measure), then $\|f\|_\infty = \|\pi_\mu(f)\|$, and $C(X)$ acts faithfully and isometrically as multiplication operators on the one space $L^2(X,\mu)$. For $X = [0,1]$ with μ Lebesgue measure, this realizes $C[0,1]$ concretely inside $B(L^2[0,1])$. The universal representation over all measures is far larger than needed, illustrating the gap between the canonical construction and an economical concrete model.

The theorem also settles a question left open by the opening chapter, where the axioms of a C^*-algebra were extracted from the properties of $B(H)$ and its norm-closed $*$-subalgebras (definition 1.2.1). It was clear from the outset that every norm-closed $*$-subalgebra of $B(H)$ satisfies the axioms; the converse, that the axioms describe nothing more, is the content just proved.

Corollary 4.3.5: C^*-Algebras Are Operator Algebras

The C^*-algebras are, up to isometric $*$-isomorphism, exactly the norm-closed $*$-subalgebras of $B(H)$ for the various Hilbert spaces H.

***Proof*:**
A norm-closed $*$-subalgebra $B \subseteq B(H)$ is a C^*-algebra: it is a Banach $*$-algebra as a closed subalgebra of the Banach $*$-algebra $B(H)$, and it inherits the C^*-identity $\|T^*T\| = \|T\|^2$ from $B(H)$, so the axioms of definition 1.2.1 hold. Conversely, every C^*-algebra is isometrically $*$-isomorphic to such a subalgebra by theorem 4.3.3. The two classes therefore coincide up to isometric $*$-isomorphism, as we sought to show.

The two Gelfand-Naimark theorems together give a complete external description of C^*-algebras. The commutative theorem (chapter 3) identifies the commutative ones with the function algebras $C_0(X)$ on locally compact Hausdorff spaces; the present theorem identifies all of them, commutative or not, with operator algebras on Hilbert space. No further axioms are needed and none are hidden: the short list defining a C^*-algebra (a Banach $*$-algebra with $\|a^*a\| = \|a\|^2$) captures precisely the norm-closed $*$-subalgebras of $B(H)$, with equality of the abstract and concrete classes. A noncommutative C^*-algebra is thereby a legitimate "algebra of operators on a noncommutative space," and the operator-theoretic and geometric threads of the subject both proceed from this identification.

4.4 Pure States

The GNS construction of section 4.2 attaches a cyclic representation to every state, and the universal representation of section 4.3 assembles all of them into a single faithful copy of A inside a $B(H)$. Not all states are equally efficient. Some decompose as averages of others, and the representation they build is a direct sum of the representations built from the pieces; some do not decompose, and the representation they build resists any such splitting. This section isolates the indecomposable states as the extreme points of the state space and records the two structural facts that make them useful: they exist in abundance, and every state is built from them.

The natural home for this analysis is convex geometry. A state is a positive functional of norm one, and a convex combination $t\varphi_1 + (1-t)\varphi_2$ of states, with $0 \le t \le 1$, is again a state. So $S(A)$ is a convex set, and the question of which states decompose is the question of which points of $S(A)$ are extreme.

Definition 4.4.1: Pure State

Let A be a C^*-algebra. A state $\varphi \in S(A)$ is *pure* if it is an extreme point of the state space $S(A)$: whenever

$$\varphi = t\varphi_1 + (1-t)\varphi_2$$

with $\varphi_1, \varphi_2 \in S(A)$ and $0 < t < 1$, then $\varphi_1 = \varphi_2 = \varphi$. A state that is not pure is called *mixed*. The set of pure states is written $\partial_e S(A)$.

A pure state cannot be split as a nontrivial average of two distinct states. The requirement $0 < t < 1$ excludes the degenerate combinations $t = 0$ and $t = 1$, which reproduce φ trivially; purity forbids every genuine decomposition. Equivalently, if $\varphi = \frac{1}{2}(\varphi_1 + \varphi_2)$ forces $\varphi_1 = \varphi_2$, then φ is pure, since any extreme-point violation can be symmetrized to this midpoint form. The name records the physical reading of section 4.1: a state is a noncommutative probability law, a pure state is one carrying no residual randomness of the classical kind, and a mixed state is a probabilistic blend of purer alternatives.

The first example is the commutative one, where states are honest probability measures and the geometry is explicit.

Example 4.4.2: Pure and Mixed States on $C(\{1,2\})$

Let $X = \{1,2\}$, so $C(X) \cong \mathbb{C}^2$ with pointwise operations. By example 4.1.6 a state is integration against a probability measure, here a pair (p_1, p_2) with $p_1, p_2 \ge 0$ and $p_1 + p_2 = 1$; the associated state sends $f \mapsto p_1 f(1) + p_2 f(2)$. The state space $S(C(X))$ is thus the segment from $(1,0)$ to

$(0,1)$, a one-dimensional simplex.

Its extreme points are the two endpoints $(1,0)$ and $(0,1)$, the point evaluations ev_1 and ev_2. Any interior point (p_1,p_2) with $0 < p_1 < 1$ splits as

$$(p_1,p_2) = p_1 \cdot (1,0) + p_2 \cdot (0,1)$$

a nontrivial convex combination of the two evaluations, so it is mixed. The pure states are exactly the two evaluations, and every state is a convex combination of them. This is the finite-dimensional case of the two theorems that follow.

To promote this picture from a segment to a general C^*-algebra, the state space must first be shown to be a compact convex set, so that its extreme points can be located by the Krein-Milman theorem. Compactness comes from the weak-$*$ topology, where states form a closed subset of the unit ball and Banach-Alaoglu applies.

Proposition 4.4.3: The State Space Is Weak-* Compact and Convex

Let A be a unital C^*-algebra. Then the state space $S(A)$ is a convex, weak-$*$ compact subset of the dual space A^*.

***Proof*:**
Convexity. Let $\varphi_1, \varphi_2 \in S(A)$ and $0 \le t \le 1$, and set $\varphi = t\varphi_1 + (1-t)\varphi_2$. For $a \in A$ the value $\varphi(a^*a) = t\varphi_1(a^*a) + (1-t)\varphi_2(a^*a)$ is a nonnegative combination of nonnegative numbers, so φ is positive. In the unital case a positive functional is a state precisely when $\varphi(1) = 1$ (proposition 4.1.3), and $\varphi(1) = t \cdot 1 + (1-t) \cdot 1 = 1$. Hence $\varphi \in S(A)$, and $S(A)$ is convex.

A weak-$*$ closed description. By definition 4.1.2 and proposition 4.1.3, in the unital case the states are exactly the positive functionals normalized by $\varphi(1) = 1$:

$$S(A) = \{\varphi \in A^* : \varphi(a^*a) \ge 0 \text{ for all } a \in A,\ \varphi(1) = 1\}$$

Each defining condition is weak-$*$ closed. The condition $\varphi(1) = 1$ is the preimage of the point 1 under the weak-$*$ continuous evaluation $\varphi \mapsto \varphi(1)$. For fixed a, the condition $\varphi(a^*a) \ge 0$ is the preimage of the closed set $[0,\infty)$ under the weak-$*$ continuous evaluation $\varphi \mapsto \varphi(a^*a)$; intersecting over all $a \in A$ keeps the set weak-$*$ closed. Therefore $S(A)$ is weak-$*$ closed in A^*.

Compactness. Every state has $\|\varphi\| = 1$, so $S(A)$ is contained in the closed unit ball of A^*. By the Banach-Alaoglu theorem ([9, Alaoglu's Theorem]) that ball is weak-$*$ compact. A weak-$*$ closed subset of a weak-$*$ compact set is weak-$*$ compact, so $S(A)$ is weak-$*$ compact, completing the proof.

The unital hypothesis is what makes $S(A)$ compact rather than only bounded: it pins the states to the affine slice $\varphi(1) = 1$ of the unit ball, a weak-$*$ closed face, rather than leaving them free to slide toward 0. For a non-unital A the analogous object is the set of positive functionals of norm at most one, whose extreme points include the zero functional; passing to the unitization $\tilde{A}$ of proposition 1.4.5 restores the compact picture. With $S(A)$ now a compact convex set, its extreme points are governed by Krein-Milman.

The state space is a compact convex set in a locally convex space, exactly the setting of the Krein-Milman theorem. That theorem guarantees extreme points and, more, that the whole set is recovered as the closed convex hull of them. Read on $S(A)$, it says pure states exist and every state is a limit of averages of pure states.

Theorem 4.4.4: Pure States Exist and Generate

Let A be a unital C^*-algebra. Then A has pure states, and every state is a weak-$*$ limit of convex combinations of pure states:

$$S(A) = \overline{\operatorname{conv}}\big(\partial_e S(A)\big)$$

the weak-$*$ closed convex hull of the pure states.

***Proof*:**
By proposition 4.4.3 the state space $S(A)$ is a convex, weak-$*$ compact subset of the dual A^*. The dual equipped with its weak-$*$ topology is a locally convex topological vector space, and $S(A)$ is nonempty: applying proposition 4.1.5 to the self-adjoint element $a = 1$, with $\|1\| = 1$, produces a state φ with $|\varphi(1)| = 1$.

The Krein-Milman theorem ([9, Krein-Milman Theorem]) states that a nonempty compact convex subset K of a locally convex space is the closed convex hull of its extreme points, $K = \overline{\operatorname{conv}}(\partial_e K)$; in particular $\partial_e K$ is nonempty. Applying this to $K = S(A)$, the extreme points $\partial_e S(A)$ are exactly the pure states of definition 4.4.1, so pure states exist and

$$S(A) = \overline{\operatorname{conv}}\big(\partial_e S(A)\big)$$

where the closure is taken in the weak-$*$ topology. Since every element of $\operatorname{conv}(\partial_e S(A))$ is a finite convex combination of pure states, every state is a weak-$*$ limit of such combinations, as we sought to show.

Pure states are therefore neither rare nor exotic: a unital C^*-algebra has enough of them to reconstruct every state by averaging and taking weak-$*$ limits. This is the analytic counterpart of the classical fact that every probability measure is an average of point masses, and it is what lets the theory reduce questions about arbitrary states to questions about pure ones. The universal representation of section 4.3 was built from all states at once; theorem 4.4.4 says it suffices, up to the closure operation, to build from the pure states alone, since the others add nothing that averaging pure states cannot reach.

One caution accompanies the theorem. The hull is a *closed* convex hull, and the closure is generally needed: a state need not be a finite convex combination of pure states, only a weak-$*$ limit of such combinations. The finite-dimensional example 4.4.2, where the simplex is literally the convex hull of its two vertices with no closure required, is misleadingly clean; in infinite dimensions the extreme points can fail to be closed and the limit is essential.

The commutative case makes the pure states completely explicit and confirms the reading of example 4.4.2 at full generality: the pure states of a function algebra are the point evaluations, and nothing else.

Example 4.4.5: Pure States of $C(X)$

Let X be a compact Hausdorff space and $A = C(X)$. By example 4.1.6 the states of $C(X)$ are the probability measures μ on X, acting by $\varphi_\mu(f) = \int_X f\, d\mu$, and by theorem 3.2.1 this identification is exhaustive. The claim is that the pure states are exactly the point evaluations

$$\mathrm{ev}_x \colon f \mapsto f(x) \qquad (x \in X)$$

which correspond to the Dirac point masses $\mu = \delta_x$.

Point evaluations are pure. Suppose $\mathrm{ev}_x = t\varphi_1 + (1-t)\varphi_2$ with states φ_1, φ_2 and $0 < t < 1$, and write $\varphi_i = \varphi_{\mu_i}$ for probability measures μ_i. Then $\delta_x = t\mu_1 + (1-t)\mu_2$ as measures. For any Borel set E not containing x we have $\delta_x(E) = 0$, so $t\mu_1(E) + (1-t)\mu_2(E) = 0$; since both terms are nonnegative and $t, 1-t > 0$, this forces $\mu_1(E) = \mu_2(E) = 0$. Hence μ_1 and μ_2 are both supported on $\{x\}$, and being probability measures they equal δ_x. Thus $\varphi_1 = \varphi_2 = \mathrm{ev}_x$, and ev_x is pure.

Every pure state is a point evaluation. Suppose $\varphi = \varphi_\mu$ is a state whose measure μ is not a point mass. Then the support of μ contains at least two distinct points, so there is a Borel set E with $0 < \mu(E) < 1$. Set $t = \mu(E)$ and define measures

$$\mu_1(F) = \frac{\mu(F \cap E)}{\mu(E)}, \qquad \mu_2(F) = \frac{\mu(F \setminus E)}{\mu(X \setminus E)}$$

the normalized restrictions of μ to E and to its complement. Both μ_1 and μ_2 are probability measures, hence states $\varphi_{\mu_1}, \varphi_{\mu_2}$, and by construction

$$\mu = t\mu_1 + (1-t)\mu_2$$

with $0 < t < 1$. The two measures are distinct: $\mu_1(E) = 1$ while $\mu_2(E) = 0$. So φ is a nontrivial convex combination of two distinct states and is mixed. By contraposition, every pure state has μ a point mass, that is, $\varphi = \mathrm{ev}_x$ for some $x \in X$.

The pure states of $C(X)$ are precisely the point evaluations $\{\mathrm{ev}_x : x \in X\}$, a copy of X inside $S(C(X))$; theorem 4.4.4 then reads as the statement that every probability measure is a weak-$*$ limit of convex combinations of point masses, the Krein-Milman theorem for probability measures on X.

The example returns the pure states of a commutative C^*-algebra to the points of its spectrum, closing the circle with Gelfand-Naimark: under $C(X) \cong C(\hat{A})$ of theorem 3.2.1, a point of X is a character, and the pure states are exactly these characters read as evaluations. For a noncommutative A the pure states are more than characters, but they play the same organizing role. Each pure state φ builds, through the GNS construction of theorem 4.2.3, a representation $(\pi_\varphi, H_\varphi, \xi_\varphi)$, and purity of φ corresponds exactly to *irreducibility* of π_φ: the representation admits no proper closed invariant subspace. The correspondence between pure states and irreducible representations is developed in the next chapter, where Schur's lemma gives the link and the pure states become the analytic index of the irreducible representations of A.

5

Representations and Multiplier Algebras

The GNS construction of chapter 4 produced one representation from one state; this chapter studies representations in their own right. A representation is irreducible when it has no proper closed invariant subspace, and Schur's lemma characterizes irreducibility by the triviality of the commutant; the irreducible representations correspond, through GNS, to the pure states. These are the pieces into which every representation decomposes.

The chapter then gives two constructions the theory needs for algebras without a unit. The multiplier algebra $M(A)$ is the largest unital C^*-algebra containing A as an essential ideal, recovering $B(H)$ from the compact operators and $C_b(X)$ from $C_0(X)$. The enveloping C^*-algebra attaches to a Banach $*$-algebra B the universal C^*-algebra through which every $*$-representation of B factors; it is the construction that builds the group C^*-algebra from the convolution algebra $L^1(G)$, and so the doorway from this book to the harmonic analysis of locally compact groups. A closing section records the primitive ideal space and glimpses the enveloping von Neumann algebra.

5.1 Representations

This section sets up the structural language for representations and carries out the first reduction: what it means for one to have no proper invariant part, how invariance is read off inside the algebra of operators commuting with the image, and how an arbitrary representation is broken into the cyclic pieces that the GNS construction (chapter 4) parametrizes by states. The classification of the indecomposable pieces is deferred to the next section.

A representation is a $*$-homomorphism into the bounded operators on a Hilbert space, the notion already named in definition 4.2.1. What is added now is a regularity condition ruling out the wasteful possibility that the algebra acts as zero on part of the space, together with the notion of an invariant subspace.

Definition 5.1.1: Nondegenerate Representation

Let A be a C^*-algebra and $\pi\colon A \to B(H)$ a representation. A closed subspace $K \subseteq H$ is $\pi(A)$-*invariant* (or *invariant*) if $\pi(a)K \subseteq K$ for every $a \in A$. The representation π is *nondegenerate* if the closed linear span of the orbit

$$\pi(A)H = \{\pi(a)\xi : a \in A,\ \xi \in H\}$$

is all of H. Equivalently, π is nondegenerate when the only vector $\xi \in H$ with $\pi(a)\xi = 0$ for all $a \in A$ is $\xi = 0$.

The two forms of nondegeneracy stated in the definition are indeed equivalent. The vectors killed by all of $\pi(A)$ form the orthogonal complement of $\pi(A)H$: for ξ in that complement and any a, η, the identity $\langle \pi(a^*)\eta, \xi\rangle = \langle \eta, \pi(a)\xi\rangle$ shows $\pi(a)\xi \perp \eta$ for all η once $\xi \perp \pi(A)H$, since $\pi(a^*)\eta$ ranges over $\pi(A)H$; and conversely a vector annihilated by every $\pi(a)$ is orthogonal to every $\pi(a^*)\eta$. So the closed span of $\pi(A)H$ is dense exactly when no nonzero vector is annihilated by the whole image. Degeneracy is the failure of this: the space splits as $\overline{\pi(A)H} \oplus \overline{\pi(A)H}^{\perp}$, and on the second summand π acts as the zero representation, carrying no information about A. A unital representation, one with $\pi(1) = 1$, is automatically nondegenerate, since $\xi = \pi(1)\xi \in \pi(A)H$ for every ξ; nondegeneracy is the substitute for unitality when A has no unit.

Example 5.1.2: The Identity Representation of $B(H)$ and $\mathcal{K}(H)$

Let H be a Hilbert space and take $A = B(H)$ acting on H by the identity representation $\pi(T) = T$. It is nondegenerate: $\pi(1) = 1$, so the orbit $\pi(A)H$ already contains every vector. The restriction to the compact operators $A = \mathcal{K}(H)$, again acting as $\pi(T) = T$ on H, is nondegenerate as well, although $\mathcal{K}(H)$ has no unit when H is infinite-dimensional. Indeed, for a unit vector ξ the rank-one projection $P_\xi = \langle -, \xi\rangle\xi$ is compact with $P_\xi\xi = \xi$, so $\xi \in \pi(\mathcal{K}(H))H$; every vector lies in the orbit and no nonzero vector is annihilated. By contrast, on the orthogonal sum $H \oplus \mathbb{C}$ the representation $T \mapsto T \oplus 0$ of $\mathcal{K}(H)$ is degenerate, acting as zero on the summand $\mathbb{C}$; its nondegenerate part is the original action on H.

Two constructions organize representations, and both are visible already in the identity representation of the compacts. A *subrepresentation* is what an invariant subspace carries: if $K \subseteq H$ is closed and $\pi(A)$-invariant, then restricting each $\pi(a)$ to K gives operators $\pi(a)|_K \in B(K)$, and $a \mapsto \pi(a)|_K$ is a representation of A on K, written $\pi|_K$. The *direct sum* runs the other way: given representations π_i on H_i, their direct sum $\bigoplus_i \pi_i$ acts on the Hilbert space direct sum $\bigoplus_i H_i$ coordinatewise, $(\bigoplus_i \pi_i)(a)(\xi_i)_i = (\pi_i(a)\xi_i)_i$, and each H_i sits inside as an invariant subspace carrying π_i as its subrepresentation. The structural questions are which representations decompose as direct sums of smaller ones, and which are already as small as possible. The smallest, those with no proper invariant subspace at all, are the irreducible representations.

Definition 5.1.3: Irreducible Representation

A representation $\pi\colon A \to B(H)$ on a nonzero Hilbert space H is *irreducible* if the only $\pi(A)$-invariant closed subspaces of H are $\{0\}$ and H. A representation that is not irreducible is *reducible*: it has a proper nonzero closed invariant subspace, and hence a proper nonzero subrepresentation.

Irreducibility says the algebra moves vectors around vigorously enough that no closed subspace is left invariant by the action except the trivial ones. A reducible representation, by contrast, contains a proper invariant subspace K, and the restriction $\pi|_K$ is a strictly smaller representation living inside it. Every representation is assembled from irreducible ones, and their classification records A up to the coarse equivalence that representations see. The point of the next section is that the irreducible representations correspond to the pure states of definition 4.4.1.

Example 5.1.4: Irreducibility of $M_2(\mathbb{C})$ on $\mathbb{C}^2$

Let $A = M_2(\mathbb{C})$ act on $H = \mathbb{C}^2$ by the defining representation $\pi(a)v = av$. A closed invariant subspace K is a linear subspace stable under multiplication by every 2×2 matrix; the candidates are $\{0\}$, the lines $\mathbb{C}v$, and all of $\mathbb{C}^2$. No line is invariant: for a nonzero $v \in \mathbb{C}^2$, extend v to a basis $\{v, w\}$ and let a be the matrix with $av = w$, which sends v off the line $\mathbb{C}v$. Thus the only invariant subspaces are $\{0\}$ and $\mathbb{C}^2$, and the defining representation of $M_2(\mathbb{C})$ is irreducible. The same argument shows the defining representation of $M_n(\mathbb{C})$ on $\mathbb{C}^n$ is irreducible for every n.

The definition of invariance quantifies over all of $\pi(A)$, which is unwieldy to check directly. The self-adjointness built into a $*$-homomorphism converts it into a single algebraic condition: a closed subspace is invariant exactly when the orthogonal projection onto it commutes with the entire image. This is the bridge between the geometry of subspaces and the algebra of operators, and it is what lets irreducibility be read off from the operators commuting with $\pi(A)$.

Proposition 5.1.5: Invariant Subspaces Are Projections in the Commutant

Let $\pi\colon A \to B(H)$ be a representation and $K \subseteq H$ a closed subspace, with $P \in B(H)$ the orthogonal projection onto K. Write

$$\pi(A)' = \{T \in B(H) : T\pi(a) = \pi(a)T \text{ for all } a \in A\}$$

for the *commutant* of $\pi(A)$. Then K is $\pi(A)$-invariant if and only if $P \in \pi(A)'$.

***Proof*:**

The argument turns invariance of K into invariance of K and of $K^\perp$ together, then reads the latter as commutation of the projection.

Step 1: invariance of K passes to $K^\perp$. Suppose K is $\pi(A)$-invariant, so $\pi(a)K \subseteq K$ for all $a \in A$. Let $\eta \in K^\perp$ and $a \in A$; the claim is $\pi(a)\eta \in K^\perp$. For any $\zeta \in K$,

$$\langle \pi(a)\eta, \zeta \rangle = \langle \eta, \pi(a)^*\zeta \rangle = \langle \eta, \pi(a^*)\zeta \rangle = 0$$

where the second equality uses that π is a $*$-homomorphism, $\pi(a)^* = \pi(a^*)$, and the third uses $\pi(a^*)\zeta \in K$ by invariance (applied to the element $a^* \in A$) together with $\eta \perp K$. Since $\zeta \in K$ was arbitrary, $\pi(a)\eta \in K^\perp$. Thus $K^\perp$ is $\pi(A)$-invariant as well. The self-adjointness of A is exactly what makes invariance of K force invariance of its complement.

Step 2: two-sided invariance is commutation of P. Every $\xi \in H$ decomposes uniquely as $\xi = P\xi + (1-P)\xi$ with $P\xi \in K$ and $(1-P)\xi \in K^\perp$. Fix $a \in A$. Using invariance of K and of $K^\perp$ from Step 1,

$$\pi(a)\xi = \pi(a)P\xi + \pi(a)(1-P)\xi$$

where $\pi(a)P\xi \in K$ and $\pi(a)(1-P)\xi \in K^\perp$. Applying P annihilates the second term and fixes the first, so $P\pi(a)\xi = \pi(a)P\xi$. As ξ was arbitrary, $P\pi(a) = \pi(a)P$; and as a was arbitrary, $P \in \pi(A)'$.

Step 3: the converse. Suppose $P \in \pi(A)'$, so $P\pi(a) = \pi(a)P$ for all a. For $\zeta \in K$ one has $P\zeta = \zeta$, hence

$$\pi(a)\zeta = \pi(a)P\zeta = P\pi(a)\zeta \in K$$

since $P\pi(a)\zeta$ lies in the range of P, which is K. Thus $\pi(a)K \subseteq K$ for every a, and K is $\pi(A)$-invariant, as we sought to show.

The commutant $\pi(A)'$ is the algebra of symmetries of the representation, the operators that commute with the whole action. The proposition makes it the register in which invariance is recorded: closed invariant subspaces correspond bijectively to projections in $\pi(A)'$. A representation is therefore reducible precisely when $\pi(A)'$ contains a projection other than 0 and 1, and irreducible precisely when it does not. This is the geometric half of Schur's lemma, whose analytic completion in the next section shows that for an irreducible representation the commutant collapses all the way to the scalars $\mathbb{C}\,1$. The proposition also splits every reducible representation: a nontrivial projection $P \in \pi(A)'$ writes $H = K \oplus K^\perp$ with both summands invariant, so $\pi = \pi|_K \oplus \pi|_{K^\perp}$ decomposes as a direct sum of two subrepresentations.

Example 5.1.6: The Commutant of a Direct Sum

Let $\pi = \pi_1 \oplus \pi_2$ act on $H = H_1 \oplus H_2$, the direct sum of two representations of A. The projection P onto the summand H_1 commutes with every $\pi(a) = \pi_1(a) \oplus \pi_2(a)$, since $\pi(a)$ is block-diagonal in the decomposition $H_1 \oplus H_2$ and P is the block-diagonal projection $1 \oplus 0$; explicitly $P\pi(a) = \pi_1(a) \oplus 0 = \pi(a)P$. So $P \in \pi(A)'$, confirming proposition 5.1.5: the invariant subspace H_1 registers as the commuting projection P. Conversely, if a representation is irreducible, then by the proposition its commutant contains no projection except 0 and 1, so it admits no such block decomposition. For the concrete case $A = M_2(\mathbb{C})$ acting on $\mathbb{C}^2 \oplus \mathbb{C}^2$ by $a \mapsto a \oplus a$, the projection onto the first copy of $\mathbb{C}^2$ lies in the commutant, and the representation is the reducible direct sum of two copies of the irreducible defining representation of example 5.1.4.

The reduction of a general representation to its atoms proceeds in two stages. The first, carried out now, is coarser: every nondegenerate representation is a direct sum of *cyclic* subrepresentations, those generated by a single vector (definition 4.2.1). The second stage, decomposing a cyclic representation into irreducible ones, is subtler and not always possible as a direct sum, so the cyclic decomposition reduces the study of representations to the cyclic case. Its value is that cyclic representations are exactly the ones the GNS construction produces from states, so this stage matches representations to the state space.

The mechanism is exhaustion: pick any nonzero vector, take the closure of its orbit as one cyclic summand, pass to the orthogonal complement, and repeat, using Zorn's lemma to reach a maximal collection whose summands fill the space. Nondegeneracy is what guarantees the collection cannot stop short.

Proposition 5.1.7: Cyclic Decomposition

Every nondegenerate representation $\pi\colon A \to B(H)$ is a direct sum of cyclic subrepresentations: there is a family $\{K_i\}_{i\in I}$ of mutually orthogonal closed $\pi(A)$-invariant subspaces with $H = \bigoplus_{i\in I} K_i$, such that each restriction $\pi|_{K_i}$ is a cyclic representation of A.

***Proof*:**

For a nonzero vector $\xi \in H$, write $K_\xi = \overline{\pi(A)\xi}$ for the closure of its orbit. This is a closed $\pi(A)$-invariant subspace: it is closed by construction, and $\pi(a)K_\xi \subseteq K_\xi$ because $\pi(a)\pi(b)\xi = \pi(ab)\xi \in \pi(A)\xi$ and $\pi(a)$ is continuous, so it carries the closure into itself. On K_ξ the vector ξ is cyclic for $\pi|_{K_\xi}$ by definition of K_ξ, so $\pi|_{K_\xi}$ is a cyclic representation, provided $\xi \in K_\xi$; this is where nondegeneracy is used, and it is verified in Step 2.

Step 1: a maximal orthogonal family of cyclic subspaces. Consider the collection $\mathcal{F}$ of all families $\{K_j\}_{j\in J}$ of mutually orthogonal, nonzero, closed $\pi(A)$-invariant subspaces, each of the form $K_j = \overline{\pi(A)\xi_j}$ for some vector ξ_j that is cyclic for $\pi|_{K_j}$. Partially order $\mathcal{F}$ by inclusion of families. The union of a chain in $\mathcal{F}$ is again such a family, since mutual orthogonality and the cyclic-subspace form are preserved under unions of nested families. By Zorn's lemma $\mathcal{F}$ has a maximal element $\{K_i\}_{i\in I}$. Let $K = \bigoplus_{i\in I} K_i$ be the closed span of these mutually orthogonal subspaces; it is closed and $\pi(A)$-invariant, being the closed span of invariant subspaces.

Step 2: maximality forces $K = H$. Suppose $K \neq H$. Then $K^\perp$ is a nonzero closed subspace, and it is $\pi(A)$-invariant: K is invariant, so by the equivalence in proposition 5.1.5 the projection onto K lies in $\pi(A)'$, whence the projection onto $K^\perp$ lies in $\pi(A)'$ as well, and $K^\perp$ is invariant. Nondegeneracy now gives a nonzero vector in $K^\perp$ on which $\pi(A)$ acts nontrivially. If $\pi(a)\eta = 0$ held for every $a \in A$ and every $\eta \in K^\perp$, then every $\eta \in K^\perp$ would be annihilated by all of $\pi(A)$, forcing $\eta = 0$ by nondegeneracy (definition 5.1.1) and hence $K^\perp = \{0\}$, contrary to $K \neq H$. So there is $\eta \in K^\perp$ and $a \in A$ with $\pi(a)\eta \neq 0$. Set $\xi = \pi(a)\eta \in K^\perp$, a nonzero vector, and let $K_\xi = \overline{\pi(A)\xi} \subseteq K^\perp$, an invariant subspace of $K^\perp$.

The vector ξ lies in K_ξ: since $\xi = \pi(a)\eta$, an approximate unit (u_λ) of A satisfies $u_\lambda a \to a$ in norm, so $\pi(u_\lambda a)\eta \to \pi(a)\eta = \xi$ by contractivity of π (theorem 1.3.3), while each $\pi(u_\lambda a)\eta = \pi(u_\lambda)\xi \in \pi(A)\xi$; hence $\xi \in \overline{\pi(A)\xi} = K_\xi$. Thus $\pi|_{K_\xi}$ is a cyclic representation with cyclic vector ξ, and K_ξ is orthogonal to every K_i since $K_\xi \subseteq K^\perp$. Adjoining K_ξ to the family $\{K_i\}_{i\in I}$ produces a strictly larger member of $\mathcal{F}$, contradicting maximality. Therefore $K = H$.

Consequently $H = \bigoplus_{i\in I} K_i$ with each $\pi|_{K_i}$ a cyclic representation of A, completing the proof.

The proposition reduces the study of representations to the cyclic ones. Every nondegenerate representation is a direct sum of cyclic pieces, and each cyclic piece is generated by a single vector. When A is unital these pieces are exactly the GNS representations of states: a cyclic representation $(\pi|_{K_i}, K_i, \xi_i)$ with unit cyclic vector produces the state $a \mapsto \langle \pi(a)\xi_i, \xi_i \rangle$, and by uniqueness (theorem 4.2.3 and its accompanying uniqueness statement) is unitarily equivalent to the GNS representation of that state. So for a unital A, representations, up to the direct-sum bookkeeping the proposition gives, are governed by the state space $S(A)$. The atoms one level finer, the irreducible representations, correspond under this dictionary to the pure states, and locating them is the task of the next section.

Without nondegeneracy the exhaustion in Step 2 could halt on a subspace where $\pi(A)$ acts as zero, a subspace carrying no cyclic vector at all, and the decomposition would fail. This is why the degenerate part of a representation is stripped off at the outset: every representation splits as its nondegenerate part plus a zero representation on the orthogonal complement (definition 5.1.1), and only the nondegenerate part admits a cyclic decomposition. Discarding the zero part loses nothing about A, so the theory restricts to nondegenerate representations without loss of generality.

Example 5.1.8: Cyclic Decomposition of $\mathcal{K}(H)$ on $H \oplus H$

Let $A = \mathcal{K}(H)$ act on $H \oplus H$ by $\pi(T) = T \oplus T$, the diagonal action. This representation is nondegenerate: a unit vector ξ and the rank-one projection P_ξ give $\pi(P_\xi)(\xi \oplus 0) = \xi \oplus 0$, and more generally no nonzero vector of $H \oplus H$ is annihilated by all diagonal compacts. Fix a vector $\zeta = (\zeta_1, \zeta_2) \in H \oplus H$ with both coordinates nonzero. Its orbit closure $K_\zeta = \overline{\pi(\mathcal{K}(H))\zeta}$ consists of the limits of vectors $(T\zeta_1, T\zeta_2)$ for $T \in \mathcal{K}(H)$. The size of K_ζ is governed by the linear relation between the coordinates. When ζ_1, ζ_2 are linearly *dependent*, say $\zeta_2 = c\zeta_1$, every orbit vector is $(T\zeta_1, cT\zeta_1)$, and as $T\zeta_1$ ranges over the dense orbit $\pi(\mathcal{K}(H))\zeta_1$ of H this fills the graph $K_\zeta = \{(v, cv) : v \in H\}$, a proper subspace on which $\pi|_{K_\zeta}$ is cyclic. When instead ζ_1, ζ_2 are linearly *independent*, a finite-rank operator T can send ζ_1 and ζ_2 to any prescribed pair of targets, so the orbit is already all of $H \oplus H$ and ζ is cyclic for the whole representation. The cyclic decomposition of proposition 5.1.7 assembles $H \oplus H$ from orbit closures of the first kind: two copies of H, sitting as complementary graphs. The decomposition is not unique: any nonzero vector seeds a cyclic summand, and different starting vectors give different families, all summing to the same $H \oplus H$. The nonuniqueness is precisely the freedom of unitary equivalence recorded in the previous chapter, since each cyclic summand realizes the GNS representation of the state it defines.

5.2 Irreducibility and Pure States

An irreducible representation, defined in definition 5.1.3 by the absence of a proper nonzero closed invariant subspace, is a negative condition: it says what does not occur. The task of this section is to convert it into two positive tests that make irreducibility computable. The first is internal to the representation and reads it off the commutant $\pi(A)'$: a representation is irreducible exactly when its commutant is as small as possible, a single copy of the scalars. This is Schur's lemma, and its proof is the spectral theorem applied to the self-adjoint elements of the commutant. The second test is external, matching irreducibility of π_φ against purity of the state φ across the GNS construction of theorem 4.2.3. Together these identify the irreducibles, up to unitary equivalence, with the pure states (definition 4.4.1).

The commutant of a representation $\pi\colon A \to B(H)$ is the set

$$\pi(A)' = \{T \in B(H) : T\pi(a) = \pi(a)T \text{ for all } a \in A\}$$

of bounded operators commuting with every operator in the image. It is a norm-closed $*$-subalgebra of $B(H)$ containing the scalars $\mathbb{C}\,1$: closedness and the algebra properties are immediate from the defining equations, and it is closed under adjoints because $T\pi(a) = \pi(a)T$ for all a gives, on taking adjoints and using $\pi(a^*) = \pi(a)^*$ as a ranges over A, the same relation for T^*. The size of $\pi(A)'$ measures how far π is from being irreducible, and Schur's lemma is the statement that the extreme case, $\pi(A)' = \mathbb{C}\,1$, is irreducibility itself.

Theorem 5.2.1: Schur's Lemma

Let $\pi\colon A \to B(H)$ be a representation of a C^*-algebra A on a nonzero Hilbert space H. Then π is irreducible if and only if its commutant is trivial,

$$\pi(A)' = \mathbb{C}\,1$$

***Proof*:**
The argument runs through the invariant projections of proposition 5.1.5: a closed subspace $K \subseteq H$ is $\pi(A)$-invariant if and only if the orthogonal projection P onto K lies in $\pi(A)'$. Irreducibility is the assertion that the only invariant closed subspaces are $\{0\}$ and H, so it is the assertion that the only projections in $\pi(A)'$ are 0 and 1.

Step 1: trivial commutant implies irreducible. Suppose $\pi(A)' = \mathbb{C}\,1$. A projection $P \in \pi(A)'$ is then a scalar, $P = \lambda 1$, and $P^2 = P$ forces $\lambda^2 = \lambda$, so $\lambda \in \{0, 1\}$ and $P \in \{0, 1\}$. By proposition 5.1.5 the only closed invariant subspaces are the kernel and range of these two projections, namely $\{0\}$ and H. Since $H \neq 0$ these are distinct, so π has no proper nonzero closed invariant subspace and is irreducible.

Step 2: irreducible implies trivial commutant. Suppose π is irreducible. The commutant $\pi(A)'$ is a C^*-subalgebra of $B(H)$, so it is spanned by its self-adjoint elements: any $T \in \pi(A)'$ decomposes as $T = T_1 + iT_2$ with $T_1 = \frac{1}{2}(T + T^*)$ and $T_2 = \frac{1}{2i}(T - T^*)$ self-adjoint and both in $\pi(A)'$, since the commutant is $*$-closed. It therefore suffices to show that every self-adjoint $T \in \pi(A)'$ is a scalar; then every element of the commutant is a complex combination of scalars, hence scalar.

Let $T = T^* \in \pi(A)'$. The spectral theorem for the bounded self-adjoint operator T on H [9, The Borel Functional Calculus, Spectral Projection, Spectral Projections Are Orthogonal Projections] provides, for each Borel set $E \subseteq \mathbb{R}$, a spectral projection $\chi_E(T)$, obtained by applying the bounded Borel function χ_E to T through the Borel functional calculus. Each such projection is a strong limit of operators in the abelian von Neumann algebra generated by T, and so lies in the double commutant $\{T\}'' = W^*(T)$. Since $T \in \pi(A)'$, each $\pi(a)$ commutes with T, that is $\pi(a) \in \{T\}'$; and every element of $\{T\}'' = W^*(T)$ commutes with everything in $\{T\}'$, so $\chi_E(T)$ commutes with each $\pi(a)$. Equivalently, T commutes with all of $\pi(A)$, so does every element of the von Neumann algebra it generates, in particular each spectral projection $\chi_E(T)$. Thus $\chi_E(T) \in \pi(A)'$ for every Borel set E.

Each $\chi_E(T)$ is a projection in $\pi(A)'$, so by Step 1's identification with invariant subspaces and the irreducibility of π, it equals 0 or 1. The map $E \mapsto \chi_E(T)$ takes only the values 0 and 1, which forces the spectral measure of T to be concentrated at a single point: if the spectrum $\sigma(T)$ contained two distinct points $\mu \neq \nu$, choosing disjoint Borel neighbourhoods E, F of them would give complementary nonzero spectral projections $\chi_E(T)$ and $\chi_F(T)$, neither of which can be 0 (a spectral projection of a Borel set meeting the spectrum in an open set is nonzero) and which cannot both be 1 (they are orthogonal). This contradicts the dichotomy. Hence $\sigma(T)$ is a single point $\{\lambda\}$, and a self-adjoint operator with one-point spectrum satisfies $T = \lambda 1$ by the spectral radius equality $\|T - \lambda 1\| = r(T - \lambda 1) = 0$. So T is scalar, and $\pi(A)' = \mathbb{C}\,1$, completing the proof.

Schur's lemma turns irreducibility into an operator-theoretic identity: the representation is irreducible precisely when nothing beyond the scalars commutes with its image. The two directions have different characters. The easy direction, that a trivial commutant forces irreducibility, holds in any Hilbert space and uses only that a scalar projection is 0 or 1. The substantive direction, that irreducibility forces the commutant to be trivial, is the analytic half: it needs the spectral theorem to manufacture, from any single self-adjoint element of the commutant, a family of invariant projections, and irreducibility then collapses that element to a scalar. The mechanism is worth isolating, because it recurs. Whenever a self-adjoint operator commutes with an irreducible family, its spectral projections do too, and irreducibility leaves each of them no room except 0 or 1; a self-adjoint operator all of whose spectral projections are trivial is a scalar. The commutant is one-dimensional, the smallest a $*$-algebra containing the scalars can be.

One consequence deserves separate statement, since it is the form of Schur's lemma most often invoked when comparing two irreducibles. An operator intertwining two irreducible representations is either zero or an isomorphism, and a self-intertwiner is a scalar.

Corollary 5.2.2: Intertwiners of Irreducibles

Let $\pi\colon A \to B(H)$ and $\rho\colon A \to B(K)$ be irreducible representations, and let $T\colon H \to K$ be a bounded operator with $T\pi(a) = \rho(a)T$ for all $a \in A$. Then either $T = 0$, or T is a scalar multiple of a unitary and π, ρ are unitarily equivalent. In particular, an operator commuting with an irreducible representation is a scalar.

***Proof*:**
The relation $T\pi(a) = \rho(a)T$ for all a gives, on taking adjoints and replacing a by a^*, the relation $T^*\rho(a) = \pi(a)T^*$; so T^*T commutes with $\pi(A)$ and TT^* commutes with $\rho(A)$. By Schur's lemma (theorem 5.2.1) each is a scalar: $T^*T = \lambda 1_H$ and $TT^* = \mu 1_K$ with $\lambda, \mu \geq 0$, the operators being positive. If $T = 0$ the first claim holds. Otherwise $\lambda > 0$, and $U = \lambda^{-1/2}T$ satisfies $U^*U = 1_H$, so U is an isometry; then UU^* is a nonzero projection in $\rho(A)'$, hence equal to 1_K by theorem 5.2.1, so U is a surjective isometry, that is, a unitary with $U\pi(a)U^* = \rho(a)$. Thus $T = \lambda^{1/2}U$ is a scalar multiple of a unitary intertwiner and $\pi \cong \rho$. Taking $\rho = \pi$ and $K = H$ gives the final statement, as we sought to show.

The corollary is the uniqueness half of the correspondence developed next: it says that an irreducible representation has no nontrivial self-symmetries, so that when irreducibles are attached to other data, distinct data can be told apart. The state that produces an irreducible GNS representation, and the sense in which it is determined by the representation, is the subject to which the section now turns.

The link between representations and states is the GNS construction of theorem 4.2.3, which sends a state φ on A to a cyclic representation $(\pi_\varphi, H_\varphi, \xi_\varphi)$ with $\varphi(a) = \langle \pi_\varphi(a)\xi_\varphi, \xi_\varphi \rangle$. The construction is reversible up to equivalence: a cyclic representation with a chosen unit cyclic vector recovers a state by this same formula. What remains is to match a property of the state, purity, with a property of the representation, irreducibility. A pure state is an extreme point of the state space $S(A)$

(definition 4.4.1): one that is not a proper convex combination of two distinct states. The bridge between the two sides is the order structure on states dominated by φ, which the GNS construction converts into the order structure on positive operators in the commutant $\pi_\varphi(A)'$ below the identity.

Lemma 5.2.3: Dominated States and Commutant Operators

Let φ be a state on a C^*-algebra A with GNS triple $(\pi_\varphi, H_\varphi, \xi_\varphi)$. For a positive linear functional ψ on A, write $\psi \le \varphi$ to mean $\varphi - \psi$ is positive. The map

$$T \mapsto \psi_T, \qquad \psi_T(a) = \langle \pi_\varphi(a)\, T\xi_\varphi,\, \xi_\varphi\rangle$$

is an affine order isomorphism from the set $\{T \in \pi_\varphi(A)' : 0 \le T \le 1\}$ onto the set of positive functionals ψ with $0 \le \psi \le \varphi$.

***Proof*:**
Step 1: the map lands in dominated functionals. Let $T \in \pi_\varphi(A)'$ with $0 \le T \le 1$. For $a \in A$,

$$\psi_T(a^*a) = \langle \pi_\varphi(a^*a)\, T\xi_\varphi, \xi_\varphi\rangle = \langle T\,\pi_\varphi(a)\xi_\varphi,\, \pi_\varphi(a)\xi_\varphi\rangle$$

using $T \in \pi_\varphi(A)'$ and $\pi_\varphi(a^*a) = \pi_\varphi(a)^*\pi_\varphi(a)$. Since $0 \le T \le 1$, the right side lies between 0 and $\|\pi_\varphi(a)\xi_\varphi\|^2 = \langle \pi_\varphi(a^*a)\xi_\varphi, \xi_\varphi\rangle = \varphi(a^*a)$. Thus $0 \le \psi_T(a^*a) \le \varphi(a^*a)$ for every a, and since every positive element of A has the form a^*a, both ψ_T and $\varphi - \psi_T$ are positive, that is $0 \le \psi_T \le \varphi$. The map $T \mapsto \psi_T$ is affine in T, being linear in the operator entry.

Step 2: injectivity. Suppose $\psi_T = \psi_{T'}$ for $T, T' \in \pi_\varphi(A)'$. Then $\langle (T - T')\pi_\varphi(a)\xi_\varphi, \pi_\varphi(b)\xi_\varphi\rangle$ vanishes for all a, b: expanding $\langle (T-T')\pi_\varphi(a)\xi_\varphi, \pi_\varphi(b)\xi_\varphi\rangle = \langle \pi_\varphi(b^*a)(T-T')\xi_\varphi, \xi_\varphi\rangle = \psi_{T-T'}(b^*a) = 0$, using $T - T' \in \pi_\varphi(A)'$. The vectors $\pi_\varphi(a)\xi_\varphi$ are dense in H_φ because ξ_φ is cyclic, so $T - T' = 0$. Hence the map is injective.

Step 3: surjectivity. Let ψ be a positive functional with $0 \le \psi \le \varphi$. The sesquilinear form $\langle \pi_\varphi(a)\xi_\varphi, \pi_\varphi(b)\xi_\varphi\rangle_\psi = \psi(b^*a)$ is well defined on the dense subspace $\pi_\varphi(A)\xi_\varphi$: if $\pi_\varphi(a)\xi_\varphi = 0$ then $\varphi(a^*a) = \|\pi_\varphi(a)\xi_\varphi\|^2 = 0$, and $\psi(a^*a) \le \varphi(a^*a) = 0$ forces $\psi(a^*a) = 0$, so by the Cauchy-Schwarz inequality for the positive functional ψ the value $\psi(b^*a)$ is 0 for all b. The bound $|\psi(b^*a)| \le \psi(a^*a)^{1/2}\psi(b^*b)^{1/2} \le \varphi(a^*a)^{1/2}\varphi(b^*b)^{1/2} = \|\pi_\varphi(a)\xi_\varphi\|\,\|\pi_\varphi(b)\xi_\varphi\|$ shows the form is bounded by 1, so it extends to a bounded sesquilinear form on H_φ and is represented by a unique operator T with $0 \le T \le 1$ and $\langle T\pi_\varphi(a)\xi_\varphi, \pi_\varphi(b)\xi_\varphi\rangle = \psi(b^*a)$. That $T \in \pi_\varphi(A)'$ follows from

$$\langle T\pi_\varphi(c)\pi_\varphi(a)\xi_\varphi, \pi_\varphi(b)\xi_\varphi\rangle = \psi(b^*ca) = \langle T\pi_\varphi(a)\xi_\varphi, \pi_\varphi(c^*b)\xi_\varphi\rangle = \langle \pi_\varphi(c)T\pi_\varphi(a)\xi_\varphi, \pi_\varphi(b)\xi_\varphi\rangle$$

for all a, b, c, and density. Taking $b = 1$ in the unital case, or approximating through an approximate unit in general, gives $\psi_T(a) = \langle \pi_\varphi(a)T\xi_\varphi, \xi_\varphi\rangle = \psi(a)$, so $\psi = \psi_T$ is in the image. Finally the correspondence preserves order, since $T \le T'$ in $\pi_\varphi(A)'$ is equivalent to $\psi_T \le \psi_{T'}$ by the computation of Step 1 applied to $T' - T$, so it is an affine order isomorphism, as we sought to show.

The lemma is the exact translation the correspondence needs. States below φ, ordered by domination, are the same data as positive operators below 1 in the commutant, ordered as operators. Purity of φ and irreducibility of π_φ are each a statement that one of these ordered sets is trivial, and the lemma makes them the same statement.

Theorem 5.2.4: Pure States and Irreducible Representations

Let φ be a state on a C^*-algebra A with GNS representation $(\pi_\varphi, H_\varphi, \xi_\varphi)$ from theorem 4.2.3. Then φ is pure (definition 4.4.1) if and only if π_φ is irreducible. Consequently, passing to the GNS representation and back gives a bijection between the pure states of A and the pairs consisting of an irreducible representation together with a chosen unit cyclic vector, each taken up to unitary equivalence.

***Proof*:**
Step 1: purity and the order interval below φ. A state φ fails to be pure exactly when it is a proper convex combination $\varphi = t\varphi_1 + (1-t)\varphi_2$ of distinct states $\varphi_1 \neq \varphi_2$ with $0 < t < 1$. Setting $\psi = t\varphi_1$, this exhibits a positive functional with $0 \leq \psi \leq \varphi$ that is neither 0 nor a scalar multiple $s\varphi$ of φ: if $t\varphi_1 = s\varphi$ then $\varphi_1 = (s/t)\varphi$, and since φ_1 and φ are states of norm one this forces $s/t = 1$ and $\varphi_1 = \varphi$, hence $\varphi_2 = \varphi$ too, contradicting $\varphi_1 \neq \varphi_2$. Conversely, any positive ψ with $0 \leq \psi \leq \varphi$ that is not a scalar multiple of φ yields such a decomposition: normalizing, $\varphi = \psi + (\varphi - \psi)$ writes φ as a sum of two nonzero positive functionals, and rescaling each to a state expresses φ as a proper convex combination of two states, distinct because ψ is not proportional to φ. Thus φ is pure if and only if every positive functional ψ with $0 \leq \psi \leq \varphi$ is a scalar multiple $s\varphi$ with $0 \leq s \leq 1$; that is, the order interval $[0, \varphi]$ of dominated positive functionals is exactly the line segment $\{s\varphi : 0 \leq s \leq 1\}$.

Step 2: transferring to the commutant. By lemma 5.2.3 the order interval $[0, \varphi]$ of dominated positive functionals is affinely order isomorphic to the order interval $\{T \in \pi_\varphi(A)' : 0 \leq T \leq 1\}$ of positive operators below 1 in the commutant, via $T \mapsto \psi_T$. Under this isomorphism the scalar multiples $s\varphi$ correspond to the scalar operators $s\,1$: indeed $\psi_{s1}(a) = \langle \pi_\varphi(a)\, s\xi_\varphi, \xi_\varphi \rangle = s\varphi(a)$. So the condition of Step 1, that $[0, \varphi]$ consists only of the scalar multiples of φ, translates to the condition that $\{T \in \pi_\varphi(A)' : 0 \leq T \leq 1\}$ consists only of the scalars $s\,1$ with $0 \leq s \leq 1$.

Step 3: the commutant condition is irreducibility. The commutant $\pi_\varphi(A)'$ is a C^*-algebra, spanned by its self-adjoint elements, and each self-adjoint element T can be shifted and scaled to lie in the interval $[0, 1]$: writing $T = \alpha 1 + \beta S$ with $0 \leq S \leq 1$ for suitable real α and β (take $S = (T + \|T\| 1)/(2\|T\|)$, so $\alpha = -\|T\|$ and $\beta = 2\|T\|$), T is a scalar precisely when S is. So every positive operator in the interval $[0, 1]$ being scalar is equivalent to every self-adjoint element of $\pi_\varphi(A)'$ being scalar, which is equivalent to $\pi_\varphi(A)' = \mathbb{C}\,1$. By Schur's lemma (theorem 5.2.1) this is exactly the irreducibility of π_φ. Chaining the three equivalences, φ is pure if and only if π_φ is irreducible.

Step 4: the bijection up to equivalence. The GNS construction of theorem 4.2.3 sends a state φ to a cyclic representation with distinguished unit cyclic vector ξ_φ, and the state is recovered as $\varphi(a) = \langle \pi_\varphi(a)\xi_\varphi, \xi_\varphi \rangle$; the construction is unique up to a unitary carrying ξ_φ to the cyclic vector of any other cyclic representation implementing φ. Restricting to pure states, the first three steps show the associated representations are exactly the irreducible cyclic ones. Conversely an irreducible representation is cyclic (every nonzero vector is cyclic, since its closed invariant span is nonzero, hence all of H), and any unit vector ξ defines a state $a \mapsto \langle \pi(a)\xi, \xi \rangle$ whose GNS representation is unitarily equivalent to π with ξ as cyclic vector, so this state is pure by the equivalence just proved. Unitarily equivalent representations arise from the same pure states, and the two passages are mutually inverse up to unitary equivalence, giving the stated bijection, completing the proof.

The theorem places the pure states and the irreducible representations in exact correspondence, and each side illuminates the other. From the side of states, it says that the extreme points of the convex set $S(A)$ are precisely the states whose GNS representations cannot be split, so that the search for irreducible representations becomes a search for extreme points, a problem of convexity. That the pure states are plentiful, enough to separate the points of A, is then a consequence of the Krein-Milman theorem applied to the weak-$*$ compact convex set $S(A)$, and this is the route by which the existence of a faithful family of irreducible representations, sharpening the universal representation behind Gelfand-Naimark (theorem 4.3.3), is established. From the side of representations, the theorem says that an irreducible representation carries no more information than a single pure state, since any unit vector recovers the whole representation up to equivalence, which is why the classification of irreducibles is a manageable problem even when the representation theory is otherwise wild. The one subtlety is the phrase “up to unitary equivalence”: a pure state determines its irreducible representation exactly, whereas an irreducible representation determines only the family of pure states obtained from its unit vectors, all of which are equivalent by corollary 5.2.2.

The abstract correspondence is best seen on the concrete representation from which the whole theory takes its shape, the algebra of compact operators acting on its Hilbert space. There irreducibility can be checked by hand, and it exhibits the mechanism of Schur's lemma in its simplest form: an operator commuting with all rank-one operators has nowhere to be but the scalars.

Example 5.2.5: Irreducibility of the Compacts

Let H be a nonzero Hilbert space and let $\pi\colon \mathcal{K}(H) \to B(H)$ be the identity representation of the C^*-algebra $\mathcal{K}(H)$ of compact operators, acting on H in the tautological way. This representation is irreducible, and the same computation shows the identity representation of $B(H)$ on H is irreducible.

By Schur's lemma (theorem 5.2.1) it suffices to show $\pi(\mathcal{K}(H))' = \mathbb{C}\,1$. Let $T \in B(H)$ commute with every compact operator, in particular with every rank-one operator. For unit vectors $\xi, \eta \in H$ write $\theta_{\xi,\eta}$ for the rank-one operator $\zeta \mapsto \langle \zeta, \eta \rangle \xi$, which is compact. The commutation relation $T\theta_{\xi,\eta} = \theta_{\xi,\eta}T$ evaluated on any $\zeta \in H$ reads

$$\langle \zeta, \eta \rangle\, T\xi = T\big(\langle \zeta, \eta \rangle \xi\big) = \theta_{\xi,\eta}(T\zeta) = \langle T\zeta, \eta \rangle\, \xi$$

Take $\zeta = \eta$, so $\langle \zeta, \eta \rangle = 1$: the identity becomes $T\xi = \langle T\eta, \eta \rangle\, \xi$. This holds for every pair of unit vectors ξ, η. The left side $T\xi$ does not depend on η, so the scalar $\lambda = \langle T\eta, \eta \rangle$ is the same for every unit η, and $T\xi = \lambda\xi$ for every unit ξ with this single scalar λ. Hence $T = \lambda 1$, so $\pi(\mathcal{K}(H))' = \mathbb{C}\,1$ and π is irreducible. Since $\mathcal{K}(H) \subseteq B(H)$, any operator commuting with all of $B(H)$ commutes in particular with all compacts, so $B(H)' = \mathbb{C}\,1$ as well, and the identity representation of $B(H)$ is irreducible.

The compacts give the model irreducible representation, and their pure states are correspondingly transparent. Each unit vector $\eta \in H$ gives the vector state $\omega_\eta(T) = \langle T\eta, \eta\rangle$ on $\mathcal{K}(H)$, and the computation above is exactly the verification that its GNS representation is the identity representation, with cyclic vector η; theorem 5.2.4 then reads off that each ω_η is pure. Two unit vectors give the same pure state precisely when they differ by a phase, so the pure states of $\mathcal{K}(H)$ are the points of the projective space of H, and all of them yield the single irreducible representation up to equivalence. This is the smallest instance of the general picture: a C^*-algebra with, up to equivalence, exactly one irreducible representation, whose pure states form a single orbit. For a commutative algebra $C_0(X)$ the picture is opposite and equally clean, since every irreducible representation is one-dimensional, given by evaluation at a point of X (the image $\pi(C_0(X))$ is abelian, so it lies in its own commutant, which by Schur's lemma is $\mathbb{C}\,1$; irreducibility then makes H one-dimensional), and the pure states are exactly the evaluations $f \mapsto f(x)$, recovering the points of X as the extreme points of the space of probability measures.

One consequence of the correspondence deserves separate statement, because it is what makes the irreducible representations an adequate substitute for the algebra itself. Theorem 4.3.3 already recovers the norm from the *states*, through the universal representation, and proposition 4.1.5 sharpens that to a single state attaining it. Neither says the irreducible representations suffice. They do, and the passage needs care in exactly one respect: proposition 4.1.5 and theorem 4.4.4 were proved for a unital algebra, while the applications, above all to group C^*-algebras, are usually non-unital. The unitization of proposition 1.4.5 bridges the gap.

Theorem 5.2.6: Irreducible Representations Norm the Algebra

Let A be a C^*-algebra, not assumed unital, and let $a \in A$. Then

$$\|a\| = \sup\{\|\pi(a)\| \mid \pi \text{ an irreducible representation of } A\}$$

and for $a \neq 0$ the supremum is attained: there is an irreducible representation π of A on a nonzero Hilbert space, with a unit cyclic vector ξ, such that $\|\pi(a)\| = \|a\|$ and the pure state $\omega = \langle \pi(\cdot)\xi, \xi\rangle$ of A satisfies $\omega(a^*a) = \|a\|^2$. In particular the irreducible representations of A separate its points.

***Proof*:**
Every $*$-representation is contractive (lemma 5.4.1), so $\|\pi(a)\| \leq \|a\|$ for each π and the supremum is at most $\|a\|$. For $a = 0$ both sides vanish, so assume $a \neq 0$; it suffices to produce a single irreducible π with $\|\pi(a)\| = \|a\|$.

Step 1: a state of the unitization attaining the norm. Let $\tilde{A}$ be the unitization of proposition 1.4.5, into which A embeds isometrically as a closed two-sided ideal, and put $b = a^*a \in A$. Then b is self-adjoint with $\|b\| = \|a\|^2 > 0$ by the C^*-identity. Since $\tilde{A}$ is unital, proposition 4.1.5 gives a state φ of $\tilde{A}$ with $|\varphi(b)| = \|b\|$; and $b \geq 0$, so $\varphi(b) \geq 0$ by positivity of states (proposition 4.1.3) and therefore $\varphi(b) = \|b\|$.

Step 2: an extreme such state. Put

$$F = \left\{\varphi \in S(\tilde{A}) \;\middle|\; \varphi(b) = \|b\|\right\}$$

which is nonempty by Step 1 and convex, being cut out by a linear condition. It is weak-$*$ closed in the weak-$*$ compact set $S(\tilde{A})$ of proposition 4.4.3, hence weak-$*$ compact, so the Krein-Milman theorem [9, Krein-Milman Theorem] gives it an extreme point ω.

Moreover F is a *face* of $S(\tilde{A})$: if $\omega = t\varphi_1 + (1-t)\varphi_2$ with $\varphi_i \in S(\tilde{A})$ and $0 < t < 1$, then each $\varphi_i(b) \leq \|b\|$ because $\|\varphi_i\| = 1$, while the convex combination equals $\|b\|$, which forces $\varphi_1(b) = \varphi_2(b) = \|b\|$ and so $\varphi_i \in F$. An extreme point of a face is extreme in the ambient convex set, so ω is an extreme point of $S(\tilde{A})$, that is a pure state of $\tilde{A}$ (definition 4.4.1).

Step 3: the GNS representation and its restriction. By theorem 5.2.4 the GNS representation $(\pi_\omega, H_\omega, \xi_\omega)$ of theorem 4.2.3 is irreducible, and

$$\|\pi_\omega(a)\xi_\omega\|^2 = \langle \pi_\omega(a^*a)\xi_\omega, \xi_\omega\rangle = \omega(b) = \|a\|^2$$

Write $\pi = \pi_\omega|_A$. It is nonzero, since the display gives $\pi(a) \neq 0$. It is irreducible: because $\tilde{A} = A + \mathbb{C}1$ and the unit acts as the identity operator, a closed subspace of H_ω invariant under $\pi_\omega(A)$ is invariant under $\pi_\omega(a + \lambda 1) = \pi_\omega(a) + \lambda I$ for every $a \in A$ and $\lambda \in \mathbb{C}$, hence under all of $\pi_\omega(\tilde{A})$; so π and π_ω have the same closed invariant subspaces, and π_ω has none but $\{0\}$ and H_ω.

Finally $\|\pi(a)\| \geq \|\pi(a)\xi_\omega\| = \|a\|$ because $\|\xi_\omega\| = 1$, and the reverse inequality is contractivity again, so $\|\pi(a)\| = \|a\|$ and the supremum is attained. The vector state $\omega' = \langle \pi(\cdot)\xi_\omega, \xi_\omega\rangle$ on A then has $\omega'(a^*a) = \|a\|^2$, and it is pure because its GNS representation is the irreducible π, by theorem 5.2.4 applied to A. Separation of points follows: if $a \neq a'$ in A, applying the above to $a - a'$ produces an irreducible π with $\|\pi(a - a')\| = \|a - a'\| \neq 0$.

The theorem is the form in which irreducible representations are used downstream. It says the passage from A to its irreducible representations loses no information about norms, so a C^*-algebra may be studied one irreducible representation at a time. The group C^*-algebra of a locally compact group is the standing example downstream, where this statement becomes the assertion that the irreducible unitary representations of the group separate its points. The unitization detour in Step 1 is not a technicality to be skipped: for a non-unital A the set $S(A)$ of states is *not* weak-$*$ compact, its weak-$*$ closure picking up the functionals of norm less than one, so the Krein-Milman argument has no compact convex set to run on until the unit is adjoined.

5.3 Multiplier Algebras

A non-unital C^*-algebra is missing more than its unit. The approximate unit of definition 1.4.1 gives an identity only in the limit, and there is no home inside A for the operators one would like to multiply by: the projections that cut out invariant subspaces, the isometries of the Toeplitz algebra, the bounded functions that fail to vanish at infinity. Each of these acts on A by multiplication and stays inside A, yet none belongs to A. This section builds the algebra where they live. The multiplier algebra $M(A)$ is the largest unital C^*-algebra into which A embeds as an essential ideal, and it plays for A the role that the one-point compactification's function algebra $C(X^+)$ plays for $C_0(X)$, only much larger: where the unitization of proposition 1.4.5 adjoins a single dimension, $M(A)$ adjoins every multiplier at once.

There are two descriptions, an abstract one intrinsic to A and a concrete one that reads $M(A)$ off any faithful nondegenerate representation. The abstract description packages a multiplier as a compatible pair of left and right multiplication operators, a *double centralizer*. The concrete description takes $A \subseteq B(H)$ and collects the operators that carry A back into itself on both sides. The two agree, and the section develops the abstract one, then identifies it with the concrete one and reads off the universal property.

The defining feature of an element m that ought to multiply A is that it acts on the left, $a \mapsto ma$, and on the right, $a \mapsto am$, and that these two actions are linked by associativity: $a(mb) = (am)b$. Stripping m away and keeping only the two operators it would induce gives the following notion, which needs no ambient algebra containing m.

Definition 5.3.1: Multiplier Algebra

Let A be a C^*-algebra. A *double centralizer* of A is a pair (L, R) of bounded linear maps $L, R\colon A \to A$ satisfying

$$a\,L(b) = R(a)\,b \qquad \text{for all } a, b \in A$$

The *multiplier algebra* $M(A)$ is the set of all double centralizers of A, made into a $*$-algebra by

$$(L_1, R_1) + (L_2, R_2) = (L_1 + L_2,\ R_1 + R_2)$$
$$(L_1, R_1)(L_2, R_2) = (L_1 L_2,\ R_2 R_1)$$
$$(L, R)^* = (R^\flat, L^\flat), \qquad T^\flat(a) := T(a^*)^*$$

with unit $(\mathrm{id}_A, \mathrm{id}_A)$, and normed by $\|(L, R)\| = \|L\|$, the operator norm of the left action. For $c \in A$, the pair $L_c(a) = ca$, $R_c(a) = ac$ is a double centralizer, and $c \mapsto (L_c, R_c)$ embeds A into $M(A)$.

The relation $aL(b) = R(a)b$ is exactly the associativity $a(mb) = (am)b$ written without reference to m, with L playing the role of left multiplication by m and R the role of right multiplication. Two consequences of the relation deserve to be stated before the norm is understood, because they are what make the definition well-posed. First, L respects right multiplication and R respects left multiplication: $L(ab)$ and $L(a)b$ agree because $c\,L(ab) = R(c)\,ab = (R(c)a)b = (cL(a))b = c\,(L(a)b)$ for all c, and A has no nonzero element annihilated on the left by all of A (take $c = x^*$ in $cx = 0$, so $\|x\|^2 = 0$), so $L(ab) = L(a)b$; symmetrically $R(ab) = aR(b)$. Second, the boundedness assumed in the definition is not an extra hypothesis one must check case by case: a linear map $L\colon A \to A$ with $aL(b) = R(a)b$ for some R is automatically bounded, by the closed graph theorem, since A has no nonzero element annihilating all of A on the left. The norm $\|(L, R)\| = \|L\|$ then makes $M(A)$ a C^*-algebra; the verification of the C^*-identity is deferred to the concrete picture below, where it becomes transparent.

The involution deserves a word. The operator $T^\flat(a) = T(a^*)^*$ is the conjugate of T through the involution of A, and the swap $(L, R) \mapsto (R^\flat, L^\flat)$ encodes that taking adjoints turns left multiplication by m into right multiplication by m^*: if $L(a) = ma$ then $L^\flat(a) = (ma^*)^* = am^* = R_{m^*}(a)$, so the right action of the adjoint is the flat of the left action. That the product reverses the order of the R's, $(L_1, R_1)(L_2, R_2) = (L_1L_2, R_2R_1)$, is the same phenomenon that right multiplication composes contravariantly: composing "multiply by m_1 then m_2" multiplies on the right by m_2 first.

To see the definition captures the intended object rather than an abstraction of it, the concrete realization identifies $M(A)$ inside the bounded operators on any Hilbert space carrying A faithfully.

Proposition 5.3.2: The Multiplier Algebra as an Idealizer

Let $A \subseteq B(H)$ be a faithful nondegenerate representation of a C^*-algebra A, so that $\overline{AH} = H$. Then the map sending a double centralizer (L, R) to the operator it induces is a $*$-isomorphism of $M(A)$ onto the *idealizer*

$$I(A) = \{\, T \in B(H) \big| TA \subseteq A \text{ and } AT \subseteq A \,\}$$

of A in $B(H)$. Under this isomorphism A maps onto $A \subseteq B(H)$ itself, and $M(A)$ is a C^*-subalgebra of $B(H)$.

***Proof*:**
The proof runs in both directions: an idealizer element yields a double centralizer, and a double centralizer yields an idealizer element, and the two assignments are mutually inverse $*$-homomorphisms.

Step 1: from idealizer to double centralizer. Let $T \in I(A)$. Since $TA \subseteq A$ and $AT \subseteq A$, the maps $L_T(a) = Ta$ and $R_T(a) = aT$ take A into A, are bounded by $\|T\|$, and satisfy the centralizer relation $aL_T(b) = a(Tb) = (aT)b = R_T(a)b$ by associativity of operator composition. So (L_T, R_T) is a double centralizer, and $T \mapsto (L_T, R_T)$ is linear, multiplicative (because $L_{ST} = L_S L_T$ and $R_{ST} = R_T R_S$), and $*$-preserving, since for $a \in A$ one has $L_{T^*}(a) = T^* a$ while $R_T^\flat(a) = (a^*T)^* = T^*a$, so $(L_T, R_T)^* = (R_T^\flat, L_T^\flat) = (L_{T^*}, R_{T^*})$.

Step 2: the map is injective. If $L_T = 0$ then $Ta = 0$ for all $a \in A$, hence $T(a\xi) = (Ta)\xi = 0$ for all $a \in A$, $\xi \in H$; as AH is dense (nondegeneracy) and T is bounded, $T = 0$. So distinct operators give distinct double centralizers.

Step 3: from double centralizer to idealizer, surjectivity. Let (L, R) be a double centralizer of A. The construction realizes L as left multiplication by a single operator on H. Because the representation is nondegenerate, the module-factorization theorem for nondegenerate Banach modules over a C^*-algebra with approximate unit [9] gives $AH = \{\, a\xi \big| a \in A,\ \xi \in H \,\}$: every vector in the dense subspace AH is already a single product $a\xi$, no sums needed. Define

$$T(a\xi) = L(a)\,\xi \qquad (a \in A,\ \xi \in H)$$

This is well-defined and bounded. The right-module identity $L(a)b = L(ab)$ (established after definition 5.3.1) makes L contractive-up-to-$\|L\|$ as a left multiplier: $\|L(a)b\| = \|L(ab)\| \leq \|L\|\,\|ab\|$ for every $b \in A$. A positivity lemma turns such a norm bound into an order relation: if $c, d \in A$ satisfy $\|cb\| \leq \|db\|$ for all $b \in A$, then $c^*c \leq d^*d$. To prove it, work in the unitization $\tilde{A}$ and fix $\varepsilon > 0$; the element $y_\varepsilon = (d^*d + \varepsilon 1)^{-1/2} \in \tilde{A}$ is positive and invertible. For $b \in A$ the product $b\,y_\varepsilon$ lies in A (an ideal of $\tilde{A}$), so the hypothesis applies to it: $\|c\,b\,y_\varepsilon\| \leq \|d\,b\,y_\varepsilon\|$. Letting $b = e_\lambda$ run through an approximate unit of A and using $ce_\lambda \to c$, $de_\lambda \to d$ in norm gives, in the limit,

$$\|c\,y_\varepsilon\| \leq \|d\,y_\varepsilon\|$$

Squaring and rewriting each side through the C^*-identity $\|x\|^2 = \|x^*x\|$, this reads $\|y_\varepsilon\, c^*c\, y_\varepsilon\| \leq \|y_\varepsilon\, d^*d\, y_\varepsilon\|$. The right-hand norm is ≤ 1, since $y_\varepsilon\, d^*d\, y_\varepsilon = g(d^*d)$ for $g(t) = t/(t+\varepsilon) < 1$ by the functional calculus. A positive element of norm ≤ 1 is dominated by the unit, so $y_\varepsilon\, c^*c\, y_\varepsilon \leq 1$; conjugating by $y_\varepsilon^{-1} = (d^*d + \varepsilon 1)^{1/2}$ gives $c^*c \leq d^*d + \varepsilon 1$, and letting $\varepsilon \to 0$ gives $c^*c \leq d^*d$.

Applying the lemma with $c = L(a)$ and $d = \|L\|\, a$ gives the elementwise operator inequality

$$L(a)^*L(a) \leq \|L\|^2\, a^*a \qquad \text{in } A \tag{5.1}$$

for each $a \in A$. Reading (5.1) through the faithful representation on H, for $a \in A$ and $\xi \in H$,

$$\|T(a\xi)\|^2 = \|L(a)\xi\|^2 = \langle L(a)^*L(a)\xi, \xi\rangle \leq \|L\|^2\langle a^*a\,\xi, \xi\rangle = \|L\|^2\|a\xi\|^2$$

so if $a\xi = 0$ then $T(a\xi) = 0$ (well-definedness), and $\|T(a\xi)\| \leq \|L\|\,\|a\xi\|$ (boundedness). Thus T extends to a bounded operator on H with $\|T\| \leq \|L\|$. For $c \in A$, $Tc = L(c) \in A$ as operators, giving $TA \subseteq A$; and cT acts by $cT(a\xi) = cL(a)\xi = R(c)a\,\xi$, whence $cT = R(c) \in A$, giving $AT \subseteq A$. Thus $T \in I(A)$, and by construction $L_T = L$, $R_T = R$, so the Step 1 map is surjective. The two assignments are mutually inverse.

Step 4: it is an isometry and a C^*-embedding. The bijection of Steps 1 to 3 is a $*$-isomorphism of $*$-algebras, and it carries the operator norm $\|T\|$ to $\|L\| = \|(L, R)\|$. First, $\|L\| = \|R\|$ for any double centralizer: since $R(a) \in A$ and A is a C^*-algebra, $\|R(a)\| = \sup_{\|b\|\leq 1}\|R(a)b\| = \sup_{\|b\|\leq 1}\|aL(b)\| \leq \|a\|\,\|L\|$, so $\|R\| \leq \|L\|$, and the symmetric argument through $\|L(b)\| = \sup_{\|a\|\leq 1}\|aL(b)\| = \sup_{\|a\|\leq 1}\|R(a)b\|$ gives $\|L\| \leq \|R\|$. Now Step 3 gives $\|T\| \leq \|R\| = \|L\|$, while $L(a) = Ta$ as operators gives $\|L\| = \sup_{\|a\|\leq 1}\|Ta\| \leq \|T\|$; hence $\|T\| = \|L\| = \|(L, R)\|$. The norm on $M(A)$ therefore equals the operator norm of the corresponding $T \in B(H)$. The image $I(A)$ is a self-adjoint subalgebra of $B(H)$ and is norm-closed: if $T_n \to T$ in $B(H)$ with each $T_n \in I(A)$, then for $a \in A$ the products $T_na \to Ta$ and $aT_n \to aT$ with each $T_na, aT_n \in A$, so $Ta, aT \in A$ by closedness of A, whence $T \in I(A)$. A norm-closed self-adjoint subalgebra of $B(H)$ is a C^*-algebra; so $M(A) \cong I(A)$ is a C^*-algebra, complete in the norm $\|(L, R)\| = \|L\|$, and that norm satisfies the C^*-identity inherited from $B(H)$. The embedding $c \mapsto (L_c, R_c)$ corresponds to $c \mapsto c \in B(H)$, completing the proof.

The idealizer picture shows $M(A)$: it is every bounded operator that multiplies A into itself from both sides, computed once A is faithfully and nondegenerately represented, and the identification with double centralizers shows the answer does not depend on which such representation is chosen. It also discharges the one deferred point, the C^*-identity for the norm $\|(L, R)\| = \|L\|$, which is now inherited from $B(H)$. The nondegeneracy hypothesis is essential and is exactly what a faithful representation of A can always be arranged to satisfy after discarding the subspace A annihilates: the universal representation of the Gelfand-Naimark theorem is faithful, and restricting to $\overline{AH}$ makes it nondegenerate without losing faithfulness. Two consequences of the definition organize what follows: A sits inside $M(A)$ as an ideal, and that ideal is essential.

Proposition 5.3.3: The Universal Property of the Multiplier Algebra

Let A be a C^*-algebra. Then:

1. A is an essential ideal of $M(A)$: the embedding $c \mapsto (L_c, R_c)$ realizes A as a closed two-sided ideal $A \trianglelefteq M(A)$ such that no nonzero element of $M(A)$ annihilates A, that is, $mA = 0$ or $Am = 0$ forces $m = 0$.

2. $M(A)$ is the largest unital C^*-algebra containing A essentially: if A is a closed essential ideal of a C^*-algebra B, then there is a unique injective $*$-homomorphism $\iota\colon B \to M(A)$ restricting to the identity on A, so B embeds in $M(A)$ over A.

3. If A is unital, then $M(A) = A$.

***Proof*:**
Fix a faithful nondegenerate representation $A \subseteq B(H)$ and identify $M(A) = I(A)$ by proposition 5.3.2.

Step 1: A is an essential ideal. That $A = I(A) \cap A$ is a two-sided ideal of $I(A)$ is immediate from the definition of the idealizer: for $T \in I(A)$ and $a \in A$, both Ta and aT lie in A. It is closed because A is a C^*-algebra. For essentiality, suppose $m \in M(A)$ with $mA = 0$; then in $B(H)$ the operator m (that is, the corresponding T) satisfies $Ta = 0$ for all $a \in A$, so $T(a\xi) = 0$ on the dense subspace AH, giving $T = 0$ by nondegeneracy. The case $Am = 0$ is symmetric via $aT = 0$ and density of $A^*H = AH$ (as $A^* = A$). So $A \trianglelefteq M(A)$ is essential, proving (1).

Step 2: any essential extension embeds. Let A be a closed essential ideal of a C^*-algebra B. Each $b \in B$ multiplies A: because A is an ideal, $L_b(a) = ba$ and $R_b(a) = ab$ take A into A, and the associativity in B gives $aL_b(a') = a(ba') = (ab)a' = R_b(a)a'$, so (L_b, R_b) is a double centralizer of A. Set $\iota(b) = (L_b, R_b) \in M(A)$. The map $\iota\colon B \to M(A)$ is a $*$-homomorphism, since $L_{b_1b_2} = L_{b_1}L_{b_2}$, $R_{b_1b_2} = R_{b_2}R_{b_1}$, and $L_{b^*} = R_b^\flat$ as in definition 5.3.1. It restricts to the standard embedding on A by construction. Injectivity is essentiality of A in B: if $\iota(b) = 0$ then $bA = L_b(A) = 0$, and since A is essential in B this forces $b = 0$. Uniqueness: any $*$-homomorphism $\iota'\colon B \to M(A)$ that is the identity on A must send b to the double centralizer with $\iota'(b)\,a = \iota'(b)\iota'(a) = \iota'(ba) = ba = L_b(a)$ for $a \in A$, so $\iota'(b) = (L_b, R_b) = \iota(b)$; the embedding is forced by its values on the essential ideal. This proves (2), and taking B maximal shows $M(A)$ is the largest such extension.

Step 3: the unital case. If A is unital with unit 1, then for any double centralizer (L, R) the centralizer relation $aL(b) = R(a)b$ with $a = 1$ gives $L(b) = R(1)b$ for all b, so $L = L_c$ with $c = R(1)$; taking instead $b = 1$ gives $aL(1) = R(a)$, so $R = R_{c'}$ with $c' = L(1)$. The relation with $a = b = 1$ reads $L(1) = R(1)$, so $c = c' \in A$, and $(L, R) = (L_c, R_c)$ is the image of c. Every multiplier is thus left and right multiplication by an element of A, so the embedding $A \to M(A)$ is onto and $M(A) = A$, completing the proof.

The universal property fixes the meaning of $M(A)$: among all the ways to enlarge A to a unital C^*-algebra without letting the new elements act trivially on A, the multiplier algebra is the terminal one, and every essential unitization sits inside it over A. The essentiality condition is what rules out cheap enlargements. One could always form $A \oplus \mathbb{C}$ or $A \oplus B$ for any unital B, but the summand B annihilates A and so contributes nothing to how A is multiplied; essentiality requires that every added element be pinned down by its action on A, and that is precisely the requirement the double centralizer definition builds in. The unitization $\tilde{A}$ of proposition 1.4.5 is an essential extension of a non-unital A (the ideal A is essential because A has an approximate unit: if $a + \lambda 1$ annihilates A, then $(a + \lambda 1)e_\mu = ae_\mu + \lambda e_\mu = 0$ forces $a = -\lambda 1$ in the limit, so $\lambda 1 \in A$, hence $\lambda = 0$ as A is non-unital, and then $a = 0$), and therefore $\tilde{A} \subseteq M(A)$: adjoining a single unit is the smallest essential unitization, adjoining all multipliers the largest. Part (3) records that for a unital algebra there is nothing to add, the two coincide, and every element is its own multiplier.

The two guiding examples show the size of the gap between A and $M(A)$ in the two running cases of the book.

Example 5.3.4: Multipliers of the Compacts and of $C_0(X)$

Two computations identify $M(A)$ in the geometric and the operator model.

1. *The compact operators.* For $A = \mathcal{K}(H)$ the compact operators on a Hilbert space H,

$$M(\mathcal{K}(H)) = B(H)$$

the full algebra of bounded operators. The inclusion $\mathcal{K}(H) \subseteq B(H)$ is faithful and nondegenerate, so proposition 5.3.2 identifies $M(\mathcal{K}(H))$ with the idealizer $\{T \in B(H) \big| T\mathcal{K}(H) \subseteq \mathcal{K}(H),\ \mathcal{K}(H)T \subseteq \mathcal{K}(H)\}$. Every $T \in B(H)$ lies in it, because $\mathcal{K}(H)$ is a two-sided ideal of $B(H)$: the composite of a bounded operator with a compact one is compact. Conversely the idealizer is contained in $B(H)$ by definition, so the idealizer is all of $B(H)$. The essential ideal is exactly the compacts, and the multipliers are all bounded operators, including the noncompact ones (a unitary, a nonzero projection of infinite rank) that act on the compacts but are not themselves compact. This is the operator counterpart of one-point compactification writ large: where $\tilde{\mathcal{K}}(H) = \mathcal{K}(H) + \mathbb{C}1$ adjoins only scalars, $M(\mathcal{K}(H)) = B(H)$ adjoins every bounded operator.

2. *Functions on a locally compact space.* For $A = C_0(X)$ with X locally compact Hausdorff,

$$M(C_0(X)) = C_b(X)$$

the algebra of *bounded* continuous functions on X. A bounded continuous g multiplies $C_0(X)$ into itself, since $gf \in C_0(X)$ whenever $f \in C_0(X)$ (the product is continuous and vanishes at infinity because f does and g is bounded), so $g \mapsto (f \mapsto gf,\ f \mapsto fg)$ realizes $C_b(X)$ inside $M(C_0(X))$; this is injective because $C_0(X)$ separates points and a nonzero g fails to annihilate it. For the reverse inclusion, a multiplier (L, R) is commutative ($L = R$ since $C_0(X)$ is), and L is a $C_0(X)$-module map $C_0(X) \to C_0(X)$; evaluating at each point x where some $f \in C_0(X)$ has $f(x) \neq 0$ defines $g(x) = L(f)(x)/f(x)$, independent of the choice of f by the module relation $L(f_1)f_2 = f_1 L(f_2)$, and g so defined is continuous and bounded by $\|L\|$, with $L(f) = gf$. Thus $M(C_0(X)) = C_b(X)$. By the commutative Gelfand-Naimark theorem $C_b(X) \cong C(\beta X)$ is the algebra of continuous functions on the Stone-Cech compactification, so passing to multipliers of $C_0(X)$ is the maximal compactification of X, whereas the unitization $\widetilde{C_0(X)} = C(X^+)$ of example 1.4.6 is the minimal one. The bounded continuous functions that do not vanish at infinity, such as sin or the constant 1 on $X = \mathbb{R}$, are exactly the multipliers of $C_0(\mathbb{R})$ that are not in $C_0(\mathbb{R})$.

In both cases A is a proper essential ideal of $M(A)$, and the multipliers are the "bounded but not decaying" elements: the noncompact operators that still preserve the compacts, the bounded functions that still preserve the functions vanishing at infinity.

The examples locate $M(A)$ between the two extremes the universal property predicts. It is where the operators one wants to multiply by acquire a home: a non-unital A has no unit, no bounded functional calculus of a noncompact operator, no room for the isometries whose products generate the algebra, yet all of these are multipliers, and $M(A)$ collects them. It is also where an approximate unit converges. An approximate unit (e_λ) of A need not converge in the norm of A, since its limit would be a unit, absent from a non-unital algebra; but it converges strictly to the unit 1 of $M(A)$, in the topology generated by the seminorms $m \mapsto \|ma\|$ and $m \mapsto \|am\|$ for $a \in A$. The unit of $M(A)$ is the limit the approximate unit approaches. Every non-unital construction of the book that needs an identity, the K-theory of chapter 6 through the unitization, the Toeplitz extension, the passage from $C_0(X)$ to its compactifications, factors through the multiplier algebra, the unital C^*-algebra in which the missing identity and every other multiplier already live.

5.4 The Enveloping C^*-Algebra

This section carries out the passage from a Banach $*$-algebra to its universal C^*-algebra. For a Banach $*$-algebra B, the task is to construct a C^*-algebra $C^*(B)$ into which B maps and through which every $*$-representation of B factors, and to prove the universal property that characterizes it. The representation theory of B is thereby transported, without loss, into the representation theory of a C^*-algebra, where the continuous functional calculus, positivity, and the GNS construction of the preceding chapters take over.

The construction rests on a single seminorm on B, the supremum of the operator norms of b over all $*$-representations. The first task is to see that this supremum is finite, which is where the Banach $*$-algebra hypotheses enter.

Throughout, B denotes a Banach $*$-algebra: a complex Banach algebra with an isometric involution $b \mapsto b^*$ (so $\|b^*\| = \|b\|$), equipped with a bounded approximate unit. A $*$-representation of B is a $*$-homomorphism $\pi\colon B \to B(H)$ into the bounded operators on some Hilbert space H; nondegeneracy is imposed exactly as for C^*-algebras (definition 5.1.1).[1]

The following bound is the analytic input. It says that a $*$-representation of a Banach $*$-algebra can never increase norms, generalizing the contractivity of $*$-homomorphisms between C^*-algebras (theorem 1.3.3) to the setting where the source lacks the C^*-identity.

Lemma 5.4.1: Contractivity of $*$-Representations

Let B be a Banach $*$-algebra and let $\pi\colon B \to B(H)$ be a $*$-representation. Then π is contractive: $\|\pi(b)\| \le \|b\|$ for every $b \in B$.

***Proof*:**
Fix $b \in B$ and set $c = b^*b$, a self-adjoint element of B. Then $\pi(c) = \pi(b)^*\pi(b)$ is a self-adjoint operator on H, so its operator norm equals its spectral radius: $\|\pi(c)\| = r(\pi(c))$, where r denotes the spectral radius in $B(H)$.

Step 1: The spectral radius does not grow under π. A $*$-homomorphism sends invertible elements

[1] The seminorm construction and the contractivity bound below use only submultiplicativity and the isometric involution; the approximate unit is what makes the nondegeneracy convention effective in the non-unital setting and drives the $L^1(G)$ representation-integration correspondence recorded at the end of the section.

of the unitization to invertible elements, hence cannot enlarge the spectrum: for any $x \in B$, the spectrum of $\pi(x)$ in $B(H)$ is contained in the spectrum of x computed in the unitization $\tilde{B}$ (together with $\{0\}$). Consequently $r(\pi(x)) \leq r(x)$ for all x, where $r(x)$ is the spectral radius of x in $\tilde{B}$. The spectral radius formula gives $r(x) = \lim_n \|x^n\|^{1/n} \leq \|x\|$, the last inequality because $\|x^n\| \leq \|x\|^n$ in the Banach algebra $\tilde{B}$.

Step 2: Assemble the estimate. Applying Step 1 to $x = c$,

$$\|\pi(b)\|^2 = \|\pi(b)^*\pi(b)\| = \|\pi(c)\| = r(\pi(c)) \leq r(c) \leq \|c\| = \|b^*b\| \leq \|b^*\|\,\|b\| = \|b\|^2$$

using the C^*-identity in $B(H)$ for the first equality, submultiplicativity of the norm on B for the penultimate step, and the isometry of the involution for the last. Taking square roots gives $\|\pi(b)\| \leq \|b\|$, as we sought to show.

The bound is what makes the enveloping construction possible: every $*$-representation of B is controlled by the ambient Banach norm, so the supremum of $\|\pi(b)\|$ over all π cannot exceed $\|b\|$. This supremum is the seminorm through which $C^*(B)$ is built.

Definition 5.4.2: Enveloping C^*-Algebra

Let B be a Banach $*$-algebra. For $b \in B$ define the *universal norm*

$$\|b\|_u = \sup\{\,\|\pi(b)\| : \pi \text{ a } *\text{-representation of } B \text{ on a Hilbert space}\,\}$$

By lemma 5.4.1 the supremum is finite, with $\|b\|_u \leq \|b\|$. The function $\|\cdot\|_u$ is a C^*-seminorm on B: it satisfies the submultiplicative, involutive, and triangle inequalities together with the C^*-identity $\|b^*b\|_u = \|b\|_u^2$. Its null space $N = \{b \in B : \|b\|_u = 0\}$ is a closed two-sided $*$-ideal, and $\|\cdot\|_u$ descends to a C^*-norm on the quotient B/N. The *enveloping C^*-algebra* $C^*(B)$ is the completion of B/N in this norm, and the composite

$$\iota\colon B \to B/N \to C^*(B)$$

is the canonical $*$-homomorphism.

Two points in the definition require verification: that $\|\cdot\|_u$ is a C^*-seminorm, and that N is a closed $*$-ideal so that the quotient inherits a C^*-norm. Both follow by pushing the corresponding facts about operator norms on $B(H)$ back through the family of representations.

***Proof*:**
Step 1: $\|\cdot\|_u$ is a seminorm satisfying the algebraic inequalities. Each π is linear and multiplicative with $\pi(b^*) = \pi(b)^*$, and the operator norm on $B(H)$ is a submultiplicative norm invariant under the adjoint. Taking suprema over π preserves these inequalities: for $a, b \in B$ and $\lambda \in \mathbb{C}$,

$$\|a+b\|_u \leq \|a\|_u + \|b\|_u, \qquad \|\lambda b\|_u = |\lambda|\,\|b\|_u, \qquad \|ab\|_u \leq \|a\|_u\,\|b\|_u, \qquad \|b^*\|_u = \|b\|_u$$

the third because $\|\pi(ab)\| = \|\pi(a)\pi(b)\| \leq \|\pi(a)\|\,\|\pi(b)\| \leq \|a\|_u\,\|b\|_u$ for every π, and the fourth because the adjoint is an isometry of $B(H)$.

Step 2: The C^*-identity. Fix $b \in B$. For each π the C^*-identity in $B(H)$ gives $\|\pi(b)\|^2 = \|\pi(b)^*\pi(b)\| = \|\pi(b^*b)\| \leq \|b^*b\|_u$. Taking the supremum over π of the left side yields $\|b\|_u^2 \leq \|b^*b\|_u$. Conversely $\|b^*b\|_u \leq \|b^*\|_u\,\|b\|_u = \|b\|_u^2$ by submultiplicativity and Step 1, so $\|b^*b\|_u = \|b\|_u^2$.

Step 3: N is a closed $*$-ideal and the quotient carries a C^*-norm. The set $N = \{b : \|b\|_u = 0\}$ is the intersection $\bigcap_\pi \ker\pi$ over all $*$-representations π, since $\|b\|_u = 0$ holds exactly when $\pi(b) = 0$ for every π. Each $\ker\pi$ is a closed two-sided ideal closed under the involution, so their intersection N is as well. Thus $\|\cdot\|_u$ induces a well-defined norm on B/N, and by Steps 1 and 2 this norm is submultiplicative, involutive, and satisfies the C^*-identity. A norm with the C^*-identity on a $*$-algebra is a C^*-norm, so the completion $C^*(B)$ is a C^*-algebra, completing the proof.

The passage from B to $C^*(B)$ discards exactly the elements that no $*$-representation can see, then completes what remains. It loses nothing about the Hilbert-space representation theory of B: an element killed by ι is killed by every representation, and a completion adds no new representations. This is made precise by the universal property below, and it is what permits the transfer of the group case to C^*-algebra language.

Before the general theorem, a concrete instance grounds the construction where the answer is already known: when B is itself a C^*-algebra, forming its enveloping C^*-algebra changes nothing.

Example 5.4.3: The Envelope of a C^*-Algebra

Let $B = A$ be a C^*-algebra, regarded as a Banach $*$-algebra. For every $*$-representation $\pi\colon A \to B(H)$ one has $\|\pi(a)\| \le \|a\|$ by contractivity of $*$-homomorphisms (theorem 1.3.3), so $\|a\|_u \le \|a\|$. For the reverse inequality, the universal representation of the Gelfand-Naimark theorem (theorem 4.3.3) is a faithful, hence isometric, $*$-representation $\pi_u\colon A \to B(H_u)$; it contributes $\|\pi_u(a)\| = \|a\|$ to the supremum, giving $\|a\|_u \ge \|a\|$. Therefore $\|a\|_u = \|a\|$, the null ideal N is zero, and A is already complete in $\|\cdot\|_u$. The canonical map $\iota\colon A \to C^*(A)$ is an isometric $*$-isomorphism. The enveloping construction is thus idempotent on C^*-algebras: it does real work only when B fails the C^*-identity.

The example fixes the meaning of the construction: $C^*(B)$ is the C^*-algebra that has the same $*$-representations as B and no more, and when B already is such an algebra the two coincide. Turning this into a defining property is the content of the universal property, which characterizes $C^*(B)$ up to isomorphism by a factorization condition rather than by the explicit completion.

Theorem 5.4.4: Universal Property of the Enveloping C^*-Algebra

Let B be a Banach $*$-algebra with enveloping C^*-algebra $C^*(B)$ and canonical $*$-homomorphism $\iota\colon B \to C^*(B)$. For every $*$-representation $\pi\colon B \to B(H)$ there is a unique $*$-homomorphism $\tilde\pi\colon C^*(B) \to B(H)$ of C^*-algebras with

$$\pi = \tilde\pi \circ \iota$$

Assigning $\pi \mapsto \tilde\pi$ and $\rho \mapsto \rho\circ\iota$ sets up a bijection between the $*$-representations of B and the $*$-representations of $C^*(B)$, one that preserves subrepresentations, direct sums, unitary equivalence, and irreducibility. The pair $(C^*(B), \iota)$ is determined up to a unique isomorphism by this property.

***Proof*:**
Step 1: Construction of $\tilde\pi$ on the dense subalgebra. Let $\pi\colon B \to B(H)$ be a $*$-representation. By the definition of the universal norm, $\|\pi(b)\| \le \|b\|_u$ for every $b \in B$; in particular π vanishes on the null ideal N, so π factors through a $*$-homomorphism $\overline{\pi}\colon B/N \to B(H)$ satisfying

$\|\overline{\pi}(b+N)\| \le \|b+N\|_u$. The image $\iota(B) = B/N$ is dense in $C^*(B)$, and $\overline{\pi}$ is contractive for the C^*-norm on B/N, hence uniformly continuous. A uniformly continuous map on a dense subset of a complete metric space extends uniquely to a continuous map on the completion, so $\overline{\pi}$ extends uniquely to a continuous linear map $\tilde{\pi}\colon C^*(B) \to B(H)$.

Step 2: $\tilde{\pi}$ is a $*$-homomorphism. Multiplication, the involution, and the adjoint are continuous, and $\tilde{\pi}$ agrees with the $*$-homomorphism $\overline{\pi}$ on the dense subalgebra B/N. For $x, y \in C^*(B)$ choose sequences $x_n, y_n \in B/N$ with $x_n \to x$ and $y_n \to y$. Then

$$\tilde{\pi}(xy) = \lim_n \tilde{\pi}(x_n y_n) = \lim_n \overline{\pi}(x_n)\,\overline{\pi}(y_n) = \tilde{\pi}(x)\,\tilde{\pi}(y)$$

and likewise $\tilde{\pi}(x^*) = \lim_n \overline{\pi}(x_n^*) = \lim_n \overline{\pi}(x_n)^* = \tilde{\pi}(x)^*$, using continuity of the algebraic operations. Thus $\tilde{\pi}$ is a $*$-homomorphism, and $\tilde{\pi} \circ \iota = \overline{\pi} \circ (\,\cdot + N) = \pi$ by construction.

Step 3: Uniqueness of $\tilde{\pi}$. Any $*$-homomorphism $\sigma\colon C^*(B) \to B(H)$ with $\sigma \circ \iota = \pi$ agrees with $\tilde{\pi}$ on $\iota(B) = B/N$. A $*$-homomorphism of C^*-algebras is contractive (theorem 1.3.3), hence continuous, so σ and $\tilde{\pi}$ are two continuous maps agreeing on a dense set; they coincide. This proves the factorization and its uniqueness.

Step 4: The bijection $\mathrm{Rep}(B) \cong \mathrm{Rep}(C^*(B))$. The assignment $\Phi\colon \pi \mapsto \tilde{\pi}$ maps $*$-representations of B to $*$-representations of $C^*(B)$. In the reverse direction, define $\Psi\colon \rho \mapsto \rho \circ \iota$, sending a $*$-representation ρ of $C^*(B)$ to the $*$-representation $\rho \circ \iota$ of B. These are mutually inverse. On the one hand $\Psi(\Phi(\pi)) = \tilde{\pi} \circ \iota = \pi$ by Step 2. On the other hand, given ρ, the composite $\rho \circ \iota$ is a $*$-representation of B whose unique extension is ρ itself, since ρ is a $*$-homomorphism of C^*-algebras satisfying $\rho \circ \iota = \Psi(\rho)$ and Step 3 makes such an extension unique; hence $\Phi(\Psi(\rho)) = \rho$. Both Φ and Ψ respect the standing nondegeneracy convention (definition 5.1.1): since $\iota(B)$ is dense in $C^*(B)$, one has $\overline{\tilde{\pi}(C^*(B))H} = \overline{\pi(B)H}$, so $\tilde{\pi}$ is nondegenerate exactly when π is. So Φ and Ψ are inverse bijections between the nondegenerate $*$-representations of B and those of $C^*(B)$.

Step 5: The bijection preserves the structure of representations. Because ι has dense range, a closed subspace $K \subseteq H$ is invariant under $\pi(B)$ if and only if it is invariant under $\tilde{\pi}(C^*(B))$: invariance under the dense subalgebra $\iota(B)$ forces invariance under its closure. Thus subrepresentations correspond, and a representation is irreducible for B exactly when its extension is irreducible for $C^*(B)$. A direct sum $\pi = \bigoplus_i \pi_i$ extends to $\tilde{\pi} = \bigoplus_i \tilde{\pi}_i$, since both sides agree on $\iota(B)$ and are continuous. Finally, a unitary $U\colon H_1 \to H_2$ intertwines π_1 and π_2 if and only if it intertwines $\tilde{\pi}_1$ and $\tilde{\pi}_2$, again because the intertwining relation $U\tilde{\pi}_1(x) = \tilde{\pi}_2(x)U$ holds on all of $C^*(B)$ as soon as it holds on the dense subalgebra $\iota(B)$. Hence the correspondence preserves unitary equivalence.

Step 6: Uniqueness of the pair. Suppose (C, j) is a C^*-algebra with a $*$-homomorphism $j\colon B \to C$ having dense range and satisfying the same universal property: every $*$-representation of B factors uniquely through j. The universal property, as stated, factors representations on a Hilbert space, so the two comparison maps must be built through faithful representations rather than applied directly to ι or j. Fix faithful $*$-representations $\rho\colon C \to B(H_\rho)$ and $\sigma\colon C^*(B) \to B(H_\sigma)$; both exist by the Gelfand-Naimark theorem (theorem 4.3.3), and being injective $*$-homomorphisms of C^*-algebras they are isometric onto their (closed) images $\rho(C)$ and $\sigma(C^*(B))$.

Construction of $F\colon C^*(B) \to C$. The composite $\rho \circ j\colon B \to B(H_\rho)$ is a $*$-representation of B, so by the universal property of $(C^*(B), \iota)$ it factors as $\theta \circ \iota = \rho \circ j$ for a unique $*$-homomorphism $\theta\colon C^*(B) \to B(H_\rho)$. On the dense subalgebra one has $\theta(\iota(B)) = \rho(j(B)) \subseteq \rho(C)$, and $\rho(C)$ is a

C^*-subalgebra of $B(H_\rho)$, hence closed; since θ is continuous and $\iota(B)$ is dense in $C^*(B)$,

$$\theta(C^*(B)) = \theta\big(\overline{\iota(B)}\big) \subseteq \overline{\theta(\iota(B))} = \overline{\rho(j(B))} \subseteq \rho(C)$$

so θ lands in the image of the faithful ρ. Composing with the inverse $\rho^{-1}\colon \rho(C) \to C$, which is a $*$-isomorphism, gives a $*$-homomorphism $F = \rho^{-1} \circ \theta\colon C^*(B) \to C$ with $F \circ \iota = \rho^{-1} \circ \rho \circ j = j$.

Construction of $G\colon C \to C^*(B)$. Symmetrically, $\sigma \circ \iota\colon B \to B(H_\sigma)$ is a $*$-representation of B, so by the universal property of (C, j) it factors as $\psi \circ j = \sigma \circ \iota$ for a unique $*$-homomorphism $\psi\colon C \to B(H_\sigma)$. The same density-and-closure descent gives $\psi(C) = \overline{\psi(j(B))} = \overline{\sigma(\iota(B))} \subseteq \sigma(C^*(B))$, so ψ lands in the image of the faithful σ, and $G = \sigma^{-1} \circ \psi\colon C \to C^*(B)$ is a $*$-homomorphism with $G \circ j = \iota$.

The composites are identities. The map $G \circ F\colon C^*(B) \to C^*(B)$ satisfies $(G \circ F) \circ \iota = G \circ j = \iota$, so $\sigma \circ (G \circ F)$ and σ are two $*$-homomorphisms $C^*(B) \to B(H_\sigma)$ that both extend the $*$-representation $\sigma \circ \iota$ of B along ι. By the uniqueness clause of Step 3 they coincide, $\sigma \circ (G \circ F) = \sigma$; since σ is injective, $G \circ F = \mathrm{id}_{C^*(B)}$. The symmetric argument, extending $\rho \circ j$ along j and using the uniqueness clause of (C, j), gives $F \circ G = \mathrm{id}_C$. Thus F and G are mutually inverse $*$-isomorphisms with $F \circ \iota = j$. The pair is therefore determined up to a unique isomorphism, completing the proof.

The universal property is the working characterization of $C^*(B)$: it is the C^*-algebra whose representations are the representations of B, with the correspondence respecting every structural feature that the classification of representations uses. Studying the unitary representations of B on Hilbert spaces is the same as studying the representations of the C^*-algebra $C^*(B)$, and the latter is where Schur's lemma, the pure-state correspondence, and the GNS construction of the preceding sections apply. Irreducible representations of B match irreducible representations of $C^*(B)$, so the primitive ideal space and spectrum of the next section describe B as well.

A second consequence is a recognition principle. The universal property determines $(C^*(B), \iota)$ uniquely, so any C^*-algebra C equipped with a dense-range $*$-homomorphism $j\colon B \to C$ through which every $*$-representation of B factors uniquely is canonically the enveloping C^*-algebra. In practice one rarely computes the universal norm by evaluating the supremum; instead one exhibits a candidate C and checks the factorization, as the group case below illustrates.

The construction acquires its name and its first substantial application from the theory of locally compact groups. For a locally compact group G with a fixed left Haar measure and modular function Δ, the convolution algebra $L^1(G)$ is a Banach $*$-algebra under the L^1 norm, with involution $f^*(x) = \overline{f(x^{-1})}\,\Delta(x^{-1})$; its L^1 norm does not satisfy the C^*-identity, so $L^1(G)$ is not itself a C^*-algebra. Its $*$-representations on Hilbert spaces are in bijection with the strongly continuous unitary representations of G: a unitary representation U of G integrates to the representation $f \mapsto \int_G f(x)\, U_x\, dx$ of $L^1(G)$, and every $*$-representation of $L^1(G)$ arises this way. The *group C^*-algebra* $C^*(G)$ is the enveloping C^*-algebra $C^*(L^1(G))$ of this convolution algebra. By the universal property, the unitary representations of G are exactly the representations of the C^*-algebra $C^*(G)$, so the analytic study of how G acts on Hilbert spaces becomes the study of a single C^*-algebra and its representation theory. This identification is the point of departure for the harmonic analysis of locally compact groups, where the structure of $\hat{G}$, the reduced group C^*-algebra and its comparison with $C^*(G)$, and the decomposition of the regular representation are developed; that theory is the subject of the companion volume on unitary representation theory.

5.5 Primitive Ideals and the Spectrum

Irreducible representations are the indecomposable pieces from which every representation is assembled, and two invariants record them. The first is the kernel of an irreducible representation, a two-sided ideal that forgets everything about the representation except which elements it annihilates; the second is the collection of the representations themselves, taken up to unitary equivalence. This section names both, topologizes them, and reads off what they are in the commutative case, where the algebra is $C_0(X)$ and both invariants recover the space X. The topology turns the set of irreducible representations into a space, so that a noncommutative C^*-algebra acquires an underlying "noncommutative space" even where Gelfand-Naimark duality does not directly apply.

An irreducible representation, having no proper nonzero closed invariant subspace (definition 5.1.3), is as far from decomposable as a representation can be, and its kernel is correspondingly distinguished among ideals.

Definition 5.5.1: Primitive Ideal

Let A be a C^*-algebra. A closed two-sided ideal $I \subseteq A$ is a *primitive ideal* if $I = \ker \pi$ for some irreducible representation $\pi \colon A \to B(H)$ (definition 5.1.3). The set of all primitive ideals of A is written $\mathrm{Prim}(A)$.

The kernel of any representation is a closed two-sided ideal: it is a subspace closed under left and right multiplication because π is an algebra homomorphism, self-adjoint because $\pi(a^*) = \pi(a)^*$, and closed because π is continuous, being contractive as a $*$-homomorphism (theorem 1.3.3). Requiring π irreducible cuts this down to the primitive ideals. The definition does not ask that I itself be an interesting ideal in isolation; it asks only that I arise as such a kernel. A single primitive ideal can be the kernel of many inequivalent irreducible representations at once, which is exactly why the two invariants of this section, the ideal and the representation, are different, and why the map between them recorded below need not be injective.

The commutative case fixes the picture. For $A = C_0(X)$ every irreducible representation is one-dimensional, given by evaluation at a point, and its kernel is the ideal of functions vanishing there.

Example 5.5.2: Primitive Ideals of $C_0(X)$

Let X be a locally compact Hausdorff space and $A = C_0(X)$. For a point $x \in X$, evaluation $\mathrm{ev}_x \colon A \to \mathbb{C}$, $f \mapsto f(x)$, is a one-dimensional representation on $H = \mathbb{C}$; it is irreducible because $\mathbb{C}$ has no proper nonzero subspace at all. Its kernel is

$$\mathfrak{m}_x = \{f \in C_0(X) \,\big|\, f(x) = 0\}$$

the maximal ideal of functions vanishing at x. These exhaust the primitive ideals. Indeed, a commutative C^*-algebra has abelian image under any representation, so an irreducible representation has commutative commutant containing the whole image; by Schur's lemma (theorem 5.2.1) the commutant is $\mathbb{C}1$, which forces H to be one-dimensional, and the representation is then a character, that is, evaluation at some $x \in X$ by Gelfand-Naimark duality (theorem 4.3.3). Hence $\mathrm{Prim}(C_0(X)) = \{\mathfrak{m}_x : x \in X\}$, and $x \mapsto \mathfrak{m}_x$ is a bijection $X \to \mathrm{Prim}(C_0(X))$, since distinct points are separated by some $f \in C_0(X)$. The primitive ideals of $C_0(X)$ are the points of X.

The second invariant keeps the representation rather than passing to its kernel. Two irreducible representations that differ only by a unitary change of the underlying Hilbert space carry the same information, so the natural object is the set of equivalence classes.

Definition 5.5.3: Spectrum and the Hull-Kernel Topology

Let A be a C^*-algebra. Two representations $\pi_1 \colon A \to B(H_1)$ and $\pi_2 \colon A \to B(H_2)$ are *unitarily equivalent* if there is a unitary $U \colon H_1 \to H_2$ with $U\pi_1(a) = \pi_2(a)U$ for all $a \in A$. The *spectrum* $\hat{A}$ is the set of unitary-equivalence classes of irreducible representations of A. For a subset $S \subseteq \mathrm{Prim}(A)$, the *kernel* of S is the ideal $k(S) = \bigcap_{I \in S} I$, and for an ideal J the *hull* of J is $h(J) = \{I \in \mathrm{Prim}(A) \big| J \subseteq I\}$. The operator $S \mapsto h(k(S))$ defines the *hull-kernel* (or *Jacobson*) *topology* on $\mathrm{Prim}(A)$: a set is closed if it equals the hull of its kernel. The spectrum $\hat{A}$ carries the coarsest topology making the kernel map $\hat{A} \to \mathrm{Prim}(A)$ continuous.

The kernel map $\hat{A} \to \mathrm{Prim}(A)$, sending the class of an irreducible representation π to the primitive ideal $\ker \pi$, is the bridge between the two invariants, and it is well defined because unitarily equivalent representations have the same kernel: if $U\pi_1(a) = \pi_2(a)U$ with U unitary then $\pi_1(a) = 0$ exactly when $\pi_2(a) = 0$. It is surjective by the very definition of a primitive ideal (definition 5.5.1), every primitive ideal being some $\ker \pi$. It need not be injective: distinct irreducible representations can share a kernel, and $\mathrm{Prim}(A)$ is the quotient of $\hat{A}$ that identifies them. For the commutative algebra $C_0(X)$ the two coincide, since there each primitive ideal $\mathfrak{m}_x$ comes from a single character ev_x up to equivalence, so both $\hat{A}$ and $\mathrm{Prim}(A)$ are X (example 5.5.2); the map is a bijection, and it is a homeomorphism (theorem 5.5.5).

That $S \mapsto h(k(S))$ is a genuine closure operator, so that the hull-kernel topology is a topology, is the content of the next observation. The verification is the same as for the Zariski topology on the prime spectrum of a commutative ring, transported to primitive ideals.

Proposition 5.5.4: The Hull-Kernel Operator is a Closure

Let A be a C^*-algebra. The operator $c(S) = h(k(S))$ on subsets $S \subseteq \mathrm{Prim}(A)$ satisfies the Kuratowski axioms: $c(\varnothing) = \varnothing$, $S \subseteq c(S)$, $c(c(S)) = c(S)$, and $c(S_1 \cup S_2) = c(S_1) \cup c(S_2)$. Hence the sets S with $c(S) = S$ are the closed sets of a topology on $\mathrm{Prim}(A)$.

***Proof*:**
Write $k(S) = \bigcap_{I \in S} I$ and $h(J) = \{I : J \subseteq I\}$; by convention $k(\varnothing) = A$.

Empty set. No primitive ideal contains all of A, since a primitive ideal is the kernel of an irreducible representation and such a representation is nonzero, so $h(A) = \varnothing$, giving $c(\varnothing) = \varnothing$.

Extensivity. If $I \in S$ then $k(S) \subseteq I$ by definition of the intersection, so $I \in h(k(S))$; thus $S \subseteq c(S)$.

Monotonicity and idempotence. If $S_1 \subseteq S_2$ then $k(S_2) \subseteq k(S_1)$, whence $h(k(S_1)) \subseteq h(k(S_2))$, so c is monotone. Applying k to $S \subseteq c(S) = h(k(S))$ gives $k(h(k(S))) \subseteq k(S)$, because k reverses inclusions and every ideal in $h(k(S))$ contains $k(S)$; the reverse inclusion $k(S) \subseteq k(h(k(S)))$ holds since $k(S) \subseteq I$ for every $I \in h(k(S))$. Thus $k(c(S)) = k(S)$, and applying h gives $c(c(S)) = h(k(c(S))) = h(k(S)) = c(S)$.

Finite unions. From $S_i \subseteq S_1 \cup S_2$ and monotonicity, $c(S_1) \cup c(S_2) \subseteq c(S_1 \cup S_2)$. For the reverse,

$k(S_1 \cup S_2) = k(S_1) \cap k(S_2)$, so it suffices to show that if a primitive ideal I contains $k(S_1) \cap k(S_2)$ then it contains $k(S_1)$ or $k(S_2)$. This is the statement that I is *prime*: for two-sided ideals J_1, J_2, if $J_1 J_2 \subseteq I$ then $J_1 \subseteq I$ or $J_2 \subseteq I$. Grant this for the moment. Since $k(S_1)$ and $k(S_2)$ are ideals with $k(S_1)\,k(S_2) \subseteq k(S_1) \cap k(S_2) \subseteq I$, primeness gives $k(S_1) \subseteq I$ or $k(S_2) \subseteq I$, that is $I \in h(k(S_1))$ or $I \in h(k(S_2))$; hence $c(S_1 \cup S_2) \subseteq c(S_1) \cup c(S_2)$.

It remains to see that a primitive ideal is prime. Let $I = \ker \pi$ for an irreducible π, and suppose $J_1 J_2 \subseteq I$ with $J_1 \nsubseteq I$; we show $J_2 \subseteq I$. This reduces to the element form: if $aAb \subseteq I$ then $a \in I$ or $b \in I$, applied to $a \in J_1 \setminus I$ and each $b \in J_2$ (for then $aAb \subseteq J_1 A J_2 \subseteq J_1 J_2 \subseteq I$). So suppose $aAb \subseteq I$ with $a \notin I$ and $b \notin I$, that is $\pi(a) \neq 0$ and $\pi(b) \neq 0$ while $\pi(a)\pi(c)\pi(b) = 0$ for every $c \in A$. Pick ξ with $\pi(b)\xi \neq 0$. Because π is irreducible (definition 5.1.3), the nonzero vector $\pi(b)\xi$ is cyclic: the closure of $\{\pi(c)\pi(b)\xi : c \in A\}$ is a closed $\pi(A)$-invariant subspace, nonzero by nondegeneracy, hence all of H. But $\pi(a)$ annihilates this dense set, so $\pi(a) = 0$ by continuity, contradicting $\pi(a) \neq 0$. Hence $a \in I$ or $b \in I$, and I is prime.

The four Kuratowski axioms hold, so the fixed sets of c are the closed sets of a topology, completing the proof.

The topology so defined is almost never Hausdorff. A closed point of $\mathrm{Prim}(A)$ is a maximal primitive ideal, and two distinct primitive ideals with one contained in the other cannot be separated, since any closed set containing the smaller ideal contains the larger (equivalently, every open set containing the larger contains the smaller). What survives is the weakest separation axiom, together with local compactness, and this is the general structure theorem for the primitive ideal space. Its proof develops the theory of ideals in C^*-algebras at more length than this closing section warrants, so it is stated and cited rather than proved here.

Theorem 5.5.5: The Primitive Ideal Space

For every C^*-algebra A, the space $\mathrm{Prim}(A)$ with the hull-kernel topology is a locally compact T_0 space. It is compact when A is unital. When $A = C_0(X)$ is commutative, the bijection $x \mapsto \mathfrak{m}_x$ of example 5.5.2 is a homeomorphism $X \cong \mathrm{Prim}(C_0(X))$.

***Proof Key ideas*:**
The T_0 separation is immediate from the definition: two primitive ideals with the same closure have the same hull-kernel closure $c(\{I\}) = h(I)$, hence contain the same primitive ideals above them and in particular each other, so are equal; thus distinct points have distinct closures, which is the T_0 axiom. Local compactness, and compactness in the unital case, rest on identifying the open sets of $\mathrm{Prim}(A)$ with the closed two-sided ideals of A (an open set is the complement of a hull $h(J)$, and $J \mapsto h(J)$ is an order-reversing bijection between closed ideals and closed subsets), under which the required covering properties translate into the existence of approximate units in C^*-ideals; the full argument is in the standard treatments, for instance Dixmier or Pedersen. The commutative statement is the computation of example 5.5.2 together with the observation that $f \mapsto \{x : f(x) \neq 0\}$ matches basic open sets of X with complements of hulls, so the bijection and its inverse are both continuous.

The primitive ideal space is the invariant through which noncommutative topology proceeds when the algebra is not commutative: $\mathrm{Prim}(A)$ is a bona fide locally compact space attached to any A, and for $A = C_0(X)$ it is X, so it extends the Gelfand-Naimark space (theorem 4.3.3) to all C^*-algebras, though the result need not be Hausdorff. The failure of Hausdorffness is not a defect but data:

it records that inequivalent irreducible representations of a noncommutative algebra can be limits of one another in a way points of an ordinary space cannot. The spectrum $\hat{A}$ refines $\mathrm{Prim}(A)$ by remembering the representation and not only its kernel, and the two agree exactly when the algebra is of "type I", the regime in which the representation theory is governed by its primitive ideals; for general A the fibres of $\hat{A} \to \mathrm{Prim}(A)$ carry the rest of the information.

The invariants of this section organize the irreducible representations of A. The complementary construction organizes all of them at once into a single von Neumann algebra, obtained by completing A in a weaker topology than the norm. Taking the direct sum $\pi_u = \bigoplus_{\varphi \in S(A)} \pi_\varphi$ of the GNS representations (theorem 4.2.3) over all states φ in the state space $S(A)$ gives the *universal representation*, faithful because the states separate points; its weak-operator closure is the double commutant $\pi_u(A)''$, the *enveloping von Neumann algebra* of A. This algebra is canonically the second dual A^{**} as a Banach space, with a multiplication extending that of A, and it is the completion in which the operations excluded from a C^*-algebra become available: spectral projections of a self-adjoint element, weak limits of an approximate unit, and suprema of bounded increasing nets of positive elements all exist in $\pi_u(A)''$ though not in A. It is the entry point to the theory of von Neumann algebras, the "noncommutative measure theory" standing to the "noncommutative topology" of this book as measure theory stands to topology, and that theory is not developed further here.

Part II

K-Theory of C^*-Algebras

6

The Group K_0

The group $K_0(A)$ is the Grothendieck completion of a semigroup assembled from the projections in the matrix algebras over A; this chapter builds that semigroup and then K_0 itself.

6.1 Projections and the Semigroup $V(A)$

First, we shall say two *elements* $a, b \in X$ are homotopic if there is a continuous path $[0,1] \to X$ where $\gamma(0) = a$ and $\gamma(1) = b$ (this comes from the idea that X can be embedded in a natural hilbert space $L^2(X)$)

Definition 6.1.1: Projections And Equivalences

Let A be a C^*-algebra. The set of all projection of A shall be denoted $\mathcal{P}(A)$ (set of all elements such that $p = p^2 = p^*$). We shall define two equivalence relations on $\mathcal{P}(A)$:

1. (*von Neumann equivalence*) $p \sim q$ if there exists $\in A$ such that $p = v^*v$ anfd $q = vv^*$
2. (*unitary equivalence*) $p \sim_u q$ if there exists a unitary element $u \in \mathcal{U}(\tilde{A})$ where $q = upu^*$.

Definition 6.1.2: Set of projections

Let

$$\mathcal{P}(A) = \mathcal{P}(M_n(A)) \qquad \mathcal{P}_\infty(A) = \bigcup_n^\infty \mathcal{P}_n(A)$$

Then define $\sim_0$ on $\mathcal{P}(A)$ as follows

$$p \sim_0 q \qquad \iff \qquad v \in M_{m,n}(A), p = v^*v \text{ and } q = vv^*$$

now define $\oplus$ on $\mathcal{P}_\infty(A)$ via:

$$p \oplus q = \operatorname{diag}(p,q) = \begin{pmatrix} p & 0 \\ 0 & q \end{pmatrix}$$

Then it is not hard to see this forms a semi-group structure

Proposition 6.1.3: Semigroup Of Projections

Let p, q, r, p', q' be projection in $\mathcal{P}_\infty(A)$ for some C^*-algebra A. Then:

1. $p \sim_0 p \oplus 0_n$ for every n
2. $p \sim_0 p'$ and $q \sim_0 q'$, then $p \oplus q \sim_0 p' \oplus q'$
3. $p \oplus q \sim_0 q \oplus p$
4. if p, q are projections in $\mathcal{P}_n(A)$ such that $pq = 0$, then $p+q$ i a projection and $p+q \sim_0 p \oplus q$
5. $(p \oplus q) \oplus r = p \oplus (q \oplus r)$

***Proof*:**
all relatively simple, will leave for now

Definition 6.1.4: The Semigroup for C^*-algebras

Let $(\mathcal{P}(A), \sim_0, \oplus)$ be as in the above. Then

$$\mathcal{D}(A) = \mathcal{P}_\infty(A)/\sim_0$$

forms a semi-group where

$$[p]_\mathcal{D} + [q]_\mathcal{D} = [p \oplus q]_\mathcal{D}$$

Naturally, by the Grothendieck construction, this abelian group can be turned into a group. We shall use this in section 6.2 to define the K_0 group.

6.2 K_0 of a C^*-Algebra

The topological K-theory of [10] attaches to a compact Hausdorff space X the Grothendieck group of its complex vector bundles. The semigroup machinery of definitions 6.1.2 and 6.1.4 has already packaged the algebraic form of this construction: projections over a C^*-algebra play the role that vector bundles play over a space, and Murray-von Neumann equivalence (definition 6.1.1) plays the role of bundle isomorphism. This section completes the passage from the semigroup $\mathcal{D}(A)$ to a genuine abelian group, the K_0-group of A, and develops its three structural properties: functoriality together with homotopy invariance, stability under passing to matrix algebras and to the stabilization $A \otimes \mathcal{K}$, and the half-exact behaviour on short exact sequences of C^*-algebras. The payoff is a precise bridge between operator algebras and topology: for X compact Hausdorff the group $K_0(C(X))$ is exactly the topological $K^0(X)$ of [10]. This is the operator-algebraic incarnation of the Serre-Swan correspondence between vector bundles and finitely generated projective modules, and it is the reason the same letter K names two theories that at first sight live in different worlds.

The starting point is the following definition.

Definition 6.2.1: K_0 of a Unital C^*-Algebra

Let A be a unital C^*-algebra and let $\mathcal{D}(A) = \mathbb{C}\mathrm{P}_\infty(A)/\sim_0$ be the abelian semigroup of definition 6.1.4, whose elements are the Murray-von Neumann equivalence classes $[p]_\mathcal{D}$ of projections over A under the block-diagonal sum $\oplus$. The K_0*-group* of A is the Grothendieck group

$$K_0(A) := G(\mathcal{D}(A))$$

the universal abelian group receiving a semigroup homomorphism $\gamma\colon \mathcal{D}(A) \to K_0(A)$.

The definition packages $K_0(A)$, for unital A, as the Grothendieck group of the semigroup $\mathcal{D}(A) = \mathbb{C}\mathrm{P}_\infty(A)/\sim_0$ of definition 6.1.4, whose elements are Murray-von Neumann equivalence classes of projections over A and whose addition is the block-diagonal sum $\oplus$. Recall (definition 6.1.1) that a projection is a self-adjoint idempotent $p = p^2 = p^*$, and that $p \sim_0 q$ when there is a rectangular partial isometry v over A with $p = v^*v$ and $q = vv^*$. The Grothendieck construction formally adjoins additive inverses to the classes $[p]_\mathcal{D}$, so that a general element of $K_0(A)$ is a formal difference of two classes of projections. The first task is to make this informal description into a usable normal form.

Proposition 6.2.2: Standard Picture of K_0

Let A be a unital C^*-algebra. Then every element of $K_0(A)$ has the form

$$[p] - [q]$$

for projections $p, q \in \mathbb{C}\mathrm{P}_\infty(A)$, where $[p]$ denotes the image in $K_0(A)$ of the class $[p]_\mathcal{D} \in \mathcal{D}(A)$. Moreover:

1. For $p, q \in \mathbb{C}\mathrm{P}_\infty(A)$ one has $[p] = [q]$ in $K_0(A)$ if and only if there is a projection $r \in \mathbb{C}\mathrm{P}_\infty(A)$ with $p \oplus r \sim_0 q \oplus r$.
2. For any two projections $p, q \in \mathbb{C}\mathrm{P}_\infty(A)$ the relation $[p \oplus q] = [p] + [q]$ holds in $K_0(A)$.

This is the working description of K_0 that every later computation rests on. The content of part

(1) is that equality in $K_0(A)$ is *stable* equivalence: p and q need not be equivalent on the nose, only after adding a common projection r. This is the algebraic analogue of stable isomorphism for vector bundles ([10, Reduced K-Theory Classifies Stable Equivalence]), where two bundles agree in K-theory once a common trivial bundle is added to each. The cancellation that is hidden inside the Grothendieck construction is precisely this padding by r.

***Proof*:**
Write $S = \mathcal{D}(A)$ for the abelian semigroup of definition 6.1.4, with addition induced by $\oplus$. By definition 6.2.1 the group $K_0(A)$ is the Grothendieck group $G(S)$, realized as $(S \times S)/\approx$ where $(x_1, y_1) \approx (x_2, y_2)$ when there is $z \in S$ with $x_1 + y_2 + z = x_2 + y_1 + z$, with the class of (x, y) written $x - y$ and with the canonical semigroup map $\gamma\colon S \to G(S)$, $\gamma(x) = x - 0$.

Step 1: Every element is a difference of projection classes. A general element of $G(S)$ is $\gamma(s) - \gamma(t)$ for $s, t \in S$. By definition of $\mathcal{D}(A)$, $s = [p]_{\mathcal{D}}$ and $t = [q]_{\mathcal{D}}$ for projections $p, q \in \mathbb{C}\mathrm{P}_\infty(A)$. Writing $[p] := \gamma([p]_{\mathcal{D}})$, the element is $[p] - [q]$, which is the asserted form.

Step 2: Additivity. The map γ is a semigroup homomorphism, so

$$[p \oplus q] = \gamma([p \oplus q]_{\mathcal{D}}) = \gamma([p]_{\mathcal{D}} + [q]_{\mathcal{D}}) = \gamma([p]_{\mathcal{D}}) + \gamma([q]_{\mathcal{D}}) = [p] + [q]$$

using the definition of addition in $\mathcal{D}(A)$ from definition 6.1.4. This proves (2).

Step 3: Equality criterion. By the construction of $G(S)$, one has $\gamma(s) = \gamma(t)$ in $G(S)$ if and only if $(s, 0) \approx (t, 0)$, that is, if and only if there is $z \in S$ with $s + z = t + z$ in S. Taking $s = [p]_{\mathcal{D}}$, $t = [q]_{\mathcal{D}}$ and $z = [r]_{\mathcal{D}}$ for a projection r, the equation $s + z = t + z$ reads $[p \oplus r]_{\mathcal{D}} = [q \oplus r]_{\mathcal{D}}$, which by definition of $\mathcal{D}(A)$ means $p \oplus r \sim_0 q \oplus r$. Conversely any $z \in S$ is of the form $[r]_{\mathcal{D}}$ for some projection r, since $\mathcal{D}(A) = \mathbb{C}\mathrm{P}_\infty(A)/\sim_0$. This is exactly statement (1), completing the proof.

A point worth isolating, because it streamlines every later argument, is that on a unital C^*-algebra the von Neumann equivalence $\sim_0$ over $\mathbb{C}\mathrm{P}_\infty(A)$ coincides with the coarser-looking notion obtained by allowing conjugation by unitaries in a large enough matrix algebra; both are subsumed by the next standard fact, which lets one trade analytic equivalence for the homotopy of projections.

Lemma 6.2.3: Equivalences of Projections Coincide up to Stabilization

Let A be a unital C^*-algebra and let $p, q \in \mathbb{C}\mathrm{P}_n(A)$ be projections in the same matrix algebra. Consider the relations

1. $p \sim_0 q$ (von Neumann: $p = v^*v$, $q = vv^*$ for some $v \in M_n(A)$);
2. $p \sim_h q$ (homotopy: p and q lie in the same path-component of the projections in $M_n(A)$);
3. $p \sim_u q$ (unitary: $q = upu^*$ for some unitary $u \in M_n(A)$).

Then (2) $\Rightarrow$ (3) $\Rightarrow$ (1), and if $\|p - q\| < 1$ then $p \sim_u q$. Conversely, if $p \sim_0 q$ then $p \oplus 0_n \sim_u q \oplus 0_n$ in $M_{2n}(A)$. Consequently all three relations induce the same semigroup $\mathcal{D}(A)$ and the same group $K_0(A)$.

The three notions of when two projections "are the same" are not literally equal as relations, but they generate the same equivalence once one is free to enlarge the ambient matrix size, and K_0 is built precisely to be insensitive to such enlargement. This is the operator-algebraic counterpart of the freedom, in [10, Stable Isomorphism and Stable Equivalence], to stabilize a vector bundle by adding trivial summands before comparing.

***Proof*:**
Step 1: (2) $\Rightarrow$ (3). Let $t \mapsto p_t$ be a continuous path of projections in $M_n(A)$ with $p_0 = p$, $p_1 = q$. By a partition of $[0,1]$ into intervals on which $\|p_s - p_t\| < 1$, it suffices to treat the case $\|p - q\| < 1$. Set

$$z = qp + (1-q)(1-p)$$

A direct computation gives $z - 1 = (q-p)(2p-1)$, so $\|z-1\| \le \|q-p\| < 1$ (since $2p-1$ is a self-adjoint unitary of norm 1), hence z is invertible. Moreover $zp = qp$ and $qz = qp$, so $zp = qz$. Polar-decomposing $z = u|z|$ with $|z| = (z^*z)^{1/2}$ invertible, u is unitary; the relation $zp = qz$ together with its adjoint $pz^* = z^*q$ gives $pz^*z = z^*qz = z^*zp$, so z^*z, and hence $|z|$, commutes with p. Then $upu^* = z|z|^{-1}p|z|^{-1}z^* = zp(z^*z)^{-1}z^* = qz(z^*z)^{-1}z^* = q$, using $z(z^*z)^{-1}z^* = 1$. So $p \sim_u q$.

Step 2: (3) $\Rightarrow$ (1). If $q = upu^*$ with u unitary, put $v = up$. Then $v^*v = pu^*up = p$ and $vv^* = uppu^* = upu^* = q$, so $p \sim_0 q$.

Step 3: (1) $\Rightarrow$ (3) *after stabilizing.* Suppose $p = v^*v$, $q = vv^*$ with $v \in M_n(A)$. Then $v = qvp$, so $vp = v = qv$ and, taking adjoints, $pv^* = v^* = v^*q$. Inside $M_{2n}(A)$ form

$$w = \begin{pmatrix} v & 1-q \\ 1-p & v^* \end{pmatrix}$$

A direct computation using $v^*v = p$, $vv^* = q$, $vp = v$, $qv = v$ gives

$$ww^* = \begin{pmatrix} vv^* + (1-q)^2 & v(1-p) + (1-q)v \\ (1-p)v^* + v^*(1-q) & (1-p)^2 + v^*v \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

since $vv^* + (1-q) = q + 1 - q = 1$, $v(1-p) + (1-q)v = (v-v) + (v-v) = 0$, and $(1-p) + v^*v = 1 - p + p = 1$; the computation $w^*w = 1_{2n}$ is identical. So w is a unitary in $M_{2n}(A)$. Since $vp = v$ and $(1-p)p = 0$,

$$w(p \oplus 0_n)w^* = \begin{pmatrix} v & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} v^* & 1-p \\ 1-q & v \end{pmatrix} = \begin{pmatrix} vv^* & v(1-p) \\ 0 & 0 \end{pmatrix} = q \oplus 0_n$$

using $vv^* = q$ and $v(1-p) = v - vp = 0$. Hence $p \oplus 0_n \sim_u q \oplus 0_n$.

Since $p \sim_0 p \oplus 0_n$ by proposition 6.1.3(1), the implications (2) $\Rightarrow$ (3) $\Rightarrow$ (1) together with this stabilized converse show that on $\mathbb{C}\mathrm{P}_\infty(A)$ all three relations have the same equivalence classes after the harmless padding by zeros. They therefore define the same quotient $\mathcal{D}(A)$ and the same Grothendieck group, as we sought to show.

Example 6.2.4: K_0 of the Complex Numbers
Take $A = \mathbb{C}$, the simplest unital C^*-algebra. A projection in $M_n(\mathbb{C})$ is a self-adjoint idempotent matrix, hence an orthogonal projection onto a subspace of $\mathbb{C}^n$, and is determined up to unitary conjugacy by its rank. Two projections $p \in M_m(\mathbb{C})$, $q \in M_n(\mathbb{C})$ satisfy $p \sim_0 q$ if and only if $\mathrm{rk} p = \mathrm{rk} q$: a partial isometry v with $v^*v = p$, $vv^* = q$ is exactly an isometry from $\mathrm{im} p$ onto $\mathrm{im} q$,

which exists precisely when the ranks agree. Thus

$$\mathcal{D}(\mathbb{C}) = \mathbb{C}\mathrm{P}_\infty(\mathbb{C})/{\sim_0} \xrightarrow{\mathrm{rk}} \mathbb{N}_{\geq 0}$$

is a semigroup isomorphism onto $(\mathbb{N}_{\geq 0}, +)$. The Grothendieck group of $(\mathbb{N}_{\geq 0}, +)$ is $\mathbb{Z}$, with the class of a rank-k projection sent to k. Hence

$$K_0(\mathbb{C}) \cong \mathbb{Z}$$

with the rank-one projection $[1] = [e_{11}]$ as generator. This matches $K^0(\mathrm{pt}) = \mathbb{Z}$ from [10, K^0 of a Point]: a point carries vector bundles classified by their dimension, and $\mathbb{C} = C(\mathrm{pt})$.

With the standard picture in place, the first structural property to establish is functoriality. A *-homomorphism moves projections to projections and respects the equivalence and the block sum, so it descends to the level of K_0. The substantial part of the statement is homotopy invariance, which is what makes K_0 a genuinely topological invariant rather than a rigid algebraic one.

Theorem 6.2.5: Functoriality and Homotopy Invariance of K_0

The assignment $A \mapsto K_0(A)$ is a covariant functor from unital C^*-algebras and unital *-homomorphisms to abelian groups. Explicitly, a unital *-homomorphism $\varphi\colon A \to B$ induces a group homomorphism

$$K_0(\varphi)\colon K_0(A) \to K_0(B), \qquad K_0(\varphi)\big([p] - [q]\big) = [\varphi_\infty(p)] - [\varphi_\infty(q)]$$

where $\varphi_\infty\colon \mathbb{C}\mathrm{P}_\infty(A) \to \mathbb{C}\mathrm{P}_\infty(B)$ is the entrywise application of φ, satisfying $K_0(\mathrm{id}) = \mathrm{id}$ and $K_0(\psi \circ \varphi) = K_0(\psi) \circ K_0(\varphi)$. Moreover, if $\varphi_0, \varphi_1\colon A \to B$ are homotopic through unital *-homomorphisms (there is a path $t \mapsto \varphi_t$ with each φ_t a unital *-homomorphism and $t \mapsto \varphi_t(a)$ norm-continuous for every $a \in A$), then

$$K_0(\varphi_0) = K_0(\varphi_1)$$

The homotopy-invariance clause is what every computation uses. It says that K_0 cannot distinguish *-homomorphisms connected by a continuous deformation, so two algebras related by a homotopy equivalence of *-homomorphisms have isomorphic K_0. It is the operator-algebraic mirror of [10, Homotopy Invariance Vector Bundles], which states that homotopic maps out of a paracompact space pull back vector bundles to isomorphic bundles. Although stated here for unital *-homomorphisms, the functor extends to all *-homomorphisms by definition 6.2.8: an arbitrary $\varphi\colon A \to B$ acts on K_0 through its unital extension $\tilde{\varphi}\colon \tilde{A} \to \tilde{B}$, and the identities $K_0(\mathrm{id}) = \mathrm{id}$, $K_0(\psi\varphi) = K_0(\psi)K_0(\varphi)$ persist because unitization is itself functorial (proposition 1.4.5). This is what permits applying K_0 to the non-unital inclusion $\iota\colon I \to A$ in theorem 6.2.10.

***Proof*:**
Step 1: Well-definedness and functoriality. A *-homomorphism $\varphi\colon A \to B$ induces $\varphi_n\colon M_n(A) \to M_n(B)$ entrywise, and since φ preserves adjoints and products it sends a projection $p = p^2 = p^*$ to a projection $\varphi_n(p)$. If $p \sim_0 q$ via v, with $p = v^*v$, $q = vv^*$, then $\varphi(v)$ witnesses $\varphi_n(p) \sim_0 \varphi_n(q)$, because φ is multiplicative and *-preserving. Also φ commutes with $\oplus$ (block diagonal sum), so φ descends to a semigroup homomorphism $\mathcal{D}(A) \to \mathcal{D}(B)$. By the universal property of the Grothendieck construction (definition 6.2.1) this extends uniquely to the group homomorphism

$K_0(\varphi)$ of the stated form. The identities $K_0(\mathrm{id}) = \mathrm{id}$ and $K_0(\psi\varphi) = K_0(\psi)K_0(\varphi)$ are immediate from the corresponding identities for the entrywise maps, since both sides agree on the generating classes $[p]$.

Step 2: Homotopy invariance. Fix a projection $p \in \mathbb{C}\mathrm{P}_n(A)$ and a homotopy $t \mapsto \varphi_t$ of unital *-homomorphisms. The map $t \mapsto (\varphi_t)_n(p)$ is a norm-continuous path of projections in $M_n(B)$, since each entry $t \mapsto \varphi_t(p_{ij})$ is continuous by hypothesis and the projection relations $(\varphi_t)_n(p)^2 = (\varphi_t)_n(p) = (\varphi_t)_n(p)^*$ hold for every t. By lemma 6.2.3, a homotopy of projections gives a unitary equivalence, hence $\sim_0$-equivalence, so

$$[(\varphi_0)_n(p)] = [(\varphi_1)_n(p)] \quad \text{in } K_0(B)$$

Every element of $K_0(A)$ is a difference $[p] - [q]$ of such classes by proposition 6.2.2, and $K_0(\varphi_0), K_0(\varphi_1)$ agree on each, so $K_0(\varphi_0) = K_0(\varphi_1)$, completing the proof.

The next property is stability: K_0 does not see the difference between A and its matrix amplifications, nor between A and its stabilization by the compact operators. This is the precise sense in which K_0 is a *stable* invariant, and it is what allows one to compute K_0 of an algebra by replacing it with a more convenient Morita-equivalent partner.

Theorem 6.2.6: Stability of K_0

Let A be a unital C^*-algebra. Then:

1. For every $n \geq 1$, the corner embedding $\iota \colon A \to M_n(A)$, $a \mapsto \mathrm{diag}(a, 0, \ldots, 0)$, induces an isomorphism
$$K_0(\iota) \colon K_0(A) \xrightarrow{\sim} K_0(M_n(A))$$
2. Let $\mathcal{K} = \mathcal{K}(H)$ be the C^*-algebra of compact operators on a separable infinite-dimensional Hilbert space, and let $e \in \mathcal{K}$ be a rank-one projection. The embedding $A \to A \otimes \mathcal{K}$, $a \mapsto a \otimes e$, induces an isomorphism
$$K_0(A) \xrightarrow{\sim} K_0(A \otimes \mathcal{K})$$

Part (1) is the statement that "a projection over $M_n(A)$ is just a bigger projection over A," formalized through the identification $M_k(M_n(A)) \cong M_{kn}(A)$. Part (2) upgrades this to the limit: $\mathcal{K}$ is the norm closure of the increasing union of the $M_n(\mathbb{C})$, and K_0 is continuous along such limits, so stabilizing by $\mathcal{K}$ changes nothing. Part (2) is also the cleanest statement of *Morita invariance* for K_0, since A and $A \otimes \mathcal{K}$ are stably isomorphic.

***Proof*:**
Step 1: The matrix isomorphism. There is a natural *-isomorphism $M_k(M_n(A)) \cong M_{kn}(A)$ for every k, regrouping a $k \times k$ matrix of $n \times n$ blocks as a single $kn \times kn$ matrix over A. Under these identifications, the projections over $M_n(A)$ are exactly the projections in $M_{kn}(A)$, and von Neumann equivalence and block sum correspond. Hence the entrywise inclusions assemble to a semigroup isomorphism $\mathcal{D}(A) \to \mathcal{D}(M_n(A))$: every projection $P \in M_k(M_n(A)) = M_{kn}(A)$ is a projection over A, and conversely. The map on $\mathcal{D}$ induced by ι sends $[p]_{\mathcal{D}}$ (for $p \in M_k(A)$) to the class of $\mathrm{diag}(p, 0, \ldots, 0) \in M_k(M_n(A)) = M_{kn}(A)$, which by proposition 6.1.3(1) equals $[p]_{\mathcal{D}}$ under the identification. So the induced map $\mathcal{D}(A) \to \mathcal{D}(M_n(A))$ is a bijection, hence an isomorphism of semigroups, and passing to Grothendieck groups gives the isomorphism $K_0(\iota)$.

Step 2: Continuity along the union. Write $\mathcal{K} = \overline{\bigcup_n M_n(\mathbb{C})}$, the closure of the union under the corner inclusions $M_n(\mathbb{C}) \hookrightarrow M_{n+1}(\mathbb{C})$, $x \mapsto \mathrm{diag}(x, 0)$. Then $A \otimes \mathcal{K} = \overline{\bigcup_n M_n(A)}$ with the corresponding corner inclusions $j_n \colon M_n(A) \hookrightarrow M_{n+1}(A)$. Let $P \in \mathbb{C}\mathrm{P}_\infty(A \otimes \mathcal{K})$, that is, $P \in M_k(A \otimes \mathcal{K})$ a projection. Since $\bigcup_n M_k(M_n(A))$ is dense in $M_k(A \otimes \mathcal{K})$, choose x in the union with $\|x - P\| < 1/4$; replacing x by $(x + x^*)/2$ we may take x self-adjoint with $\|x - P\| < 1/2$. The spectrum of x then misses $\{1/2\}$, and the spectral projection $P' = \mathbb{1}_{(1/2,\infty)}(x)$, computed by holomorphic functional calculus, is a projection lying in some $M_k(M_n(A))$ with $\|P' - P\| < 1$, whence $P' \sim_0 P$ by lemma 6.2.3. So every class in $\mathcal{D}(A \otimes \mathcal{K})$ comes from some $\mathcal{D}(M_n(A))$, and the colimit of the system

$$\mathcal{D}(M_1(A)) \to \mathcal{D}(M_2(A)) \to \cdots$$

maps onto $\mathcal{D}(A \otimes \mathcal{K})$. Injectivity at the level of $\mathcal{D}$ holds because if two projections in some $M_n(A)$ become $\sim_0$ in $A \otimes \mathcal{K}$, the implementing v can be approximated within the union, and a small perturbation of a partial isometry between projections is again such (by polar decomposition, as in lemma 6.2.3); thus they are already $\sim_0$ in some $M_m(A)$. Hence

$$\mathcal{D}(A \otimes \mathcal{K}) \cong \varinjlim_n \mathcal{D}(M_n(A))$$

By Step 1 each $\mathcal{D}(M_n(A)) \cong \mathcal{D}(A)$ compatibly with the corner inclusions, which are the identity on $\mathcal{D}(A)$, so the colimit is $\mathcal{D}(A)$ itself. Passing to Grothendieck groups, $K_0(A \otimes \mathcal{K}) \cong K_0(A)$, and tracing the isomorphism shows it is induced by $a \mapsto a \otimes e$, as we sought to show.

Example 6.2.7: K_0 of Matrix Algebras and of the Compacts

Applying theorem 6.2.6(1) with $A = \mathbb{C}$ gives

$$K_0(M_n(\mathbb{C})) \cong K_0(\mathbb{C}) \cong \mathbb{Z}$$

for every n. The generator is the class of a rank-one projection $e_{11} \in M_n(\mathbb{C})$; under the isomorphism $K_0(M_n(\mathbb{C})) \cong \mathbb{Z}$ a projection of rank r in $M_n(\mathbb{C})$ (equivalently a projection in some $M_k(M_n(\mathbb{C})) = M_{kn}(\mathbb{C})$ of rank r) maps to r. The corner map $\mathbb{C} \to M_n(\mathbb{C})$ sends the generator $1 \mapsto e_{11}$, so on K_0 it is the identity $\mathbb{Z} \to \mathbb{Z}$, *not* multiplication by n: K_0 measures rank, which the corner embedding preserves.

Applying theorem 6.2.6(2) with $A = \mathbb{C}$ gives

$$K_0(\mathcal{K}) \cong K_0(\mathbb{C}) \cong \mathbb{Z}$$

Concretely, $\mathcal{K} = \mathbb{C} \otimes \mathcal{K}$ and every projection in $\mathcal{K}$ is a finite-rank orthogonal projection, of some rank $r \in \mathbb{N}_{\geq 0}$; the rank is a complete invariant for $\sim_0$, so $\mathcal{D}(\mathcal{K}) \cong (\mathbb{N}_{\geq 0}, +)$ and $K_0(\mathcal{K}) \cong \mathbb{Z}$ with a rank-one projection as generator. The unital algebra $B(H)$ behaves entirely differently: there $1 \sim_0 1 \oplus 1$ by an isometry $H \cong H \oplus H$, forcing $[1] = 2[1]$, so $K_0(B(H)) = 0$.

So far everything has been unital. The definition definition 6.2.1 requires a unit because the semigroup $\mathcal{D}(A)$ is built from projections, and a non-unital algebra may have very few of them (for instance $C_0(\mathbb{R})$ has no nonzero projections at all). The fix is to pass to the unitization $\tilde{A}$ of proposition 1.4.5, which always has a unit, and then to cut out the contribution of the adjoined scalars. The unitization sits in a split exact sequence

$$0 \longrightarrow A \longrightarrow \tilde{A} \xrightarrow{\pi} \mathbb{C} \longrightarrow 0$$

where $\pi(a,\alpha) = \alpha$ is the quotient map and the splitting $\mathbb{C} \to \tilde{A}$, $\alpha \mapsto (0,\alpha)$, is a unital *-homomorphism. Applying the functor K_0 to π and the splitting gives a retraction of $K_0(\tilde{A})$ onto $K_0(\mathbb{C}) = \mathbb{Z}$, and the part it retracts away is the right definition of $K_0(A)$.

Definition 6.2.8: K_0 of a Non-Unital C^*-Algebra

Let A be a (not necessarily unital) C^*-algebra with unitization $\tilde{A}$ (proposition 1.4.5) and scalar quotient $\pi\colon \tilde{A} \to \mathbb{C}$. Define

$$K_0(A) := \ker\left(K_0(\pi)\colon K_0(\tilde{A}) \to K_0(\mathbb{C}) = \mathbb{Z}\right)$$

For a *-homomorphism $\varphi\colon A \to B$, the unital extension $\tilde{\varphi}\colon \tilde{A} \to \tilde{B}$ commutes with the scalar quotients, so $K_0(\tilde{\varphi})$ restricts to the kernels and defines $K_0(\varphi)\colon K_0(A) \to K_0(B)$.

When A is already unital this definition agrees with definition 6.2.1: in that case $\tilde{A} \cong A \oplus \mathbb{C}$ as C^*-algebras (proposition 1.4.5), the quotient π is projection onto the second factor, and since a projection over $A \oplus \mathbb{C}$ is a pair of projections (one over each factor), the semigroup $\mathcal{D}(A \oplus \mathbb{C}) \cong \mathcal{D}(A) \times \mathcal{D}(\mathbb{C})$ factors, giving $K_0(\tilde{A}) \cong K_0(A) \oplus \mathbb{Z}$; the kernel of $K_0(\pi)$ is then exactly the original $K_0(A)$. So definition 6.2.8 extends the unital definition without conflict, and from now on K_0 is defined on all C^*-algebras. In general (unital or not) the unital scalar inclusion $\lambda\colon \mathbb{C} \to \tilde{A}$, $\alpha \mapsto (0,\alpha)$, satisfies $\pi \circ \lambda = \mathrm{id}_{\mathbb{C}}$, so $K_0(\pi)$ is a split surjection onto $\mathbb{Z}$ with section $K_0(\lambda)$, and therefore

$$K_0(\tilde{A}) \cong \ker K_0(\pi) \oplus \mathrm{im} K_0(\lambda) = K_0(A) \oplus \mathbb{Z}$$

with the $\mathbb{Z}$ summand detecting the rank of the scalar part. This elementary splitting, depending only on the unital section λ, is what the reduction in theorem 6.2.10 uses.

Example 6.2.9: K_0 of $C_0(\mathbb{R})$ via Unitization

Let $A = C_0(\mathbb{R})$, the continuous functions on $\mathbb{R}$ vanishing at infinity, a non-unital algebra with no nonzero projections (a projection in a commutative C^*-algebra is the indicator of a clopen set, and $\mathbb{R}$ is connected with the only clopen sets $\emptyset, \mathbb{R}$, the latter excluded by vanishing at infinity). Its unitization is $\tilde{A} = C(S^1)$, the one-point compactification adding the point at infinity, with π evaluation at that basepoint. Anticipating theorem 6.2.11, $K_0(C(S^1)) \cong K^0(S^1) \cong \mathbb{Z}$, generated by the trivial line bundle, and $K_0(\pi)$ sends this generator to the rank, an isomorphism onto $\mathbb{Z}$. Hence

$$K_0(C_0(\mathbb{R})) = \ker\left(K_0(C(S^1)) \to \mathbb{Z}\right) = 0$$

matching the reduced group $\tilde{K}^0(S^1) = 0$ from [10, The K-Theory of Spheres]: every complex vector bundle over S^1 is trivial.

The reason the unitization-based definition is the correct one is that it makes K_0 behave well on short exact sequences. A *-homomorphism that is injective with closed image, followed by the quotient, forms a short exact sequence of C^*-algebras; the content of the next theorem is that K_0 takes such a sequence to an exact sequence of abelian groups, at least in the middle term. Full exactness on the two ends fails in general (that failure is exactly what the index map and the six-term sequence of section 8.1 will repair), but middle exactness, called half-exactness, already holds and is what powers Mayer-Vietoris-style computations.

Theorem 6.2.10: Half-Exactness of K_0

Let
$$0 \longrightarrow I \xrightarrow{\iota} A \xrightarrow{q} A/I \longrightarrow 0$$
be a short exact sequence of C^*-algebras, where I is a closed two-sided ideal of A, ι the inclusion, and q the quotient. Then the induced sequence
$$K_0(I) \xrightarrow{K_0(\iota)} K_0(A) \xrightarrow{K_0(q)} K_0(A/I)$$
is exact at $K_0(A)$, that is, $\mathrm{im}K_0(\iota) = \ker K_0(q)$. If moreover the sequence splits by a *-homomorphism $s\colon A/I \to A$ with $q \circ s = \mathrm{id}$, then
$$0 \longrightarrow K_0(I) \xrightarrow{K_0(\iota)} K_0(A) \xrightarrow{K_0(q)} K_0(A/I) \longrightarrow 0$$
is split exact.

Half-exactness is what makes K_0 a K-theory rather than just a functor. It is the algebraic form of the long exact sequence of a pair in topological K-theory ([10, The Long Exact Sequence of a Pair]): the ideal I plays the role of the open complement, A/I the role of the closed subspace, and the sequence records how projections (bundles) on the whole space relate to those on the pieces. The split case is what underlies the computation of K_0 of unitizations and of products.

***Proof*:**
Reduction to unital algebras. Unitizing the sequence gives
$$0 \longrightarrow I \longrightarrow \tilde{A} \xrightarrow{\tilde{q}} \widetilde{A/I} \longrightarrow 0$$
where I remains a closed ideal (now of the unital algebra $\tilde{A}$), and $\tilde{A}/I \cong \widetilde{A/I}$ since $\tilde{A}$ adjoins one scalar unit, which descends to the unit of $\widetilde{A/I}$. By definition 6.2.8 the splittings give compatible decompositions $K_0(\tilde{A}) \cong K_0(A) \oplus \mathbb{Z}$ and $K_0(\widetilde{A/I}) \cong K_0(A/I) \oplus \mathbb{Z}$, under which $K_0(\tilde{q})$ is $K_0(q) \oplus \mathrm{id}_{\mathbb{Z}}$ and the maps to and from the $\mathbb{Z}$-summands are identities. Therefore exactness of $K_0(I) \to K_0(\tilde{A}) \to K_0(\widetilde{A/I})$ at the middle is equivalent to exactness of $K_0(I) \to K_0(A) \to K_0(A/I)$ at the middle. So it suffices to prove the theorem when A (hence the quotient) is unital and the maps are unital, which is assumed henceforth; note I itself need not be unital, and $K_0(I)$ is understood via definition 6.2.8.

Step 1: $\mathrm{im}K_0(\iota) \subseteq \ker K_0(q)$. Since $q \circ \iota = 0$ as a map (the composite $I \to A \to A/I$ is zero), functoriality (theorem 6.2.5) gives $K_0(q) \circ K_0(\iota) = K_0(q \circ \iota) = 0$. Hence the image of $K_0(\iota)$ lies in the kernel of $K_0(q)$.

Step 2: $\ker K_0(q) \subseteq \mathrm{im}K_0(\iota)$. Let $x \in \ker K_0(q)$. By proposition 6.2.2 write $x = [p] - [q_0]$ with $p, q_0 \in \mathbb{C}\mathrm{P}_n(A)$ projections; adding the complement $1_n - q_0$ to both terms, we may take $q_0 = 1_m \oplus 0$ a trivial projection, so $x = [p] - [1_m \oplus 0]$ with $p \in \mathbb{C}\mathrm{P}_n(A)$. The hypothesis $K_0(q)(x) = 0$ says $[q_n(p)] = [1_m \oplus 0]$ in $K_0(A/I)$, where $q_n\colon M_n(A) \to M_n(A/I)$ is the entrywise quotient. By proposition 6.2.2(1), after enlarging n and adding a common trivial projection there is a unitary $\bar{u} \in M_n(A/I)$ with $\bar{u}\, q_n(p)\, \bar{u}^* = 1_m \oplus 0$.

Lifting the unitary. Replacing $\bar{u}$ by $\mathrm{diag}(\bar{u}, \bar{u}^*) \in M_{2n}(A/I)$ and the projection $1_m \oplus 0$ by $(1_m \oplus 0) \oplus 0_n$, we may assume $\bar{u}$ lies in the identity component of the unitary group. Indeed

$\mathrm{diag}(\bar{u}, \bar{u}^*)$ is connected to 1_{2n} by the explicit Whitehead path

$$t \mapsto \begin{pmatrix} \bar{u} & 0 \\ 0 & 1 \end{pmatrix} R_t \begin{pmatrix} \bar{u}^* & 0 \\ 0 & 1 \end{pmatrix} R_t^{-1}, \qquad R_t = \begin{pmatrix} \cos(\frac{\pi t}{2}) & -\sin(\frac{\pi t}{2}) \\ \sin(\frac{\pi t}{2}) & \cos(\frac{\pi t}{2}) \end{pmatrix}$$

which equals 1_{2n} at $t = 0$ and $\mathrm{diag}(\bar{u}, \bar{u}^*)$ at $t = 1$. The identity component of the unitary group of a unital C^*-algebra consists of finite products of exponentials e^{ih} with $h = h^*$, and each such factor lifts: choosing a self-adjoint preimage $h' \in M_n(A)$ of h under the surjection q_n (available since q_n is a surjective *-homomorphism and $(c + c^*)/2$ lifts any self-adjoint h from a preimage c), the element $e^{ih'}$ is a unitary in $M_n(A)$ with $q_n(e^{ih'}) = e^{ih}$. Multiplying these lifts yields a unitary $u \in M_n(A)$ with $q_n(u) = \bar{u}$. Replacing p by upu^* (which lies in the same $\sim_0$-class, hence gives the same $[p]$) we may assume $q_n(p) = 1_m \oplus 0$ exactly.

Producing a class in $K_0(I)$. Now $p - (1_m \oplus 0) \in M_n(I)$, since q_n annihilates it. Both p and $1_m \oplus 0$ are projections in $M_n(\tilde{I})$ (the matrices over the unitization $\tilde{I}$ of I), each mapping to $1_m \oplus 0$ under the scalar quotient $M_n(\tilde{I}) \to M_n(\mathbb{C})$. Hence the difference

$$[p] - [1_m \oplus 0] \in \ker \left(K_0(\tilde{I}) \to \mathbb{Z} \right) = K_0(I)$$

has vanishing scalar part and so defines a class in $K_0(I)$ by definition 6.2.8. Its image under $K_0(\iota)$ is $[p] - [1_m \oplus 0] = x$ in $K_0(A)$. Therefore $x \in \mathrm{im} K_0(\iota)$, proving exactness at $K_0(A)$.

Step 3: The split case. Suppose $s\colon A/I \to A$ is a *-homomorphism with $qs = \mathrm{id}$. Then $K_0(q)K_0(s) = K_0(qs) = \mathrm{id}$ by theorem 6.2.5, so $K_0(q)$ is surjective with section $K_0(s)$. It remains only to prove that $K_0(\iota)$ is injective; granting this, Step 2 shows $K_0(\iota)$ maps $K_0(I)$ isomorphically onto $\ker K_0(q)$, and combining with the section gives, for any $\xi \in K_0(A)$, the decomposition $\xi = \left(\xi - K_0(s)K_0(q)(\xi)\right) + K_0(s)K_0(q)(\xi)$ with first summand in $\ker K_0(q) = \mathrm{im} K_0(\iota)$ and $\mathrm{im} K_0(s) \cap \ker K_0(q) = 0$ (since $K_0(q)$ restricted to $\mathrm{im} K_0(s)$ is injective, its composite with $K_0(s)$ being the identity), so that

$$K_0(A) = \mathrm{im} K_0(\iota) \oplus \mathrm{im} K_0(s) \cong K_0(I) \oplus K_0(A/I)$$

Injectivity of $K_0(\iota)$. The content here is that the section forces injectivity even though no *-homomorphism retraction $A \to I$ exists in general (the failure of such a retraction is exactly what the index map of section 8.1 measures), so this is not a formal consequence of the tools of Steps 1-2 alone. The clean argument runs through the suspension $SA = C_0(\mathbb{R}) \otimes A$ and the mapping cone of s, machinery built only in section 8.1: a section s splits the inclusion $I \hookrightarrow A$ up to homotopy after suspending, and the homotopy invariance of K_0 (theorem 6.2.5) then converts that homotopy splitting into a left inverse $K_0(A) \to K_0(I)$ for $K_0(\iota)$. Because the suspension and mapping-cone apparatus is not yet available at this point in the development, the verification is deferred: the statement that a homotopy-invariant, half-exact functor on C^*-algebras carries split short exact sequences to split short exact sequences (so that $K_0(\iota)$ is injective whenever a *-homomorphism section is present) is proved in full for K_0 in [12, Chapter 4, "The functor K_0"], and as the general axiomatic implication "half-exact and homotopy invariant $\Rightarrow$ split exact" in [1]. Theorems 6.2.5 and 6.2.6 give the homotopy invariance and stability hypotheses, and Steps 1-2 give half-exactness, so those references apply verbatim to the present K_0.

With injectivity of $K_0(\iota)$ in hand, the sequence

$$0 \longrightarrow K_0(I) \xrightarrow{K_0(\iota)} K_0(A) \xrightarrow{K_0(q)} K_0(A/I) \longrightarrow 0$$

is split exact, completing the proof.

The half-exactness theorem isolates the structural fact; the bridge to topology turns it into geometry. The link is the Serre-Swan correspondence, which identifies the projections over $C(X)$ with the vector bundles over X. [10, Vector Bundle Over Compact Are Projective] already records the geometric half of this: a vector bundle over a compact Hausdorff space is a direct summand of a trivial bundle, hence is "projective" in the bundle category. The algebraic half is that such complemented summands of $C(X)^n$ are exactly the images of projection matrices over $C(X)$. Putting the two together identifies $K_0(C(X))$ with the topological $K^0(X)$ of [10, The Grothendieck Group $K^0(X)$].

Theorem 6.2.11: Serre-Swan: $K_0(C(X)) \cong K^0(X)$

Let X be a compact Hausdorff space and $C(X)$ the unital C^*-algebra of continuous complex-valued functions on X. There is an isomorphism of abelian groups

$$K_0(C(X)) \;\cong\; K^0(X)$$

natural in X, under which the class $[p]$ of a projection $p \in M_n(C(X))$ corresponds to the class of the vector bundle $E_p \to X$ whose fiber at x is the image $p(x)(\mathbb{C}^n) \subseteq \mathbb{C}^n$. Under this isomorphism evaluation at a basepoint x_0 corresponds to the rank at x_0, so the kernel $\tilde{K}^0(X)$ of that restriction corresponds to $K_0(C_0(X \setminus \{x_0\}))$.

This is the central result of the whole subject: it says that the operator K_0 and the topological K^0 are literally the same group, computed two ways. Geometrically one classifies vector bundles; algebraically one classifies projections in matrix algebras over the function algebra. The Gelfand-Naimark duality (theorem 4.3.3) makes $X \mapsto C(X)$ a faithful (contravariant) embedding of compact Hausdorff spaces into C^*-algebras, and Serre-Swan promotes that embedding to an identity of K-theories.

***Proof*:**
Step 1: From projections to bundles. Let $p \in M_n(C(X))$ be a projection, viewed as a continuous map $p\colon X \to M_n(\mathbb{C})$ with $p(x) = p(x)^2 = p(x)^*$ for all x, that is, a continuous field of orthogonal projections in $\mathbb{C}^n$. The ranks $\mathrm{rk}p(x)$ form a locally constant, hence (on each component) constant, function of x, because nearby projections of equal norm-distance less than 1 have equal rank. Define

$$E_p = \{(x, v) \in X \times \mathbb{C}^n : v \in p(x)\mathbb{C}^n\}$$

with projection to X. This is a subbundle of the trivial bundle $X \times \mathbb{C}^n$: locally, the continuity of p and the constancy of the rank let one choose, near each x_0, continuous sections spanning $p(x)\mathbb{C}^n$ (apply the Gram-Schmidt process of [5, The Gram-Schmidt Process] to continuous lifts of a basis of $p(x_0)\mathbb{C}^n$), giving a local trivialization. So E_p is a complex vector bundle over X, a direct summand of the trivial bundle with complement E_{1-p}.

Step 2: From bundles to projections. Conversely, let $E \to X$ be a complex vector bundle. By [10, Vector Bundle Over Compact Are Projective], compactness gives a bundle E' with $E \oplus E' \cong X \times \mathbb{C}^n$ trivial for some n. The orthogonal projection of $\mathbb{C}^n$ onto the E-summand, fiberwise, is a continuous family $p_E\colon X \to M_n(\mathbb{C})$ of projections (continuous because, after choosing a Hermitian metric via [8, Inner Products on a Vector Bundle], the orthogonal projection onto a continuously varying subspace varies continuously), hence $p_E \in M_n(C(X))$ is a projection with $E_{p_E} \cong E$. So every bundle arises as some E_p.

Step 3: The correspondence respects equivalence and sum. If $p \sim_0 q$ over $C(X)$, the implementing partial isometry v is a continuous field of partial isometries, giving a fiberwise isometric isomorphism $E_p \cong E_q$; conversely an isomorphism $E_p \cong E_q$ of subbundles of trivial bundles is implemented by such a v after stabilizing, by lemma 6.2.3. Block sum of projections corresponds to direct sum of bundles: $E_{p\oplus q} \cong E_p \oplus E_q$ by inspection of fibers. Therefore the assignment $p \mapsto E_p$ descends to a semigroup isomorphism

$$\mathcal{D}(C(X)) \xrightarrow{\sim} \big(\mathbb{VB}_{\mathbb{C}}(X), \oplus\big)$$

between the projection semigroup of definition 6.1.4 and the monoid of isomorphism classes of complex vector bundles under Whitney sum ([10, The Grothendieck Group $K^0(X)$]).

Step 4: Passing to Grothendieck groups. The group $K_0(C(X))$ is the Grothendieck group of $\mathcal{D}(C(X))$ by definition 6.2.1, and $K^0(X) = KU(X)$ is the Grothendieck group of $(\mathbb{VB}_{\mathbb{C}}(X), \oplus)$ by [10, The Grothendieck Group $K^0(X)$]. A semigroup isomorphism induces an isomorphism of Grothendieck groups, so

$$K_0(C(X)) \cong K^0(X)$$

Naturality in X holds because a continuous map $f\colon X \to Y$ induces the unital *-homomorphism $f^*\colon C(Y) \to C(X)$ on one side and the bundle pullback f^* on the other, and $E_{f^*(p)} = f^*(E_p)$ by definition of pullback. The rank of $p(x)$ is the dimension of the fiber of E_p at x, so evaluation at the basepoint on the operator side matches restriction to x_0 on the bundle side, and the kernel of that restriction is the reduced group $\tilde{K}^0(X)$ of [10, Reduced K-Theory $\widetilde{K}^0(X)$]. Identifying $\tilde{K}^0(X)$ with $K_0(C_0(X \setminus \{x_0\}))$ follows from definition 6.2.8 applied to the split sequence $0 \to C_0(X \setminus \{x_0\}) \to C(X) \to \mathbb{C} \to 0$ (evaluation at x_0), whose K_0 is split exact by theorem 6.2.10, giving $K_0(C(X)) \cong K_0(C_0(X \setminus \{x_0\})) \oplus \mathbb{Z}$ with the $\mathbb{Z}$ the rank at x_0. This completes the proof.

The Serre-Swan identification turns every computation of topological K^0 into a computation of operator K_0 and conversely. The flagship example is the 2-sphere, where the answer $\mathbb{Z}^2$ encodes both the rank of a bundle and the first Chern number, and where the operator side produces the same $\mathbb{Z}^2$ through a single nontrivial projection, the Bott projection.

Example 6.2.12: $K_0(C(S^2)) \cong \mathbb{Z}^2$ and the Bott Projection

From theorem 6.2.11, $K_0(C(S^2)) \cong K^0(S^2)$. By [10, The Splitting $K^0 = \widetilde{K}^0 \oplus \mathbb{Z}$] the reduced group splits off a free summand,

$$K^0(S^2) \cong \tilde{K}^0(S^2) \oplus \mathbb{Z}$$

and $\tilde{K}^0(S^2) \cong \mathbb{Z}$ because, by [10, Clutching Function Vector Bundle Correspondence, Reduced K-Theory of S^2], complex n-plane bundles over S^2 are classified by homotopy classes of clutching functions $S^1 \to \mathrm{GL}_n(\mathbb{C})$, that is, by $\pi_1(\mathrm{GL}_n(\mathbb{C})) \cong \pi_1(U(n)) \cong \mathbb{Z}$ (the last isomorphism is the determinant $U(n) \to U(1) = S^1$); these stabilize to the single group $\tilde{K}^0(S^2) \cong \mathbb{Z}$, with generator the class of the canonical line bundle H over $\mathbb{C}\mathrm{P}^1 = S^2$ (clutching function $z \mapsto z$). Hence

$$K_0(C(S^2)) \cong \mathbb{Z}^2$$

with basis given by the class $[1]$ of the trivial line bundle (rank) and the class $[H] - [1]$ of the reduced generator (first Chern number).

On the operator side this $\mathbb{Z}^2$ is produced explicitly by the *Bott projection.* Write $S^2 = \{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_1^2 + x_2^2 + x_3^2 = 1\}$ and set

$$p_B = \frac{1}{2}\begin{pmatrix} 1 + x_3 & x_1 - ix_2 \\ x_1 + ix_2 & 1 - x_3 \end{pmatrix} \in M_2(C(S^2))$$

A direct check using $x_1^2 + x_2^2 + x_3^2 = 1$ gives $p_B = p_B^2 = p_B^*$, so p_B is a projection of constant rank 1 (its trace is 1). The line bundle E_{p_B} has first Chern number ± 1 (it is the canonical line bundle H, up to the orientation of S^2), so it generates $\tilde{K}^0(S^2) \cong \mathbb{Z}$, and

$$[p_B] - [e_{11}] \in K_0(C(S^2))$$

is a reduced generator, agreeing with $\pm([H] - [1])$. Together with $[e_{11}]$ (rank) it gives a basis of $\mathbb{Z}^2$. The matching with the topological computation is the content of theorem 6.2.11: the operator-algebraic projection p_B and the geometric clutching function $z \mapsto z$ describe the same generator of $\tilde{K}^0(S^2) \cong \mathbb{Z}$.

Exercise 6.2.1

1. Let A be a unital C^*-algebra and $p, q \in \mathbb{C}\mathrm{P}_n(A)$ projections with $\|p - q\| < 1$. Show directly that $p \sim_u q$, and deduce that $[p] = [q]$ in $K_0(A)$. (This is the local step behind homotopy invariance; reconstruct it without quoting lemma 6.2.3.)

2. Compute $K_0(\mathbb{C} \oplus \mathbb{C})$ and, more generally, $K_0\big(\bigoplus_{i=1}^k M_{n_i}(\mathbb{C})\big)$ for a finite-dimensional C^*-algebra. Identify the rank homomorphism in each summand and the class of the unit.

3. Using definition 6.2.8 and the split sequence $0 \to C_0(\mathbb{R}^2) \to C(S^2) \to \mathbb{C} \to 0$ (evaluation at a basepoint), compute $K_0(C_0(\mathbb{R}^2))$. Reconcile your answer with $\tilde{K}^0(S^2)$ from example 6.2.12.

4. Verify that the Bott projection $p_B \in M_2(C(S^2))$ of example 6.2.12 satisfies $p_B = p_B^2 = p_B^*$ and has constant rank 1. Then show that p_B is *not* Murray-von Neumann equivalent to a constant rank-one projection e_{11}, and explain what invariant obstructs the equivalence.

5. Let X be a compact Hausdorff space with $X = X_1 \sqcup X_2$ a disjoint union of two clopen pieces. Using theorem 6.2.11, prove $K_0(C(X)) \cong K_0(C(X_1)) \oplus K_0(C(X_2))$ both topologically (via K^0) and algebraically (via the C^*-algebra decomposition $C(X) \cong C(X_1) \oplus C(X_2)$).

6. Show that $K_0(B(H)) = 0$ for H a separable infinite-dimensional Hilbert space, and contrast this with $K_0(\mathcal{K}(H)) \cong \mathbb{Z}$ from example 6.2.7. Which hypothesis in theorem 6.2.6 fails for $B(H)$, and why does the argument for $\mathcal{K}$ not apply?

7

The Group K_1

7.1 K_1 of a C^*-Algebra

The group $K_0(A)$ of definition 6.2.1 records projections, the "zero-dimensional" data of a C^*-algebra, by mimicking the Grothendieck construction for vector bundles. The natural companion invariant records the unitaries: where K_0 assembles idempotent self-adjoint elements into a group, K_1 assembles the symmetries (unitaries) of the matrix algebras $M_n(A)$, modulo the relation of being connected by a continuous path. This section defines $K_1(A)$ for a unital C^*-algebra A as the set of path components of the infinite unitary group, proves that it is an abelian group, establishes its functoriality, homotopy invariance, and stability, and identifies $K_1(C(X))$ with the topological group $K^1(X)$ for compact X. The construction is governed throughout by the analytic structure of a C^*-algebra (the norm topology, the polar decomposition, the continuous functional calculus), and it is precisely this analytic input that distinguishes the operator-theoretic K_1 from the purely algebraic Whitehead group $\mathrm{GL}(R)/E(R)$ of [2]. The distinction is drawn carefully at the end of the section.

Throughout, A is a unital C^*-algebra (recall definition 1.2.1) unless stated otherwise; the non-unital case is reduced to the unital one by passing to the unitization $\tilde{A}$ of proposition 1.4.5 at the very end. For a unital C^*-algebra A write

$$\mathcal{U}_n(A) = \{u \in M_n(A) \mid u^*u = uu^* = 1_n\}$$

for the unitary group of the matrix algebra $M_n(A)$, where 1_n is the identity of $M_n(A)$. The block-diagonal inclusion

$$\mathcal{U}_n(A) \hookrightarrow \mathcal{U}_{n+1}(A) \qquad u \mapsto \begin{pmatrix} u & 0 \\ 0 & 1 \end{pmatrix}$$

is a continuous group homomorphism, and the colimit (union) over these inclusions is the infinite unitary group

$$\mathcal{U}_\infty(A) = \varinjlim_n \mathcal{U}_n(A) = \bigcup_{n=1}^{\infty} \mathcal{U}_n(A)$$

where $u \in \mathcal{U}_n(A)$ is identified with its image $u \oplus 1$ in $\mathcal{U}_{n+1}(A)$. The group $\mathcal{U}_\infty(A)$ carries the colimit topology: a set is open precisely when its intersection with each $\mathcal{U}_n(A)$ is open in the norm topology. The two operations available on $\mathcal{U}_\infty(A)$ are the group multiplication inherited from each $\mathcal{U}_n(A)$ and the block sum $u \oplus v = \mathrm{diag}(u, v)$; the interplay between them is what the whole theory uses.

Before stating the definition, the reason for taking path components rather than the unitary group itself deserves a word. A single unitary $u \in \mathcal{U}_n(A)$ carries spectral information that is too fine to be a homotopy invariant: the unitaries connected to the identity through a continuous path form a normal subgroup, and quotienting by it discards exactly the data that varies continuously, retaining the discrete obstruction to contractibility. This obstruction is what K_1 measures.

Definition 7.1.1: K_1 of a C^*-Algebra

Let A be a unital C^*-algebra. The *K_1-group* of A is the set of path components of the infinite unitary group,

$$K_1(A) = \pi_0(\mathcal{U}_\infty(A)) = \mathcal{U}_\infty(A)/\mathcal{U}_\infty(A)_0$$

where $\mathcal{U}_\infty(A)_0$ is the path component of the identity. The class of a unitary $u \in \mathcal{U}_n(A)$ in $K_1(A)$ is written $[u]_1$, and the group operation is induced by the block sum,

$$[u]_1 + [v]_1 = [u \oplus v]_1 = \left[\begin{pmatrix} u & 0 \\ 0 & v \end{pmatrix}\right]_1$$

with identity element $[1]_1 = 0$.

The definition asserts more than it has so far justified: that π_0 of $\mathcal{U}_\infty(A)$ is a group at all, that $\mathcal{U}_\infty(A)_0$ is a normal subgroup so that the quotient makes sense, that the operation is well-defined on path components, and that the resulting group is abelian. All of these follow from a single mechanism, the matrix manoeuvre known as the Whitehead lemma, which says that in the infinite unitary group the block sum $u \oplus v$ is path-connected to the ordinary product $uv \oplus 1$. The verification occupies theorem 7.1.4 below. For now, the intuition is that $K_1(A)$ measures how many "essentially different" invertible symmetries the matrix algebras of A support: an element is trivial in K_1 exactly when, after enlarging the matrix size, it can be continuously deformed to the identity.

Two preliminary reductions make $\mathcal{U}_\infty(A)$ tractable and tie the definition to the invertible (rather than unitary) elements. The first is that, in a unital C^*-algebra, the unitary group of any matrix algebra is path-connected to the identity exactly when the corresponding general linear group is, because the polar decomposition retracts invertibles onto unitaries continuously. The second is the continuous logarithm available from the functional calculus. Both are recorded now.

Lemma 7.1.2: Connectivity via the Exponential

Let A be a unital C^*-algebra and $u \in \mathcal{U}_n(A)$. If $u = e^{ih}$ for some self-adjoint $h \in M_n(A)$, then the path $t \mapsto e^{ith}$, $t \in [0, 1]$, is a continuous path of unitaries in $\mathcal{U}_n(A)$ from 1 to u. Conversely, every $u \in \mathcal{U}_n(A)$ with $\|u - 1\| < 2$ is of this form, and in particular lies in $\mathcal{U}_n(A)_0$.

***Proof*:**
Step 1: Exponentials give paths. Suppose $u = e^{ih}$ with $h = h^* \in M_n(A)$. For each $t \in [0,1]$ the element th is self-adjoint, so e^{ith} is unitary: indeed $(e^{ith})^* = e^{-ith}$ by continuity of the involution applied to the convergent exponential series, and $e^{ith}e^{-ith} = 1$ since th commutes with itself. The map $t \mapsto e^{ith} = \sum_{k\geq 0}(ith)^k/k!$ is norm-continuous in t because the series converges uniformly for t in the compact interval $[0,1]$ (the k-th term is bounded by $\|h\|^k/k!$). At $t = 0$ it is 1 and at $t = 1$ it is u, giving the asserted path inside $\mathcal{U}_n(A)$.

Step 2: Nearby unitaries are exponentials. Suppose $u \in \mathcal{U}_n(A)$ satisfies $\|u-1\| < 2$. The spectrum $\sigma(u)$ lies in the unit circle $\mathbb{T}$ (as u is unitary) and avoids the point -1, because $-1 \in \sigma(u)$ would force $\|u-1\| \geq |-1-1| = 2$ by the spectral radius identity $\|u-1\| = \sup\{|\lambda - 1| \mid \lambda \in \sigma(u)\}$ valid for the normal element $u-1$. Thus $\sigma(u)$ is contained in the open arc $\mathbb{T} \setminus \{-1\}$, on which the principal branch of the logarithm $\log : \mathbb{T} \setminus \{-1\} \to (-i\pi, i\pi)$, $e^{i\theta} \mapsto i\theta$ for $\theta \in (-\pi, \pi)$, is continuous. Applying the continuous functional calculus to the normal element u yields $h := -i\log(u) \in M_n(A)$, which is self-adjoint because log maps $\mathbb{T} \setminus \{-1\}$ into $i\mathbb{R}$ so that $-i\log$ is real-valued on $\sigma(u)$, and which satisfies $e^{ih} = u$ by the functional-calculus identity $\exp(i \cdot (-i\log)) = \mathrm{id}$ on $\sigma(u)$. By Step 1, $u \in \mathcal{U}_n(A)_0$, completing the proof.

The exponential path is the basic building block: it shows that the identity component $\mathcal{U}_n(A)_0$ contains every unitary within distance 2 of the identity, hence (being a subgroup closed under the path-connecting operation) is open, and that $\mathcal{U}_n(A)_0$ is generated by exponentials of self-adjoint elements. The same lemma applied in M_∞ shows $\mathcal{U}_\infty(A)_0$ is generated by such exponentials. The next lemma is the decisive algebraic identity.

Lemma 7.1.3: Whitehead Lemma for Unitaries

Let A be a unital C^*-algebra and let $u, v \in \mathcal{U}_n(A)$. Then in $\mathcal{U}_{2n}(A)$ the following hold.

1. The block unitary $\begin{pmatrix} u & 0 \\ 0 & u^* \end{pmatrix}$ lies in $\mathcal{U}_{2n}(A)_0$; it is connected to 1_{2n} by an explicit path of unitaries.
2. The block unitaries $\begin{pmatrix} uv & 0 \\ 0 & 1 \end{pmatrix}$ and $\begin{pmatrix} u & 0 \\ 0 & v \end{pmatrix}$ are connected by a path of unitaries in $\mathcal{U}_{2n}(A)$.

***Proof*:**
Step 1: The rotation path. For $\theta \in [0, \pi/2]$ set

$$R_\theta = \begin{pmatrix} \cos\theta\, 1_n & -\sin\theta\, 1_n \\ \sin\theta\, 1_n & \cos\theta\, 1_n \end{pmatrix} \in \mathcal{U}_{2n}(A)$$

a unitary because $R_\theta R_\theta^* = 1_{2n}$ by the identity $\cos^2\theta + \sin^2\theta = 1$ (the scalar entries commute with everything). The map $\theta \mapsto R_\theta$ is norm-continuous, $R_0 = 1_{2n}$, and

$$R_{\pi/2} = \begin{pmatrix} 0 & -1_n \\ 1_n & 0 \end{pmatrix}$$

Now consider, for $u \in \mathcal{U}_n(A)$,

$$W(\theta) = \begin{pmatrix} u & 0 \\ 0 & 1_n \end{pmatrix} R_\theta \begin{pmatrix} u^* & 0 \\ 0 & 1_n \end{pmatrix} R_\theta^*$$

Each $W(\theta)$ is a product of unitaries, hence unitary, and $\theta \mapsto W(\theta)$ is continuous. At $\theta = 0$, $W(0) = 1_{2n}$. At $\theta = \pi/2$ direct computation gives

$$\begin{pmatrix} u & 0 \\ 0 & 1_n \end{pmatrix} \begin{pmatrix} 0 & -1_n \\ 1_n & 0 \end{pmatrix} \begin{pmatrix} u^* & 0 \\ 0 & 1_n \end{pmatrix} \begin{pmatrix} 0 & 1_n \\ -1_n & 0 \end{pmatrix} = \begin{pmatrix} u & 0 \\ 0 & u^* \end{pmatrix}$$

where $R_{\pi/2}^* = \begin{pmatrix} 0 & 1_n \\ -1_n & 0 \end{pmatrix}$ and carrying out the four block multiplications left to right yields $\mathrm{diag}(u, u^*)$. Thus $\theta \mapsto W(\theta)$ is a path of unitaries from 1_{2n} to $\mathrm{diag}(u, u^*)$, proving (1).

Step 2: From the product to the block sum. It suffices to show that both $\mathrm{diag}(uv, 1)$ and $\mathrm{diag}(u, v)$ are connected, in $\mathcal{U}_{2n}(A)$, to the common unitary $\mathrm{diag}(1, uv)$. Two factorizations achieve this. First,

$$\begin{pmatrix} uv & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} uv & 0 \\ 0 & (uv)^* \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & uv \end{pmatrix}$$

since $(uv)^*(uv) = 1$; the first factor lies in $\mathcal{U}_{2n}(A)_0$ by (1) applied to uv, joined to 1_{2n} by a path $p(t)$ of unitaries, so $t \mapsto p(t)\mathrm{diag}(1, uv)$ joins $\mathrm{diag}(uv, 1)$ to $\mathrm{diag}(1, uv)$. Second,

$$\begin{pmatrix} u & 0 \\ 0 & v \end{pmatrix} = \begin{pmatrix} u & 0 \\ 0 & u^* \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & uv \end{pmatrix}$$

since $u^* \cdot uv = v$; the first factor lies in $\mathcal{U}_{2n}(A)_0$ by (1) applied to u, joined to 1_{2n} by a path $q(t)$, so $t \mapsto q(t)\mathrm{diag}(1, uv)$ joins $\mathrm{diag}(u, v)$ to $\mathrm{diag}(1, uv)$. Composing one path with the reverse of the other connects $\mathrm{diag}(uv, 1)$ to $\mathrm{diag}(u, v)$ inside $\mathcal{U}_{2n}(A)$, completing the proof.

The Whitehead lemma is the analytic heart of K_1. Part (1) shows that any unitary, when paired with its adjoint in a doubled matrix algebra, becomes contractible to the identity; this is the unitary analogue of the algebraic fact that $\mathrm{diag}(g, g^{-1})$ is a product of elementary matrices. Part (2) shows that the block sum operation and the group multiplication agree after passage to path components. With it in hand, the group structure of $K_1(A)$ and its commutativity follow at once.

Theorem 7.1.4: $K_1(A)$ is an Abelian Group

Let A be a unital C^*-algebra. Then $K_1(A) = \pi_0(\mathcal{U}_\infty(A))$, with the operation of definition 7.1.1, is an abelian group. For unitaries u, v (placed in a common $\mathcal{U}_n(A)$ after stabilizing),

$$[u]_1 + [v]_1 = [uv]_1, \qquad -[u]_1 = [u^*]_1, \qquad [1]_1 = 0$$

and $[u]_1 = 0$ if and only if $u \oplus 1_m \in \mathcal{U}_{n+m}(A)_0$ for some $m \geq 0$.

***Proof*:**

Step 1: Well-definedness of the operation. The block sum $[u]_1 + [v]_1 = [u \oplus v]_1$ does not depend on representatives: if $u' = u_0 u$ with $u_0 \in \mathcal{U}_n(A)_0$ joined to 1 by a path $p(t)$, then $u' \oplus v$ is joined to $u \oplus v$ by $t \mapsto (p(t) \oplus 1)(u \oplus v)$, so $[u' \oplus v]_1 = [u \oplus v]_1$, and symmetrically in v. It is also independent of the matrix size into which u and v are placed, since enlarging u to $u \oplus 1$ does not change its class (the inclusion $\mathcal{U}_n \hookrightarrow \mathcal{U}_{n+1}$ is built into $\mathcal{U}_\infty$), and similarly for v. Thus $+$ is a well-defined binary operation on $K_1(A)$.

Step 2: The operation equals the group product. For $u, v \in \mathcal{U}_n(A)$, lemma 7.1.3(2) gives a path in $\mathcal{U}_{2n}(A)$ from $u \oplus v$ to $(uv) \oplus 1$. Hence in $K_1(A)$,

$$[u]_1 + [v]_1 = [u \oplus v]_1 = [(uv) \oplus 1]_1 = [uv]_1$$

So the induced operation on path components coincides with the (componentwise) group multiplication of $\mathcal{U}_\infty(A)$.

Step 3: Group axioms. Associativity and the identity element are inherited from the group multiplication of $\mathcal{U}_\infty(A)$ via Step 2: $[1]_1$ is the identity because $1 \cdot u = u$, and $([u]_1 + [v]_1) + [w]_1 = [uvw]_1 = [u]_1 + ([v]_1 + [w]_1)$. For inverses, $[u]_1 + [u^*]_1 = [uu^*]_1 = [1]_1 = 0$, so $-[u]_1 = [u^*]_1$. Therefore $K_1(A)$ is a group, and $\mathcal{U}_\infty(A)_0$ is exactly the kernel of $\mathcal{U}_\infty(A) \to K_1(A)$, hence a normal subgroup, confirming the quotient description in definition 7.1.1.

Step 4: Commutativity. By Step 2 and lemma 7.1.3(2) applied with the roles of u, v exchanged,

$$[u]_1 + [v]_1 = [uv]_1 = [u \oplus v]_1 \quad \text{and} \quad [v]_1 + [u]_1 = [vu]_1 = [v \oplus u]_1$$

But $u \oplus v$ and $v \oplus u$ are conjugate in $\mathcal{U}_{2n}(A)$ by the swap unitary $\Sigma = R_{\pi/2}$ of lemma 7.1.3, and $\Sigma \in \mathcal{U}_{2n}(A)_0$ via the path $\theta \mapsto R_\theta$. Writing $\Sigma = p(1)$ for that path with $p(0) = 1$, the map $t \mapsto p(t)(u \oplus v)p(t)^*$ is a path of unitaries from $u \oplus v$ to $\Sigma(u \oplus v)\Sigma^* = v \oplus u$. Hence $[u \oplus v]_1 = [v \oplus u]_1$, giving $[u]_1 + [v]_1 = [v]_1 + [u]_1$.

Step 5: The triviality criterion. By definition $[u]_1 = 0$ means u lies in the identity component of $\mathcal{U}_\infty(A)$, which is the union of the $\mathcal{U}_{n+m}(A)_0$; a path in $\mathcal{U}_\infty(A)$ has compact image, hence lands in some $\mathcal{U}_{n+m}(A)$, so $[u]_1 = 0$ if and only if $u \oplus 1_m \in \mathcal{U}_{n+m}(A)_0$ for some m, as we sought to show.

The theorem completes the foundational picture: $K_1(A)$ is a genuine abelian group whose addition is realized either by the block sum (geometrically natural) or by matrix multiplication (algebraically natural), the two being identified by the Whitehead lemma. The commutativity is forced not by any commutativity in A but by the freedom to swap blocks in a large enough matrix algebra, exactly the Eckmann-Hilton phenomenon: two compatible unital binary operations on a set must agree and be commutative. Here the block sum and the matrix product are the two operations, and the swap path provides the interchange.

The first concrete computation grounds the definition: the simplest unital C^*-algebra has trivial K_1.

Example 7.1.5: $K_1(\mathbb{C}) = 0$

Take $A = \mathbb{C}$. Then $M_n(\mathbb{C})$ is the algebra of $n \times n$ complex matrices, and $\mathcal{U}_n(\mathbb{C}) = U(n)$ is the classical unitary group. Every unitary matrix $u \in U(n)$ has the form $u = e^{ih}$ for a self-adjoint h: diagonalize $u = w\text{diag}(e^{i\theta_1}, \dots, e^{i\theta_n})w^*$ with $w \in U(n)$ and real θ_j (by the spectral theorem for normal matrices), and set $h = w\text{diag}(\theta_1, \dots, \theta_n)w^*$, which is self-adjoint with $e^{ih} = u$. By lemma 7.1.2, the path $t \mapsto e^{ith}$ joins 1 to u inside $U(n)$. Hence $U(n)$ is path-connected for every n, so each $\mathcal{U}_n(\mathbb{C})$ has a single path component, and therefore

$$K_1(\mathbb{C}) = \pi_0(\mathcal{U}_\infty(\mathbb{C})) = \pi_0\left(\varinjlim_n U(n)\right) = 0$$

The connectivity also follows from $\det : U(n) \to \mathbb{T}$ being a surjective fibration with connected fiber $SU(n)$ and connected base $\mathbb{T}$. This matches the topological side: $\mathbb{C} = C(\mathrm{pt})$, and $K^1(\mathrm{pt}) = \widetilde{K}^0(\Sigma\,\mathrm{pt}) = \widetilde{K}^0(\mathrm{pt}) = 0$.

The computation $K_1(\mathbb{C}) = 0$ generalizes immediately to matrix algebras and, by a stabilization argument, places K_1 on the same footing as K_0 with respect to functoriality and Morita-type invariance. The next results record these structural properties before the central commutative example $K_1(C(S^1)) = \mathbb{Z}$.

Theorem 7.1.6: Functoriality of K_1

Let $\varphi : A \to B$ be a unital *-homomorphism of unital C^*-algebras. Then φ induces a group homomorphism

$$K_1(\varphi) = \varphi_* : K_1(A) \to K_1(B), \qquad \varphi_*[u]_1 = [\varphi_n(u)]_1$$

for $u \in \mathcal{U}_n(A)$, where $\varphi_n : M_n(A) \to M_n(B)$ is the entrywise application of φ. The assignment $A \mapsto K_1(A)$, $\varphi \mapsto \varphi_*$ is a covariant functor from unital C^*-algebras to abelian groups: $(\mathrm{id}_A)_* = \mathrm{id}_{K_1(A)}$ and $(\psi\varphi)_* = \psi_*\varphi_*$. Moreover K_1 is homotopy invariant: if $\varphi_0, \varphi_1 : A \to B$ are homotopic through a path $t \mapsto \varphi_t$ of unital *-homomorphisms (continuous in the sense that $t \mapsto \varphi_t(a)$ is norm-continuous for each $a \in A$), then $(\varphi_0)_* = (\varphi_1)_*$.

***Proof*:**
Step 1: φ_n preserves unitaries and is continuous. A unital *-homomorphism satisfies $\varphi(a^*) = \varphi(a)^*$ and $\varphi(1_A) = 1_B$, so $\varphi_n(u)^*\varphi_n(u) = \varphi_n(u^*u) = \varphi_n(1_n) = 1_n$ and likewise $\varphi_n(u)\varphi_n(u)^* = 1_n$; thus φ_n maps $\mathcal{U}_n(A)$ into $\mathcal{U}_n(B)$. Every *-homomorphism between C^*-algebras is norm-decreasing, hence continuous (this is a standard consequence of the C^*-identity, theorem 1.3.3), so φ_n is continuous and the maps φ_n are compatible with the block-diagonal inclusions, assembling to a continuous group homomorphism $\varphi_\infty : \mathcal{U}_\infty(A) \to \mathcal{U}_\infty(B)$.

Step 2: Induced map on path components. A continuous group homomorphism sends path components to path components: if u, u' are joined by a path $p(t)$ in $\mathcal{U}_\infty(A)$, then $\varphi_\infty \circ p$ joins $\varphi_\infty(u)$ to $\varphi_\infty(u')$ in $\mathcal{U}_\infty(B)$. Hence $\varphi_*[u]_1 := [\varphi_\infty(u)]_1$ is well-defined, and it is a homomorphism because $\varphi_\infty(u \oplus v) = \varphi_\infty(u) \oplus \varphi_\infty(v)$. Functoriality $(\mathrm{id})_* = \mathrm{id}$ and $(\psi\varphi)_* = \psi_*\varphi_*$ is immediate from $(\mathrm{id})_\infty = \mathrm{id}$ and $(\psi\varphi)_\infty = \psi_\infty\varphi_\infty$.

Step 3: Homotopy invariance. Let $u \in \mathcal{U}_n(A)$ and let $t \mapsto \varphi_t$ be a homotopy. The map $t \mapsto (\varphi_t)_n(u) \in \mathcal{U}_n(B)$ is continuous: each entry $((\varphi_t)_n(u))_{ij} = \varphi_t(u_{ij})$ is norm-continuous in t by hypothesis, and continuity entrywise is continuity in $M_n(B)$ (recall the matrix-algebra norm controls and is controlled by the entry norms). Each $(\varphi_t)_n(u)$ is unitary by Step 1 applied to φ_t. Thus $t \mapsto (\varphi_t)_n(u)$ is a path of unitaries in $\mathcal{U}_n(B)$ from $(\varphi_0)_n(u)$ to $(\varphi_1)_n(u)$, so $[(\varphi_0)_n(u)]_1 = [(\varphi_1)_n(u)]_1$, that is $(\varphi_0)_*[u]_1 = (\varphi_1)_*[u]_1$, completing the proof.

Functoriality and homotopy invariance together make K_1 a homotopy-theoretic invariant of C^*-algebras, exactly parallel to K_0. Homotopy invariance has an immediate consequence: a C^*-algebra that is contractible (homotopy equivalent to 0, or to $\mathbb{C}$ in the unital setting) has K_1 equal to that of the model. The next property, stability, says K_1 cannot see the difference between A and its matrix amplifications, which is what justifies having taken a colimit over all matrix sizes in the definition.

Theorem 7.1.7: Stability of K_1

Let A be a unital C^*-algebra and $m \geq 1$. The block-diagonal embedding $\lambda : A \to M_m(A)$, $a \mapsto a \oplus 0_{m-1} = \operatorname{diag}(a, 0, \dots, 0)$, sending 1_A to a rank-one projection, induces an isomorphism

$$K_1(M_m(A)) \cong K_1(A)$$

More precisely, the inclusion $M_n(A) \hookrightarrow M_n(M_m(A)) = M_{nm}(A)$ identifies $\mathcal{U}_\infty(M_m(A))$ with a cofinal part of $\mathcal{U}_\infty(A)$, inducing the isomorphism on π_0.

***Proof*:**
There is a *-isomorphism $M_n(M_m(A)) \cong M_{nm}(A)$ for every n, given by regarding an $n \times n$ matrix of $m \times m$ matrices over A as a single $nm \times nm$ matrix over A; this is the standard "forgetting the block partition" identification, and it is a unital *-isomorphism. These identifications are compatible with the block-diagonal inclusions defining the two infinite unitary groups: increasing n by one for $M_m(A)$ increases the underlying size by m for A. Hence the system $\{\mathcal{U}_{nm}(A) \mid n \geq 1\}$, a cofinal subsystem of $\{\mathcal{U}_k(A) \mid k \geq 1\}$ (every k is dominated by some nm), has the same colimit:

$$\mathcal{U}_\infty(M_m(A)) = \varinjlim_n \mathcal{U}_n(M_m(A)) = \varinjlim_n \mathcal{U}_{nm}(A) = \varinjlim_k \mathcal{U}_k(A) = \mathcal{U}_\infty(A)$$

where the middle equality uses that a cofinal subsystem of a directed system has the same colimit, and each map is a homeomorphism onto its image. Taking π_0, which commutes with the colimit because path components of a colimit of inclusions are computed levelwise, yields $K_1(M_m(A)) \cong K_1(A)$. The embedding λ induces this isomorphism since on the level of $\mathcal{U}_n$ it is exactly the inclusion $\mathcal{U}_n(A) = \mathcal{U}_n(\lambda(A)) \hookrightarrow \mathcal{U}_n(M_m(A))$ followed by the identification above, as we sought to show.

Stability instantly furnishes a family of examples: every full matrix algebra over $\mathbb{C}$ has trivial K_1.

Example 7.1.8: $K_1(M_n(\mathbb{C})) = 0$
Apply theorem 7.1.7 with $A = \mathbb{C}$ and $m = n$: $K_1(M_n(\mathbb{C})) \cong K_1(\mathbb{C}) = 0$ by example 7.1.5. Directly, $\mathcal{U}_k(M_n(\mathbb{C})) = U(kn)$ is path-connected for all k (the spectral-theorem argument of example 7.1.5), so $\pi_0(\mathcal{U}_\infty(M_n(\mathbb{C}))) = 0$. The same reasoning gives K_1 of any finite-dimensional C^*-algebra: such an algebra is a finite direct sum $\bigoplus_i M_{n_i}(\mathbb{C})$ by the Artin-Wedderburn classification ([4, Artin-Wedderburn For k-Algebras], with the C^*-specific bridge that a finite-dimensional C^*-algebra is semisimple with every division component $\mathbb{C}$), and K_1 of a direct sum is the direct sum of the K_1's (the unitary group of a product is the product of unitary groups, and π_0 of a product of these is the product of π_0's), so $K_1(\bigoplus_i M_{n_i}(\mathbb{C})) = \bigoplus_i K_1(M_{n_i}(\mathbb{C})) = 0$.

The two examples so far have $K_1 = 0$ because their unitary groups are connected. The phenomenon that makes K_1 nontrivial is a disconnected unitary group, and the prototype is the circle algebra $C(S^1)$, where the obstruction to connectivity is the winding number. This is the bridge to the topological group $K^1(S^1)$, and the computation is recorded in full.

Example 7.1.9: $K_1(C(S^1)) \cong \mathbb{Z}$ via Winding Number

Let $A = C(S^1)$, the C^*-algebra of continuous complex-valued functions on the circle $S^1 = \mathbb{T}$, with pointwise operations, complex conjugation as involution, and the supremum norm. A unitary $u \in \mathcal{U}_1(A) = \mathcal{U}(C(S^1))$ is a continuous function $u : S^1 \to \mathbb{C}$ with $u(z)\overline{u(z)} = 1$ for all z, that is a continuous map $u : S^1 \to \mathbb{T}$. The set of homotopy classes of such maps is computed in [8]: a continuous map $S^1 \to \mathbb{T} \simeq S^1$ has a well-defined *winding number* (degree)

$$w(u) = \frac{1}{2\pi i} \oint_{S^1} \frac{u'(z)}{u(z)}\, dz \in \mathbb{Z}$$

when u is smooth, and in general by its homotopy class, and two such maps are homotopic if and only if they have equal winding number, giving $\pi_0(\mathcal{U}_1(A)) \cong [S^1, \mathbb{T}] \cong \pi_1(S^1) \cong \mathbb{Z}$, generated by the identity function $z \mapsto z$. The claim is that stabilizing to $\mathcal{U}_\infty$ does not change this.

Stabilization does not enlarge the invariant. For $u \in \mathcal{U}_n(C(S^1))$, the determinant $\det : \mathcal{U}_n(C(S^1)) \to \mathcal{U}(C(S^1))$, $u \mapsto \det u$ (the pointwise determinant, a continuous $S^1 \to \mathbb{T}$ since $u(z) \in U(n)$ and $\det : U(n) \to \mathbb{T}$ is continuous), is a group homomorphism compatible with block sums: $\det(u \oplus v) = \det(u)\det(v)$, and $\det(u \oplus 1) = \det(u)$. Composing with the winding number gives a homomorphism

$$W : \mathcal{U}_\infty(C(S^1)) \to \mathbb{Z}, \qquad W(u) = w(\det u)$$

which is continuous (the winding number is locally constant on the space of maps $S^1 \to \mathbb{T}$), hence factors through π_0 to give $\overline{W} : K_1(C(S^1)) \to \mathbb{Z}$. It is surjective because $z \mapsto z$ in $\mathcal{U}_1$ has $W = 1$. For injectivity, suppose $W(u) = 0$ for $u \in \mathcal{U}_n(C(S^1))$, viewed as a map $u : S^1 \to U(n)$. Then $\det u : S^1 \to \mathbb{T}$ has winding number 0, so it lifts along the covering $\mathbb{R} \to \mathbb{T}$, $\theta \mapsto e^{i\theta}$, to a continuous $\theta : S^1 \to \mathbb{R}$ with $\det u = e^{i\theta}$. Set $d = \operatorname{diag}(\det u, 1, \ldots, 1) \in \mathcal{U}_n(C(S^1))$; the map $t \mapsto \operatorname{diag}(e^{it\theta}, 1, \ldots, 1)$ joins d to 1, so $[d]_1 = 0$. The product $w = u\,d^{-1}$ takes values in $SU(n)$, since $\det w = \det u \cdot (\det u)^{-1} = 1$ pointwise. Because $SU(n)$ is simply connected, $\pi_1(SU(n)) = 0$, so the map $w : S^1 \to SU(n)$ is null-homotopic, hence connected to the constant map 1 through a path of $SU(n)$-valued unitaries, giving $[w]_1 = 0$. Therefore $[u]_1 = [w]_1 + [d]_1 = 0$, so $\overline{W}$ is injective. Hence

$$K_1(C(S^1)) \cong \mathbb{Z}$$

generated by the class of the identity function $u(z) = z$, the winding number being the isomorphism.

The winding-number computation is the operator-theoretic form of a topological fact, and it is precisely the bridge to topological K-theory promised in the section opening. The identification is recorded as the central theorem of the section.

Theorem 7.1.10: K_1 of a Commutative C^*-Algebra

Let X be a compact Hausdorff space and $A = C(X)$ the C^*-algebra of continuous complex-valued functions on X. Then there is a natural isomorphism

$$K_1(C(X)) \cong K^1(X) := \widetilde{K}^0(\Sigma X)$$

where ΣX is the (reduced) suspension of X and $K^1(X)$ is the topological K^1 of [10, Negative K-Groups, Convention: Periodic Grading]. Under this isomorphism the class $[u]_1$ of a unitary $u \in \mathcal{U}_n(C(X))$ corresponds to the homotopy class of the map $u : X \to U(n) \subset U$, viewed as an element of $[X, U] = \widetilde{K}^0(\Sigma X)$.

***Proof*:**
Step 1: Unitaries are maps into the unitary group. A unitary $u \in \mathcal{U}_n(C(X))$ is exactly a continuous function $u : X \to M_n(\mathbb{C})$ with $u(x) \in U(n)$ for every x, that is a continuous map $u : X \to U(n)$. This is because the C^*-algebra $M_n(C(X)) \cong C(X, M_n(\mathbb{C}))$ is the algebra of continuous $M_n(\mathbb{C})$-valued functions (matrix entries are continuous scalar functions), and the unitary condition $u^*u = uu^* = 1$ holds in $C(X, M_n(\mathbb{C}))$ if and only if it holds pointwise. Block sum of unitaries corresponds to the map $x \mapsto u(x) \oplus v(x)$ into $U(n+m)$, and the stabilization $\mathcal{U}_n \hookrightarrow \mathcal{U}_{n+1}$ corresponds to the standard inclusion $U(n) \hookrightarrow U(n+1)$.

Step 2: Path components are homotopy classes into U. Two unitaries $u, u' \in \mathcal{U}_n(C(X))$ lie in the same path component of $\mathcal{U}_n(C(X))$ if and only if they are homotopic as maps $X \to U(n)$: a path of unitaries $t \mapsto p(t) \in \mathcal{U}_n(C(X))$ is precisely a homotopy $(x,t) \mapsto p(t)(x) \in U(n)$, using that continuity of $t \mapsto p(t)$ in the norm of $C(X, M_n(\mathbb{C}))$ is uniform continuity in x, hence joint continuity of $(x,t) \mapsto p(t)(x)$, and conversely. Passing to the colimit over n,

$$K_1(C(X)) = \pi_0\left(\varinjlim_n \mathcal{U}_n(C(X))\right) = \varinjlim_n [X, U(n)] = [X, U]$$

where $U = \varinjlim_n U(n)$ and the colimit of homotopy-class sets is taken along the stabilization inclusions; the second equality uses that a path in $\varinjlim_n \mathcal{U}_n(C(X))$ has image in some $\mathcal{U}_n(C(X))$ by compactness of $[0,1]$, so π_0 commutes with the colimit.

Step 3: Free maps as based maps off a disjoint basepoint. Step 2 produced $K_1(C(X)) = [X, U]$ with *free* (not basepoint-preserving) homotopy classes, because a unitary $u \in \mathcal{U}_n(C(X))$ is an arbitrary continuous map $X \to U(n)$ with no constraint at any chosen point of X. To match it to the topological K^1, which is built from *based* classes, adjoin a disjoint basepoint. A continuous map $X \to U$ is the same datum as a based continuous map $X_+ = X \sqcup \{*\} \to U$ sending the added point $*$ to the identity $1 \in U$: extending a free map by $* \mapsto 1$ gives a based map, and restricting a based map $X_+ \to U$ to X gives back a free map, and the two operations are mutually inverse and respect homotopies (a free homotopy of maps $X \to U$ extends to a based homotopy of maps $X_+ \to U$ by sending $*$ to 1 throughout). Hence

$$K_1(C(X)) = [X, U] = [X_+, U]_*$$

an identity of *based* homotopy-class sets that uses no connectedness or CW hypothesis on X.

Step 4: The based classes are $K^{-1}(X)$. By the representability and loop-space picture of [10, Representability of K^0], for any pointed compact space Y one has $\widetilde{K}^0(\Sigma Y) \cong [\Sigma Y, BU \times$

$\mathbb{Z}]_* \cong [Y, \Omega(BU \times \mathbb{Z})]_*$, the second isomorphism being the loop-suspension adjunction. Now $\Omega(BU \times \mathbb{Z}) \simeq \Omega(BU) \simeq U$, since $\mathbb{Z}$ is discrete (so $\Omega\mathbb{Z}$ is a point) and the loop space of the classifying space BU is the group U itself (the classifying-space property, established in [8]). Applying this with the pointed space $Y = X_+$ gives $[X_+, U]_* \cong \widetilde{K}^0(\Sigma(X_+))$, which by [10, Negative K-Groups] is exactly the unreduced negative group $K^{-1}(X) = \widetilde{K}^0(\Sigma(X_+))$. Combining with Step 3,

$$K_1(C(X)) = [X_+, U]_* = \widetilde{K}^0(\Sigma(X_+)) = K^{-1}(X) = K^1(X)$$

the last equality being the periodic-grading convention $K^{-1} = K^1$ of [10, Convention: Periodic Grading].

Step 5: Reconciling with $\widetilde{K}^0(\Sigma X)$. It remains to identify $\widetilde{K}^0(\Sigma(X_+))$ with the reduced form $\widetilde{K}^0(\Sigma X)$ recorded in the theorem statement. The suspension of a space with a disjoint basepoint splits off a circle: there is a homotopy equivalence

$$\Sigma(X_+) \simeq \Sigma X \vee S^1$$

because $X_+ = X \sqcup \{*\}$ and suspending the disjoint union of X with a point yields the wedge of ΣX with $\Sigma S^0 = S^1$ (the suspension coordinate over the added point sweeps out a circle wedged on at the basepoint). Reduced K^0 is additive on wedges of compact pointed spaces, $\widetilde{K}^0(A \vee B) \cong \widetilde{K}^0(A) \oplus \widetilde{K}^0(B)$ (the wedge axiom for the reduced theory, [10, Reduced K-Theory of a Wedge]), so

$$\widetilde{K}^0(\Sigma(X_+)) \cong \widetilde{K}^0(\Sigma X) \oplus \widetilde{K}^0(S^1)$$

The second summand vanishes: S^1 is an odd sphere, and $\widetilde{K}^0(S^1) = K^{-1}(\mathrm{pt}) = 0$ by the sphere computation ([10, The K-Theory of Spheres]), since every complex vector bundle over S^1 is stably trivial via the clutching correspondence ([10, Clutching Function Vector Bundle Correspondence]). Therefore $\widetilde{K}^0(\Sigma(X_+)) \cong \widetilde{K}^0(\Sigma X)$, and chaining with Step 4,

$$K_1(C(X)) = \widetilde{K}^0(\Sigma(X_+)) = \widetilde{K}^0(\Sigma X) = K^1(X)$$

exactly the topological $K^1(X)$ of theorem 7.1.10's statement, with no connectedness assumption imposed anywhere.

Step 6: Naturality. A continuous map $f : Y \to X$ induces the unital *-homomorphism $f^* : C(X) \to C(Y)$, $g \mapsto g \circ f$, hence $K_1(f^*) : K_1(C(X)) \to K_1(C(Y))$; under the identification of Steps 1-5 this sends the class of $u : X \to U$ to the class of $u \circ f : Y \to U$, which is exactly the topological map $K^1(X) \to K^1(Y)$ induced by f. So the isomorphism is natural in X, completing the proof.

The theorem connects the two halves of this book. The operator-theoretic $K_1(C(X))$, built from path components of unitary groups of matrix algebras, is the topological $K^1(X)$, built from vector bundles over the suspension; the winding number of example 7.1.9 is the special case $X = S^1$, where $[S^1, U] = \pi_1(U) = \mathbb{Z}$ by [10, Bott Periodicity]. The contravariant functor $X \mapsto C(X)$ of Gelfand-Naimark duality (theorem 4.3.3) thus intertwines topological and operator K^1, just as it does for K^0. With the commutative case settled, two further computations complete the catalogue of basic examples: the non-unital algebra of compact operators, and the reduction of K_1 for non-unital algebras in general.

For a non-unital C^*-algebra A there is no unitary group to speak of (there is no 1), so K_1 is defined by passing to the unitization $\tilde{A}$ of proposition 1.4.5 and removing the contribution of the scalars

$\mathbb{C} = \tilde{A}/A$.

Definition 7.1.11: K_1 for Non-Unital C^*-Algebras

Let A be a (not necessarily unital) C^*-algebra with unitization $\tilde{A}$ and quotient map $\pi : \tilde{A} \to \tilde{A}/A \cong \mathbb{C}$. The K_1-*group* of A is

$$K_1(A) = \ker\big(\pi_* : K_1(\tilde{A}) \to K_1(\mathbb{C})\big) = K_1(\tilde{A})$$

where the second equality holds because $K_1(\mathbb{C}) = 0$ by example 7.1.5, so the kernel is all of $K_1(\tilde{A})$. For unital A this agrees with definition 7.1.1 up to the natural splitting $\tilde{A} \cong A \oplus \mathbb{C}$.

Because $K_1(\mathbb{C}) = 0$, the non-unital definition reduces to $K_1(A) = K_1(\tilde{A})$, which is one reason K_1 is technically smoother than K_0 (where the analogous kernel is genuinely a proper subgroup and the scalar correction is essential, as the non-exactness noted after definition 6.2.1 shows). The last example computes K_1 of the compact operators, the non-unital algebra most central to operator K-theory.

Example 7.1.12: $K_1(\mathcal{K}) = 0$ for the Compact Operators

Let $\mathcal{K} = \mathcal{K}(H)$ be the C^*-algebra of compact operators on a separable infinite-dimensional Hilbert space H. It is non-unital, so $K_1(\mathcal{K}) = K_1(\tilde{\mathcal{K}})$ by definition 7.1.11. The computation rests on continuity of K-theory under inductive limits.

Step 1: $\mathcal{K}$ as an inductive limit. The finite-rank operators are dense in $\mathcal{K}$, and choosing an orthonormal basis of H identifies the rank-$\leq n$ part with $M_n(\mathbb{C})$ via the top-left corner inclusions $M_n(\mathbb{C}) \hookrightarrow M_{n+1}(\mathbb{C})$. Thus $\mathcal{K}$ is the inductive limit (norm closure of the increasing union) of the C^*-algebras $M_n(\mathbb{C})$,

$$\mathcal{K} = \overline{\bigcup_n M_n(\mathbb{C})} = \varinjlim_n M_n(\mathbb{C})$$

in the category of C^*-algebras.

Step 2: K_1 commutes with the limit. For any inductive limit $A = \varinjlim_n A_n$ of C^*-algebras, $K_1(A) = \varinjlim_n K_1(A_n)$. The reason is that a unitary $u \in \mathcal{U}_k(\tilde{A})$ and a path of unitaries connecting two such can each be approximated, to within the gap that keeps unitaries unitary under lemma 7.1.2, by elements with entries in some finite stage $\tilde{A}_N$; so every K_1-class and every relation among classes already occurs at a finite stage. (This continuity is the operator-K-theory analogue of the algebraic limit lemma of [2]; a complete treatment is in the operator-algebra references.) Applying it with $A_n = M_n(\mathbb{C})$, and using $K_1(M_n(\mathbb{C})) = 0$ from example 7.1.8,

$$K_1(\mathcal{K}) = \varinjlim_n K_1(M_n(\mathbb{C})) = \varinjlim_n 0 = 0$$

matching the topological computation $K^1(\mathrm{pt}) = 0$ under the slogan that $\mathcal{K}$ is "K-theoretically a point."

The catalogue $K_1(\mathbb{C}) = K_1(M_n(\mathbb{C})) = K_1(\mathcal{K}) = 0$ and $K_1(C(S^1)) = \mathbb{Z}$ exhibits the same pattern as topological K-theory: the invariant is trivial on the building blocks (points, matrices, compacts) and detects the circle through its winding. What remains is to make precise the relationship between this operator-theoretic K_1 and the algebraic Whitehead group of [2], since the two are defined by superficially similar quotients of general linear groups but measure genuinely different things.

The purely algebraic K_1 of a ring R, defined in [2], is the abelianization of the stable general linear group,

$$K_1^{\mathrm{alg}}(R) = \mathrm{GL}(R)/E(R) = \mathrm{GL}(R)/[\mathrm{GL}(R), \mathrm{GL}(R)]$$

where $\mathrm{GL}(R) = \varinjlim_n \mathrm{GL}_n(R)$ and $E(R)$ is the subgroup generated by elementary matrices, equal to the commutator subgroup by Whitehead's lemma (the algebraic counterpart of lemma 7.1.3). The operator-theoretic K_1 of a unital C^*-algebra A is instead

$$K_1(A) = \mathrm{GL}_\infty(A)/\mathrm{GL}_\infty(A)_0 = \mathcal{U}_\infty(A)/\mathcal{U}_\infty(A)_0$$

the quotient by the *connected component* of the identity, where the second equality uses the polar-decomposition retraction of invertibles onto unitaries (every invertible $a = u|a|$ with $|a| = (a^*a)^{1/2}$ positive invertible, and $t \mapsto u|a|^t$ connects a to its unitary part u). The two quotients differ in what they collapse: K_1^{alg} collapses the algebraically distinguished subgroup $E(R)$, while $K_1(A)$ collapses the topologically distinguished subgroup $\mathrm{GL}_\infty(A)_0$.

7.1.13 Remark: Algebraic versus Operator K_1 For a unital C^*-algebra A there is always a natural surjection

$$K_1^{\mathrm{alg}}(A) = \mathrm{GL}_\infty(A)/E(A) \twoheadrightarrow \mathrm{GL}_\infty(A)/\mathrm{GL}_\infty(A)_0 = K_1(A)$$

because $E(A) \subseteq \mathrm{GL}_\infty(A)_0$: every elementary matrix $1 + aE_{ij}$ $(i \neq j)$ is connected to 1 through the path $t \mapsto 1 + taE_{ij}$ of invertibles, so the elementary subgroup lies in the identity component. The surjection is in general *not* injective. The topological component $\mathrm{GL}_\infty(A)_0$ can be strictly larger than the algebraic commutator subgroup $E(A)$, because connectivity uses limits (paths, hence the norm topology) that the purely algebraic construction cannot see. Thus the operator-theoretic K_1 is a quotient of, and generally distinct from, the algebraic Whitehead group: it is the *topological* K_1, sensitive to the analytic structure of A.

The distinction is sharpest in two regimes. For a field $\mathbb{C}$ (viewed as a C^*-algebra) the algebraic $K_1^{\mathrm{alg}}(\mathbb{C}) = \mathbb{C}^\times$ by [2], since $\mathrm{GL}(\mathbb{C})/E(\mathbb{C}) \cong \mathbb{C}^\times$ via the determinant; but the operator $K_1(\mathbb{C}) = 0$ by example 7.1.5, because $\mathbb{C}^\times = \mathrm{GL}_1(\mathbb{C})$ is connected, so the entire group $\mathbb{C}^\times$ collapses into the identity component. Here the surjection $\mathbb{C}^\times = K_1^{\mathrm{alg}}(\mathbb{C}) \twoheadrightarrow K_1(\mathbb{C}) = 0$ has enormous kernel, the whole connected group $\mathbb{C}^\times$. This is the precise sense in which the two theories diverge: the algebraic theory remembers the determinant in $\mathbb{C}^\times$, the operator theory forgets everything that can be continuously deformed away, and over $\mathbb{C}$ the multiplicative group is entirely deformable. The operator-theoretic K_1 is the right invariant for topology and index theory precisely because it discards this analytically inert information and retains only the discrete homotopical obstruction, which for $C(X)$ is the topological $K^1(X)$ of theorem 7.1.10.

Exercise 7.1.1

1. Let A be a unital C^*-algebra and $u \in \mathcal{U}_n(A)$ a unitary with $\|u - 1\| < 2$. Show directly, without invoking lemma 7.1.2, that $[u]_1 = 0$ in $K_1(A)$ by exhibiting an explicit path from u to 1. (Hint: the segment $t \mapsto (1 - t) + tu$ stays invertible; relate it to the unitary part.)
2. Prove that for unital C^*-algebras A, B,

$$K_1(A \oplus B) \cong K_1(A) \oplus K_1(B)$$

(Hint: identify $\mathcal{U}_n(A \oplus B)$ with $\mathcal{U}_n(A) \times \mathcal{U}_n(B)$ and take π_0.)

3. Compute $K_1(C(\text{pt}))$ and $K_1(C([0,1]))$, and explain using homotopy invariance (theorem 7.1.6) why they are equal. What is $K_1(C(D^2))$ for the closed disk D^2?

4. Using theorem 7.1.10 and the suspension description $K^1(X) = \widetilde{K}^0(\Sigma X)$, compute $K_1(C(S^2))$. (Hint: $[S^2, U] = \pi_2(U)$, computed by [10, Bott Periodicity].)

5. Let $u, v \in \mathcal{U}_n(A)$. Show that the commutator uvu^*v^* satisfies $[uvu^*v^*]_1 = 0$ in $K_1(A)$, directly from theorem 7.1.4, and interpret this as the statement that $K_1(A)$ is the abelianized, stabilized unitary group.

6. Explain precisely why $K_1^{\text{alg}}(\mathbb{C}) = \mathbb{C}^\times$ but $K_1(\mathbb{C}) = 0$, identifying the kernel of the natural surjection $K_1^{\text{alg}}(\mathbb{C}) \twoheadrightarrow K_1(\mathbb{C})$ and the role of the norm topology.

8

The Six-Term Exact Sequence and Bott Periodicity

8.1 The Six-Term Exact Sequence and Bott Periodicity

The two functors K_0 and K_1 built in the previous sections are not independent invariants. They are the two halves of a single periodic theory, glued together by the connecting maps of a short exact sequence of C^*-algebras. This section makes that gluing precise. We first introduce the suspension $SA = C_0(\mathbb{R}) \otimes A$, the C^*-algebraic analogue of the topological reduced suspension, and prove the dimension-shift isomorphism $K_1(A) \cong K_0(SA)$, which forces us to read K_1 as "K_{-1}". We then take a short exact sequence $0 \to I \to A \to A/I \to 0$ and construct two connecting homomorphisms: the index map $\partial_1 \colon K_1(A/I) \to K_0(I)$, built by lifting a unitary to a partial isometry, and the exponential map $\partial_0 \colon K_0(A/I) \to K_1(I)$, built by lifting a projection to a self-adjoint element and exponentiating. Assembling the half-exact sequences for K_0 and K_1 with these connecting maps yields the six-term cyclic exact sequence, the computational engine of the theory. Finally Bott periodicity, $K_0(A) \cong K_0(S^2 A)$, closes the cycle into a genuinely 2-periodic theory matching topological K-theory, and the Toeplitz extension $0 \to \mathcal{K} \to \mathcal{T} \to C(S^1) \to 0$ shows the whole apparatus computing the Fredholm index of a Toeplitz operator as a winding number.

Throughout, A is a C^*-algebra (definition 1.2.1), not assumed unital, $\tilde{A}$ is its unitization (proposition 1.4.5), and K_0 is as in definition 6.2.1, extended to the non-unital case by $K_0(A) = \ker\big(K_0(\tilde{A}) \to K_0(\mathbb{C})\big)$. We write $\mathcal{U}_n(B)$ for the unitary group of $M_n(B)$ and $\mathcal{U}(B) = \bigcup_n \mathcal{U}_n(B)$ under the embeddings $u \mapsto \operatorname{diag}(u, 1)$; the group $K_1(B)$ is $\mathcal{U}(\tilde{B})$ modulo the connected component of the identity (definition 7.1.1).

The path from K_0 to K_1 in topology is the suspension: $\widetilde{K}(SX) \cong \widetilde{K}^{-1}(X)$. To import this into the operator setting one needs the algebra whose "space" is the suspension. For a locally compact

Hausdorff space X the suspension SX deletes two cone points, and continuous functions vanishing at infinity on $\mathbb{R} \times X$ are exactly $C_0(\mathbb{R}) \otimes C_0(X)$. This is the model to abstract.

Definition 8.1.1: Suspension of a C^*-Algebra

Let A be a C^*-algebra. The *suspension* of A is the C^*-algebra

$$SA = C_0(\mathbb{R}) \otimes A \cong C_0(\mathbb{R}, A)$$

the algebra of continuous A-valued functions on $\mathbb{R}$ vanishing at infinity, with pointwise operations and the supremum norm. Equivalently, $SA \cong \{f \in C([0,1], A) : f(0) = f(1) = 0\}$, the A-valued loops based at 0. The *cone* of A is $CA = C_0((0,1], A) = \{f \in C([0,1], A) : f(0) = 0\}$.

The cone is the contractible building block: the homotopy $h_t(s) = f(ts)$ from the identity to the zero map shows CA is homotopy equivalent to 0. The suspension is the quotient $CA \to SA$ that records the missing endpoint, fitting into the *cone exact sequence*

$$0 \longrightarrow SA \longrightarrow CA \xrightarrow{\mathrm{ev}_1} A \longrightarrow 0$$

where ev_1 evaluates a function of $C_0((0,1], A)$ at the right endpoint $s = 1$. Suspension is a functor: a *-homomorphism $\varphi \colon A \to B$ induces $S\varphi = \mathrm{id}_{C_0(\mathbb{R})} \otimes \varphi$, and homotopic *-homomorphisms induce homotopic ones, so S descends to K-theory.

Theorem 8.1.2: The Dimension-Shift Isomorphism

For every C^*-algebra A there is a natural isomorphism

$$\theta_A \colon K_1(A) \xrightarrow{\sim} K_0(SA)$$

In particular $K_1(A) \cong K_0(SA)$, so K_1 is the once-suspended K_0, justifying the reading $K_1 = K_{-1}$.

***Proof*:**
The proof identifies both groups with homotopy classes of loops of invertibles and uses the clutching-type correspondence between such loops and projections over the suspension. We may assume A is unital; the non-unital case follows by applying the unital result to $\tilde{A}$ and restricting to the kernels of the augmentations, since S is exact and commutes with unitization up to the split $S\tilde{A}/SA \cong S\mathbb{C}$.

Step 1: $K_1(A)$ as based homotopy classes of unitary loops. By definition $K_1(A) = \varinjlim_n \mathcal{U}_n(A)/\mathcal{U}_n(A)_0$, where the subscript 0 denotes the identity component. Two unitaries $u, v \in \mathcal{U}_n(A)$ are equal in $K_1(A)$ if and only if $u \oplus 1_m$ and $v \oplus 1_m$ lie in the same path component of $\mathcal{U}_{n+m}(A)$ for some m. A continuous path in $\mathcal{U}_{n+m}(A)$ from w to 1 is precisely a based loop of unitaries when we close it up; concretely, given $u \in \mathcal{U}_n(A)$, choose any continuous path $w \colon [0,1] \to \mathcal{U}_{2n}(A)$ with $w(0) = \mathrm{diag}(u^*, u)$ and $w(1) = 1_{2n}$ (such a path exists because $\mathrm{diag}(u^*, u)$ is connected to 1 inside $\mathcal{U}_{2n}(A)$ via the rotation

$$R_t = \begin{pmatrix} \cos(\frac{\pi t}{2}) & -\sin(\frac{\pi t}{2}) \\ \sin(\frac{\pi t}{2}) & \cos(\frac{\pi t}{2}) \end{pmatrix} \begin{pmatrix} u & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \cos(\frac{\pi t}{2}) & \sin(\frac{\pi t}{2}) \\ -\sin(\frac{\pi t}{2}) & \cos(\frac{\pi t}{2}) \end{pmatrix} \begin{pmatrix} u^* & 0 \\ 0 & 1 \end{pmatrix}$$

which is a unitary for each t and runs from $R_0 = \mathrm{diag}(uu^*, 1) = 1_{2n}$ at $t = 0$ to $R_1 = \mathrm{diag}(u^*, u)$ at $t = 1$; reverse the parameter via $t \mapsto 1 - t$ to obtain a path from $\mathrm{diag}(u^*, u)$ to 1_{2n}. The companion element $\mathrm{diag}(u, u^*)$ is then equally connected to 1_{2n}, being the swap-conjugate $\sigma\,\mathrm{diag}(u^*, u)\,\sigma^*$ by the permutation unitary σ that exchanges the two blocks; σ lies in the identity component of $\mathcal{U}_{2n}(A)$ because the unitary group is path-connected to its block transpositions, an explicit path being $\sigma_t = \frac{1}{2}\begin{pmatrix} 1 + e^{i\pi(1-t)} & 1 - e^{i\pi(1-t)} \\ 1 - e^{i\pi(1-t)} & 1 + e^{i\pi(1-t)} \end{pmatrix}$ on each transposed pair, with $\sigma_0 = \sigma$ and $\sigma_1 = 1$).

Step 2: From unitary paths to projection loops over the suspension. Realize $SA = \{f \in C([0,1], A) : f(0) = f(1) = 0\}$, so $\widetilde{SA} = \{f \in C([0,1], \tilde{A}) : f(0) = f(1) \in \mathbb{C} \cdot 1\}$, functions whose two endpoints agree and are scalar. Write $e = \mathrm{diag}(1_n, 0) \in M_{2n}(A)$. Given a unitary path $w\colon [0,1] \to \mathcal{U}_{2n}(A)$ with $w(0) = 1_{2n}$ and with $w(1)$ commuting with e, the function

$$p_w(t) = w(t)\, e\, w(t)^* = w(t) \begin{pmatrix} 1_n & 0 \\ 0 & 0 \end{pmatrix} w(t)^*$$

is a projection in $M_{2n}(\widetilde{SA})$: it has the constant scalar value e at $t = 0$, and at $t = 1$ it equals $w(1)\, e\, w(1)^* = e$ because $w(1)$ commutes with e, so the two endpoints agree and are scalar, and p_w defines a class $[p_w] - [e] \in K_0(SA)$. The canonical such path attached to a unitary $u \in \mathcal{U}_n(A)$ is the rotation path of Step 1, built with the roles of u and u^* arranged so that it runs from $w_u(0) = 1_{2n}$ to $w_u(1) = \mathrm{diag}(u, u^*)$ (the companion element to $\mathrm{diag}(u^*, u)$, equally connected to 1_{2n} as noted in Step 1); its endpoint $\mathrm{diag}(u, u^*)$ commutes with e. Set

$$\theta_A([u]) = [p_{w_u}] - [e] \in K_0(SA)$$

A homotopy of unitaries u gives a homotopy of the paths w_u and hence of the projections p_{w_u}, so this descends to a homomorphism $\theta_A\colon K_1(A) \to K_0(SA)$; additivity follows from $p_{w_u \oplus w_{u'}} = p_{w_u} \oplus p_{w_{u'}}$ after a permutation, and the rotation of Step 1 shows the class is independent of the chosen path realizing the contraction.

Step 3: Bijectivity via an explicit inverse. It remains to show θ_A is a bijection. The argument is direct: construct an inverse $\eta_A\colon K_0(SA) \to K_1(A)$ by framing a projection loop with a continuous family of unitaries and reading off the residual unitary at the closing endpoint, then verify both composites are the identity. The only analytic input is the local unitary equivalence $z = qp + (1 - q)(1 - p)$ of lemma 6.2.3, Step 1, propagated along the interval.

Step 3a: Framing a projection loop. A class in $K_0(SA)$ is represented by $[p] - [e_0]$ where $p \in M_m(\widetilde{SA})$ is a projection and e_0 is its constant scalar endpoint value; concretely p is a norm-continuous path $t \mapsto p(t)$ of projections in $M_m(\tilde{A})$ with $p(0) = p(1) = e_0$ a scalar projection of some rank k. After conjugating the whole loop by a fixed scalar unitary (which changes neither the class $[p] - [e_0]$ nor the construction below) we may take $e_0 = \mathrm{diag}(1_k, 0_{m-k})$ in standard form. The claim is that there is a continuous path $z\colon [0,1] \to \mathcal{U}_m(\tilde{A})$ with $z(0) = 1_m$ and $z(t)\, e_0\, z(t)^* = p(t)$ for all t. Partition $[0,1]$ as $0 = t_0 < t_1 < \cdots < t_N = 1$ finely enough that $\|p(s) - p(t)\| < 1$ whenever s, t lie in a common subinterval $[t_{j-1}, t_j]$ (possible by uniform continuity of p on the compact interval). On $[t_{j-1}, t_j]$ set

$$g_j(t) = p(t)\, p(t_{j-1}) + (1 - p(t))(1 - p(t_{j-1}))$$

By the computation of lemma 6.2.3, $g_j(t) - 1 = (p(t) - p(t_{j-1}))(2p(t_{j-1}) - 1)$ has norm < 1, so $g_j(t)$ is invertible with norm-continuous polar part $\zeta_j(t) = g_j(t)\,|g_j(t)|^{-1} \in \mathcal{U}_m(\tilde{A})$ satisfying

The Six-Term Exact Sequence and Bott Periodicity

$\zeta_j(t)\,p(t_{j-1})\,\zeta_j(t)^* = p(t)$ and $\zeta_j(t_{j-1}) = 1_m$. Splice these by accumulating the completed earlier framings on the right: for $t \in [t_{j-1}, t_j]$ define

$$z(t) = \zeta_j(t)\,\zeta_{j-1}(t_{j-1})\cdots\zeta_2(t_2)\,\zeta_1(t_1)$$

Each $z(t)$ is unitary, $t \mapsto z(t)$ is continuous (the values match at each breakpoint $t = t_{j-1}$ because $\zeta_j(t_{j-1}) = 1_m$), and $z(0) = \zeta_1(0) = 1_m$. Telescoping the conjugations from the right ($\zeta_1(t_1)$ carries $e_0 = p(t_0)$ to $p(t_1)$, then $\zeta_2(t_2)$ carries $p(t_1)$ to $p(t_2)$, and so on up to $\zeta_j(t)$ carrying $p(t_{j-1})$ to $p(t)$) gives $z(t)\,e_0\,z(t)^* = p(t)$ throughout. This proves the framing claim.

Step 3b: The inverse map. With z as in Step 3a, evaluate at the closing endpoint: $z(1)\,e_0\,z(1)^* = p(1) = e_0$, so $z(1)$ commutes with $e_0 = \mathrm{diag}(1_k, 0_{m-k})$ and is therefore block-diagonal, $z(1) = \mathrm{diag}(a, d)$ with $a \in \mathcal{U}_k(\tilde{A})$ and $d \in \mathcal{U}_{m-k}(\tilde{A})$. Define

$$\eta_A\big([p] - [e_0]\big) = [a]_1 \in K_1(\tilde{A}) = K_1(A)$$

using $K_1(\tilde{A}) \cong K_1(A)$ for unital A (theorem 7.1.4, since $K_1(\mathbb{C}) = 0$). This is well-defined. Any two framings z, z' of the same loop p differ by $r(t) = z'(t)^* z(t)$, a path in $\mathcal{U}_m(\tilde{A})$ with $r(0) = 1_m$ and $r(t)\,e_0\,r(t)^* = e_0$ for all t (both framings conjugate e_0 to $p(t)$), so $r(t) = \mathrm{diag}(a_r(t), d_r(t))$ is block-diagonal throughout and $t \mapsto a_r(t)$ is a path in $\mathcal{U}_k(\tilde{A})$ from 1_k to the upper block $a_r(1) = (a')^* a$ of $r(1) = z'(1)^* z(1)$; hence $a = a'\,a_r(1)$ with $a_r(1) \in \mathcal{U}_k(\tilde{A})_0$, giving $[a]_1 = [a']_1$. A homotopy of projection loops p_s frames to a homotopy of unitaries $z_s(1)$ by running Step 3a with a partition refined uniformly over the compact square (s, t), so $[a]_1$ is unchanged under homotopy; and η_A is additive because the block-diagonal sum of loops frames by the block-diagonal sum of the z's, giving upper block $a \oplus a'$.

Step 3c: $\eta_A \circ \theta_A = \mathrm{id}$. Start from $[u] \in K_1(A)$ with $u \in \mathcal{U}_n(A)$, so by Step 2 the projection loop is $p_{w_u}(t) = w_u(t)\,e\,w_u(t)^*$ with $e = e_0 = \mathrm{diag}(1_n, 0)$, $k = n$, $m = 2n$, $w_u(0) = 1_{2n}$, $w_u(1) = \mathrm{diag}(u, u^*)$. The path $z(t) = w_u(t)$ is itself a framing of p_{w_u}: it satisfies $z(0) = 1_{2n}$ and $z(t)\,e_0\,z(t)^* = w_u(t)\,e\,w_u(t)^* = p_{w_u}(t)$. Its closing value is $z(1) = w_u(1) = \mathrm{diag}(u, u^*)$, block-diagonal with respect to e_0 with upper-left $n \times n$ block u. By Step 3b the class η_A produces is independent of the framing, so $\eta_A(\theta_A[u]) = [u]_1 = [u]$, and therefore $\eta_A \circ \theta_A = \mathrm{id}_{K_1(A)}$.

Step 3d: Surjectivity of θ_A. Step 3c gives $\eta_A \circ \theta_A = \mathrm{id}_{K_1(A)}$, so θ_A is injective and the retraction η_A is surjective; it remains to see θ_A is surjective, equivalently that η_A is injective. The framing already does most of this. Given $[p] - [e_0]$ with framing z and $z(1) = \mathrm{diag}(a, d)$, the path z exhibits $z(1) \in \mathcal{U}_m(\tilde{A})_0$ (it is joined to 1_m by z itself), so $[a]_1 + [d]_1 = [z(1)]_1 = 0$ and hence $[d]_1 = -[a]_1$. Thus the lower block contributes nothing new, and the candidate preimage of $[p] - [e_0]$ under θ_A is $[a]_1$. The verification that $\theta_A([a]_1) = [p] - [e_0]$ amounts to constructing a homotopy, through projection loops over SA, from p (stabilized) to the standard Bott-type loop p_{w_a} of Step 2: one slides the framing z to the Whitehead framing w_a keeping the closing block a fixed. The construction is the operator-algebraic transcription of [10, Vector Bundles and Suspension], that the based-homotopy class of a clutching function determines the bundle over a suspension and conversely; it is a single direct but lengthy homotopy bookkeeping with no conceptual content beyond Steps 3a-3c, and the full details are carried out in [12, Chapters 7-8] and [1, Chapter 9]. Granting it, θ_A is surjective.

Steps 3c and 3d show θ_A is injective with surjective image, so θ_A is a bijection, and being a homomorphism (Step 2) it is an isomorphism, with inverse η_A. Injectivity and the explicit inverse η_A rest only on the framing construction of Steps 3a-3c, with no appeal to the six-term exact sequence developed later; this is what makes the later derivations that invoke theorem 8.1.2 non-circular. For commutative $A = C(X)$ the whole statement is the topological clutching correspondence of [10, Vector Bundles and Suspension] read through Gelfand duality, and the framing above is the operator-algebraic form of trivializing the clutched bundle over each cone of the suspension.

Step 4: Naturality. For a *-homomorphism $\varphi\colon A \to B$, the diagram comparing θ_A and θ_B commutes because $S\varphi$ acts on p_w by φ entrywise and $K_1(\varphi)$ acts on $[w]$ by φ entrywise, and both are computed from the same loop w. Thus θ is a natural isomorphism, completing the proof.

The dimension-shift isomorphism is the structural reason the operator theory is periodic rather than an unbounded tower. It says K_1 contains no information beyond K_0 of a derived algebra: every odd invariant is an even invariant of the suspension. Combined with Bott periodicity below, which identifies K_0 of the double suspension with K_0 itself, the two groups K_0 and K_1 exhaust the theory. The same isomorphism defines the negative-index groups uniformly by iterating the suspension, and shows they collapse modulo 2.

Definition 8.1.3: Higher and Negative K-Groups

For $n \geq 0$ define $K_n(A) = K_0(S^n A)$, where $S^0 A = A$ and $S^n A = S(S^{n-1}A)$. By theorem 8.1.2 this agrees for $n = 1$ with the K_1 of definition 7.1.1, and Bott periodicity (theorem 8.1.10) gives $K_n(A) \cong K_{n+2}(A)$, so the only groups are K_0 and K_1.

Example 8.1.4: The Suspension of $\mathbb{C}$ and $K_*(\mathbb{C})$

Take $A = \mathbb{C}$. Then $S\mathbb{C} = C_0(\mathbb{R})$, the continuous functions on the line vanishing at infinity. Via the one-point compactification $\mathbb{R}^+ = S^1$ (so $\mathbb{R} \cong S^1 \setminus \{1\}$), $C_0(\mathbb{R}) \cong \{f \in C(S^1) : f(1) = 0\}$, the kernel of evaluation at the basepoint, and $K_0(C_0(\mathbb{R})) = \widetilde{K}^0(\mathbb{R}^+) = \widetilde{K}^0(S^1)$. The dimension-shift isomorphism gives

$$K_1(\mathbb{C}) \cong K_0(S\mathbb{C}) = K_0(C_0(\mathbb{R})) = \widetilde{K}^0(S^1)$$

Now $\mathcal{U}_n(\mathbb{C}) = \mathrm{U}(n)$ is connected for every n, so $K_1(\mathbb{C}) = 0$, and indeed $\widetilde{K}^0(S^1) = 0$ because every complex vector bundle over S^1 is trivial (the clutching function $S^0 \to \mathrm{U}(n)$ lands in a connected group, [10, Vector Bundles On Spheres]). On the other side $K_0(\mathbb{C}) = \mathbb{Z}$, generated by [1]. Iterating, $K_0(C_0(\mathbb{R}^2)) = K_0(S^2\mathbb{C}) = \widetilde{K}^0(S^2) = \mathbb{Z}$, generated by the Bott class, while $K_1(C_0(\mathbb{R}^2)) = 0$. Thus $K_*(\mathbb{C})$ is $\mathbb{Z}$ in even degree and 0 in odd degree, matching $K_*(\mathrm{pt})$ exactly.

A short exact sequence of C^*-algebras

$$0 \longrightarrow I \xrightarrow{\iota} A \xrightarrow{\pi} A/I \longrightarrow 0$$

(with I a closed two-sided ideal, ι the inclusion, π the quotient map) does not generally split, and K_0, K_1 individually are only half-exact: applying either functor produces a three-term exact sequence in the middle but no map back. The connecting maps repair this. The index map descends from K_1 of the quotient to K_0 of the ideal; its construction is the operator-algebraic incarnation of the Fredholm index, as the Toeplitz example will show.

Definition 8.1.5: The Index Map

Let $0 \to I \to A \xrightarrow{\pi} A/I \to 0$ be a short exact sequence of C^*-algebras and $u \in \mathcal{U}_n(\widetilde{A/I})$ a unitary representing a class $[u] \in K_1(A/I)$. The *index map*

$$\partial_1 \colon K_1(A/I) \longrightarrow K_0(I)$$

is defined as follows. Lift the unitary $\mathrm{diag}(u^*, u) \in \mathcal{U}_{2n}(\widetilde{A/I})$ to a unitary $w \in \mathcal{U}_{2n}(\tilde{A})$ with $\tilde{\pi}(w) = \mathrm{diag}(u^*, u)$ (possible because $\mathrm{diag}(u^*, u)$ lies in the identity component, hence lifts to a unitary by surjectivity of $\tilde{\pi}$ on identity components). Then

$$\partial_1[u] = \left[w \begin{pmatrix} 1_n & 0 \\ 0 & 0 \end{pmatrix} w^* \right] - \left[\begin{pmatrix} 1_n & 0 \\ 0 & 0 \end{pmatrix} \right] \in K_0(I)$$

where the difference of projections lies in $K_0(I)$ because $\tilde{\pi}$ sends it to $\mathrm{diag}(u^*, u)\mathrm{diag}(1_n, 0)\mathrm{diag}(u, u^*) - \mathrm{diag}(1_n, 0) = \mathrm{diag}(1_n, 0) - \mathrm{diag}(1_n, 0) = 0$, so the projection difference has entries in I. When the unitary u admits a lift to a partial isometry $v \in M_n(A)$ (an element with v^*v and vv^* projections and $\pi(v) = u$), this formula collapses to

$$\partial_1[u] = [1 - vv^*] - [1 - v^*v] \in K_0(I)$$

the difference of the two defect projections, both of which lie in $M_n(I)$.

The mechanism is the Halmos picture of a partial isometry. A unitary u in the quotient lifts only to a contraction $a \in A$ with $\pi(a) = u$ and $a^*a, aa^* \equiv 1$ modulo I; the defect operators $1 - a^*a$ and $1 - aa^*$ measure the failure to be unitary and live in I. Doubling to $\mathrm{diag}(u^*, u)$ and lifting to an unitary w converts these defects into a projection difference, and that difference is the $K_0(I)$-valued index. When $A = \mathcal{T}$ is the Toeplitz algebra and $A/I = C(S^1)$, the lift is a Fredholm Toeplitz operator T_u and the projection difference $[1 - T_u T_u^*] - [1 - T_u^* T_u]$ is $[\ker T_u^*] - [\ker T_u]$, the negative of the analytic Fredholm index. The exponential map is the even-to-odd companion, built from lifting a projection.

Definition 8.1.6: The Exponential Map

Let $0 \to I \to A \xrightarrow{\pi} A/I \to 0$ be a short exact sequence of C^*-algebras and let $p \in M_n(\widetilde{A/I})$ be a projection representing a class $[p] - [s] \in K_0(A/I)$ (with s the scalar part of p). The *exponential map*

$$\partial_0 \colon K_0(A/I) \longrightarrow K_1(I)$$

is defined by lifting p to a self-adjoint element $a \in M_n(\tilde{A})$ with $\tilde{\pi}(a) = p$ and then setting

$$\partial_0([p] - [s]) = \big[\exp(2\pi i\, a) \big] \in K_1(I)$$

The element $\exp(2\pi i\, a)$ is a unitary in $M_n(\tilde{A})$, and since $\tilde{\pi}(a) = p$ is a projection, $\tilde{\pi}(\exp(2\pi i a)) = \exp(2\pi i p) = 1$ (as $p^2 = p$ forces $\exp(2\pi i p) = 1 + (e^{2\pi i} - 1)p = 1$), so $\exp(2\pi i a)$ has entries congruent to 1 modulo I, hence lies in $\mathcal{U}_n(\tilde{I})$ and defines a class in $K_1(I)$.

The name records the formula: the connecting map in the even-to-odd direction literally exponentiates a self-adjoint lift of a projection. The two maps ∂_0 and ∂_1 are the two connecting homomorphisms of the same six-term sequence, and under the dimension-shift isomorphism they are identified with each other applied to the suspended sequence, which is why a single periodicity statement governs both. The exactness proof below requires that both maps are well-defined homomorphisms, recorded first.

Proposition 8.1.7: The Connecting Maps Are Well-Defined Homomorphisms

The maps $\partial_1 \colon K_1(A/I) \to K_0(I)$ and $\partial_0 \colon K_0(A/I) \to K_1(I)$ of definitions 8.1.5 and 8.1.6 are well-defined group homomorphisms, independent of all choices of lift.

***Proof*:**
Independence of the lift, index map. Suppose w, w' are two unitary lifts in $\mathcal{U}_{2n}(\tilde{A})$ of $\operatorname{diag}(u^*, u)$. Then $w'w^* \in \mathcal{U}_{2n}(\tilde{A})$ projects to 1 under $\tilde{\pi}$, so $w'w^* \in \mathcal{U}_{2n}(\tilde{I})$. Writing $e = \operatorname{diag}(1_n, 0)$, the projections wew^* and $w'e(w')^*$ are unitarily equivalent in $M_{2n}(\tilde{I})$ via $w'w^*$, hence equal in $K_0(I)$; and the difference with e is unchanged. Existence of a unitary lift: $\operatorname{diag}(u^*, u)$ is connected to 1 by the rotation path of theorem 8.1.2, Step 1, applied in $\widetilde{A/I}$; lifting that path of unitaries through the surjection $\tilde{\pi}$ (using that the unitary group surjects on identity components, a consequence of the fact that for b self-adjoint in $\widetilde{A/I}$ any self-adjoint lift $\tilde{b}$ gives $\exp(i\tilde{b})$ lifting $\exp(ib)$) produces w with $\tilde{\pi}(w) = \operatorname{diag}(u^*, u)$.

Independence of the representative, index map. If $u \sim u'$ are homotopic unitaries in $\mathcal{U}_n(\widetilde{A/I})$, lift the homotopy to a path of unitaries w_t in $\mathcal{U}_{2n}(\tilde{A})$; then $w_t e w_t^*$ is a homotopy of projections in $M_{2n}(\tilde{I})$ (after subtracting e), giving equal $K_0(I)$ classes. Additivity: a lift of $u \oplus u'$ may be taken to be $w \oplus w'$ after reordering coordinates, and ∂_1 of a direct sum is the sum of the projection differences.

Exponential map. If a, a' are two self-adjoint lifts of the projection p, then $a - a' \in M_n(\tilde{I})_{sa}$, and the straight-line homotopy $a_s = (1-s)a + sa'$ is self-adjoint with $\tilde{\pi}(a_s) = p$ throughout, so $\exp(2\pi i a_s)$ is a homotopy in $\mathcal{U}_n(\tilde{I})$, giving equal $K_1(I)$ classes. If $p \sim p'$ are homotopic projections, lift the homotopy to a path of self-adjoint elements and exponentiate to get a homotopy of unitaries. For additivity, $a \oplus a'$ is a self-adjoint lift of $p \oplus p'$, and $\exp(2\pi i(a \oplus a')) = \exp(2\pi i a) \oplus \exp(2\pi i a')$, whose K_1 class is the sum. Adding a scalar projection s changes a by a scalar self-adjoint with $\exp(2\pi i s) = 1$, leaving the class unchanged, so ∂_0 depends only on $[p] - [s]$. This establishes both maps are well-defined homomorphisms, as we sought to show.

With the connecting maps in hand, the half-exact sequences for K_0 and K_1 assemble into a single cyclic six-term sequence. This is the central computational tool of the subject: it lets one compute the K-theory of A from that of an ideal I and a quotient A/I, exactly as the long exact sequence of a pair computes cohomology in topology.

Theorem 8.1.8: The Six-Term Exact Sequence

Let $0 \to I \xrightarrow{\iota} A \xrightarrow{\pi} A/I \to 0$ be a short exact sequence of C^*-algebras. Then the diagram

$$\begin{tikzcd}
K_0(I) \arrow[r, "K_0(\iota)"] & K_0(A) \arrow[r, "K_0(\pi)"] & K_0(A/I) \arrow[d, "\partial_0"] \\
K_1(A/I) \arrow[u, "\partial_1"] & K_1(A) \arrow[l, "K_1(\pi)"] & K_1(I) \arrow[l, "K_1(\iota)"]
\end{tikzcd}$$

is exact at every one of its six vertices, where ∂_0 is the exponential map (definition 8.1.6) and ∂_1 is the index map (definition 8.1.5).

***Proof*:**

The proof has two layers. First, the horizontal three-term pieces (exactness at $K_0(A)$ and at $K_1(A)$) are the half-exactness of each functor, which we prove in full. Second, exactness at the four corners involving the connecting maps requires the index/exponential constructions; we prove exactness at $K_1(A/I)$ in full and at $K_0(A/I)$ in full apart from one path-lifting verification deferred to the references, then deduce the remaining two corners from the dimension-shift isomorphism of theorem 8.1.2 applied to the suspended sequence. The use of theorem 8.1.2 is legitimate and non-circular: that isomorphism was proved earlier and independently of this theorem (its injectivity and explicit inverse rest only on the framing construction of theorem 8.1.2, Steps 3a-3c, which never invoke the six-term sequence).

Step 1: Half-exactness of K_0 at $K_0(A)$. The composite $K_0(\pi) \circ K_0(\iota)$ is induced by $\pi \circ \iota = 0$, hence is 0, so $\mathrm{im} K_0(\iota) \subseteq \ker K_0(\pi)$. Conversely, let $x \in \ker K_0(\pi)$. Represent $x = [p] - [q]$ with $p, q \in M_n(\tilde{A})$ projections having equal scalar parts. The hypothesis gives $[\tilde{\pi}(p)] = [\tilde{\pi}(q)]$ in $K_0(A/I)$, so after enlarging n and adding complementary projections there is a unitary $\bar{u} \in \mathcal{U}_n(\widetilde{A/I})$ with $\bar{u}\,\tilde{\pi}(p)\,\bar{u}^* = \tilde{\pi}(q)$. The doubled unitary $\mathrm{diag}(\bar{u}, \bar{u}^*)$ lies in the identity component, hence lifts through $\tilde{\pi}$ to a unitary $w \in \mathcal{U}_{2n}(\tilde{A})$. Set $p' = \mathrm{diag}(p, 0)$ and $q' = \mathrm{diag}(q, 0)$, projections in $M_{2n}(\tilde{A})$ with $[p] - [q] = [p'] - [q']$. Then

$$\tilde{\pi}(wp'w^*) = \mathrm{diag}(\bar{u}, \bar{u}^*)\,\tilde{\pi}(p')\,\mathrm{diag}(\bar{u}^*, \bar{u}) = \mathrm{diag}(\bar{u}\,\tilde{\pi}(p)\,\bar{u}^*,\, 0) = \tilde{\pi}(q')$$

so $wp'w^* - q' \in M_{2n}(I)$. Hence $wp'w^*$ and q' are projections agreeing modulo I with the same scalar part, and $[wp'w^*] - [q']$ is a projection difference over I, lying in $\mathrm{im} K_0(\iota)$. Since $wp'w^*$ is unitarily equivalent to p' in $M_{2n}(\tilde{A})$, $[wp'w^*] = [p']$ in $K_0(A)$, so $[p] - [q] = [p'] - [q'] = [wp'w^*] - [q'] \in \mathrm{im} K_0(\iota)$.

Step 2: Half-exactness of K_1 at $K_1(A)$. Identical in structure using unitaries: $K_1(\pi)K_1(\iota) = 0$ since $\pi\iota = 0$. For the reverse inclusion, let $[u] \in \ker K_1(\pi)$ with $u \in \mathcal{U}_n(\tilde{A})$. Then $\tilde{\pi}(u)$ is null-homotopic in $\mathcal{U}_n(\widetilde{A/I})$; lift the contracting path to a path of unitaries u_t in $\mathcal{U}_n(\tilde{A})$ (possible by the identity-component lifting used above) with $u_0 = u$ and $\tilde{\pi}(u_1) = 1$, so $u_1 \in \mathcal{U}_n(\tilde{I})$. Since $u \sim u_1$ in $\mathcal{U}_n(\tilde{A})$, we get $[u] = [u_1] \in \mathrm{im} K_1(\iota)$.

Step 3: Exactness at $K_0(A/I)$. We show $\mathrm{im} K_0(\pi) = \ker \partial_0$.

$\mathrm{im} K_0(\pi) \subseteq \ker \partial_0$: if $x = K_0(\pi)(y)$ with $y = [r] - [s] \in K_0(A)$, $r \in M_n(\tilde{A})$ a projection, then r is itself a self-adjoint lift of $\tilde{\pi}(r)$, and $\exp(2\pi i\, r) = 1$ since r is a projection. Hence $\partial_0(x) = [\exp(2\pi i r)] = [1] = 0$.

$\ker \partial_0 \subseteq \mathrm{im} K_0(\pi)$: let $[p] - [s] \in K_0(A/I)$ with $\partial_0([p] - [s]) = 0$. Choose a self-adjoint lift $a \in M_n(\tilde{A})$ of p with spectrum in $[0, 1]$, possible because p is a projection: take any self-adjoint lift and apply the functional calculus with the function $t \mapsto \max(0, \min(1, t))$, which fixes the spectrum $\{0, 1\}$ of p and hence does not change $\tilde{\pi}(a) = p$. The strategy is to replace a by a self-adjoint lift a'' of (a stabilization of) p whose exponential $\exp(2\pi i a'')$ is exactly 1; such an a'' has spectrum in $\mathbb{Z}$, and the functional calculus with the indicator of $(1/2, \infty)$ then converts a'' into a genuine projection $r \in M_m(\tilde{A})$ with $\tilde{\pi}(r) = p \oplus 0$ (the scalar 0 part coming from the stabilization), whence $K_0(\pi)([r] - [s]) = [p] - [s]$ exhibits the class in the image. The hypothesis $\partial_0([p] - [s]) = 0$ says exactly that $\exp(2\pi i a)$ is null-homotopic in $\mathcal{U}_m(\tilde{I})$ after stabilizing, and it is this hypothesis that allows a to be corrected to such an a''. The correction must be made multiplicatively, by deforming the unitary $\exp(2\pi i a)$ to 1 through a path of unitaries in $\mathcal{U}_m(\tilde{I})$ and lifting that path to a self-adjoint deformation of the lift; one cannot subtract a self-adjoint $b \in M_m(\tilde{I})_{sa}$ with $\exp(2\pi i b) = \exp(2\pi i a)$ and conclude $\exp(2\pi i(a \oplus 0 - b)) = 1$, since a and b do not commute and the exponential is not additive on non-commuting self-adjoints. The path-lifting argument is a several-page verification with no extra conceptual content beyond the constructions already given; it is deferred to [12, Chapter 12] and [1, Chapter 9].

Step 4: Exactness at $K_1(A/I)$. We show $\mathrm{im} K_1(\pi) = \ker \partial_1$.

$\mathrm{im} K_1(\pi) \subseteq \ker \partial_1$: if $[u'] = K_1(\pi)([v])$ with $v \in \mathcal{U}_n(\tilde{A})$, then $\mathrm{diag}(v^*, v)$ is a unitary lift of $\mathrm{diag}(\tilde{\pi}(v)^*, \tilde{\pi}(v)) = \mathrm{diag}((u')^*, u')$, and $\mathrm{diag}(v^*, v)\mathrm{diag}(1_n, 0)\mathrm{diag}(v, v^*) - \mathrm{diag}(1_n, 0) = \mathrm{diag}(1_n, 0) - \mathrm{diag}(1_n, 0) = 0$ in $K_0(I)$ (using $v^* v = 1$). Hence $\partial_1[u'] = 0$.

$\ker \partial_1 \subseteq \mathrm{im} K_1(\pi)$: let $[u]$ with $\partial_1[u] = 0$. Let $w \in \mathcal{U}_{2n}(\tilde{A})$ lift $\mathrm{diag}(u^*, u)$ and set $e = \mathrm{diag}(1_n, 0)$. The hypothesis says $[wew^*] - [e] = 0$ in $K_0(I)$, so after stabilizing wew^* and e are homotopic projections in $M_{2n}(\tilde{I})$ via a path f_t with $f_0 = wew^*$, $f_1 = e$. Lift this projection homotopy to a unitary path: there is $g_t \in \mathcal{U}(\tilde{I})$ with $g_0 = 1$ and $g_t(wew^*)g_t^* = f_t$, so $g_1 wew^* g_1^* = e$, meaning $g_1 w$ commutes with e up to the ideal and, after adjusting, $g_1 w \in \mathcal{U}_{2n}(\tilde{A})$ has $\tilde{\pi}(g_1 w) = \mathrm{diag}(u^*, u)$ while commuting with e. A unitary commuting with $e = \mathrm{diag}(1_n, 0)$ is block-diagonal $\mathrm{diag}(x, y)$, so $\tilde{\pi}(x) = u^*$, whence $\tilde{\pi}(x^*) = u$ exhibits $[u] = K_1(\pi)([x^*]) \in \mathrm{im} K_1(\pi)$.

Step 5: The two remaining corners. Exactness at $K_0(I)$ (that is, $\mathrm{im}\partial_1 = \ker K_0(\iota)$) and exactness at $K_1(I)$ (that is, $\mathrm{im}\partial_0 = \ker K_1(\iota)$) follow from the previous four corners by the dimension-shift isomorphism of theorem 8.1.2. This appeal is not circular: theorem 8.1.2 is established before this theorem and independently of it (its injectivity and explicit inverse come from the framing construction of theorem 8.1.2, Steps 3a-3c, which never invoke the six-term sequence), so it is legitimate to use it here. Suspending the short exact sequence gives

$$0 \longrightarrow SI \longrightarrow SA \longrightarrow S(A/I) \longrightarrow 0$$

an exact sequence because the suspension functor $C_0(\mathbb{R}) \otimes (-)$ preserves exactness (tensoring a short exact sequence by the nuclear $C_0(\mathbb{R})$ is exact). The strategy is to read each ideal corner of the original sequence as a quotient corner of the suspended sequence under the natural identification $\theta_B : K_1(B) \cong K_0(SB)$. The three-term K_0-exactness of the suspended extension at $K_0(SI), K_0(SA), K_0(S(A/I))$, together with θ, recovers the K_1-row of the original (Steps 1-2 again), while the exponential and index maps of the suspended extension correspond under θ to the index and exponential maps of the original by naturality of θ (theorem 8.1.2, Step 4). Splicing the K_0-six-term sequences of the extension and of its suspension along these identifications places the two ideal corners $K_0(I)$ and $K_1(I)$ of the original into positions where exactness is exactly one of the quotient-corner statements (theorem 8.1.8, Steps 3-4) for the suspended extension, which has already been proved. The diagram chase intertwining ∂_0, ∂_1 with θ across the suspension is direct but lengthy bookkeeping with no content beyond the naturality already in hand; it is carried out in full in [12, Chapters 8-9] and [1, Chapter 9].

Exactness has now been established at all six vertices, completing the proof.

The six-term sequence collapses the entire homological machinery of the subject into a hexagon. Its power is leverage: knowing K_* of any two of I, A, A/I pins down K_* of the third up to extension data carried by the connecting maps. The two quotient corners worked out above, $K_0(A/I)$ and $K_1(A/I)$, show the mechanism transparently: an obstruction to lifting a projection (respectively a unitary) is exactly a class in $K_1(I)$ (respectively $K_0(I)$). The remaining two corners are the same statements read through the suspension, which is why no genuinely new argument is needed there.

Example 8.1.9: The Six-Term Sequence of a Quotient by a Hereditary Ideal

Consider the unitization sequence $0 \to A \to \tilde{A} \xrightarrow{q} \mathbb{C} \to 0$ for a non-unital A, where q is the augmentation $(a, \lambda) \mapsto \lambda$. This sequence splits via $\lambda \mapsto (0, \lambda)$, so all connecting maps vanish and the six-term sequence breaks into two split short exact sequences:

$$0 \to K_0(A) \to K_0(\tilde{A}) \to K_0(\mathbb{C}) \to 0, \qquad 0 \to K_1(A) \to K_1(\tilde{A}) \to K_1(\mathbb{C}) \to 0$$

Since $K_0(\mathbb{C}) = \mathbb{Z}$ and $K_1(\mathbb{C}) = 0$, the split gives $K_0(\tilde{A}) \cong K_0(A) \oplus \mathbb{Z}$ and $K_1(\tilde{A}) \cong K_1(A)$. This recovers the definition $K_0(A) = \ker(K_0(\tilde{A}) \to K_0(\mathbb{C}))$ used for non-unital algebras and shows it is consistent with the six-term sequence: the splitting forces $\partial_0 = \partial_1 = 0$ exactly when the extension is split, the operator-algebraic form of the topological fact that a split cofibration contributes no boundary.

The dimension-shift isomorphism reduces all higher groups to K_0 of iterated suspensions, but on its own it does not close the cycle: a priori $K_0(S^2A)$ could be a new group. Bott periodicity is the statement that the double suspension returns to the start. In topology this is $\Omega^2(BU \times \mathbb{Z}) \simeq BU \times \mathbb{Z}$ ([10, Note: The Loop-Space Form]); the C^*-algebraic version is its exact translation under Gelfand duality (theorem 4.3.3).

Theorem 8.1.10: Bott Periodicity for C^*-Algebras

For every C^*-algebra A the *Bott map*

$$\beta_A : K_0(A) \longrightarrow K_0(S^2A) = K_2(A), \qquad \beta_A(x) = b \cdot x$$

given by external multiplication with the Bott generator $b \in K_0(S^2\mathbb{C}) = \widetilde{K}^0(S^2) \cong \mathbb{Z}$ (explicitly $\beta_A([p] - [s]) = [\beta(p)] - [\beta(s)]$ for the Bott projection $\beta(p)$ over S^2 defined in the proof) is an isomorphism, natural in A. Equivalently, via theorem 8.1.2, $K_1(SA) \cong K_0(A)$, and hence

$$K_n(A) \cong K_{n+2}(A) \qquad \text{for all } n \geq 0$$

so the theory is 2-periodic with the only groups $K_0(A)$ and $K_1(A)$.

***Proof Key ideas*:**
The proof is the analytic core of the subject and is genuinely beyond the elementary scope of this section; the full argument is given in the named references below, and the idea is sketched here per the deferral policy.

The Bott map. For a projection $p \in M_n(\tilde{A})$ one builds a projection-valued function on S^2 (equivalently on the double suspension) by the formula

$$\beta(p)(z) = \begin{pmatrix} p & 0 \\ 0 & 0 \end{pmatrix} + (\text{rank-one rotation in } z), \qquad z \in S^2 = \mathbb{C} \cup \{\infty\}$$

concretely the Bott projection $\beta(p)(z) = \frac{1}{1+|z|^2} \begin{pmatrix} 1 & \bar{z} \\ z & |z|^2 \end{pmatrix} \otimes p + \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \otimes (1-p)$ over the Riemann sphere; this is the generator of $\widetilde{K}^0(S^2)$ tensored with p. The map $[p] \mapsto [\beta(p)]$ is the external product with the Bott generator $b \in K_0(S^2\mathbb{C}) = \widetilde{K}^0(S^2) = \mathbb{Z}$.

Why it is an isomorphism. For commutative $A = C(X)$ the statement is the topological Bott periodicity $\widetilde{K}^0(X) \cong \widetilde{K}^0(S^2 X)$, which over a point is $\widetilde{K}^0(S^2) \cong \mathbb{Z}$ and in general is proved directly by the clutching and linearization argument of [10, Bott Periodicity Isomorphism], which establishes the product theorem $K^0(X) \otimes K^0(S^2) \cong K^0(X \times S^2)$ without passing through the loop-space form.

Granting the scalar case and the multiplicativity of the external product, the general isomorphism follows, completing the argument.

Bott periodicity is what makes operator K-theory a genuine generalized cohomology theory in the sense of [8, Generalized Cohomology Theory]: the two functors K_0 and K_1, the suspension isomorphism, and the six-term sequence together satisfy exactness, homotopy invariance, and 2-periodicity, the operator analogues of the Eilenberg-Steenrod axioms ([10, K-Theory is a Generalized Cohomology Theory] records the topological side). The periodicity collapses the infinite ladder of connecting maps from successive suspensions into a single recurring hexagon, so that the index and exponential maps are, up to the Bott isomorphism, the only two connecting maps the theory ever produces. For the commutative algebra $C(X)$ the whole structure reduces, via Gelfand-Naimark duality, to topological K-theory: $K_0(C(X)) \cong K^0(X)$ and $K_1(C(X)) \cong K^1(X) = \widetilde{K}^0(SX)$.

The Toeplitz extension is the worked example that makes every abstract construction of this section concrete: the index map becomes the Fredholm index, and the Fredholm index becomes the winding number. Let $H^2 \subseteq L^2(S^1)$ be the Hardy space, the closed span of $\{z^n : n \geq 0\}$, with orthogonal projection $P \colon L^2(S^1) \to H^2$. For $f \in C(S^1)$ the *Toeplitz operator* is $T_f = PM_f|_{H^2}$, where M_f is multiplication by f. The C^*-algebra $\mathcal{T}$ generated by all T_f is the *Toeplitz algebra*.

Example 8.1.11: The Toeplitz Extension $0 \to \mathcal{K} \to \mathcal{T} \to C(S^1) \to 0$

The commutator $T_f T_g - T_{fg}$ is compact for all $f, g \in C(S^1)$, and the compact operators $\mathcal{K} = \mathcal{K}(H^2)$ form a closed ideal of $\mathcal{T}$ with $\mathcal{T}/\mathcal{K} \cong C(S^1)$ via the *symbol map* $\sigma(T_f) = f$. This yields the short exact sequence

$$0 \longrightarrow \mathcal{K} \xrightarrow{\iota} \mathcal{T} \xrightarrow{\sigma} C(S^1) \longrightarrow 0$$

We compute its six-term sequence and identify the index map.

K-theory of the three terms. The compacts satisfy $K_0(\mathcal{K}) = \mathbb{Z}$ (generated by any rank-one projection, since $\mathcal{K} = \varinjlim M_n(\mathbb{C})$ and K_0 commutes with the colimit) and $K_1(\mathcal{K}) = 0$ (every unitary in $\widetilde{\mathcal{K}} = \mathcal{K} + \mathbb{C}1$ is a compact perturbation of a scalar, connected to it). For the circle, $C(S^1)$ gives $K_0(C(S^1)) = \mathbb{Z}$ (generated by [1], as every complex vector bundle over S^1 is trivial, [10, Vector Bundles On Spheres]) and $K_1(C(S^1)) = \mathbb{Z}$ (generated by the class of the unitary $z = \mathrm{id}_{S^1}$, since $\mathcal{U}_n(C(S^1))/\mathcal{U}_n(C(S^1))_0 \cong \pi_1(\mathrm{U}(n)) = \mathbb{Z}$ detects the winding number of det).

The six-term sequence. Substituting,

$$\begin{tikzcd} \mathbb{Z} = K_0(\mathcal{K}) \arrow[r, "K_0(\iota)"] & K_0(\mathcal{T}) \arrow[r, "K_0(\sigma)"] & K_0(C(S^1)) = \mathbb{Z} \arrow[d, "\partial_0"] \\ K_1(C(S^1)) = \mathbb{Z} \arrow[u, "\partial_1"] & K_1(\mathcal{T}) \arrow[l, "K_1(\sigma)"] & K_1(\mathcal{K}) = 0 \arrow[l] \end{tikzcd}$$

The bottom-right entry is 0, so $\partial_0 \colon \mathbb{Z} \to 0$ is zero and $K_0(\sigma)$ is surjective.

The index map is an isomorphism. The generator of $K_1(C(S^1)) = \mathbb{Z}$ is $[z]$. A lift of z to $\mathcal{T}$ is the unilateral shift $T_z = S$, which sends $z^n \mapsto z^{n+1}$ on H^2; it is an isometry, hence a partial isometry with $S^*S = 1$ and $1 - SS^* = P_0$, the rank-one projection onto the constants. By the partial-isometry formula of definition 8.1.5, for $v = S$,

$$\partial_1[z] = [1 - SS^*] - [1 - S^*S] = [P_0] - [0]$$

the class of a rank-one projection, which generates $K_0(\mathcal{K}) = \mathbb{Z}$; numerically $\partial_1[z] = +1$. The analytic Fredholm index of S is $\dim\ker S - \dim\ker S^* = 0 - 1 = -1$, so the index map records $[1 - SS^*] - [1 - S^*S] = [\ker S^*] - [\ker S]$, the negative of the Fredholm index, and $\partial_1[z] = -\operatorname{index}(S) = w(z) = 1$. Hence ∂_1 is an isomorphism $\mathbb{Z} \xrightarrow{\sim} \mathbb{Z}$.

Consequences. Since ∂_1 is injective, exactness at $K_1(C(S^1))$ forces $K_1(\sigma) = 0$, and since ∂_1 is surjective, exactness at $K_0(\mathcal{K})$ forces $K_0(\iota) = 0$. Exactness at $K_0(\mathcal{T})$ then gives $K_0(\mathcal{T}) \cong K_0(C(S^1)) = \mathbb{Z}$ (generated by $[1_{\mathcal{T}}]$), and exactness at $K_1(\mathcal{T})$ gives $K_1(\mathcal{T}) = 0$ (it injects via $K_1(\sigma) = 0$ into $K_1(C(S^1))$ with image $\ker\partial_1 = 0$, and receives 0 from $K_1(\mathcal{K}) = 0$). Thus $K_0(\mathcal{T}) = \mathbb{Z}$ and $K_1(\mathcal{T}) = 0$: the Toeplitz algebra is K-theoretically a point, the operator-algebraic statement that $\mathcal{T}$ is the "cone" on the circle.

The Toeplitz index theorem. For an invertible $f \in C(S^1)$ (equivalently a class $[f] \in K_1(C(S^1)) = \mathbb{Z}$ given by the winding number $w(f)$ of f around 0), the Toeplitz operator T_f is Fredholm and

$$\operatorname{index}(T_f) = \dim\ker T_f - \dim \operatorname{coker} T_f = -\,w(f)$$

so that the index map records $\partial_1[f] = [1 - T_f T_f^*] - [1 - T_f^* T_f] = [\operatorname{coker} T_f] - [\ker T_f] = -\operatorname{index}(T_f) = w(f)$. The index map of the Toeplitz extension is therefore the winding number itself, which is the Gohberg-Krein / Toeplitz index theorem realized as the connecting homomorphism of the six-term sequence.

The Toeplitz extension is central because it ties together three apparently separate notions: the abstract connecting map of homological K-theory, the Fredholm index of analysis, and the winding number of topology. Cuntz's proof of Bott periodicity (theorem 8.1.10) runs through exactly this computation: $K_*(\mathcal{T})$ being that of a point is the engine that drives periodicity for all A via the tensored extension $0 \to \mathcal{K} \otimes A \to \mathcal{T} \otimes A \to C(S^1) \otimes A \to 0$. The same calculation, read over $C(X)$ instead of $\mathbb{C}$, is the Atiyah-Singer index theorem for Toeplitz operators with matrix symbol, where the winding number is replaced by the degree of the symbol's homotopy class in $K^1(X)$.

Exercise 8.1.1

1. Let A be a unital C^*-algebra. Using the cone exact sequence $0 \to SA \to CA \xrightarrow{\mathrm{ev}_1} A \to 0$ and the contractibility of the cone CA, deduce from the six-term exact sequence that $K_0(SA) \cong$

$K_1(A)$ and $K_1(SA) \cong K_0(A)$. Explain how this recovers the dimension-shift isomorphism of theorem 8.1.2 and, combined with Bott periodicity, the relation $K_2(A) \cong K_0(A)$.

2. Compute K_0 and K_1 of $C(S^1)$ directly from the dimension-shift isomorphism and the values $K_0(\mathbb{C}) = \mathbb{Z}$, $K_1(\mathbb{C}) = 0$, using $C(S^1) \cong \widetilde{C_0(\mathbb{R})}$ and the split sequence $0 \to C_0(\mathbb{R}) \to C(S^1) \to \mathbb{C} \to 0$. Confirm your answer matches the values used in example 8.1.11.

3. Let S be the unilateral shift on H^2. Verify directly that $S^*S = 1$ and $1 - SS^*$ is the rank-one projection onto the constant functions, and conclude that S is a Fredholm operator of index -1. Then explain, using definition 8.1.5, why the class $[P_0]$ of this rank-one projection generates $K_0(\mathcal{K}) = \mathbb{Z}$ and equals $\partial_1[z]$.

4. Show that the exponential map $\partial_0 \colon K_0(A/I) \to K_1(I)$ vanishes whenever the extension $0 \to I \to A \to A/I \to 0$ splits by a *-homomorphism $s \colon A/I \to A$ with $\pi \circ s = \mathrm{id}$. Deduce that for the unitization sequence the six-term sequence breaks into two short exact sequences, recovering example 8.1.9.

5. Use the Toeplitz extension tensored with A, namely $0 \to \mathcal{K} \otimes A \to \mathcal{T} \otimes A \to C(S^1) \otimes A \to 0$, together with $K_*(\mathcal{T} \otimes A) \cong K_*(A)$ and $K_*(\mathcal{K} \otimes A) \cong K_*(A)$ (stability), to derive the periodicity statement $K_1(SA) \cong K_0(A)$. Identify which map in the six-term sequence becomes the Bott isomorphism.

6. Let $f \in C(S^1)$ be invertible with winding number $w(f) = 2$. Determine $\dim \ker T_f$ and $\dim \operatorname{coker} T_f$ given that one of them is zero, and state the class $\partial_1[f] \in K_0(\mathcal{K}) = \mathbb{Z}$. Then describe how the answer changes for $w(f) = -3$.